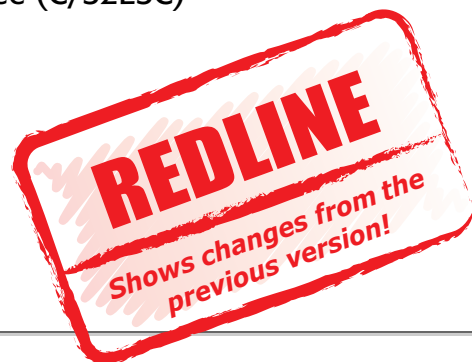


IEEE Standard for System and Software Verification and Validation

IEEE Computer Society

Sponsored by the
Software & Systems Engineering Standards Committee (C/S2ESC)



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IEEE Standard for System and Software Verification and Validation

Sponsor

Software & Systems Engineering Standards Committee (C/S2ESC)

of the

IEEE Computer Society

Approved 29 March 2012

IEEE-SA Standards Board

Abstract: Verification and validation (V&V) processes are used to determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. V&V life cycle process requirements are specified for different integrity levels. The scope of V&V processes encompasses systems, software, and hardware, and it includes their interfaces. This standard applies to systems, software, and hardware being developed, maintained, or reused [legacy, commercial off-the-shelf (COTS), nondevelopmental items]. The term *software* also includes firmware and microcode, and each of the terms *system*, *software*, and *hardware* includes documentation. V&V processes include the analysis, evaluation, review, inspection, assessment, and testing of products.

Keywords: environmental verification and validation (V&V) factors, hardware V&V, IEEE 1012, integrity level, independent V&V (IV&V), risk/hazard/security analyses, software life cycle, software V&V, system life cycle, system V&V, V&V, V&V measures, V&V of reuse software

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Introduction

This introduction is not part of IEEE Std 1012-2012, IEEE Standard for System and Software Verification and Validation.

Verification and validation (V&V) is a technical discipline of systems engineering. The purpose of V&V is to help the development organization build quality into the system during the life cycle. V&V processes provide an objective assessment of products and processes throughout the life cycle. This assessment demonstrates whether the requirements are correct, complete, accurate, consistent, and testable. The V&V processes determine whether the development products of a given activity conform to the requirements of that activity and whether the product satisfies its intended use and user needs. The determination includes the assessment, analysis, evaluation, review, inspection, and testing of products and processes. V&V is performed in parallel with all life cycle stages, not at their conclusion.

V&V is an extension of program management and systems engineering that employs a rigorous methodology to identify objective data and conclusions to provide feedback about quality, performance, and schedule to the supplier. This feedback consists of anomaly resolutions, performance improvements, and quality improvements not only for expected operating conditions but also across the full spectrum of the system and its interfaces. Early feedback results allow the development organization to modify the products in a timely fashion and thereby reduce overall project and schedule impacts. Without a proactive approach, the anomalies and associated system changes are typically delayed to later in the program schedule, resulting in greater program costs and schedule delays.

IEEE Std 1012™-2012 is a process standard that defines the V&V processes in terms of specific activities and related tasks. The standard also defines the contents of the V&V plan (VVP), including example formats.

V&V may be performed at the level of the system, software element, or hardware element, or on any combination of these. V&V may also be performed on an element of a system, including a subordinate system (i.e., subsystem). Throughout this standard, the term *hardware* means an electronic or mechanical hardware element. In each case, the V&V processes are invoked, either in parallel or recursively, across the full life cycle of the system or element.

This version of the standard is a major revision to IEEE Std 1012™-2004 [B5].^a The earliest version of this standard (1986) described the content of a software V&V plan, with subsequent versions (1998 and 2004) changing the focus from the software V&V plan to software V&V processes. This revision expands the scope of the V&V processes to include systems and hardware as well as software. It also aligns the terminology and structure to be consistent with ISO/IEC 15288:2008 [B16] and ISO/IEC 12207:2008 [B11]. The following is a summary of the changes made in this version:

- a) No new software V&V activities or tasks have been added, and the use of IEEE Std 1012-2004 [B5] is compliant with the software V&V clauses of this standard. Some software V&V activities and tasks have been rearranged to facilitate the understanding and ease of use.
- b) System V&V activities and tasks and hardware V&V activities and tasks were added to the existing software V&V standard.
- c) The standard was restructured to allow for the performance of system, software, and hardware V&V individually or in any combination. The standard contains separate clauses that address common, system, software, and hardware V&V activities.
- d) The terminology, structure, and mappings were revised to be consistent with ISO/IEC 12207:2008 [B11]. The new system V&V terminology, structure, and mappings are consistent with ISO/IEC 15288:2008 [B16].

^a The numbers in brackets correspond to those of the bibliography in Annex M.

The following key concepts are emphasized in this standard:

- *Integrity levels.* Defines four integrity levels to describe the importance of the system, software, and hardware, varying from high integrity to low integrity, to the user.
- *Minimum V&V tasks for each integrity level.* Defines the minimum V&V tasks required for each of the four integrity levels. Includes a table of optional V&V tasks for tailoring the V&V effort to address the project needs and application-specific characteristics.
- *Intensity and rigor applied to V&V tasks.* Includes the concept that the intensity and rigor applied to the V&V tasks vary according to the integrity level. Higher integrity levels require the application of greater intensity and rigor to the V&V task. Intensity includes a greater scope of analysis across all normal and abnormal system operating conditions. Rigor includes more formal techniques and recording procedures.
- *Detailed criteria for V&V tasks.* Defines specific criteria for each V&V task, including minimum criteria for correctness, consistency, completeness, accuracy, readability, and testability. The V&V task descriptions include a list of the required task inputs and outputs.
- *Systems viewpoints.* Includes minimum software and hardware V&V tasks to address system issues. These tasks include hazard analysis, security analysis, risk analysis, migration assessment, and retirement assessment. Specific system issues are contained in individual V&V task criteria.
- *Conformance to international and IEEE standards.* Defines the V&V processes to conform to life cycle process standards such as ISO/IEC 15288:2008 [B16], IEEE Std 1074™-2006 [B9], and ISO/IEC 12207:2008 [B11], as well as the entire family of IEEE software engineering standards. This standard addresses all system and software life cycle processes, including the Agreement, Organizational Project-Enabling, Project, Technical, Software Implementation, Software Support, and Software Reuse process groups. This standard is compatible with all life cycle models; however, not all life cycle models use all of the life cycle processes described in this standard.

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1. Overview

1.1 Scope

This verification and validation (V&V) standard is a process standard that addresses all system and software life cycle processes including ~~acquisition, supply, development, operation, and maintenance~~ the Agreement, Organizational Project-Enabling, Project, Technical, Software Implementation, Software Support, and Software Reuse process groups. This standard is compatible with all life cycle models (e.g., system, software, and hardware); however, not all life cycle models use all of the ~~life-cycle~~ processes listed in this standard.

~~Software~~ V&V processes determine whether the development products of a given activity conform to the requirements of that activity and whether the ~~software~~ product satisfies its intended use and user needs. This determination may include the analysis, evaluation, review, inspection, assessment, and testing of ~~software~~ products and processes.

The user of this standard may invoke those ~~software~~ life cycle processes and the associated V&V processes that apply to the project. A description of system life cycle processes may be found in ISO/IEC 15288:2008 [B16],¹ and a description of software life cycle processes may be found in ISO/IEC 12207: 2008 1995 [B1] and IEEE Std 1074™-1997 [B10], and IEEE/EIA Std 12207.0™- 2006 1996 [B9]. Annex A maps ISO/IEC 15288:2008(E) [B16] (Table A.1 and ~~IEEE Std 1074-1997 [B10]~~ Table A.2) and ISO/IEC 12207:2008 1995 [B1] (Table A.3 and Table A.4) to the V&V activities and tasks defined in this standard.

The numbers in brackets correspond to those of the bibliography in Annex M.

This standard defines the verification and validation processes that are applied to the system, software, and hardware development throughout the life cycle, including acquisition, supply, development, operations, maintenance, and retirement. This standard applies to the system, software, and hardware being acquired, developed, maintained, or reused ~~[legacy, modified, commercial off the shelf (COTS), non-developmental items (NDI)]~~. The term *software* also includes firmware and microcode (e.g., Field Programmable Gate Arrays and Programmable Logic Devices). Each of the terms system, software, and hardware includes its associated documentation.

¹The numbers in brackets correspond to those of the bibliography in Annex H.

~~Software~~ V&V processes consist of the Verification Process and Validation Process. The Verification Process provides objective evidence for whether the ~~software and its associated~~ products ~~and processes~~ perform the following:

- a) Conform to requirements (e.g., for correctness, completeness, consistency, and accuracy) for all ~~life-cycle~~ activities during each life cycle process (~~acquisition, supply, development, operation, and maintenance~~)
- b) Satisfy the standards, practices, and conventions during life cycle processes
- c) Successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities (i.e., builds the ~~software~~ product correctly)

The Validation Process provides evidence for whether the ~~software and its associated~~ products ~~and processes~~ perform the following:

- Satisfy system requirements allocated to ~~software~~ the products at the end of each life cycle activity
- Solve the right problem (e.g., correctly model physical laws, implement business rules, and use the proper system assumptions)
- Satisfy intended use and user needs in the operational environment (i.e., builds the correct product)

The Verification Process and the Validation Process are interrelated and complementary processes that use each other's process results to establish better completion criteria and analysis, evaluation, review, inspection, assessment, and test V&V tasks for each ~~software~~ life cycle activity. The V&V task criteria described in Table Ia [through Table Id explicitly](#) define the conformance requirements for V&V processes.

The development of a ~~reasonable~~ [sufficient](#) body of evidence requires a trade-off between the amount of time spent and a finite set of system conditions and assumptions against which to perform the V&V tasks. Each project should define criteria for a ~~reasonable~~ [sufficient](#) body of evidence (i.e., selecting an ~~software~~ integrity level establishes one of the basic parameters), time schedule, and scope of the V&V analysis and test tasks (i.e., range of system conditions and assumptions).

This standard does not assign the responsibility for performing the V&V tasks to any specific organization. The analysis, evaluation, and test activities may be performed by multiple organizations; however, the methods and purpose will differ for each organization's functional objectives.

ISO/~~IEC~~ [TEC 15288:2008 \[B16\]](#) and ISO/IEC [12207:1995 \[B13\]](#) or ~~IEEE/EIA 12207-0- 2008 1996~~ [B11] require that the developer perform various testing and evaluation tasks as an integral part of ~~the development process. Even though the tests and evaluations are not part of the V&V processes, the~~ [implementation. The](#) techniques described in this standard may be useful in performing the [developer's tests and evaluations.](#) Therefore, whenever this standard mentions the developer's performance of a verification or validation activity, it is to be understood that the reference applies to the ~~integral~~ test and evaluation tasks of ~~the development process~~ [implementation.](#)

1.2 Purpose

The purpose of the standard is to perform the following:

- Establish a common framework of the V&V processes, activities, and tasks in support of all system, software, and hardware life cycle processes, ~~including acquisition, supply, development, operation, and maintenance processes~~
- Define the V&V tasks, required inputs, and required outputs in each life cycle process
- Identify the minimum V&V tasks corresponding to a four-level ~~software~~ integrity scheme
- Define the content of the ~~software~~ Verification and Validation ~~V&V~~ Plan (~~SVVP~~)

1.3 Field of application

This standard applies to all applications of ~~software~~ systems. When conducting ~~the software~~ V&V ~~process~~ of a system, software, or hardware element, it is important to examine the ~~software in its~~ interactions with the system of which ~~it~~ the element is a part. This standard identifies the important system considerations that ~~software~~ V&V processes and tasks address in determining ~~software~~ correctness and other ~~software~~ V&V attributes (e.g., completeness, accuracy, consistency, and testability).

The dynamics of ~~software~~ complex systems and the multitude of different logic paths available within ~~software~~ the system in response to varying ~~system~~ stimuli and conditions demand that the ~~software~~ V&V effort examines the correctness of the ~~code~~ system for each possible variation in ~~system~~ conditions. The ability to model complex, real-world conditions will be limited, and thus, the ~~software~~ V&V effort ~~must~~ examines whether the limits of the modeling are realistic and reasonable for the desired solution. The unlimited combination of system conditions presents the ~~software~~ V&V effort with the ~~unique~~ challenge of using a finite set of analytical, test, simulation, and demonstration techniques to establish a reasonable body of evidence that the ~~software~~ system is correct.

A ~~software~~ system provides a capability to satisfy a stated need or objective by combining one or more of the following: processes, hardware, software, facilities, and people. These relationships ~~between the software and the system~~ require that ~~software~~ V&V processes consider ~~software~~ interactions ~~with~~ among all system ~~components~~. ~~Since software links together all key components of a digital system, the software V&V process examines the interactions with each of the key system components to determine the extent to which each component influences the software and is conversely influenced by the~~ elements (software and hardware). The V&V processes address the following interactions with ~~software~~ the system:

- *Environment*: Determines that the ~~solution represented in the software~~ system correctly accounts for all conditions, natural phenomena, physical laws of nature, business rules, and physical properties and the full ranges of the system operating environment.
- *Operators/users*: Determines that the ~~software~~ system communicates the proper status/condition of the ~~software~~ system to the operator/user and correctly processes all operator/user inputs to produce the required results. For incorrect operator/user inputs, ~~ensure~~ assure that the system ~~software is~~ protected from entering into a dangerous or uncontrolled state. Validate that operator/user policies and procedures (e.g., security, interface protocols, data representations, and system assumptions) are consistently applied and used across each component interface.

- *Other software, hardware and systems*: Determines that the software correctly interacts with each or hardware component interfaces ~~and provides a for controlled system response (i.e., graceful degradation)~~ for hardware ~~faults~~ in the system in accordance with requirements and that error are not propagated between components of the system.

~~Since software directly affects system behavior and performance,~~ The scope of V&V processes ~~is the software system,~~ includes the operational environment, operators and users, hardware, ~~and interfacing~~ software, data processes (e.g., processes for providing service to users), procedures (e.g., operator instructions), and facilities. The user of this standard should consider V&V as part of the ~~software~~ life cycle processes defined by industry standards, such as ISO/~~IEC~~ IEC 12207: 2008 1995 [B11], IEEE Std 1074-2006 1997 [B9], or ~~IEEE/EIA Std 12207.0 1996 [B12]~~ISO/IEC 15288:2008 [B16].

To address the systems perspective, system, software, and hardware V&V effort should provide an integrated analysis where the V&V tasks are interrelated, providing input and insight into other V&V tasks. The results from completed life cycle processes provide valuable and necessary inputs to V&V tasks in ~~later~~ other life cycle processes. The results and findings from one V&V task may cause previously completed V&V tasks to be analyzed again with the new data. This relationship among V&V tasks (including feedback to the ~~development~~ Technical and Software/Hardware Specific Processes) employing rigorous systems engineering techniques is a key approach to an integrated systems ~~and~~ software, and hardware V&V. The ~~software~~ V&V results provide the ~~development~~ other processes with an early detection of anomalies and potential process trends that may be used for ~~development~~ process improvement. The ~~software~~ V&V processes and tasks described in this standard ~~are consistent~~ may be used in conjunction with systems engineering and process improvement models, such as the ~~Capability—Maturity—Model—Integrated (CMMI).~~CMU/SEI-2010-TR-033 [B1].

1.4 V&V objectives

V&V processes provide an objective assessment of ~~software~~ products and processes throughout the ~~software~~ life cycle. This assessment demonstrates whether the ~~software~~ requirements ~~and-system requirements (i.e., those allocated to software)~~ are correct, complete, accurate, consistent, and testable. The ~~software~~ V&V processes determine whether the ~~development~~ products of a given activity conform to the requirements of that activity and whether the ~~software~~ product satisfies its intended use and user needs. The determination includes assessment, analysis, evaluation, review, inspection, and testing of ~~software~~ products and processes. ~~Software~~ V&V tasks shall be performed in parallel with ~~software-development~~ all life cycle stages, not at their conclusion ~~of the development effort~~.

The results of V&V ~~create~~ provide the following benefits to the program:

- Facilitate early detection and correction of ~~software~~ anomalies
- Enhance management insight into process and product risks
- Support the life cycle processes to ~~ensure~~ assure conformance to program performance, schedule, and budget
- Provide an early assessment of ~~software-and-system~~ performance
- Provide objective evidence of ~~software-and-system~~ conformance to support a formal certification process

- Improve the ~~software~~ [products from the acquisition, supply,](#) development, and maintenance processes
- Support ~~the~~ process improvement ~~for an integrated systems analysis model~~ [activities](#)

15 Organization of the standard

This standard is organized into clauses (Clause 1 through Clause 12), tables (Table 1, ~~Table 2,~~ and [through Table 3, and their respective subparts a-d](#)), figures (Figure 1 and Figure 2, [and their respective subparts a-d](#)), and annexes (Annex A through Annex M). Clause 2 through Clause 12 and Table 1a [through](#) Table 1d and Table 2a [through Table 2d](#) provide the mandatory V&V requirements for this standard. Table 1a [through](#) Table 1d and Table 2a [through](#) Table 2d are the focal point of this standard, containing detailed V&V ~~process,~~ activity and task requirements. [Table 1a through Table 1d and Table 2a through Table 2d are broken into four subtables:](#)

[—Common V&V tasks in Table 1a and Table 2a](#)

[—System V&V tasks in Table 1b and Table 2b](#)

[—Software V&V tasks in Table 1c and Table 2c](#)

[—Hardware V&V tasks in Table 1d and Table 2d](#)

Each clause containing V&V activities and tasks has the subset of Table 1a through Table 1d and Table 2a through Table 2d associated with the V&V requirements for that clause. Figure 1a through Figure 1d, Figure 2a through Figure 2d, and Table 3a through Table 3d contain informative material that provides examples of V&V processes and provides guidance for using this standard, and they are similarly organized. All annexes are informative.

- Clause 1 provides guidance for using this standard.
- Clause 2 is reserved for normative references; however, this standard does not prescribe any normative references.
- Clause 3 provides a definition of terms, abbreviations, and conventions.
- Clause 4 describes the use of **software** integrity levels to determine the scope and rigor of V&V processes relationships between the V&V processes and the life cycle processes from ISO/TEC 15288:2008 [B16] and ISO/IEC 12207:2008 [B11], and it describes how the V&V standard is applied recursively within the concept of a system of systems and from system to software or hardware components.
- Clause 5 explains how V&V tasks are described within this standard ~~primary software life cycle processes and lists the V&V activities associated with the life cycle process.~~
- Clause 7 describes ~~the content of a software V&V plan~~ common V&V tasks; that is, tasks that are common across application of this standard to system, software, or hardware.

- 1) Table 1a contains the V&V tasks and activities and their criteria for common V&V tasks and activities.
- 2) Table 2a contains the minimum tasks for each integrity level for common V&V tasks.
- 3) Table 3a contains the optional common V&V tasks.
- 4) Figure 1 a depicts a summary of common V&V activities and tasks.
- 5) Figure 2a depicts a summary of V&V test products and tasks.

— Clause 8 describes system V&V tasks.

- 1) Table 1b contains the V&V tasks and activities and their criteria for system V&V tasks and activities.
- 2) Table 2b contains the minimum tasks for each integrity level for system V&V tasks.
- 3) Table 3b contains the optional system V&V tasks.
- 4) Figure 1b depicts a summary of system V&V activities and tasks
- 5) Figure 2b depicts a summary of system V&V test products and tasks

— Clause 9 describes software V&V tasks.

- 1) Table 1c contains the V&V tasks and activities and their criteria for software V&V tasks and activities.
- 2) Table 2c contains the minimum tasks for each integrity level for software V&V tasks.
- 3) Table 3c contains the optional tasks.
- 4) Figure 1c depicts a summary of software V&V activities and tasks.
- 5) Figure 2c depicts a summary of software V&V test products and tasks.

— Clause 10 describes hardware V&V tasks.

- 1) Table 1d contains the V&V tasks and activities and their criteria for hardware V&V tasks and activities.
- 2) Table 2d contains the minimum tasks for each integrity level for hardware V&V tasks.
- 3) Table 3d contains the optional hardware V&V tasks.
- 4) Figure 1d depicts a summary of hardware V&V activities and tasks.

5) [Figure 2d depicts a summary of hardware V&V test products and tasks.](#)

— [Clause 11](#) describes V&V reporting, administrative, and documentation requirements.

— [Clause 12](#) describes the content of a V&V plan.

[This standard is organized so that the V&V processes \(system, software, or hardware\) may be invoked separately or in any combination. V&V processes may be accomplished in any of the following combinations: system V&V \(Clause 8\), software V&V \(Clause 9\), hardware V&V \(Clause 10\), system/software V&V \(Clause 8 and Clause 9\), system/hardware V&V \(Clause 8 and Clause 10\), software/hardware V&V \(Clause 9 and Clause 10\), and system/software/hardware V&V \(Clause 8 through Clause 10\). Some V&V tasks are common to systems, software, and hardware V&V; in order not to repeat these V&V tasks in each part, the common V&V tasks are listed once in Clause 7. The common V&V tasks are invoked for any V&V that is performed. These combinations are illustrated in Figure 3.](#)

V&V Scope	Common	System	Software	Hardware
System only	X	X		
Software only	X		X	
Hardware only	X			X
System and software	X	X	X	
System and hardware	X	X		X
Software and hardware	X		X	X
System, software, and hardware	X	X	X	X

[Figure 3— V&V effort combinations](#)

NOTE—References to Table 1 and Table 2 without a suffix a, b, c, or d are understood to include those parts of the table that correspond to the parts of the V&V scope being invoked (i.e., Common, System, Software, or Hardware). References to Figure 1 and Figure 2 that do not have a suffix a, b, c, or d are similarly understood to include those parts of the figure that correspond to the parts of the V&V scope being invoked.²

Clause 4 through Clause 6 contain guidance that is applicable to all combinations of the V&V scope. Clause 11 contains reporting requirements for all combinations of V&V scope. Clause 12 contains a sample V&V plan.

Table 1a through Table 1d provide V&V task descriptions, inputs, and outputs for each life cycle process. Table 2a through Table 2d list minimum V&V tasks required for different **software** integrity levels. Table 3a through Table 3d provides a list of optional V&V tasks and their suggested applications in the **software** system life cycle. These optional V&V tasks may be added to the minimum V&V tasks, ~~as necessary,~~ to tailor the V&V effort to project needs.

Figure 1 a through Figure 1 d provide an example of an overview of the V&V inputs, outputs, and minimum V&V tasks for **software** integrity level 4. Figure 2a through Figure 2d provide guidelines for scheduling V&V test planning, execution, and verification activities. An example of a phased life cycle model was used in Figure 1a through Figure 1d and Figure 2a through Figure 2d to illustrate a mapping of the ISO/~~IEC~~TEC 12207:2008 ~~1995~~ [B11] life cycle processes to the V&V activities and tasks described in this standard.

² Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

Annex A describes the mapping of ISO/~~IEC~~EEC 12207: ~~2008~~ 1995 [B11] and ~~IEEE Std 1074-1997 [B10]~~ ISO/TEC 15288:2008 [B16] to this standard's V&V activities and tasks. Annex B provides an example of a risk-based, four-level integrity scheme. Annex C provides a definition of independent verification and validation (IV&V). Annex D provides guidelines for conducting V&V of reuse software. Annex E describes V&V measures. Annex F illustrates an example of the V&V organizational relationship to other project responsibilities. Annex G describes optional V&V tasks. Annex H describes environmental factors that should be considered when conducting V&V. Annex I discusses potential issues with interactions among system, software, and hardware. Annex J describes hazard analysis, security analysis, and risk analysis and their role in V&V. Annex K provides an example of assigning and changing the integrity level of supporting system functions. Annex L maps the process outcomes from ISO/EEC 12207:2008 [B11] and ISO/TEC 15288:2008 [B16] to associated V&V tasks. Annex M provides a bibliography of the informative standards referenced in this standard.

1.6 Audience

The audience for this standard includes system, software, and hardware suppliers, acquirers, developers, maintainers, V&V practitioners, operators, users, and managers in both the supplier and acquirer organizations.

1.7 Conformance

The word *shall* identifies the mandatory requirements ~~strictly~~ to be followed ~~in-order~~ to conform to this standard. The words *should* and *may* indicate optional tasks that are not required to claim conformance to this standard.

Not all V&V efforts are initiated at the start of the life cycle process of acquisition and continued through the maintenance process. If a project uses only selected life cycle processes, then conformance to this standard is achieved if the minimum V&V tasks are implemented for [all of](#) the associated life cycle processes selected for the project. Any claim of conformance to this standard shall identify the applicable life cycle processes. As in all cases, the minimum V&V tasks are defined by the **software** integrity level assigned to the [system](#), software, [or hardware](#). For life cycle processes that are not used by the project, the V&V requirements and tasks for those life cycle processes are optional V&V tasks invoked as needed at the discretion of the project. Specific **software** development methods and technologies (such as automated code generation from detailed design) may eliminate development steps or combine several development steps into one; therefore, a corresponding adaptation of the minimum V&V tasks is permitted [and shall be documented in any claim of conformance to this standard](#).

When this standard is invoked for existing [systems](#), software, [or hardware](#) and the required V&V inputs are not available, then [the](#) V&V tasks may use other available project input sources or may reconstruct the needed inputs to achieve conformance to this standard.

[System, software, or hardware conformance to this standard can be achieved by demonstrating that all the "shall" requirements in the specified clauses are met as defined below:](#)

[—System V&V conformance:](#) [Clause 7 and Clause 8](#)

[—Software V&V conformance:](#) [Clause 7 and Clause 9](#)

[— Hardware V&V conformance:](#) [Clause 7 and Clause 10](#)

All conformance waivers need to be approved by the acquiring organization and shall be described in the verification and validation plan (VVP). The information required for deviations and waivers shall include the task identification, rationale, and the effect on quality.

1.8 Disclaimer

This standard establishes minimum criteria for V&V processes, activities, and tasks. However, implementing these criteria does not automatically ensure conformance to system or mission objectives, or prevent adverse consequences (e.g., loss of life, mission failure, loss of system safety or security, [or](#) financial or social loss). Conformance to this standard does not absolve any party from any social, moral, financial, or legal obligations.

1.9 Limitations

~~None.~~

2. [Normative](#) references

This standard does not require the use of any normative references. Standards useful for the implementation and interpretation of this standard are listed in Annex M.

3. Definitions, abbreviations, and acronyms

3.1 Definitions

For the purposes of this ~~standard~~ document, the following terms and definitions apply. ~~The Authoritative Dictionary~~ ISO/IEC/IEEE 24765-2010 IEEE Standard Glossary of Software Engineering Terminology [B19] and the IEEE Standards Dictionary:

Glossary of Terms, ~~Seventh Edition and Definitions~~ [B2] and IEEE Std 610.12TM-1990 [B3] should be ~~referenced~~ consulted for terms not defined in this clause.³ A note following a definition indicates the source of the definition.

acceptance testing: (A) Formal testing conducted to determine whether ~~or not~~ a system satisfies its acceptance criteria and to enable the customer to determine whether ~~or not~~ to accept the system. (B) Formal testing conducted to enable a user, customer, or other authorized entity to determine whether to accept a system or component.

NOTE—See ISO/IEC/IEEE Std 610.12-1990 [B3].² 24765-2010 [B19].

³IEEE Standard Glossary of Software Engineering Terminology and IEEE Standards Dictionary: Glossary of Terms and Definitions are available at <http://shop.ieee.org/>. Definitions of individual terms may also be obtained free of charge at <http://www.computer.org/sevocab>.

²~~Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.~~

acquirer: Stakeholder that acquires or procures a product or service from a supplier. Other terms commonly used for an acquirer are buyer, customer, owner, or purchaser.

NOTE—See ISO/IEC/IEEE 24765-2010 [B19].

anomaly: Anything observed in the documentation or operation of ~~software~~ a system that deviates from expectations based on previously verified system, software, or hardware products or reference documents.

NOTE—See ISO/IEC/IEEE ~~Std 610.12-1990~~ [B3] 24765-2010 [B19].

asset: An item (e.g., design, specifications, source code, documentation, test suites, or manual procedures) that has been designed for use in multiple contexts.

NOTE—See IEEE Std 1517™ - ~~1999~~ [B11] 2010 [B10].

baseline: Any agreement or result designated and fixed at a given time, from which changes require justification and approval.

NOTE—See ISO/IEC/IEEE 24765-2010 [B19].

component: One ~~of the~~ part that makes up a system. A component may be hardware or software and may be subdivided into other components.

NOTE 1—The terms *module*, *component*, and *unit* are often used interchangeably or defined to be subelements of one another in different ways depending ~~up~~ upon the context. The relationship of these terms is not yet standardized.

NOTE 2—See ISO/IEC/IEEE ~~Std 610.12-1990~~ [B3] 24765-2010 [B19].

component testing: Testing of individual hardware or software components ~~or groups of related components.~~

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#) [24765-2010 \[B19\]](#).

criticality: The degree of impact that a requirement, module, error, fault, failure, or other item has on the development or operation of a system.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#) [24765-2010 \[B19\]](#).

domain: A problem space.

NOTE—See IEEE Std 1517-~~1999 [B11]~~[2010 \[B10\]](#).

domain analysis: (A) The analysis of systems within a domain to discover commonalities and differences among them. (B) The process by which information used in developing software systems is identified, captured, and organized so that it can be reused to create new systems, within a domain. (C) The result of the process in (A) and (B).

NOTE—See IEEE Std 1517-~~1999 [B11]~~[2010 \[B10\]](#).

domain engineering: A reuse-based approach to defining the scope (i.e., domain definition), specifying the structure (i.e., domain architecture), and building the assets (e.g., requirements, designs, software code, [and](#) documentation) for a class of systems, subsystems, or applications. Domain engineering may include the following activities: domain definition, domain analysis, developing the domain architecture, and domain implementation.

NOTE—See IEEE Std 1517-~~1999 [B11]~~[2010 \[B10\]](#).

firmware: The combination of a hardware device and computer instructions and data that reside as read only software on that device.

NOTE 1—This term is sometimes used to refer only to the hardware device or only to the computer instructions or data, but these meanings are deprecated.

NOTE 2—The confusion surrounding this term has led some to suggest that it be avoided altogether.

NOTE 3—See [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#) [24765-2010 \[B19\]](#).

hazard: (A) An intrinsic property or condition that has the potential to cause harm or damage. (B) A source of potential harm or a situation with a potential for harm in terms of human injury, damage to health, property, or the environment, or some combination of these.

NOTE—For (A), see [ISO/IEC/IEEE ~~EIA Std 12207.0~~TM-1996 \[B12 24765-2010 \[B19\]](#).

hazard identification: The process of recognizing that a hazard exists and defining its characteristics.

NOTE—For (A), see [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

independent verification and validation (IV&V): V&V performed by an organization that is technically, managerially, and financially independent of the development organization.

NOTE—See [ISO/IEC/IEEE ~~Std 610.12-1990~~ \[B3 24765-2010 \[B19\]](#).

integration testing: Testing in which software components, hardware components, or both are combined and tested to evaluate the interaction between them.

NOTE—See [ISO/IEC/IEEE ~~Std 610.12-1990~~ \[B3 24765-2010 \[B19\]](#).

integrity level: A value representing project-unique characteristics (e.g., ~~software~~ complexity, criticality, risk, safety level, security level, desired performance, [and](#) reliability) that define the importance of the [system](#), software, [or hardware](#) to the user.

interface design document (IDD): Documentation that describes the architecture and design interfaces between system and components. These descriptions include control algorithms, protocols, data contents and formats, and performance.

NOTE—See [ISO/IEC/IEEE ~~Std 610.12-1990~~ \[B3 24765-2010 \[B19\]](#).

interface requirements specification (IRS): Documentation that specifies [the](#) requirements for [the](#) interfaces between systems and components. These requirements include constraints on formats and timing.

NOTE—See *The Authoritative Dictionary* [B2] [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

life cycle processes: A set of interrelated [or interacting](#) activities that result in the development or assessment of [system](#), software, [or hardware](#) products. Each activity consists of tasks. The life cycle processes may overlap one another. For verification and validation (V&V) purposes, no [life cycle](#) process is concluded until its development products are verified and validated according to the defined tasks in the **S**VVP. verification and validation plan (VVP).

NOTE—See *The Authoritative Dictionary* [B2] [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

microcode: A collection of microinstructions, comprising part of , all of, or [a set of](#) microprograms.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#) [24765-2010 \[B19\]](#).

microprogram: A sequence of instructions, called microinstructions, specifying the basic operations needed to carry out a machine language instruction.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#) [24765-2010 \[B19\]](#).

minimum tasks: Those verification and validation (V&V) tasks required for the **software** integrity level assigned to the [system](#), software, [or hardware](#) to be verified and validated.

~~NOTE—See *The Authoritative Dictionary* [B2].~~

optional tasks: Those verification and validation (V&V) tasks that may be added to the minimum V&V tasks to address specific application requirements.

~~NOTE—See *The Authoritative Dictionary* [B2].~~

required inputs: The set of items necessary to perform the minimum verification and validation (V&V) tasks mandated within any life cycle activity.

~~NOTE—See *The Authoritative Dictionary* [B2].~~

required outputs: The set of items produced as a result of performing the minimum V&V tasks mandated within any life cycle activity.

~~NOTE—See *The Authoritative Dictionary* [B2].~~

reusable software product: A [system](#), software, [or hardware](#) product developed for one use but having other uses, or one developed specifically to be usable on multiple projects or in multiple roles on one project. Examples include, but are not limited to, commercial off-the-shelf (COTS) software products, acquirer-furnished software products, software products in reuse libraries, and preexisting developer software products. Each use may include all or part of the software product and may involve its modification. This term can be applied to any software product (for example, requirements, architectures), not just to software itself.

NOTE—See ~~*The Authoritative Dictionary* [B2]~~[ISO/IEC/IEEE 24765-2010 \[B19\]](#).

risk: (A) The combination of the ~~probability~~ [likelihood](#) of occurrence and the consequences of a given future undesirable event. Risk can be associated with products and/or projects. (B) The combination of the ~~probability~~ [likelihood](#) of an abnormal event or failure and the consequence(s) of that event or failure to a system's components, operators, users, or environment.

NOTE—See ~~*The Authoritative Dictionary* [B2]~~[ISO/IEC 16085-2006 \[B18\]](#).

security: (A) The protection of computer hardware or software from accidental or malicious access, use, modification, destruction, or disclosure. Security also pertains to personnel, data, communications, and the physical protection of computer installations. (B) The protection of information and data so that unauthorized persons or systems cannot read or modify them and authorized persons or systems are not denied access to them.

~~NOTE—For (A), see *The Authoritative Dictionary* [B2]. For subdefinition (B), see ISO/IEC 12207:1995 [B13].~~

~~NOTE—See *The Authoritative Dictionary* [B2].~~ [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

software design description (SDD): A representation of software created to facilitate analysis, planning, implementation, and decision-making. The software design description is used as a medium for communicating software design information and may be thought of as a blueprint or model of the system.

~~NOTE—See *The Authoritative Dictionary* [B2].~~ [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

software requirements specification (SRS): Documentation of the essential requirements (functions, performance, design constraints, and attributes) of the software and its external interfaces.

~~NOTE—See *ISO/IEC/IEEE Std 610.12-1990* [B3].~~ [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

system of interest: System whose life cycle is under consideration in the application of one instance of this verification and validation (V&V) standard.

~~NOTE—See *ISO/IEC 15288:2008* [B16].~~

system testing: Testing conducted *on* a complete, integrated system to evaluate the system's compliance with its specified requirements.

~~NOTE—See *ISO/IEC/IEEE Std 610.12-1990* [B3].~~ [ISO/IEC/IEEE 24765-2010 \[B19\]](#).

test case: (A) A set of test inputs, execution conditions, and expected results developed for a particular objective, such as to exercise a particular program path or to verify compliance with a specific requirement. (B) Documentation specifying inputs, predicted results, and a set of execution conditions for a test item.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B324765-2010 \[B19\]](#).

test design: Documentation specifying the details of the test approach for a [system](#), software, [or hardware](#) feature or combination of [software](#) features and identifying the associated tests.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B324765-2010 \[B19\]](#).

test plan: (A) A document describing the scope, approach, resources, and schedule of intended test activities. It identifies test items, the features to be tested, the testing tasks, who will do each task, and any risks requiring contingency planning. (B) A document that describes the technical and management approach to be followed for testing a system or component. Typical contents identify the items to be tested, tasks to be performed, responsibilities, schedules, and required resources for the testing activity.

NOTE—See [ISO/IEC/IEEE Std 610.12-1990 \[B324765-2010 \[B19\]](#).

test procedure: (A) Detailed instructions for the setup, execution, and evaluation of results for a given test case. (B) A document containing a set of associated instructions as in (A). (C) Documentation that specifies a sequence of actions for the execution of a test.

NOTE—See IEEE [Std 982.1™-1998 \[B52005 \[B4\]](#).

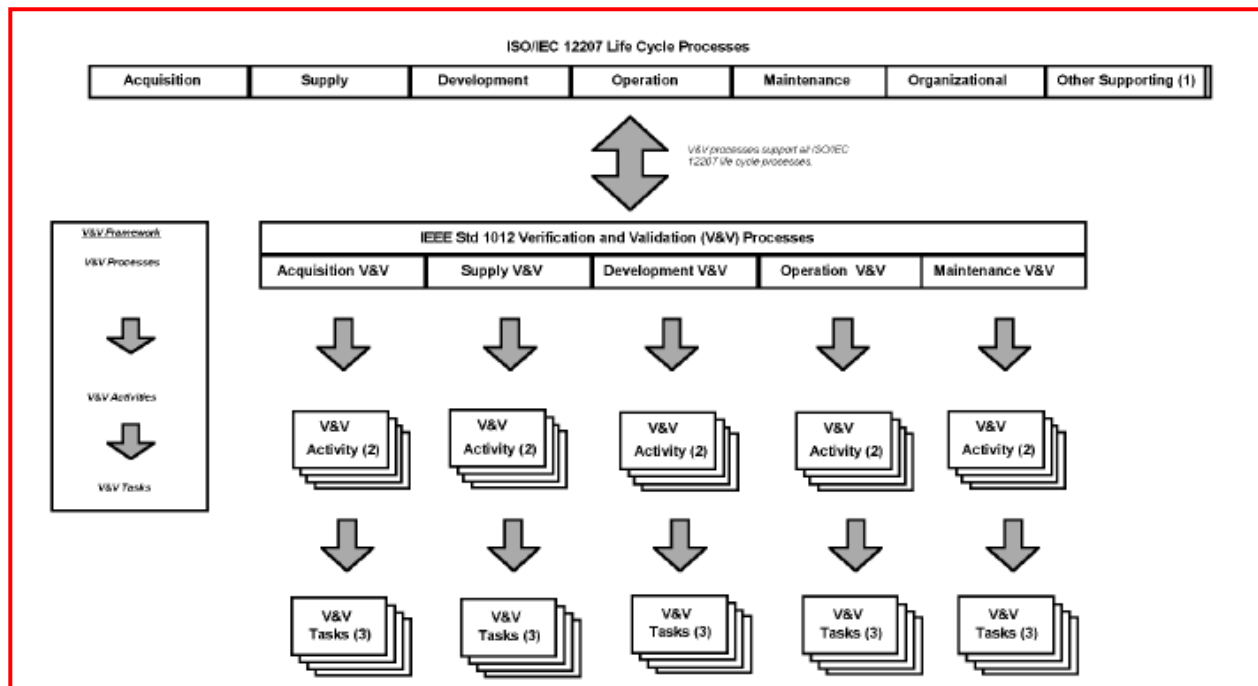
validation: (A) The process of evaluating a system or component during or at the end of the development process to determine whether it satisfies specified requirements. (B) The process of providing evidence that the [system](#), software, [or hardware](#) and its associated products satisfy [system](#) requirements allocated to [software](#) [it](#) at the end of each life cycle activity, solve the right problem (e.g., correctly model physical laws, implement business rules, [and](#) use the proper system assumptions), and satisfy intended use and user needs.

NOTE—For (A), see [ISO/IEC/IEEE Std 610.12-1990 \[B324765-2010 \[B19\]](#).

verification: (A) The process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase. (B) The process of providing objective evidence that the [system](#), software, [or hardware](#) and its associated products conform to requirements (e.g., for correctness, completeness, consistency, [and](#) accuracy) for all life cycle activities during each life cycle process (acquisition, supply, development, operation, and maintenance); satisfy standards, practices, and conventions during life cycle processes; and successfully complete each life cycle activity and satisfy all the criteria for initiating succeeding life cycle activities ~~(e.g., building the software correctly.~~ [Verification of interim work products is essential for proper understanding and assessment of the life cycle phase product\(s\).](#)

NOTE—For ~~subdefinition~~ (A), see [ISO/IEC/IEEE Std 610.12-1990 \[B3\]](#)[24765-2010 \[B19\]](#).

verification and validation (V&V) effort: The work associated with performing the V&V processes, activities, and tasks. ~~The following framework illustrates how V&V processes are subdivided into activities, which in turn have associated tasks:~~



(Deleted)

~~NOTE 1—Other supporting processes consist of documentation, configuration management, quality assurance, joint review, audit, and problem resolution.~~

~~NOTE 2—Management of V&V activity is concurrent with all V&V activities.~~

~~NOTE 3—The task description, inputs, and outputs of all V&V tasks are included in Table 1.~~

3.2 Abbreviations and acronyms

The following acronyms and abbreviations appear in this standard:

COTS commercial off-the-shelf

[CPU](#) [central processing unit](#)

[GOTS](#) [government off-the-shelf](#)

[HDD](#) [hardware design description](#)

[HRS](#) [hardware requirements specification](#)

[HW](#) [hardware](#)

~~ANSI~~ ~~American National Standards Institute~~

~~IEC~~ ~~International Electrotechnical Commission~~

~~IEEE~~ ~~Institute of Electrical and Electronic Engineers~~

IDD interface design document

IRS interface requirements specification

~~ISO~~ ~~International Organization for Standardization~~

IV&V independent verification and validation

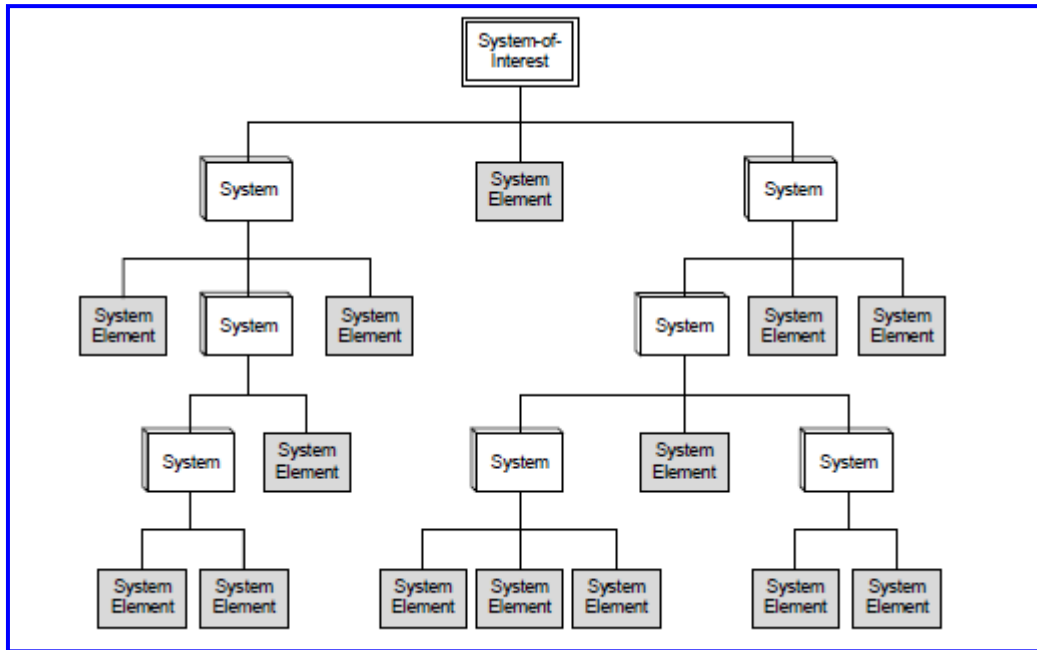
[N/A](#) [not applicable](#)

<u>OCD</u>	<u>operational concept document</u>
<u>QA</u>	<u>quality assurance</u>
NDI	non-developmental item
RFP	request for proposal (tender)
SDD	software design description
SRD	system requirements document
SRS	software requirements specification
<u>SW</u>	<u>software</u>
<u>TRA</u>	<u>threat and risk assessment</u>
<u>VVP</u>	<u>verification and validation plan</u>
<u>V&V</u>	<u>verification and validation</u>
<u>WBS</u>	<u>work breakdown structure</u>
SVVP	software V&V plan
SVVR	software V&V report
V&V	verification and validation

4. Relationships between V&V and life cycle processes

This standard is structured to enable a V&V practitioner to use Clause 7 (common V&V) and any of Clause 8 (system V&V), Clause 9 (software V&V), or Clause 10 (hardware V&V) in performing V&V for a particular product. V&V on a system is performed using Clause 7 and Clause 8 (common and system), on software using Clause 7 and Clause 9 (common and software), and on hardware using Clause 7 and Clause 10 (common and hardware).

Most systems consist of more than a single hardware or a single software element. A depiction of a more complex system with subsystems and system elements (i.e., software or hardware, shown as shaded boxes) is shown in Figure 4. The system life cycle processes are applied recursively from the system of interest to the systems at the next level, with the recursion continuing until a system element is reached. For software system elements, the software life cycle processes are applied, and for hardware system elements, the hardware life cycle processes are applied. The life cycle processes at the system, software, or hardware levels may be conducted in parallel for systems or system elements at the same level. The full set of life cycle processes (system, software, or hardware, as appropriate) are applied to each product. In the same way, system V&V is applied recursively for each system of the system of interest, and software or hardware V&V is applied to each system element. For V&V purposes, no life cycle process is concluded until its development products are verified and validated according to the defined tasks in the VVP.

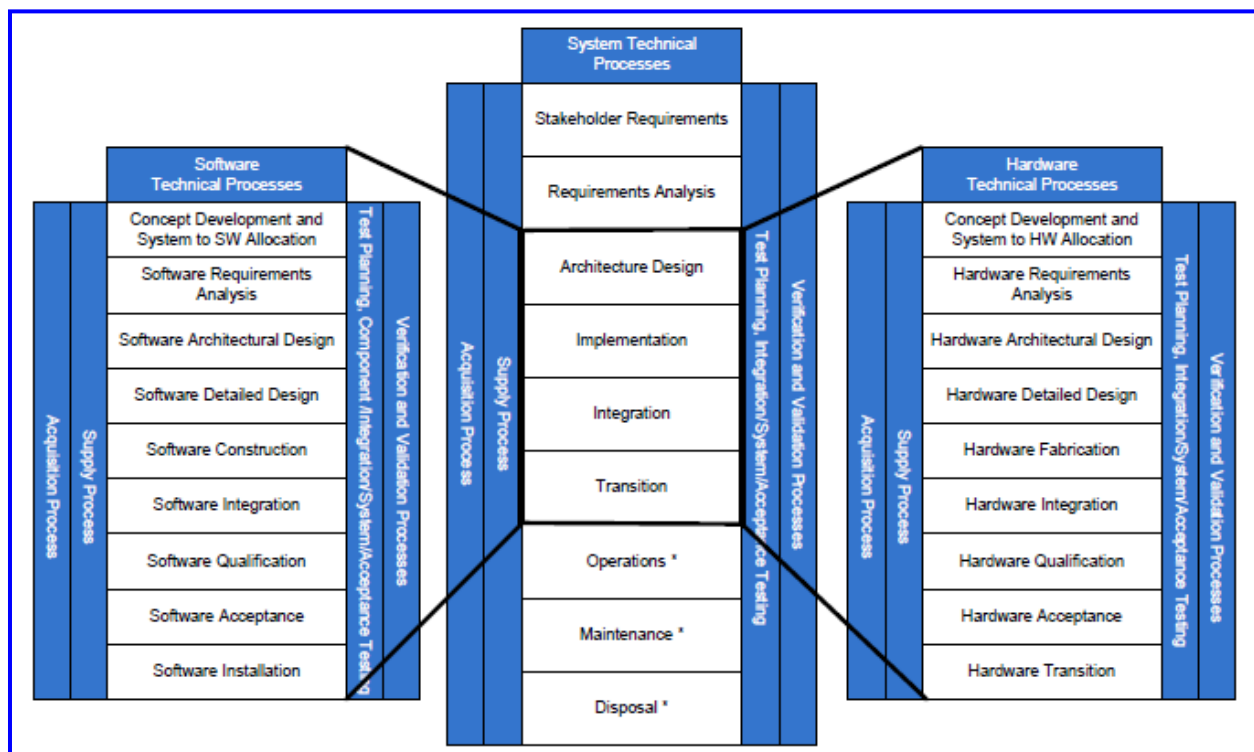


(New)

Figure 4 — Example system-of-interest structure

The Verification Process and the Validation Process are two processes within the Technical Processes group of the System Life Cycle Processes defined in ISO/IEC 15288:2008 [B16]. System V&V as described in this V&V standard is a conforming instance of the Verification and Validation processes in ISO/IEC 15288:2008 [B16] (i.e., conducting system V&V as described in this V&V standard enables the V&V practitioner to claim full conformance with those two System Life Cycle Processes, although other approaches may also be in conformance). In a similar fashion, software V&V as described in this V&V standard is a conforming instance of the Software Verification and Software Validation Processes within the Software Support Processes group of the Life Cycle Processes defined in ISO/IEC 12207:2008 [B11].

The purpose of the system Implementation Process is to produce a specified system element implemented as a software and hardware product or service. Within the System Implementation Process, each software and hardware element is developed using the full software or hardware life cycle (requirements, design, implementation/construction, integration, and test). This concept is reflected in Figure 5, which also illustrates that the Technical Processes for software and hardware overlap with the System Architecture Design Process and the System Transition Process. At the system element level, the implementation will be realized by the software Implementation Process (described in Clause 9) or the hardware Fabrication Process (described in Clause 10).



(New)

NOTE 1—The layout of the processes is not intended to specify an ordering of processes. V&V activities may be applied to any life cycle process.

NOTE 2—The Operations, Maintenance, and Disposal Processes within the System Technical Processes are marked with an asterisk (*) to indicate that there are Software and Hardware V&V activities and tasks that support these System Technical Processes but are not shown in Figure 5.

NOTE 3—Management of V&V activity is concurrent with all V&V activities.

NOTE 4—The task description, inputs, and outputs of all V&V tasks are included in Table Ia through Table Id.

Figure 5—Relationship of system, software, and hardware processes

The following cross-reference maps the V&V activities to those processes for which this V&V standard calls out specific activities and tasks.

<u>V&V Activity IEEE 1012</u>	<u>System Processes (ISO/IEC 15288:2008 [B16])</u>	<u>Software Processes (ISO/IEC 12207:2008 [B11])</u>	<u>Hardware Processes (NOTE 3)</u>
<u>Management of V&V</u>	<u>Not specified</u>	<u>Not specified</u>	<u>Not specified</u>
<u>Acquisition Support V&V</u>	<u>Acquisition Process</u>	<u>Acquisition Process</u>	<u>Acquisition</u>
<u>Supply Planning V&V</u>	<u>Supply Process</u>	<u>Supply Process</u>	<u>Supply</u>
<u>Project Planning V&V</u>	<u>Project Planning Process</u>	<u>Project Planning Process</u>	<u>Project Planning Process</u>
<u>Configuration Management V&V</u>	<u>Configuration Management Process</u>	<u>Software Configuration Management Process</u>	<u>Hardware Configuration Management</u>
<u>System Stakeholder Requirements Definition V&V</u>	<u>Stakeholder Requirements Definition Process</u>	<u>Stakeholder Requirements Definition Process</u>	<u>Stakeholders Requirements Definition Process</u>
<u>System Requirements Analysis V&V</u>	<u>System Requirements Analysis Process</u>	<u>System Requirements Analysis Process</u>	<u>System Requirements Analysis Process</u>
<u>System Architecture Design V&V</u> <u>Software Concept V&V</u> <u>Hardware Concept V&V</u>	<u>System Architectural Design Process</u>	<u>System Architectural Design Process</u>	<u>System Architectural Design Process</u>
<u>System Implementation V&V</u> <u>Software/Hardware V&V Activities</u> = <u>Concept V&V</u> = <u>Requirements V&V</u>	<u>System Implementation Process</u>	<u>Software Requirements Analysis Process</u> <u>Software</u>	<u>Hardware Requirements Analysis Process</u> <u>Hardware</u>

= Design V&V = Implementation/Fabrication V&V = Integration Test V&V = Qualification Test V&V = Acceptance Test V&V		Architectural Design Process Software Detailed Design Process Software Construction Process Software Integration Process Software Qualification Testing Process	Architectural Design Process Hardware Fabrication Process Hardware Test Process Hardware Integration Test Process Hardware Qualification Test Process
System Integration V&V	System Integration Process	Support System Integration Process	Support System Integration Process
All verification activities of the IEEE 1012 life cycle	System Verification Process	Software Verification Process	Hardware Verification Process
System Transition V&V Software Installation and Checkout V&V (Transition) Hardware Transition V&V	System Transition Process	Software Installation Process	Hardware Transition Process
All validation activities of the IEEE 1012 life cycle	System Validation Process	Software Validation Process	Hardware Validation Process
System Operation V&V Software Operation V&V Hardware Operation V&V	System Operation Process	Software Operation Process	Hardware Operation Process
System Maintenance V&V	System	Software	Hardware

Software Maintenance V&V	Maintenance	Maintenance	Maintenance
Hardware Maintenance V&V	Process	Process	Process
System Disposal V&V	System Disposal	Software Disposal	Hardware
Software Disposal V&V	Process	Process	Disposal Process
Hardware Disposal V&V			

[NOTE 1—Not all life cycle processes from ISO/IEC 15288:2008 \[B16\] and ISO/IEC 12207:2008 \[B11\] have V&V tasks specified in this standard. The V&V processes also interact with the other system and software processes not included in this table, such as providing inputs and receiving outputs that are used in the V&V effort.](#)

[NOTE 2—V&V practitioners provide feedback on noted deficiencies or areas for improvement \(e.g., a life cycle model that does not produce a product needed to perform V&V tasks and activities\).](#)

[NOTE 3—There are no applicable IEEE or International Standards for hardware lifecycle processes.](#)

[NOTE 4—Each IEEE 12207 Software Specific Process is a specialization of its corresponding process in ISO/IEC 15288:2008 \[B16\]. Users may consider claiming conformance to the ISO/IEC/IEEE 15288 process rather than the IEEE 12207 process.](#)

[NOTE 5—Each Hardware Process that corresponds to an IEEE 12207 Software Specific Process is a specialization of its corresponding process in ISO/IEC 15288.](#)

[Figure 6—Mapping of V&V activities to system, software, and hardware processes](#)

5. **Software** Integrity levels

~~Software~~ Integrity-level ~~are a range of values that represent~~ determination quantifies the ~~software~~ complexity, criticality, risk, safety level, security level, desired performance, reliability, or other system-unique characteristics ~~that define~~ based on the importance of the system ~~software~~ to the user and acquirer. Integrity levels are used to determine the V&V tasks, activities, rigor, and the level of intensity of the V&V to be performed.

~~The software is a part of the system, and its integrity level is to be determined as a part of that system. The assigned software integrity levels may change as the software evolves.~~

The following is a result of the integrity-level determination:

- a) An integrity schema is defined.
- b) Integrity levels for all system entities are assigned.
- c) Integrity assignments are used to determine V&V activities and tasks using Table 2a through Table 2d.
- d) A basis is established to review and revalidate the integrity levels throughout the V&V effort.

~~This standard uses software integrity levels to determine the V&V tasks to be performed. Some software elements and components may not require the assignment of an integrity level (i.e., not applicable) because their failure would impart no consequences on the intended system operations. The V&V effort may recommend technical and procedural mitigation approaches to reduce the integrity level.~~

~~As an example, this standard uses the following four level software integrity scheme. This example scheme is based upon the concepts of consequences and mitigation potential.~~

Description	Level
Software element must execute correctly or grave consequences (loss of life, loss of system, economic or social loss) will occur. No mitigation is possible.	4
Software element must execute correctly or the intended use (mission) of the system/software will not be realized, causing serious consequences (permanent injury, major system degradation, economic or social impact). Partial to complete mitigation is possible.	3
Software element must execute correctly or an intended function will not be realized, causing minor consequences. Complete mitigation possible.	2
Software element must execute correctly or intended function will not be realized, causing negligible consequences. Mitigation not required.	1
Another example, a scheme based on risk, is described in Annex B. This standard does not require the use of any particular software integrity level scheme described or referenced in this standard.	

The degree of rigor and intensity in performing and documenting the task shall be commensurate with the integrity level. As the integrity level decreases, so does the required scope, intensity, and degree of rigor associated with the V&V task. For example, a hazard analysis performed for integrity level 4 software might be formally documented and consider failures at the module level; a hazard analysis for integrity level 3 software may consider only significant software failures and be documented informally as part of the design review process.

~~Integrity levels shall be assigned to software elements or components as part of the criticality analysis task. The V&V effort shall specify a software integrity level scheme if one is not already defined.~~

Use the integrity schema established for the system, software, or hardware, if one exists. An integrity-level scheme shall be specified if one is not already defined. Establish gradations of criticality and integrity to assure complete partitioning of the potential effects space from "no effect" to "worst case." The ~~software~~ integrity levels established for a ~~project~~ system element by the developer should result from agreements among the acquirer and the supplier and remain consistent with the regulatory requirements ~~independent assurance authorities (e.g., a regulatory body or responsible agency).~~

This standard was developed with a four-level schema to explain the minimum normative V&V requirements. Other integrity schemas are acceptable. For any selected integrity schema, the selected integrity levels shall be mapped into this standard's four-level schema to demonstrate that the minimum V&V requirements are satisfied. Because V&V is applied recursively from the system of interest down to each of the subsystems or elements at the next level, it is not necessary to use the same integrity-level schema for all the subsystems or elements of the system of interest.

NOTE—A process for developing an integrity schema can be found in ISO/IEC 15026-2011 [B12].

The V&V criticality analysis task is used to classify the integrity level of every system (e.g., subsystem) or element in the system of interest. Integrity levels are assigned to software requirements, functions, groups of functions, or software components, or subsystems. The integrity levels characterize and quantify the potential for undesirable effects and consequences resulting from integrity lapse, unintended effects, failure mode effects, and unverified performance effects. The characteristics that used to determine software integrity level vary depending on the intended application and use of the system.

Integrity-level assignments shall be applied recursively to the components of each system element. As the development stages progressively decompose the solution into more details, the criticality analysis performed at each development stage assigns integrity levels to each function. By recursively assigning integrity levels to individual solution parts, greater rigor and intensity is applied to the higher integrity level elements. Progressive decomposition for criticality analysis and assignment of integrity levels shall be applied to segregate parts that warrant high-integrity V&V tasks from those parts that do not. This recursive assignment of integrity levels to solution parts keeps the V&V effort focused on the high-integrity elements, making the V&V effort cost efficient and technically effective.

From the system perspective, during integrity-level analysis, software and hardware are treated as functional elements. Subsystem components cannot have a higher criticality or integrity level than the parent subsystem. The software system shall be assigned the same integrity level as the highest level assigned to any individual element. The integrity level assigned to the parent shall be as high as the highest integrity level of the system elements. The system elements can have the integrity level of the system parent or lower depending on the critical functions it supports.

The integrity level assigned to reused, COTS, and government off-the-shelf (GOTS) components ~~software products~~ shall be in accordance with the integrity-level scheme adopted for the system project (see Annex D), element into which COTS or GOTS may be integrated for the project ~~and~~ The reused COTS or GOTS component ~~software product~~ shall be evaluated for use in the context of its application. The design, development, procedural, and technology features implemented in the system can raise or lower the assigned software integrity levels.

The V&V criticality analysis task is used to classify the integrity level of enabling systems. The tools that insert or translate code (e.g., optimizing compilers and auto-code generators) shall be assigned the same integrity level as the integrity level assigned to the software element that the tool affects. If the tool cannot be verified and validated, then the output of the tool shall be subject to the same level of V&V as the software element.

~~Table 2 identifies minimum V&V tasks that shall be performed for each software integrity level. To identify the minimum V&V tasks that apply to a different selected software integrity level scheme, the user of the standard shall map this standard's software integrity level scheme and associated minimum V&V tasks to their selected software integrity level scheme.~~

The mapping of the integrity-level scheme and the associated minimum V&V tasks shall be documented in the VVP. The basis for assigning integrity levels to components shall be documented in a V&V task report and V&V final report. Table 2a through Table 2d of this standard identify the necessary and sufficient V&V activities and tasks to perform for each integrity level. Once the integrity levels have been determined, Table 2a through Table 2d are used to identify the activities and tasks to adapt and plan the V&V effort. High integrity ~~software~~ requires a larger set of V&V processes and a more rigorous application of V&V tasks. The V&V processes should be tailored to specific system elements ~~requirements and applications through the selection of a software integrity level~~ with ~~its corresponding~~ a combination of the minimum V&V tasks and addition of optional V&V tasks.

The addition of optional V&V tasks allows the V&V effort to address application-specific characteristics ~~of the software~~. Table 3a through Table 3d provide a list of optional V&V tasks and their suggested applications in the life cycle of the system of interest. These optional V&V tasks may be added to the minimum V&V tasks, as necessary, to adapt the V&V effort to project needs to provide additional assurance and confidence in the operational use of the system.

The ~~software~~ integrity-level assignment shall be continually reviewed and updated by conducting the V&V criticality analysis task throughout the life cycle ~~software development process~~. Changes to assignments shall be used to reassess the V&V plan and performance for appropriate integrity assurance throughout the life cycle. If the integrity level of any component is revised, the effect of the revision on existing requirements shall be evaluated to identify additional activities and tasks to be performed on the system, software, and hardware. This includes adopting V&V tasks for the revised component integrity level and performing those V&V tasks for previous life cycle stages.

Critical functions are assigned a high system integrity level at the initiation of the project and shall not be lowered at any stage of the development process unless authorized by the system acquirer or governing regulatory organization. Other system functions may be assigned lower system integrity levels than the critical functions at the beginning stages. However, if any lower system integrity-level functions can modify or alter critical data or can create an improper condition or system state to exist causing critical system functions to take incorrect actions, then those lower system functions shall be elevated to critical system integrity levels for V&V analysis and test. For discussion purposes, these system functions shall be designated "critical supporting system functions."

During the evolution of the system through requirements, design, implementation, and integration, the selection of technology or design/implementation techniques could mitigate or eliminate the impact that a "critical supporting system function" may have on critical system functions. In such cases and only with extreme care, the integrity level for a "critical supporting system function" may be lowered for V&V analysis and test purposes. In contrast, there may be means to obtain a higher integrity level by combination of components having lower integrity levels (e.g., duplication or redundancy—see ISO/IEC 15026-2011 [B12]).

Annex K describes an example of how system integrity levels are assigned to "supporting system functions" and how system integrity levels may be changed during the system development stages for the "supporting system functions" as a result of the selection of design/implementation techniques.

6. ~~Software~~ V&V processes overview

6.1 General

V&V processes have activities and tasks defined for the processes in the Agreement Processes and Technical Processes group, the Project Planning Process, and the Configuration Management Process of ISO/IEC 15288:2008 [B16]. V&V processes have activities and tasks defined for the processes in the Software Implementation Processes group and for the Software Configuration Management Process of ISO/IEC 12207:2008 [B1 1]. Although there is no corresponding IEEE standard that addresses the hardware life cycle, V&V activities and tasks are defined for processes analogous to those for the software life cycle. ~~V&V processes support the six primary processes of ISO/IEC 12207:1995 [B13]: the management process (see 5.1), acquisition process (see 5.2), supply process (see 5.3), development process (see 5.4), operation process (see 5.5), and maintenance process (see 5.6).~~ The minimum V&V activities and tasks supporting these processes are referenced in the following clauses and are defined in Table Ia through Table Id of each part of this standard. The subclause titles in this clause are the same as the column headings in Table Ia through Table Id to correlate the requirements of the following subclauses with Table Ia through Table Id tasks. Not all ~~software~~ projects include each of the life cycle processes listed. To conform to this standard, the V&V processes shall address all those life cycle processes used by the ~~software~~ project.

V&V activities that are common across system, software, and hardware are in Clause 7 of this standard. These include the V&V Management activity, which is not specifically described by ISO/IEC 15288:2008 [B16] or ISO/IEC 12207:2008 [B1 1] but is essential to the implementation of V&V tasks. Other common V&V activities in Clause 7 are associated with the Acquisition Process, the Supply Process, the Project Planning Process, and the Configuration Management Process.

V&V activities that are associated with system life cycle processes are in Clause 8 (System V&V) of this standard. The system processes are those in the Technical Processes group of ISO/TEC 15288:2008 [B16].

V&V activities that are associated with software life cycle processes are in Clause 9 (Software V&V) of this standard. The software life cycle processes are the lower level processes under the Software Implementation Process of ISO/TEC 12207:2008 [B1 1]. The Software Design V&V activity addresses both the Software Architectural Design Process and the Software Detailed Design Process. The Software Test V&V activity addresses testing in several of the software life cycle processes, including the Software Construction Process, the Software Integration Process, and the Software Qualification Testing Process.

V&V activities that are associated with hardware life cycle processes are in Clause 10 (Hardware V&V) of this standard.

The V&V effort shall conform to the task descriptions, inputs, and outputs as described in Table 1a through Table 1d. The V&V effort shall include ~~perform~~ the minimum V&V tasks specified in Table 2a through Table 2d for the assigned ~~software~~-integrity level. If the user of this standard has selected a different ~~software~~-integrity-level scheme, then this standard's ~~software~~-integrity-level scheme and associated minimum V&V tasks of Table 2a through Table 2d shall be mapped to their selected ~~software~~-integrity-level scheme.

Optional V&V tasks may also be performed to augment the V&V effort to satisfy project needs. Optional V&V tasks are listed in Table 3a [through Table 3d](#) and described in Annex G. The list in Table 3a [through Table 3d](#) is illustrative and not exhaustive.

Some V&V activities and tasks include analysis, evaluations, and tests that may be performed by multiple organizations (e.g., ~~software~~-development, project management, quality assurance, [and](#) V&V). For example, risk analysis and hazard analysis may be performed by project management, the development organization, and the V&V effort. The V&V [organization](#) ~~effort~~ performs [or uses the outputs from](#) these tasks [from other sources](#) to develop the supporting basis of evidence showing whether the ~~software~~-product satisfies its requirements [and to mitigate risks to the project](#). These V&V analyses are complementary to other analyses and do not eliminate or replace the analyses performed by other organizations. The degree to which these analysis efforts will be coordinated with other organizations shall be documented in the organizational responsibility section of the SVVP.

~~Testing requires advance planning that spans several development activities. Test documentation and its generation at specific processes in the life cycle are shown in Figure 1 and Figure 2.~~

~~The user of this standard shall document the V&V processes in the SVVP and shall define the information and facilities necessary to manage and perform these processes, activities, and tasks, and to coordinate the V&V processes with other related aspects of the project. The results of V&V activities and tasks shall be documented in task reports, activity summary reports, anomaly reports, V&V test documents, and the V&V final report.~~

6.2 V&V testing

Testing is an activity that may be performed by various organizations within the development effort. This standard requires a minimum level for V&V testing dependent on the integrity level, as reflected in Table Ia (common V&V activities and tasks), Table Ib (system V&V activities and tasks), Table Ic (software V&V activities and tasks), and Table Id (hardware V&V activities and tasks). Figure 7 summarizes the minimum level for V&V testing for the types of testing and integrity level. The term *perform* means that the V&V organization specifies and creates its testing products (i.e., test plan, test design, test cases, and test procedures) and either conducts that testing or analyzes the results of that testing if it is conducted by another organization. The term *review* means that the V&V organization reviews the testing plans and analyzes the results of tests.

<u>Software</u>	<u>V&V testing by integrity level</u>			
	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
<u>V&V Software Component Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>
<u>V&V Software Integration Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>
<u>V&V Software Qualification Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>
<u>V&V Software Acceptance Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>

<u>Hardware</u>	<u>V&V testing by integrity level</u>			
	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
<u>V&V Hardware Component Testing</u>	<u>Review</u>	<u>Review</u>	<u>Review</u>	<u>No action</u>
<u>V&V Hardware Integration Testing</u>	<u>Review</u>	<u>Review</u>	<u>Review</u>	<u>No action</u>
<u>V&V Hardware Qualification Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>
<u>V&V Hardware Acceptance Testing</u>	<u>Perform</u>	<u>Perform</u>	<u>Review</u>	<u>No action</u>

<u>System</u>	<u>V&V testing by integrity level</u>			
	<u>4</u>	<u>3</u>	<u>2</u>	<u>1</u>
<u>V&V System Integration Testing</u>	<u>Perform</u>	<u>Review</u>	<u>Review</u>	<u>No action</u>
<u>V&V System Qualification Testing</u>	<u>Perform</u>	<u>Review</u>	<u>Review</u>	<u>No action</u>
<u>V&V System Acceptance Testing</u>	<u>Perform</u>	<u>Review</u>	<u>Review</u>	<u>No action</u>

Figure 7— Minimum level for V&V testing by integrity level

NOTE 1 (for Figure 7)—The requirement to "perform" testing means that the V&V organization specifies and creates its testing products, confirms that it is conducted appropriately, and reviews the results. The term allows but does not require that the V&V organization actually conducts the test and includes alternatives such as witnessing a test conducted by the supplier or development organization. As noted in the figure, V&V testing is mandatory for only the highest integrity levels. Having the V&V organization perform ("specify") its own testing increases the detection of system/software/hardware errors, especially for high integrity levels because V&V testing can provide the following benefits: 1) examines the stress, abnormal, and fault recovery scenarios that development testing often does not have sufficient time for such test case scenarios and 2) creates additional testing coverage of the system/software/hardware from the V&V perspective of how the system should perform with inputs and stimuli different from the development test scenarios such that those different inputs/stimuli may expose errors undetected in development test scenarios.

NOTE 2 (for Figure 7)—For V&V testing, it is ideal for the V&V processes to have their own test drivers, test simulations, test models, and system test facility. It is not the intended purpose for verification and validation processes to repeat or duplicate the same or similar testing as accomplished by the development processes. Rather, V&V testing provides the objective evidence that the high integrity level requirements perform as intended under all operational scenarios including stress and abnormal conditions, error recovery situations, and high-risk hazardous situations. Having a separate V&V test capability allows for development testing to be uninterrupted (not shared); permits special test scenarios, error injection, and test monitoring capability to be integrated into the test facility; and provides for an independent test capability to investigate erroneous and unusual scenarios without interference with the development test schedule.

However, because of budgetary considerations, the development testing and V&V testing often share the same system test capabilities (e.g., resources or facilities). When sharing the same test capabilities with the development testing, the V&V organization should establish the following procedures on common facility usage to maintain the integrity of both the V&V and development test processes: 1) maintain a controlled copy of system and hardware settings and software elements used in the test facility, such as test drivers, test simulations, interface drivers, and software items that are being tested and undergoing V&V (These configured items and descriptions are required to allow V&V testing to restore the test facility to a known state before beginning V&V testing since items could have been changed during the development testing period); 2) initialize the test facility to the V&V test configuration at the start of V&V testing and return the test facility configuration to the development test process configuration at the conclusion of V&V testing; 3) use V&V test procedures and V&V provided system test data base configurations and scenarios; and 4) monitor and record the V&V test processes and results (although V&V testing may share the test facility operators used by development testing). The V&V organization generates the input data and interfacing data inputs in accordance with the V&V test cases/procedures. The objectives of V&V testing are to surface the system/software/hardware element errors. Often, the small differences in test data and interface data inputs may cause the error source(s) to be evident. The tests of the system/software/hardware during key system transition states and at all boundary conditions are a method of exercising the key requirements of the system. Satisfactory V&V testing results and absence of major errors constitute the objective evidence for verifying and validating the system.

NOTE 3 (for Figure 7)—Hardware integration takes place at many levels. For electronics, this can range from the chip level and board level to a fully integrated computer system that can be integrated into a larger computer network. For a mechanical system, the levels of integration can range from the assembly of a nut-and-bolt to landing gear of an aircraft, or a turbine. At the lowest level of integration, the issues are so elemental or the integration processes so standardized and well understood that V&V of the integration may not be required or applicable. In some cases, as in the assembly of a board, the V&V activities may be carried out by the manufacturer during the integration of the assembled product. It would not be practical or feasible to conduct V&V on the board components. The same would apply to an internal combustion engine where the many components of the engine may be subjected to V&V by the manufacturer and the acquirer can only conduct V&V on the entire assembled product.

5.4.5 Activity: Test V&V

~~Testing includes software testing, software integration testing, software qualification testing, system integration testing, and system qualification testing. The Test V&V activity and its relationship to the software life cycle are shown in Figure 2. The objective of Test V&V is to ensure that the software requirements and system requirements allocated to software are validated by execution of integration, system, and acceptance tests.~~

~~The V&V effort shall perform, as specified in Table 2 for the selected software integrity level, the following Test V&V tasks described in Table 1:~~

~~1) Task: Traceability analysis~~

~~2) Task: Acceptance V&V test procedure generation~~

~~3) Task: Integration V&V test execution~~

~~4) Task: System V&V test execution~~

~~5) Task: Acceptance V&V test execution~~

~~6) Task: Hazard analysis~~

~~7) Task: Security analysis~~

~~8) Task: Risk analysis~~

7. Common V&V activities

7.1 Activity: V&V Management ~~of the~~ effort

The V&V management activity monitors and evaluates all V&V outputs. Management of the V&V effort is performed for all ~~software~~-life cycle processes and activities. This activity involves the following:

- ~~A~~Continual review of the V&V effort
- Revision of the ~~S~~VVP as necessary based ~~up~~on updated project schedules and development status
- Coordination of the V&V results with ~~the developer and~~ other parties performing life cycle activities ~~supporting processes, such as quality assurance and configuration management~~
- Performance of reviews and audits
- Identification of process improvement opportunities in the conduct of V&V

NOTE—V&V Management is focused on those management activities unique to V&V. Other project management activities that are common across all projects, such as cost management and human resource management should also be performed but are not addressed by this standard.

V&V Management assesses each proposed change to the system~~and~~, software, or hardware; identifies the ~~software~~-requirements that are affected by the change; and plans V&V tasks to address the change. For each proposed change, management assesses whether any new hazards or risks are introduced in the ~~software-or-system~~ development process and identifies the impact of the change on the assigned ~~software~~-integrity levels. V&V task planning is revised by adding new V&V tasks or changing the scope or intensity of existing V&V tasks if ~~software~~-integrity levels, hazards, or risks are changed. A baseline change results from changes allocated to ~~software~~ releases in an incremental~~software~~ development process (e.g., planned baseline versions).

Through the use of V&V measures and other qualitative and quantitative measures, ~~this V&V activity develops~~ program trend data and possible risk issues, ~~which~~ are ~~then~~developed and provided to the developer and acquirer to effect timely notification and resolution. At key program milestones (e.g., requirements review, design review, and test readiness), V&V Management consolidates the V&V results to establish supporting evidence of whether to proceed to the next stage of the project life cycle. ~~set of software development activities.~~ Whenever necessary, ~~the~~ V&V Management determines whether a V&V task should be repeated ~~reperformed~~ as a result of changes in the ~~software~~ program.

The V&V effort shall perform, as specified in Table 2a for the selected ~~software~~-integrity level, the following V&V Management tasks described in Table 1a:

- 1) ~~Task:~~ SVVP Generation
- 2) ~~Task:~~ Interface with Other Processes
- 3) ~~Task:~~ Proposed/Baseline Change Assessment
- 4) ~~Task:~~ Management Review of the V&V Effort

5) ~~Task:~~ Management and Technical Review Support

~~5) Task: Interface with organizational and supporting processes~~

6) ~~Task:~~ Identify Process Improvement Opportunities in the Conduct of V&V

7) V&V Final Report Generation

The V&V Management tasks listed shall be used to verify and validate the Software Qualification Testing process outcomes described in ISO/IEC 12207:2008 [B1]. The mappings of IEEE Std 1012-2004 [B5] V&V tasks to ISO/IEC 12207:2008 [B1] Software Qualification Testing process outcomes are contained in Annex L.

7.2 Activity: Acquisition support V&V

~~5.2 Process: Acquisition~~

The acquisition process begins with the definition of the need (e.g., statement of need) to acquire a system, ~~software~~ product, or ~~software~~ service. The process continues with the possible preparation and issuance of a request for proposal (RFP) (e.g., bid request or tender), selection of a supplier, and management of the acquisition process through to the acceptance of the system, ~~software~~ product, or ~~software~~ service.

The acquisition process is used to scope the V&V effort, plan interfaces with the supplier and acquirer, review the draft systems requirements to be included in the RFP, and provide the V&V task results to support acquirer acceptance of the system. Acquirer acceptance of the system culminates after acceptance testing and installation. The V&V acquisition acceptance support activities occur throughout the ~~software~~-life cycle, in conjunction with other interrelated development and V&V tasks, inputs, and outputs.

The Acquisition Support V&V activity addresses project initiation, RFP, contract preparation, supplier monitoring, and acceptance and completion.

The V&V effort shall perform, as specified in Table 2a for the selected ~~software~~-integrity level, the following Acquisition Support V&V tasks described in Table 1a:

- 1) ~~Task~~-Scoping the V&V Effort
- 2) ~~Task~~-Planning the Interface between the V&V Effort and Supplier
- 3) ~~Task~~-System Requirements Review
- 4) ~~Task~~-Acceptance Support

7.3 Activity ~~Process~~ : Supply Planning V&V

The supply process is initiated by either a decision to prepare a proposal to answer an acquirer's request for proposal or by negotiating, finalizing, and entering into a contract with the acquirer to provide the system, ~~software~~-product, or ~~software~~-service. The process continues with the determination of procedures and resources needed to manage the project, including development of project plans and execution of the plans through delivery of the system, ~~software~~-product, or ~~software~~ service to the acquirer.

The Supply V&V effort uses the supply process products to confirm that the request for proposal requirements and contract requirements are consistent and satisfy user needs before the contract is finalized. The V&V planning activity uses the contract requirements, including the program schedules, to revise and update the interface planning between the supplier and [the](#) acquirer.

~~5.3.1 Activity: Planning V&V~~

The [Supply](#) Planning V&V activity addresses the initiation, preparation of response, contract, planning, execution and control, review and evaluation, [as well as](#) ~~and~~ delivery and completion activities.

The V&V effort shall perform, as specified in Table 2a for the selected ~~software~~ integrity level, the following [Supply](#) Planning V&V tasks described in Table 1a:

- 1) ~~Task:~~ Planning the Interface between the V&V Effort and Supplier
- 2) ~~Task:~~ Contract Verification

7.4 Activity: Project Planning V&V

The Project Planning Process determines the scope of the project management and technical activities; identifies process outputs, project tasks, and deliverables; and establishes schedules for project task conduct. The V&V processes use and provide inputs to the Project Planning Process to manage the V&V effort. The V&V organization assesses the Project Planning Process to assure that the project scope is complete and that all activities are defined.

The V&V effort shall perform, as specified in Table 2a for the selected integrity level, the following Project Planning Process V&V tasks described in Table 1a:

1) Project Planning Strategy Assessment

5.4 Process: Development

~~The development process contains the activities and tasks of the developer. The process contains the activities for requirements analysis, design, coding, integration, testing, and installation and support to acceptance of software products.~~

~~The V&V activities verify and validate these software products. The V&V activities are organized into Concept V&V, Requirements V&V, Design V&V, Implementation V&V, Test V&V, and Installation and checkout V&V.~~

7.5 Activity: Configuration Management V&V

The V&V processes use and provide inputs to the Configuration Management Process throughout the project life cycle. The V&V processes assess the Configuration Management Process to assure that the following occur:

- a) A configuration management strategy is defined for the program that will assure configuration controls are in place to maintain and document configuration item baselines with unique identifiers. The strategy should include notification to the V&V effort for all changes made to the configuration item baselines.

- b) Items requiring configuration management are defined and documented. The items controlled should include enabling the systems, tools, and processes that are integral to system development and life cycle support that will be subject to V&V to demonstrate conformance to this standard.

- c) The status of items under configuration management is made available throughout the life cycle. Provisions should be included to assure that the V&V effort receives the current status for all configuration items.

If Configuration Management issues are discovered during the V&V effort that indicate configuration baselines are not established or controlled or if the configuration of released items is not controlled, the issues should be documented and provided to the system developer and to the V&V customer for resolution.

The V&V effort shall perform, as specified in Table 2a for the selected system integrity level, the following Configuration Management Process V&V tasks described in Table 1a:

1) Configuration Management Assessment

Table 1 a— V&V tasks, inputs, and outputs

7.1. Activity: <u>V&V Management (Common)</u> Management of the V&V Effort (Process: Management)		
V&V tasks	Required inputs ^a	Required outputs
(1) SVVP Generation a) Generate a SVVP for all life cycle processes. The SVVP may require updating throughout the life cycle. Outputs of other activities are inputs to the SVVP. b) Establish a baseline SVVP prior to the Requirements V&V activities, c) Identify project milestones in the SVVP. d) Schedule V&V tasks to support project management reviews and technical reviews. See Clause 7 for an example SVVP outline and content of the SVVP.	SVVP (previous update Contract Concept documentation (e.g., statement of need, advance planning report, project initiation memo, feasibility studies, system requirements, governing regulations, procedures, policies, customer acceptance criteria and requirements, acquisition documentation, business rules, <u>and</u> draft system architecture)	SVVP and updates

	Supplier development plans and schedules	
(2) Interface with Other Processes <u>a) Coordinate the V&V effort with Organizational Project-Enabling Processes and Project Processes,</u> <u>b) Identify the V&V data to be exchanged with these processes,</u> <u>c) Document the data exchange requirements in the VVP.</u>	<u>VVP</u> <u>Data identified in the VVP from the Organizational Project-Enabling Processes and Project Processes</u>	<u>Updated VVP</u>
(3) Proposed/Baseline Change Assessment a) Evaluate proposed software changes (i.e., modifications, enhancements, and additions as a result of anomaly corrections or requirement changes) for effects on the system and previously completed V&V tasks, b) Plan iteration of affected tasks or initiate new tasks to address software <u>the</u> proposed changes or baseline changes associated with an iterative development process, c) Verify and validate that the change is consistent with system requirements and does not adversely affect requirements directly or indirectly. An adverse effect is a change that could create new system hazards and risks or impact previously resolved hazards and risks.	S VVP Proposed changes Hazard analysis report Risks identified by V&V tasks Supplier development plans and schedules	Task report(s) - Proposed/base line change assessment Updated S VVP Anomaly report(s)

7.1. Activity: V&V Management of the Effort (Common) (Process: Management)		
V&V tasks	Required inputs ^a	Required outputs
	Developer products (produced to date)	
4) Management Review of the V&V Effort a) Review and summarize the V&V effort to define changes to V&V tasks or to redirect the V&V effort, b) Evaluate each anomaly for its impact on the software system and assess whether it is a criticality anomaly (e.g., IEEE Std 1044 TM -2009 1993[B78]). <u>Assess anomalies and risks for trends and tendencies.</u> The scope and application of V&V activities and tasks shall be revised to address the causes of these anomalies and risks, c) Recommend whether to proceed to the next <u>stage of the project</u> set of V&V and development life cycle activities , and provide task reports, anomaly reports, and V&V Activity Summary Reports to the organizations identified in the SVVP. d) Verify that all V&V tasks conform to task requirements defined in the SVVP. e) Verify that V&V task results have a basis of evidence supporting the results, f) Assess all V&V results and provide recommendations for program acceptance and certification as input to the V&V final report, g) Use results of review to identify process improvement opportunities in the conduct of V&V. h) Review the quality of the products and services	VVP and updates Supplier development plans and schedules Anomaly reports V&V task results [e.g., technical accomplishments, V&V reports, resource utilization, V&V measures (see Annex E), plans, and identified risks]	Task report(s) – Recommendations Updated VVP V&V activity summary reports Recommendations to the V&V final report

to assure they meet customer requirements, i) Review the program risks and initiate actions to mitigate above threshold risks, j) Review program measures to <u>assure</u> ensure the quality of products and processes. <u>NOTE</u>—The management review of V&V may use any review methodology (e.g., IEEE Std 1028TM-2008 1997 [B6]).		
(5) Management and Technical Review Support a) Support project management reviews and <u>provide V&V results at project defined</u> technical reviews (e.g., preliminary design review, and critical design review) by assessing the review materials, attending the reviews, and providing task reports and anomaly reports. <u>Provide objective evidence on the status of development products (system, software, hardware) to project management authorities, including recommendations and assessment of positive trends and anomaly trends. This evidence can be used to determine whether the criteria of major program milestones have been achieved and whether to proceed to the next stage of the project life cycle. Provide an assessment of the positive trends and anomaly trends to allow greater focus toward the functional program requirements warranting more resources and additional analysis support or to require changes in V&V resources or tasks,</u>	V&V task results Materials for review (e.g., operational concept document [OCD], architectural design documents, system requirements document (SRD), SRS, IRS, SDD, IDD, and test documents)	Task report(s) - Review results Anomaly report(s)

b) Verify timely delivery according to the approved schedule of all software products and documents. The management and technical review support may use any review methodology (e.g., IEEE Std 1028-1997 [B7]).		
(5) Interface with organizational and supporting processes a) Coordinate the V&V effort with organizational (e.g., management, improvement) and supporting processes (e.g., quality assurance, joint review, and problem resolution). b) Identify the V&V data to be exchanged with these processes. c) Document the data exchange requirements in the SVVP.	SVVP Data identified in the SVVP from organizational and supporting processes	Updated SVVP

7.1. Activity: V&V Management (<u>Common</u>)		
V&V tasks	Required inputs	Required outputs
(6) Identity Process Improvement Opportunities in the Conduct of V&V a) Gather and analyze the lessons learned, b) Gather and analyze the risks identified, c) Gather and analyze the V&V measures, d) Identify and analyze deficiencies in the V&V processes, e) Determine and implement corrective actions <u>using</u>	<u>V&V task results</u> <u>Materials for review (e.g., OCD, architectural design documents, SRD, SRS, IRS,</u>	<u>Task report(s) - Review results</u> <u>Anomaly report(s)</u>

<u>best practices</u> (e.g., repeat V&V tasks or conduct a new V&V task to address the corrective action or use a different method/technique for executing a V&V task), f) Monitor the efficacy of the corrective actions, g) Document findings in final report.	<u>SDD, IDD, and test documents)</u>	
<u>(7) V&V Final Report Generation</u> a) <u>Summarize in the V&V final report the V&V activities, tasks, and results, including status and disposition of anomalies,</u> b) <u>Provide an assessment of the overall quality and provide recommendations.</u> NOTE— <u>The V&V Final Report is intended to be produced at the completion of V&V following development, operations, maintenance, or retirement.</u>	<u>V&V Activity summary report(s)</u>	<u>Task report(s) - V&V final report</u>

5.1.1 Activity: Management of the V&V Effort (Process: Management) (continued)		
V&V tasks	Required inputs ^a	Required outputs
(6) Identify process improvement opportunities in the conduct of V&V a) Gather and analyze the lessons learned. b) Gather and analyze the risks identified. c) Gather and analyze the V&V measures. d) Identify and analyze deficiencies in the V&V process.	SVVP Results of analyses Prior end of activity	Updated SVVP Input to the end of activity report Input to the final report

e) Determine and implement corrective actions (e.g., repeat V&V tasks or conduct a new V&V task to address the corrective action or use a different method/technique for executing a V&V task). f) Monitor the efficacy of the corrective actions. g) Document findings in final report.	reports Task reports	New/updated V&V policies/procedures/reports Updated V&V infrastructure
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7.2. Activity: Acquisition Support V&V (Common , Process: Acquisition)		
V&V tasks	Required inputs	Required outputs
(1) Scoping the V&V Effort a) Determine the characteristics (e.g., complexity, criticality, risk, safety level, security level, desired performance, reliability, or other projectunique characteristics) that define the importance of the software system of interest to the user. b) Adopt the system integrity scheme assigned to the project. If no system integrity level scheme exists, then one is selected. c) Assign a software integrity level to the system of interest and the software. d) Establish the degree of independence (see Annex C), if any, required for the V&V. e) Determine the minimum V&V tasks for the software integrity level using Table 2a through	Preliminary system description Statement of need Draft RFP or tender System integrity level scheme	SVVP

Table 2d and the selected software integrity level scheme. f) Determine the extent of V&V on reuse items software selected for the program (see Annex D). g) Determine the extent of V&V for tools that insert or translate code (e.g., optimizing compilers and auto-code generators). h) Augment the minimum V&V tasks with optional V&V tasks, as necessary. i) Provide an estimate of the V&V budget, including test facilities, and tools as required.		
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7.2. Activity: Acquisition Support V&V (Common , Process: Acquisition)		
V&V tasks	Required inputs	Required outputs
(2) Planning the Interface between the V&V Effort and Supplier (Preparing the preliminary data and processes for the interface with a supplier to be selected in the supply process) a) Plan the V&V schedule for each V&V task. b) Identify the preliminary list of development processes and products to be evaluated by the V&V processes. c) Describe V&V access rights to proprietary and classified information, d) Coordinate the plan with the acquirer. e) Incorporate the project software integrity level scheme into the planning process.	SVVP Draft RFP or tender Contract	Task Report(s) - Recommendation s for RFP or tender Updated SVVP
(3) System Requirements Review a) Review the system requirements (e.g., system	Preliminary system	Task report(s) - System

requirements specification, feasibility study report, business rules description) in the RFP or tender to perform the following: 1) Verify the consistency of requirements to user needs. 2) Validate whether the requirements can be satisfied by the defined technologies, methods, and algorithms defined for the project (feasibility). 3) Verify whether objective information that can be demonstrated by testing is provided in the requirements (testability). b) Review other requirements, such as deliverable definitions, listing of appropriate compliance standards and regulations, user needs, etc., for completeness, correctness, and accuracy.	description Statement of need User needs Draft RFP or tender	requirements review Anomaly report(s)
(4) Acceptance Support (The following V&V activities support Acceptance in the acquisition process. The activities are described in various V&V tasks across the life cycle where required inputs for the V&V activities are generated to aid the understanding of the activity flow.) a) System Acceptance Test Plan V&V generation [5.4 8.2, Task 7]. b) System Acceptance Test Design V&V generation [5.4 8.3, Task 7]. c) System Acceptance Test Case V&V generation [5.4 8.4, Task 7]. d) System Acceptance Test Procedure V&V generation [5.4 8.5, Task 6 2]. e) System Acceptance Test Execution V&V [(5.4.5 8.6, Task 3 5].	Concept documentation SDD IDD SRS IRS Source code Hardware items Executable code User	Task report(s) - Acceptance V&V Acceptance V&V test plan, design(s), cases, procedures, test results Anomaly report(s)

f) Software Acceptance Test Plan V&V [9.2, Task 6]. g) Software Acceptance Test Design V&V [9.3, Task 10]. h) Software Acceptance Test Case V&V [9.4, Task 8]. i) Software Acceptance Test Procedure V&V [9.7, Task 1]. j) Software Acceptance Test Execution V&V [9.7, Task 2]. k) Hardware Acceptance Test Plan V&V [10.2, Task 6]. l) Hardware Acceptance Test Design V&V [10.3, Task 10]. m) Hardware Acceptance Test Case V&V [10.4, Task 8]. n) Hardware Acceptance Test Procedure V&V [10.7, Task 2]. o) Hardware Acceptance Test Execution V&V [10.7, Task 3].	documentation Test plans, designs, cases, procedures, results Acceptance test plan V&V task results	
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7.3. Activity: Supply Planning V&V (Common, Process: Supply)		
V&V tasks	Required inputs	Required outputs
(1) Planning the Interface between the V&V Effort and Supplier (Coordinating and documenting the interface data and processes with the selected supplier) a) Review the supplier development plans and schedules to coordinate the V&V effort with development activities. b) Establish procedures to exchange V&V data and results with the development effort. c) Coordinate the plan with the supplier.	SVVP Contract Supplier development plans and schedules	Updated SVVP
(2) Contract Verification Verify the following characteristics of the contract : a) System requirements (from RFP or tender, and contract) satisfy and are consistent with user needs. b) Procedures are documented for managing requirement	SVVP RFP or tender Contract	Task Report(s) - Contract verification Updated SVVP

changes and for identifying the management hierarchy to address problems.	User needs	Anomaly report(s)
c) Procedures for interface and cooperation among the parties are documented, including ownership, warranty, copyright, and confidentiality.	Supplier development plans and	
d) Acceptance criteria and procedures are documented in accordance with requirements.	schedules	

<u>7.4. Activity: Project Planning V&V (Common, Process: Project Planning)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
<u>(1) Project Planning Strategy Assessment</u>	<u>Project plan</u>	<u>Taskreport(s) -</u>
<u>a) Verify the project plan strategy, including objectives and constraints.</u>	<u>strategy</u>	<u>Project planning</u>
<u>b) Verify project scope includes all relevant activities.</u>	<u>Schedules and</u>	<u>Anomaly</u>
<u>c) Verify life cycle model is defined.</u>	<u>Tasks</u>	<u>report(s)</u>
	<u>Requirements</u>	
	<u>Life cycle model</u>	
	<u>Project scope</u>	

7.5. Activity: Configuration Management V&V (Common, Process: Configuration Management)		
V&V tasks	Required inputs	Required outputs
(1) Configuration Management Assessment Verify that the configuration management process is complete and adequate. The task criteria are as follows: a) Completeness Verify that there is a process for describing the system, software and hardware product functionality, tracking program versions, generating baselines (including parameters and settings), and managing changes. b) Adequacy Verify that the configuration management process is adequate for the development complexity, system size, integrity level, project plans, and user needs.	Configuration management process documentation	Task report(s) – Configuration management assessment Anomaly report(s)

NOTE (for Table 1a)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.

5.4.2 Activity: Requirements V&V (Process: Development) (continued)		
V&V tasks	Required inputs	Required outputs
(7) Configuration management assessment Verify that the configuration management process is complete and adequate. The task criteria are a) Completeness Verify that there is a process for describing the software product functionality, tracking program versions, and managing changes. b) Adequacy Verify that the configuration management process is adequate for the development complexity, software and system size, software integrity level, project plans, and user needs.	Software configuration management process documentation	Task report(s)— Configuration management assessment Anomaly report(s)
(8) Hazard analysis a) Determine software contributions to system hazards. The hazard analysis shall 1) Identify the software requirements that contribute to each system hazard. 2) Validate that the software addresses, controls, or mitigates each hazard.	SRS IRS Hazard analysis report	Task report(s)— Hazard analysis Anomaly report(s)
(9) Security analysis a) Determine that the security requirements identified in the SRS and IRS address the security risks introduced by the system	SRS IRS Preliminary TRA	Task report(s)— Security analysis Anomaly

concept: b) Verify that the system security requirements will mitigate the identified security risks to an acceptable level.		report(s)
(10) Risk analysis a) Review and update risk analysis using prior task reports. b) Provide recommendations to eliminate, reduce or mitigate the risks.	Concept documentation SRS IRS Supplier development plans and schedules Hazard analysis report Security analysis V&V task results	Task report(s) Risk analysis Anomaly report(s)

**Table 2a-Minimum V&V tasks assigned to each integrity level for
common v&v**

<u>V&V Activities</u>	<u>Activity:</u> <u>V&V</u> <u>Management</u> <u>(see 7.1)</u>				<u>Activity:</u> <u>Acquisition</u> <u>Support V&V</u> <u>(see 7.2)</u>				<u>Activity:</u> <u>Supply</u> <u>Planning</u> <u>V&V</u> <u>(see 7.3)</u>				<u>Activity:</u> <u>Project</u> <u>Planning</u> <u>V&V</u> <u>(see 7.4)</u>				<u>Activity:</u> <u>Configuration</u> <u>Management</u> <u>V&V</u> <u>(see 7.5)</u>			
<u>Integrity Levels</u>	<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>			
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>
<u>Acceptance Support</u>					X	X	X	X												
<u>Configuration Management</u> <u>Assessment</u>																	X	X	X	X
<u>Contract Verification</u>									X	X	X	X								
<u>Identify Process</u> <u>Improvement</u> <u>Opportunities in the</u> <u>Conduct of V&V</u>	X	X	X	X																
<u>Interface with other Processes</u>	X	X	X	X																
<u>Management and</u> <u>Technical Review</u> <u>Support</u>	X	X	X	X																
<u>Management Review of the</u> <u>V&V Effort</u>	X	X	X	X																
<u>Planning the Interface</u> <u>between the V&V Effort and</u> <u>Supplier</u>					X	X	X	X	X	X	X	X								
<u>Project Planning Strategy</u> <u>Assessment</u>													X	X	X	X				
<u>Proposed/Baseline Change</u> <u>Assessment</u>	X	X	X	X																
<u>Scoping the V&V Effort</u>					X	X	X	X												
<u>System Requirements Review</u>					X	X	X	X												
<u>V&V Final Report Generation</u>	X	X	X	X																
<u>VVP Generation</u>	X	X	X	X																

NOTE (for Table 2a)—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of the analysis or test. A higher integrity-level implementation requires greater rigor (e.g., formal methods or structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, or comprehensive fault and recovery scenarios) of the analysis or test than the lower integrity-level implementation.

Table 3a—Optional V&V tasks and suggested applications in System Agreement and Project Processes

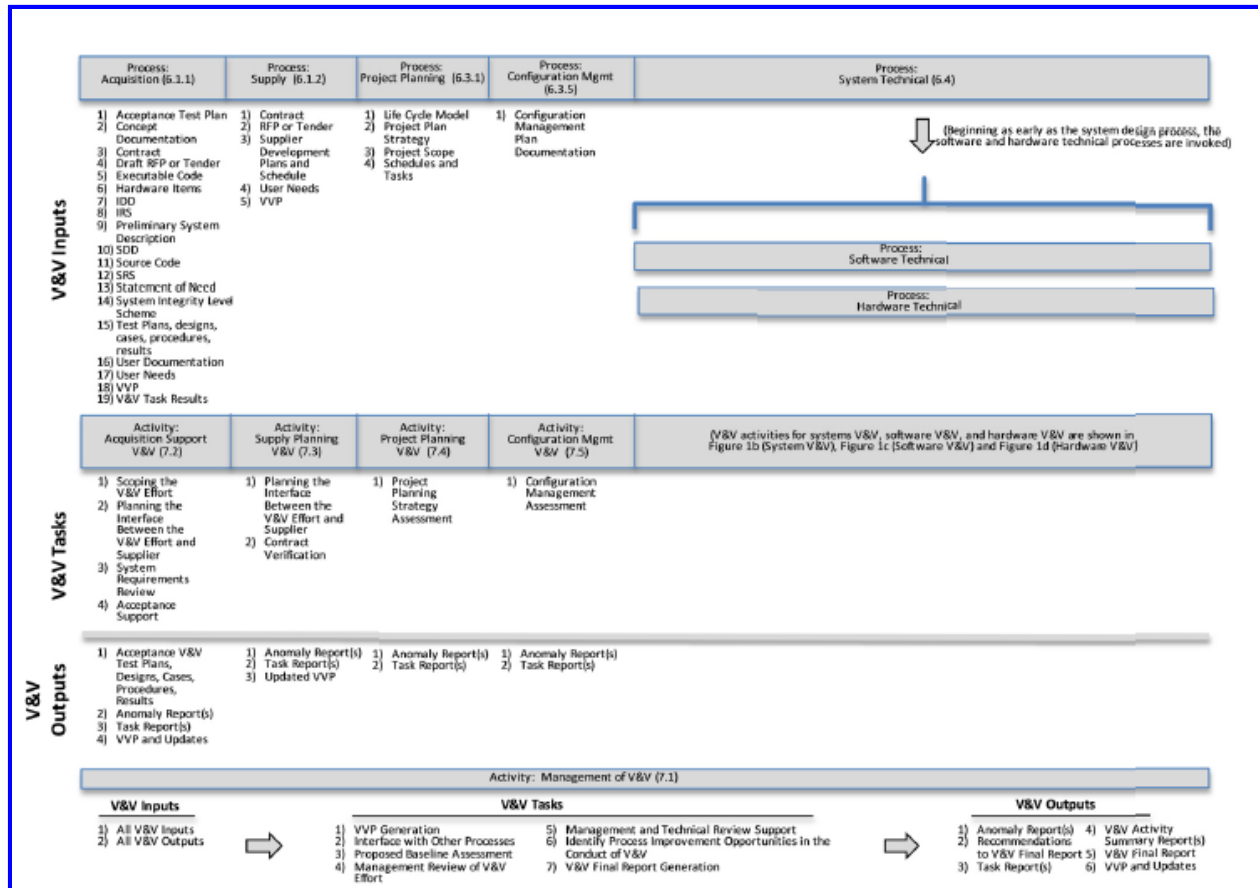
	Acquisition	Supply	Project Planning
Algorithm analysis		X	
Audit performance			
Audit support			X
Control flow analysis			
Cost analysis	X	X	X
Database analysis			
Data flow analysis			
Disaster recovery plan assessment			X
Distributed architecture assessment			
Feasibility study evaluation	X	X	X
Independent risk assessment			
Inspection			
Concept			
Requirements			
Design			
Source code			
Test plan			
Test design			
Test case			

Operational evaluation		<u>X</u>	
Performance monitoring	<u>X</u>		
Post installation validation			
Project management oversight support	X	<u>X</u>	X
Proposal evaluation support	X		
Qualification testing			
Regression analysis and testing			
Reusability analysis	X	<u>X</u>	X
Reuse analysis	X	<u>X</u>	X
Simulation analysis		<u>X</u>	
Sizing and timing analysis		<u>X</u>	
System software assessment			
Test certification			
Test evaluation			
Test witnessing			
Training document evaluation			
Usability analysis			
User documentation evaluation			<u>X</u>
User training			<u>X</u>
V&V tool plan generation	X		X
<u>V&V tool qualification</u>	<u>X</u>	<u>X</u>	<u>X</u>

	Acquisition	Supply	Project Planning
Walkthroughs			
Design			
Requirements			
Source code			
Test			
Work breakdown structure evaluation			X

Annex G contains a description of the optional V& V tasks.

Figure 1a—Summary of common V&V activities and tasks



(NEW)

NOTE 1—Clause references in the Process Definitions (top graphic bar) are ISO/IEC 15288:2008 [B16] clause numbers.

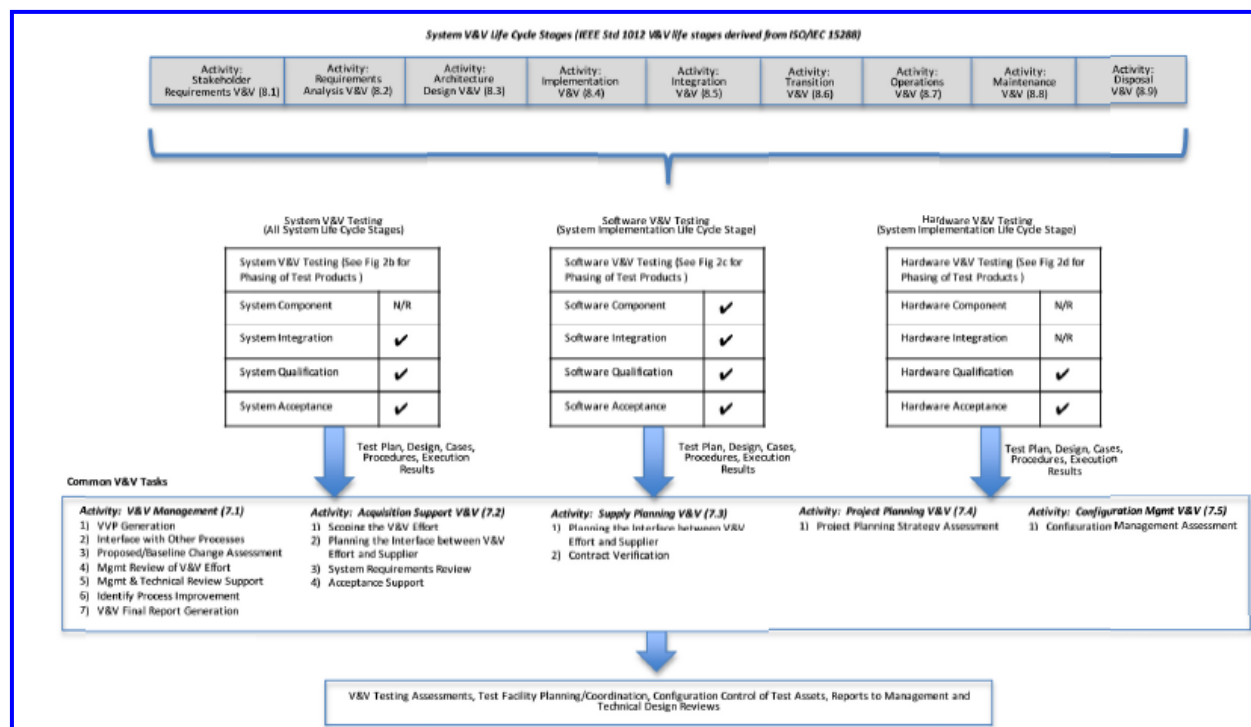
NOTE 2—The V&V tasks may be performed concurrently. The sequential (waterfall) model is shown as an example of a life cycle model.

NOTE 3—V&V tasks listed in the figure are the minimum required for integrity level 4 (highest integrity level).

NOTE 4—Table 1a (common V&V tasks), Table 1b (system V&V tasks), Table 1c (software V&V tasks), and Table 1d (hardware V&V tasks) contain a complete list of minimum V&V tasks. Table 2a (common V&V tasks), Table 2b (system V&V tasks), Table 2c (software V&V tasks), and Table 2d (hardware V&V tasks) define the minimum V&V tasks for each integrity level.

NOTE 5—The clause numbers in the Activity V&V graphic bar (s) correspond to the Clause 7 of this standard where the common V&V processes and activities can be found.

Figure 2a – Summary of V&V test products and tasks



(New)

NOTE 1—N/R = not required for integrity level 4; S = required for integrity level 4. Diagram illustrates integrity level 4 V&V testing tasks.

NOTE 2—System component consists of software or hardware elements. Because those are tested as part of software and hardware V&V testing, no system component V&V testing is specified.

NOTE 3—The V&V activity clauses shown in the diagram are IEEE 1012 clauses.

NOTE 4—System V&V testing covers test planning to execution (spanning system life cycle stages of stakeholder requirements through transition). Software/hardware V&V testing covers test planning to execution (spanning software/hardware life cycle stages of requirements analysis through transition). Software/hardware life cycle stages all occur within the system implementation life cycle stage.

Table 3—Optional V&V tasks and suggested applications in the life cycle

V&V life-cycle activities	Activity: Management of V&V (see 5.1.1)	Activity: Acquisition support V&V (see 5.2.1)	Activity: Planning V&V (see 5.3.1)	Activity: Concept V&V (see 5.4.1)	Activity: Requirements V&V	Activity: Design V&V (see 5.4.3)	Activity: Implementation V&V	Activity: Test V&V (see 5.4.5)	Activity: Installation/checkout V&V	Activity: Operation V&V (see 5.5.1)	Activity: Maintenance V&V
Algorithm analysis					*	*	*				*
Audit performance					*	*	*	*	*		*
Audit support	*				*	*	*	*	*		*
Control flow analysis					*	*	*				*
Cost analysis	*	X	X	*	*	*	*	*	*		*
Database analysis					*	*	*	*			*
Data flow analysis					*	*	*				*
Disaster recovery plan assessment	*			*	*	*	*			*	*
Distributed architecture assessment				*	*	*					*
Feasibility study evaluation	*	X	X	*	*	*					*
Independent risk assessment	*	*	*	*	*	*	*	*	*	*	*
Inspection											
Concept				*							*
Requirements					*						*
Design						*					*
Source code							*				*
Test plan					*	*	*		*		*
Test design						*	*		*		*
Test case						*	*	*	*		*
Operational evaluation										*	
Performance monitoring				*	*	*	*	*	*	*	*
Post installation validation									*	*	*
Project management oversight support	*	X	X	*	*	*	*	*	*	*	*
Proposal evaluation support		X									
Qualification testing								*	*		*
Regression analysis and testing					*	*	*	*	*		*

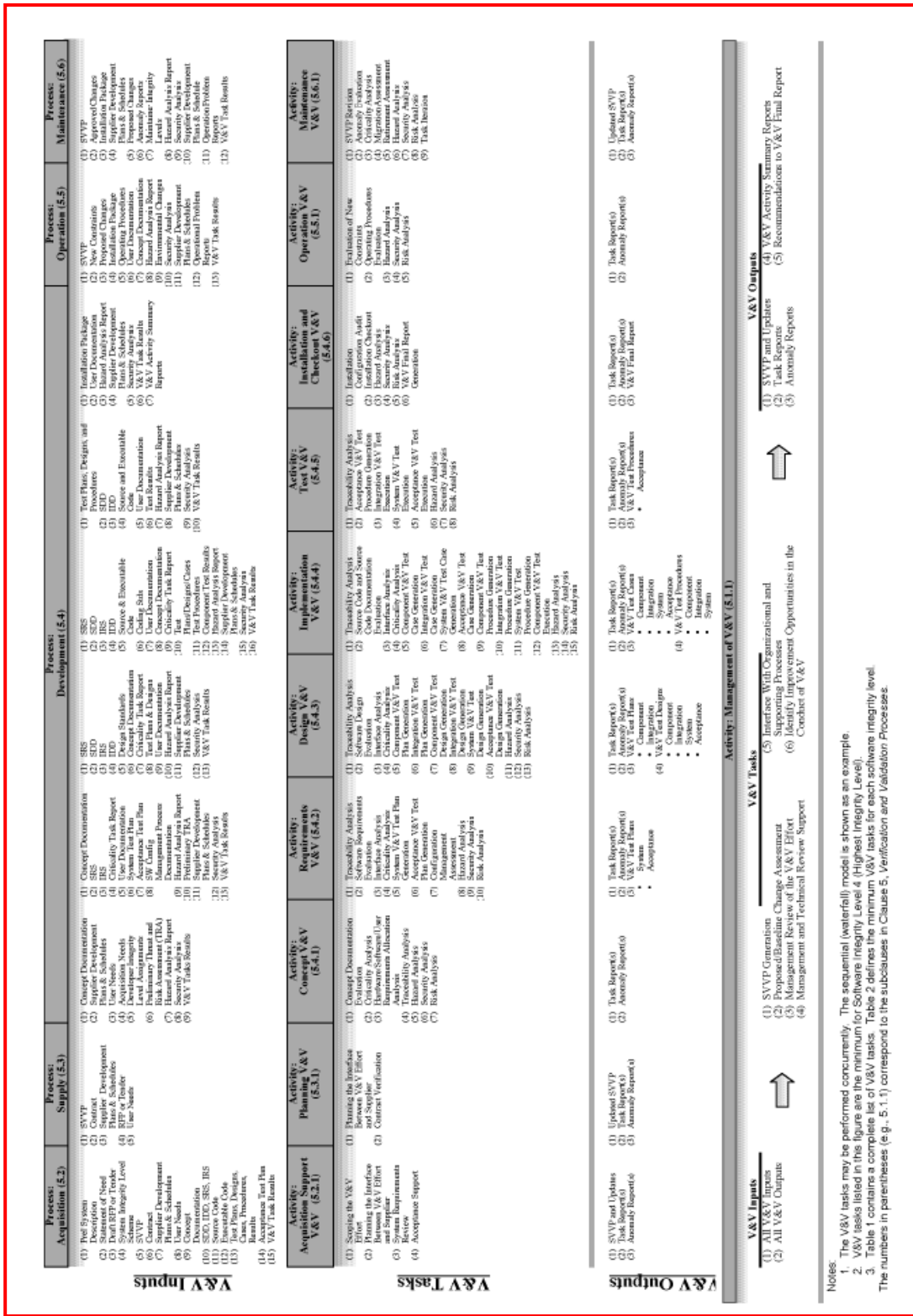
Table 3—Optional V&V tasks and suggested applications in ~~the life cycle~~
(continued)

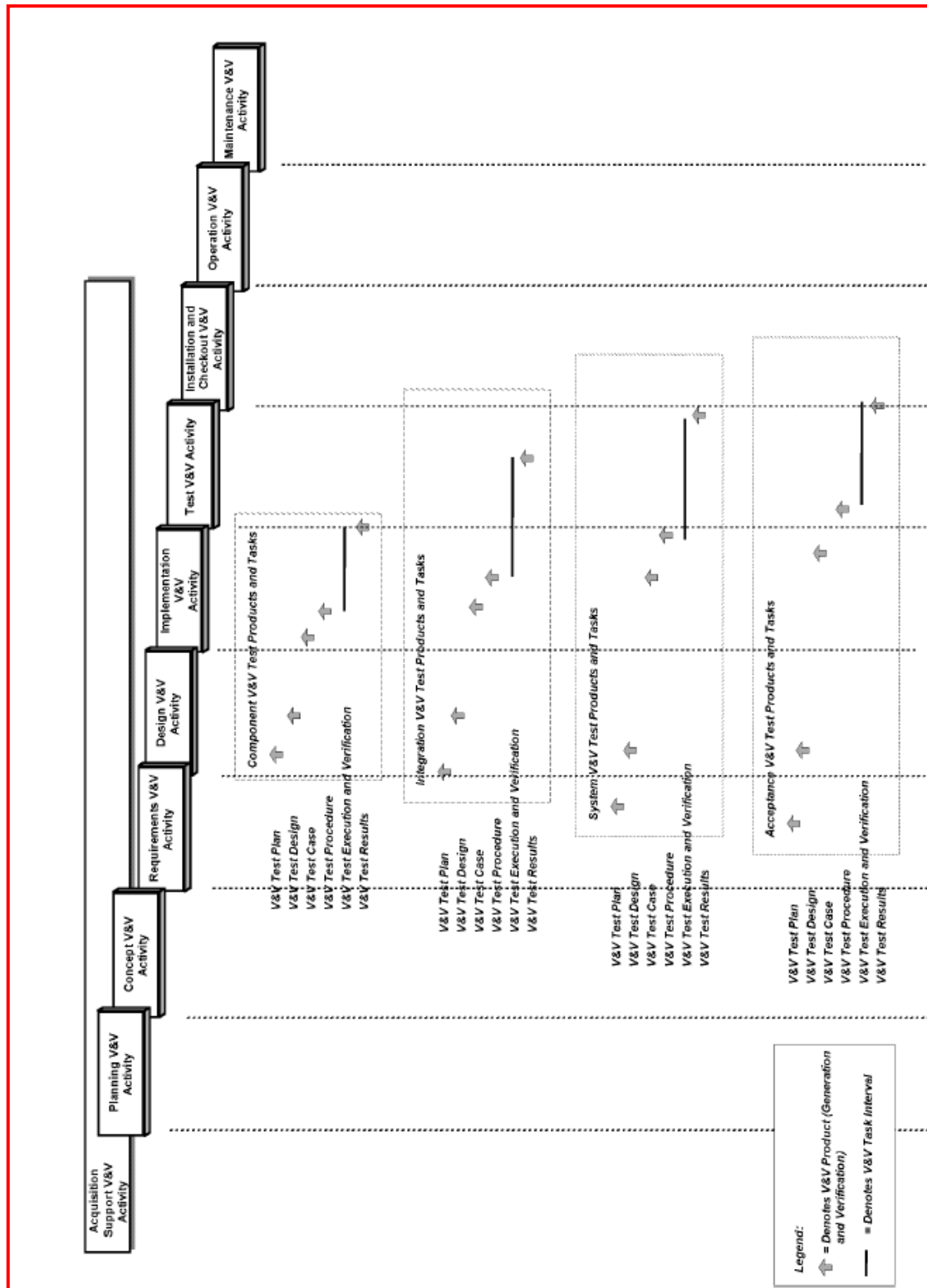
<i>V&V life cycle activities</i>	<i>Management of V&V (see 5.1.4)</i>	<i>Mission support V&V (see 5.2.1)</i>	<i>Activity Planning V&V (see 5.3.1)</i>	<i>Activity-Concept V&V (see 5.4.1)</i>	<i>Activity-Requirements V&V</i>	<i>Activity-Design V&V (see 5.4.3)</i>	<i>Activity-Implementation V&V</i>	<i>Activity-Test V&V (see 5.4.5)</i>	<i>Activity-Installation/checkout V&V</i>	<i>Activity-Operation V&V (see 5.5.1)</i>	<i>Activity-Maintenance V&V</i>
Reusability analysis	✗	X	X	✗	✗	✗	✗				✗
Reuse analysis	✗	X	X	✗	✗	✗					✗
Simulation analysis				✗	✗	✗	✗	✗	✗	✗	✗
Sizing and timing analysis					✗	✗	✗	✗			✗
System software assessment						✗	✗	✗	✗	✗	✗
Test certification								✗	✗	✗	✗
Test evaluation					✗	✗	✗	✗	✗	✗	✗
Test witnessing								✗	✗	✗	✗
Training document evaluation					✗	✗	✗	✗	✗	✗	✗
Usability analysis				✗	✗	✗	✗	✗	✗	✗	✗
User documentation evaluation	✗			✗	✗	✗	✗	✗	✗	✗	✗
User training	✗							✗	✗	✗	✗
V&V tool plan generation	✗	X	X								✗
Walk-throughs											
Design						✗					✗
Requirements					✗						✗
Source code							✗				✗
Test								✗	✗		✗

Annex G contains a description of the optional V&V tasks.

(Deleted)

Figure 1 An example of V&V processes, activities, and tasks





(Deleted)

Figure 2—An example of V&V test product and test execution task-scheduling

8. System V&V activities

8.1 Activity: Stakeholder Requirements Definition ~~Concept~~ V&V

The purpose of the Stakeholder Requirements Definition Process is to define the requirements for a system that can provide the services needed by users and other stakeholders in a defined environment. It identifies stakeholders, or stakeholder classes, involved with the system throughout its life cycle, and their needs, expectations, and desires. It analyzes and transforms these into a common set of stakeholder requirements that express the intended interaction the system will have with its operational environment and that are the reference against which each resulting operational service is validated (ISO/TEC 15288:2008 [B16]).

~~The concept activity represents the delineation of a specific implementation solution to solve the user's problem. During the concept activity, the system architecture is selected and system requirements are allocated to hardware, software, and user interface components. The Concept V&V activity addresses system architectural design and system requirements analysis. The objective of Concept V&V is to verify the allocation of system requirements, validate the selected solution, and ensure that no false assumptions have been incorporated in the solution.~~

The objectives of Stakeholder Requirements Definition V&V are to analyze and assess whether the stakeholder requirements (performance, safety, security, maintenance, interface, logistics, and user): 1) specify the required characteristics and context of use of services and operational concepts; 2) define all constraints on a system solution; 3) are traceable to the originating stakeholders; 4) are complete, unambiguous, correct, and accurate; and 5) can be validated by measurable analyses or tests.

The V&V effort shall perform, as specified in Table 2b for the selected ~~software~~ integrity level, the following [Stakeholder Requirements Definition](#) ~~Concept~~ V&V tasks described in Table 1b:

- 1) [Stakeholder Requirements Evaluation](#)
- 2) ~~Task:~~ Traceability Analysis
- ~~1) Task: Concept documentation evaluation~~
- 3) ~~Task:~~ Criticality Analysis
- 4) ~~Task:~~ Hazard Analysis
- ~~3) Task: Hardware/software/user requirements allocation analysis~~
- 5) ~~Task:~~ Security Analysis
- 6) ~~Task:~~ Risk Analysis

[The Stakeholder Requirements Definition V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 \[B16\]. The mappings of IEEE 1012 V&V tasks to ISO/IEC 15288:2008\(E\) \[B16\] Stakeholder Requirements Definition process outcomes are contained in Annex L.](#)

8.2 Activity: Requirements [Analysis](#) V&V

[The purpose of the Requirements Analysis Process is to transform the stakeholder, requirement-driven view of desired services into a technical view of a required product that could deliver those services. This process builds a representation of a future system that will meet stakeholder requirements and that, as far as constraints permit, does not imply any specific implementation. It results in measurable system requirements that specify, from the supplier's perspective, what characteristics the system is to possess and with what magnitude in order to satisfy stakeholder requirements \(ISO/IEC 15288:2008 \[B16\]\).](#)

~~The Requirements V&V activity addresses software requirements analysis of the functional and performance requirements, interfaces external to the software, and requirements for qualification, safety and security, human factors engineering, data definitions, user documentation for the software, installation and acceptance, user operation and execution, and user maintenance. V&V test planning begins during the Requirements V&V activity and spans several V&V activities.~~

The objectives of Requirements Analysis V&V are to analyze and assess whether the system requirements: 1) specify all required characteristics, attributes, functional and performance requirements, interface requirements, and requirements for qualification, safety and security, human factors engineering, and user documentation for the system; 2) specify all constraints that will affect the architectural design of the system and the means to realize the system; 3) are unique, complete, unambiguous, consistent with all other requirements, implementable, and verifiable; 4) are traceable to the stakeholder requirements; and 5) provide a basis for verifying that each system requirement can be satisfied.

~~The objective of Requirements V&V is to ensure the correctness, completeness, accuracy, testability, and consistency of the system software requirements.~~

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Requirements Analysis V&V tasks described in Table 1b:

- 1) Requirements Evaluation
- 2) ~~Task:~~ Interface Analysis
- 3) ~~Task: Software requirements evaluation~~
- 4) ~~Task:~~ Traceability Analysis
- 5) ~~Task:~~ Criticality Analysis
- 6) System Integration Test Plan V&V
- 7) System Qualification Test Plan V&V
- 8) ~~Task:~~ System Acceptance Test Plan V&V ~~generation~~

- 9) ~~Task:~~ Hazard Analysis
- 10) ~~Task:~~ Security Analysis
- 11) ~~Task:~~ Risk Analysis
- 5) ~~Task: System V&V test plan generation~~
- 7) ~~Task: Configuration management assessment~~

The Requirements Analysis V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Requirements Analysis process outcomes are contained in Annex L.

8.3 Activity: Architectural Design V&V

The purpose of the Architectural Design Process is to synthesize a solution that satisfies system requirements. This process encapsulates and defines areas of solution expressed as a set of separate problems of manageable, conceptual, and ultimately, realizable proportions. It identifies and explores one or more implementation strategies at a level of detail consistent with the system's technical and commercial requirements and risks. From this, an architectural design solution is defined in terms of the requirements for the set of system elements from which the system is configured. The specified design requirements resulting from this process are the basis for verifying the realized system and for devising an assembly and verification strategy (ISO/IEC 15288:2008 [B16]).

~~In software design, software requirements are transformed into an architecture and a detailed design for each software component. The design includes databases and system interfaces (e.g., hardware, operator/user, software components, and subsystems). The Design V&V activity addresses software architectural design and software detailed design. V&V test planning continues during the Design V&V activity.~~

The objectives of Architectural Design V&V are to assure 1) that the architecture (e.g., hardware, software, interfaces, and communication) satisfies the system requirements, 2) that the system architecture is realizable, 3) that the selected architecture is based on specified selection criteria, 4) that the basis for verifying the system elements is defined, and 5) that the basis for integration of the system elements is established.

~~The objective of Design V&V is to demonstrate that the design is a correct, accurate, and complete transformation of the software requirements and that no unintended features are introduced.~~

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Architectural Design V&V tasks described in Table 1b:

1)Design Evaluation

2)~~Task:~~ Interface Analysis

3)~~Task:~~ Traceability Analysis

4)~~Task:~~ Criticality Analysis

5)~~Task:~~ System Integration Test Design V&V ~~generation~~

6)~~Task:~~ System Qualification Test Design V&V ~~generation~~

7)~~Task:~~ System Acceptance Test Design V&V ~~generation~~

8)~~Task:~~ Hazard Analysis

9)~~Task:~~ Security Analysis

10) ~~Task:~~ Risk Analysis

~~3) Task: Software design evaluation~~

~~4) Task: Component V&V test plan generation~~

~~7) Task: Component V&V test design generation~~

The Architectural Design V&V tasks listed above shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/TEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks to ISO/TEC 15288 Architectural Design process outcomes are contained in Annex L.

8.4 Activity: Implementation V&V

The purpose of the Implementation Process is to realize a specified system element. This process transforms specified behavior, interfaces, and implementation constraints into fabrication actions that create a system element according to the practices of the selected implementation technology. The system element is constructed or adapted by processing the materials and/or information appropriate to the selected implementation technology and by employing appropriate technical specialties or disciplines. This process results in a system element that satisfies specified design requirements through verification and stakeholder requirements through validation (ISO/TEC 15288:2008 [B16]).

~~In software implementation, the system design is transformed into code, database structures, and related machine-executable representations. The Implementation V&V activity addresses software coding and testing, including the incorporation of reused software products. The objective of Implementation V&V is~~

~~to verify and validate that these transformations are correct, accurate, and complete.~~

The objectives of Implementation V&V are to 1) verify the implementation activities performed produce a system element that conforms to the system requirements, 2) verify that the system element has been implemented within the defined constraints, 3) verify the recorded evidence of implementation is complete and correct, and 4) verify that the system element has been packaged and stored in accordance with defined requirements.

The Implementation V&V tasks are to monitor, review, audit, and analyze hardware and software products. If a hardware or software V&V is being conducted, the system V&V activities will use those V&V results to perform, in part or in whole, the system V&V activities.

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Implementation V&V tasks described in Table 1b:

- 1) Implementation Strategy Assessment
- 2) System Element Implementation Analysis
- 3) ~~Task:~~ Criticality Analysis
- 4) System Integration Test Case V&V
- 5) System Qualification Test Case V&V
- 6) ~~Task:~~ System Acceptance Test Case V&V ~~generation~~
- 7) ~~Task:~~ Hazard Analysis
- 8) ~~Task:~~ Security Analysis
- 9) ~~Task:~~ Risk Analysis
- 1) ~~Task:~~ Traceability analysis
- 2) ~~Task:~~ Source code and source code documentation evaluation
- 3) ~~Task:~~ Interface analysis

- ~~5) Task: Component V&V test case generation~~
- ~~6) Task: Integration V&V test case generation~~
- ~~7) Task: System V&V test case generation~~
- ~~9) Task: Component V&V test procedure generation~~
- ~~10) Task: Integration V&V test procedure generation~~
- ~~11) Task: System V&V test procedure generation~~
- ~~12) Task: Component V&V test execution~~

The Implementation V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Implementation Process outcomes are contained in Annex L.

8.5 Activity: Integration V&V

The purpose of the Integration Process is to assemble a system that is consistent with the architectural design. This process combines system elements to form complete or partial system configurations to create a product specified in the system requirements (ISO/IEC 15288:2008 [B16]).

The objectives of Integration V&V are to assure that 1) the system architectural design is implemented correctly through integration, 2) the integrated system meets the system requirements, 3) the system integration strategy is consistent with the system architecture, 4) the integration test plan and procedures are traceable to the system architectural design, 5) the unavoidable constraints of integration that influence requirements are addressed correctly, 6) the integration of human performance into systems and their operation is correct, and 7) the nonconformances due to integration actions are recorded and addressed.

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Integration V&V tasks described in Table 1b:

- 1) System Integration Strategy Assessment

- [2\) System Integration Test Procedure V&V](#)
- [3\) System Integration Test Execution V&V](#)
- [4\) System Qualification Test Procedure V&V](#)
- [5\) System Qualification Test Execution V&V](#)
- [6\) System Acceptance Test Procedure V&V](#)

[The Integration V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 \[B16\]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Integration Process outcomes are contained in Annex L.](#)

~~5.4.5 Activity: Test V&V~~

~~Testing includes software testing, software integration testing, software qualification testing, system integration testing, and system qualification testing. The Test V&V activity and its relationship to the software life cycle are shown in Figure 2. The objective of Test V&V is to ensure that the software requirements and system requirements allocated to software are validated by execution of integration, system, and acceptance tests.~~

~~The V&V effort shall perform, as specified in Table 2 for the selected integrity level, the following Test V&V tasks described in Table 1:~~

- ~~1) Task: Traceability analysis~~
- ~~2) Task: Acceptance V&V test procedure generation~~
- ~~3) Task: Integration V&V test execution~~
- ~~4) Task: System V&V test execution~~
- ~~5) Task: Acceptance V&V test execution~~
- ~~6) Task: Hazard analysis~~

~~7) Task: Security analysis~~

~~8) Task: Risk analysis~~

8.6 Activity: Transition V&V

The purpose of the Transition Process is to establish a capability to provide the services specified by stakeholder requirements in the operational environment. This process installs a verified system, together with relevant enabling systems (e.g., operating system, support system, operator training system, and user training system) as defined in agreements. This process is used at each level in the system structure and in each stage to complete the criteria established for exiting the stage. It includes preparing applicable storage, handling, and shipping enabling systems (ISO/IEC 15288:2008 [B16]).

The objectives of Transition V&V are to assess that 1) the system transition strategy is comprehensive and explicitly documented, 2) the system is installed in its operational location in accordance with the transition plan, 3) the system delivers all specified services per its governing requirements, and 4) the system is sustainable by enabling systems.

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Transition V&V tasks described in Table 1b:

- 1) Transition Strategy Evaluation
- 2) Transition Demonstration Assessment
- 3) System Acceptance Test Execution V&V

The Transition V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Transition Process outcomes are contained in Annex L.

5.4.6 Activity: Installation and checkout V&V

~~In installation and checkout, the software product is installed and tested in the target environment. The Installation and checkout V&V activity supports the software system installation activities. The objective of Installation and checkout V&V is to verify and validate the correctness of the software installation in the target environment.~~

5.5 Process: Operation

~~The operation process involves the use of the software system by the end user in an operational environment.~~

8.7 Activity: Operation V&V

The purpose of the Operation Process is to use the system to deliver its services. This process assigns personnel to operate the system and monitors the services and operator-system performance. To sustain services, it identifies and analyzes operational problems in relation to agreements, stakeholder requirements, and organizational constraints (ISO/TEC 15288:2008 [B16]).

The objective of Operation V&V is to verify that the operation strategy satisfies stakeholder requirements and needs.

~~The Operation V&V activity evaluates the impact of changes in the operating environment; assesses the effect on the system of any proposed changes; evaluates operating procedures for adherence with the intended use; and analyzes risks affecting the user and the system. The objective of Operation V&V is to evaluate new constraints in the system, assess proposed system changes and their impact on the~~

~~software, and evaluate operating procedures for correctness and usability.~~

The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Operation V&V tasks described in Table 1b:

- 1) ~~Task:~~ Operating Procedures Evaluation
- 2) ~~Task:~~ Hazard Analysis
- 3) ~~Task:~~ Security Analysis
- 4) ~~Task:~~ Risk Analysis
- ~~1) Task: Evaluation of new constraints~~

The Operation V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Operation Process outcomes are contained in Annex L.

5.6 Process: Maintenance

~~The maintenance process is activated when the software system or associated documentation must be changed in response to a need for system maintenance. The Maintenance V&V activity addresses software system~~

- ~~— Modifications (i.e., corrective, adaptive, or perfective changes)~~
- ~~— Migration (i.e., the movement of software to a new operational environment)~~
- ~~— Retirement (i.e., the withdrawal of active support by the operation and maintenance organization, partial or total replacement by a new system, or installation of an upgraded system)~~

8.8 Activity: Maintenance V&V

The purpose of the Maintenance Process is to sustain the capability of the system to provide a service. This process monitors the system's capability to deliver services; records problems for analysis; takes

corrective, adaptive, perfective, and preventive actions; and confirms restored capability (ISO/TEC 15288:2008 [B16]).

System modifications may be derived from requirements specified to correct ~~software~~ errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the ~~software~~ system shall be treated as development efforts ~~processes~~ and shall be verified and validated by performing V&V tasks corresponding to ~~as described in~~ the ~~following:~~ modifications. Integrity- ~~Software integrity-~~ level assignments shall be assessed as described in Clause 5. The ~~software~~-integrity-level assignments shall be revised as appropriate to reflect requirements derived from the Maintenance Process.

~~—5.1 Process: Management~~

~~—5.4. Process: Development~~

The integrity level could be revised on the basis of corrective, perfective, or adaptive changes to the system if these changes modify the original integrity level of the system. If the integrity level is revised, then the effect of the revision on existing system requirements shall be evaluated to identify additional activities and tasks to be performed regressively on the system, software, and hardware. Corresponding V&V activities and tasks shall be identified to confirm the system conforms to its required integrity level.

~~For migrating software, the V&V effort shall verify that the migrated software meets the requirements of Clauses 4 and 5.~~

If the system ~~software~~ V&V were performed in accordance with this standard, then the maintenance process shall continue to conform to this standard. If the system ~~software~~ were not verified and validated using this standard and appropriate documentation is not available or adequate, then the Maintenance V&V effort shall determine whether the missing or incomplete documentation should be generated. In making this determination of whether to generate missing documentation, the minimum V&V requirements of the assigned ~~software~~-integrity level shall be taken into consideration.

The objectives of Maintenance V&V are to 1) assess that the maintenance strategy is comprehensive and explicitly documented; 2) evaluate the corrective actions to assure they resolve the issue or negative trend; 3) assess the effect of the corrective, adaptive, perfective, and preventive changes on stakeholder and system requirements, architectural design, and implementation; and 4) evaluate problem reports, corrective actions, and trends to determine the possible corrective, adaptive, perfective, and preventive actions.

~~is to assess proposed software system changes and their impact on the software, evaluate anomalies that are discovered during operation, assess migration requirements, assess retirement requirements, and reperform V&V tasks. The proposed changes are assessed by the *proposed/ baseline change assessment* task of the Management of V&V activity.~~

The V&V effort shall perform, as specified in Table 2b for the selected ~~software~~ integrity level, the following Maintenance V&V tasks described in Table 1b:

1) System Maintenance Strategy Assessment

2) System Maintenance Execution Assessment

~~1) Task: SVVP revision~~

~~2) Task: Anomaly evaluation~~

~~3) Task: Criticality analysis~~

~~4) Task: Migration assessment~~

~~5) Task: Retirement assessment~~

~~6) Task: Hazard analysis~~

~~7) Task: Security analysis~~

~~8) Task: Risk analysis~~

~~9) Task: Task iteration~~

The Maintenance V&V tasks listed shall be used to verify and validate the Stakeholder Requirements Definition process outcomes described in ISO/IEC 15288:2008 [B16]. The mappings of IEEE 1012 V&V tasks

[to ISO/IEC/IEEE 15288 Maintenance Process outcomes are contained in Annex L.](#)

8.9 Activity: Disposal V&V

[The purpose of the Disposal Process is to end the existence of a system \(ISO/TEC 15288:2008 \[B16\]\).](#)

[This process ends active support by the operation and maintenance organization, and it deactivates, disassembles, and removes the affected products, consigning them to a final condition, and leaving the environment in an acceptable condition. This process destroys or stores system elements and related products in a sound manner, in accordance with legislation, agreements, organizational constraints, and stakeholder requirements. Where required, it maintains records that may be monitored.](#)

[The V&V effort shall perform, as specified in Table 2b for the selected integrity level, the following Disposal V&V tasks described in Table 1b:](#)

[1\) Disposal Plan Evaluation](#)

[The Disposal V&V tasks listed shall be used to verify and validate the Disposal process outcomes described in ISO/TEC 15288:2008 \[B16\]. The mappings of IEEE 1012 V&V tasks to ISO/IEC/IEEE 15288 Disposal Process outcomes are contained in Annex L.](#)

Table 1 b— V&V tasks, inputs, and outputs

8.1 Activity: Stakeholder Requirements Definition V&V (System, Process: Stakeholder Requirements Definition)		
V&V tasks	Required inputs	Required outputs
(1) Stakeholder Requirements Evaluation Evaluate the stakeholder requirements for correctness.	Stakeholders requirements	Task report(s) - Stakeholder

<u>consistency, completeness, readability, and testability.</u> <u>The task criteria are as follows:</u> <u>a) Correctness</u> 1) <u>Verify and validate that the requirements satisfy the stakeholder needs for the system</u> 2) <u>Verify that the requirements comply with standards, references, regulations, policies, physical laws, and business rules.</u> 3) <u>Validate that the requirements define the intended interaction of the system with its operating environment.</u> <u>b) Consistency</u> 1) <u>Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</u>	<u>document</u> <u>Concept</u> <u>Documentation</u> <u>Organizational</u> <u>processes and</u> <u>procedures</u>	<u>requirements</u> <u>evaluation</u> <u>Anomaly</u> <u>report(s)</u>
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2) <u>Verify that there is consistency between assumptions, requirements, and between groups of requirements.</u> c) <u>Completeness</u> 1) <u>Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</u> 2) <u>Verify that all stakeholders, or stakeholder classes (used here as groupings of stakeholders), are identified.</u> 3) <u>Verify that the stakeholders requirements document satisfies specified configuration management procedures.</u> d) <u>Readability</u> 1) <u>Verify that the documentation is legible, understandable, and unambiguous to the intended audience.</u> 2) <u>Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</u> e) <u>Testability</u> <u>Verify that objective acceptance criteria can be developed to validate the requirements.</u>		
<u>(2) Traceability Analysis</u> a) <u>Establish the root of the traceability, starting with the stakeholder requirements.</u>	<u>Stakeholders requirements document</u>	<u>Task report(s) – Root of traceability tree</u> <u>Anomaly report(s)</u>

b) Verify that all requirements are traceable to one or more stakeholders or stakeholder classes.	Concept Documentation	
	requirements document Concept documentation	Root of traceability tree Anomaly report(s)

8.1 Activity: Stakeholder Requirements Definition V&V (System, Process: Stakeholder Requirements Definition)		
V&V tasks	Required inputs	Required outputs
(3) Criticality Analysis a) Determine whether system integrity levels are established for stakeholder and system requirements, detailed functions, subsystems, or other partitions. b) Verify that the assigned system integrity levels are correct. If system integrity levels are not assigned, then assign system integrity levels to the stakeholder requirements. c) Document the system integrity level assigned to stakeholder requirements. For V&V planning purposes, the system shall be assigned the same integrity level as the highest level assigned to any individual stakeholder requirement.	Stakeholders requirements document Concept documentation	Task report(s) - Criticality analysis Anomaly report(s)

<u>(4) Hazard Analysis</u> <u>Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</u> <u>a) Identify the potential system hazards.</u> <u>b) Assess the consequences of each hazard.</u> <u>c) Assess the probability of each hazard.</u> <u>d) Identify mitigation strategies for each hazard.</u>	<u>Stakeholders</u> <u>requirements</u> <u>document</u> <u>Concept</u> <u>documentation</u>	<u>Taskreport(s) -</u> <u>Hazard analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(5) Security Analysis</u> <u>a) Review the system owner's definition of an acceptable level of security risk.</u> <u>b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</u> <u>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</u>	<u>Stakeholders</u> <u>requirements</u> <u>document</u> <u>Concept</u> <u>documentation</u> <u>Preliminary threat</u> <u>and risk assessment</u> <u>(TRA)</u>	<u>Taskreport(s) -</u> <u>Security analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(6) Risk Analysis</u> <u>a) Identify the technical and management risks.</u> <u>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>Stakeholders</u> <u>requirements</u> <u>document</u> <u>Concept</u> <u>documentation</u> <u>Supplier</u>	<u>Task report(s) -</u> <u>Risk analysis</u> <u>Anomaly</u> <u>report(s)</u>

	development plans and schedules Hazard analysis report Security analysis report V&V task results	
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8.2 Activity: Requirements Analysis V&V (System, Process: Requirements Analysis)		
V&V tasks	Required inputs	Required outputs
(1) Requirements Evaluation Evaluate the system requirements for correctness, consistency, completeness, readability, and testability. The task criteria are as follows: a) Correctness 1) Verify and validate that the required characteristics, attributes, constraints, and functional and performance requirements for a product solution are correct. 2) Verify and validate that the system requirements satisfy the stakeholder requirements. 3) Verify that the system requirements comply	Stakeholders requirements document System requirements document Concept documentation Organizational processes and procedures	Task report(s) System requirements evaluation Anomaly report(s)

<u>with standards, references, regulations, policies, physical laws, and business rules.</u> 4) <u>Validate that the system requirements define the intended interaction of the system with its operating environment.</u> b) <u>Consistency</u> 1) <u>Verify that all terms, concepts, and requirements are documented consistently in accordance with accepted syntax and structure (e.g., style guides and requirements modeling structure).</u> 2) <u>Verify that there is consistency among the requirements, groups of requirements (functional interaction) and assumptions.</u> c) <u>Completeness</u> 1) <u>Validate that all stakeholder needs are satisfied by the set of system requirements.</u> 2) <u>Validate that performance criteria and functionality are described in the requirements, within the assumptions and constraints of the operating environment and system boundaries.</u> 3) <u>Verify that the System Requirements Document satisfies specified configuration management procedures.</u> d) <u>Readability</u> 1) <u>Verify that the documentation is legible, understandable, and unambiguous to the</u>		
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<u>intended audience.</u> 2) <u>Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols.</u> e) <u>Testability</u> <u>Verify that objective acceptance criteria can be developed to validate the requirements.</u>		
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<u>8.2 Activity: Requirements Analysis V&V (System, Process: Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(2) Interface Analysis</u> <u>Verify and validate that the requirements for system interfaces with other systems are correct, complete and testable. The task criteria are as follows:</u> a) <u>Correctness</u> <u>Validate the external interface requirements, including information to be exchanged.</u> b) <u>Completeness</u> <u>Verify that all external interface requirements are described.</u> c) <u>Testability</u> <u>Verify that objective acceptance criteria can be developed to validate the requirements.</u>	<u>System requirements document</u> <u>Stakeholders requirements document</u> <u>Organizational processes and procedures</u>	<u>Task report(s) - Requirements analysis</u> <u>Anomaly report(s)</u>
<u>(3) Traceability Analysis</u> <u>Verify that the traceability analysis is complete. The task</u>	<u>System requirements document</u>	<u>Task report(s) - Requirements</u>

<u>criteria are as follows:</u> a) <u>All system requirements are traceable to one or more stakeholder requirements.</u> b) <u>All stakeholder requirements are traceable to one or more system requirements.</u>	<u>Stakeholders requirements document</u>	<u>analysis</u> <u>Anomaly report(s)</u>
<u>(4) Criticality Analysis</u> <u>Review and update the existing criticality analysis results from the prior criticality task report using the stakeholder requirements and system requirements.</u>	<u>Criticality task report</u> <u>System requirements document</u> <u>Stakeholders requirements document</u>	<u>Task report(s) - Criticality analysis</u> <u>Anomaly report(s)</u>
<u>(5) System Integration Test Plan V&V</u> a) <u>System elements of integrity level 4</u> 1) <u>Plan V&V system integration testing to validate that the system element correctly implements the system requirements and design as each system element is incrementally integrated with each other.</u> 2) <u>Plan tracing of system requirements to test designs, cases, procedures, and results.</u> 3) <u>Plan documentation of test designs, cases, procedures, and results.</u> 4) <u>The V&V system integration test plan shall address the following:</u> i) <u>Conformance to increasingly larger set of functional requirements at each stage of integration.</u> ii) <u>Assessment of timing, sizing, and accuracy.</u> iii) <u>Performance at boundaries and under stress conditions.</u> iv) <u>Measures of requirements test coverage and</u>	<u>System requirements document</u> <u>Stakeholders requirements document</u> <u>System integration test plan</u>	<u>V&V system integration test plan</u> <u>Anomaly report(s)</u>

<u>system reliability.</u> 5) <u>Verify that the V&V system integration test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829™-2008 [B2]).</u> 6) <u>Validate that the V&V system integration test plan satisfies the following criteria:</u> i) <u>Traceable to the system requirements.</u>		
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<u>8.2 Activity: Requirements Analysis V&V (System, Process: Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
ii) <u>External consistency with the system requirements,</u> iii) <u>Internal consistency,</u> iv) <u>Test coverage of the system requirements,</u> v) <u>Appropriateness of test standards and methods used,</u> vi) <u>Conformance to expected results,</u> vii) <u>Feasibility of system integration testing,</u> viii) <u>Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</u> b) <u>System elements of integrity levels 2 and 3</u> 1) <u>Verify that the developer's system integration test plan conforms to the project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Validate that the developer's system integration test plan satisfies the following criteria:</u> i) <u>Traceable to the system requirements,</u>		

<u>ii) External consistency with the system requirements,</u> <u>iii) Internal consistency,</u> <u>iv) Test coverage of the system requirements,</u> <u>v) Appropriateness of test standards and methods,</u> <u>vi) Conformance to expected results,</u> <u>c) System elements of integrity level 1</u> <u>There are no system integration V&V test plan review requirements.</u>		
<u>(6) System Qualification Test Plan V&V</u> <u>a) System integrity level 4</u> <u>1) Plan V&V system qualification testing to validate the system against system requirements. The resulting V&V system qualification test plan shall address the following:</u> <u>i) Tracing of system requirements to test designs, cases, procedures, and results.</u> <u>ii) Documentation of test designs, cases, procedures, and results.</u> <u>iii) Satisfaction of all system requirements (e.g., functional, performance, security, operation, and maintenance) in the system environment.</u> <u>iv) Adequacy of user documentation (e.g., training materials, procedural changes, and user guides).</u> <u>v) Performance at boundaries (e.g., data and interfaces) and under stress conditions.</u> <u>vi) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>vii) Test coverage of system requirements,</u>	<u>System requirements document</u> <u>Stakeholders requirements document</u> <u>System qualification test plan (developer's)</u>	<u>V&V system qualification test plan</u> <u>Anomaly report(s)</u>

<u>viii) Expected results.</u> <u>b) System integrity levels 2 and 3</u> <u>1) Verify that the developer's system qualification test plan satisfies the following criteria:</u> <u>i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u>		
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<u>8.2 Activity: Requirements Analysis V&V (System, Process: Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>ii) Test coverage of system requirements.</u> <u>2) Verify that the developer's system qualification test plan addresses the following:</u> <u>i) Appropriateness of test methods and standards used.</u> <u>ii) Expected results.</u> <u>iii) Feasibility of system qualification testing.</u> <u>iv) Capability to be operated and maintained.</u> <u>c) System integrity level 1</u> <u>There are no system qualification V&V test plan generation requirements.</u>		
<u>(7) System Acceptance Test Plan V&V</u> <u>a) System integrity level 4</u> <u>1) Plan V&V system acceptance testing to validate that the system correctly implements system requirements in the intended operational environment or as close to the intended</u>	<u>System requirements</u> <u>Document</u> <u>Stakeholders requirements</u>	<u>V&V system acceptance test plan</u> <u>Anomaly report(s)</u>

<u>operational environment as possible. The resulting V&V system acceptance test plan shall address the following:</u> <u>i) Tracing of acceptance test requirements to test design, cases, procedures, and execution results.</u> <u>ii) Documentation of test tasks and results.</u> <u>iii) Conformance to acceptance requirements in the operational environment (limitations of testing).</u> <u>iv) Adequacy of user documentation (e.g., training materials, procedural changes, user guides).</u> <u>v) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>vi) Test coverage of acceptance requirements.</u> <u>vii) Expected results.</u> <u>b) System integrity levels 2 and 3</u> <u>1) Verify that the acquirer's system acceptance test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>2) Verify that the acquirer's system acceptance test plan addresses the following:</u> <u>i) Test coverage of acceptance requirements.</u> <u>ii) Expected results.</u> <u>iii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</u> <u>c) System integrity level 1</u>	<u>document</u> <u>System acceptance test plan (acquirer's)</u>	
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There are no system acceptance V&V test plan generation requirements.		
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8.2 Activity: Requirements Analysis V&V (System, Process: Requirements Analysis)

V&V tasks	Required inputs	Required outputs
(8) Hazard Analysis Analyze the potential hazards to and from the stakeholder requirements. The analysis shall perform the following: a) Identify the potential system hazards. b) Assess the consequences of each hazard. c) Assess the likelihood of each hazard. d) Identify mitigation strategies for each hazard.	System requirements document Stakeholders requirements document Concept documentation Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)
(9) Security Analysis a) Review the system owner's definition of an acceptable level of security risk. b) Analyze the stakeholder requirements from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/	System requirements document Stakeholders requirements document Concept documentation	Task report(s) - Security analysis Anomaly report(s)

<u>process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</u> <u>c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.</u>	<u>Preliminary threat and risk assessment (TRA)</u> <u>Security analysis report</u>	
<u>(10) Risk Analysis</u> <u>a) Identify the technical and management risks.</u> <u>b) Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>System requirements document</u> <u>Stakeholders requirements document</u> <u>Concept documentation</u> <u>Supplier development plans and schedules</u> <u>Hazard analysis report</u> <u>Security analysis report</u> <u>Risk analysis report</u> <u>V&V task results</u>	<u>Task report(s) -Risk analysis</u> <u>Anomaly report(s)</u>

<u>8.3 Activity: Architectural Design V&V (System, Process: Architectural Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Design Evaluation</u> Evaluate the system architectural design for correctness, consistency, completeness, and testability. The task criteria are as follows: a) <u>Correctness</u> 1) Verify the characteristics, attributes, constraints, and functional and performance requirements of the architectural design.	<u>System requirements</u> <u>Business needs</u> <u>Architectural design documents</u> <u>Architecture trade-off analyses</u>	<u>Task report(s) - System architectural design assessment</u> <u>Anomaly report(s)</u>

<u>8.3 Activity: Architectural Design V&V (System, Process: Architectural Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
2) Verify the system architectural design complies with standards, regulations, policies, physical laws, and business rules. 3) Validate the product solution(s) defined by the system architectural design satisfies the stakeholder needs. 4) Validate the system architectural design and its element interactions do not result in unnecessary, unintended, or deleterious consequences. 5) Validate the system architectural design defines the intended interaction of the system with its operating environment. b) <u>Consistency</u>	<u>Organizational processes and procedures</u>	<u>Input to final task report</u>

<u>1) Verify the selected architecture design conforms to the architectural guidance, principles, and tenets of the organization processes and procedures (e.g., service-oriented architecture, modular open-systems architecture).</u> <u>2) Verify the architectural principles, characteristics, and rules established for the system are being applied across the system architectural design.</u> <u>c) Completeness</u> <u>1) Verify the system functions are allocated to the elements of the system architectural design.</u> <u>2) Verify all system requirements are included in the system architectural design.</u>		
<u>(2) Interface Analysis</u> <u>Verify the architectural design interfaces between system elements and with other systems are correct and complete. The task criteria are</u> <u>a) Correctness</u> <u>1) Verify the system architectural design satisfies the system interface requirements between system elements.</u> <u>2) Verify the system architectural design satisfies the system interface requirements with external systems (external system boundaries).</u> <u>b) Completeness</u> <u>Verify the architectural design describes all internal and external interfaces.</u>	<u>Systems requirements document</u> <u>Architectural design documents</u>	<u>Task report(s) - System architectural design assessment</u> <u>Anomaly report(s)</u>

<u>(3) Traceability Analysis</u>	<u>Systems</u>	<u>Task report(s) -</u>
<u>Verify that the traceability analysis is complete. The task</u>	<u>requirements</u>	<u>System architectural</u>
<u>criteria are as follows:</u>	<u>document</u>	<u>design assessment</u>
<u>a) Verify that all architectural design elements are</u>	<u>Architectural design</u>	<u>Anomaly report(s)</u>
<u>traceable to one or more system requirements.</u>	<u>documents</u>	
<u>b) Verify that all system requirements are traceable to one</u>		
<u>or more architectural design elements.</u>		

<u>8.3 Activity: Architectural Design V&V (System, Process: Architectural Design)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
<u>(4) Criticality Analysis</u>	<u>Criticality task report</u>	<u>Task report(s) -</u>
<u>a) Review and update the existing criticality analysis</u>	<u>Systems</u>	<u>Criticality analysis</u>
<u>results from the prior criticality task report using the</u>	<u>requirements</u>	
<u>system architectural design to assign a system</u>	<u>document</u>	<u>Anomaly report(s)</u>
<u>integrity level to each architectural design element.</u>	<u>Architectural design</u>	
<u>b) Implementation methods and interfacing technologies</u>	<u>documents</u>	
<u>may cause previously assigned system integrity</u>		
<u>levels to be raised or lowered for a given</u>		
<u>stakeholder or system requirement. Verify that no</u>		
<u>inconsistent or undesired system integrity</u>		
<u>consequences are introduced by reviewing the revised</u>		
<u>system integrity levels.</u>		
<u>(5) System Integration Test Design V&V</u>	<u>Systems</u>	<u>V&V system</u>
<u>a) System integrity level 4</u>	<u>requirements</u>	<u>integration test</u>
<u>1) Design tests for V&V system integration testing.</u>	<u>document</u>	<u>design</u>
<u>2) Continue tracing required by the V&V system</u>	<u>Architectural design</u>	<u>Anomaly report(s)</u>
<u>integration test plan. Verify that the V&V</u>		

<u>system integration test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 3) <u>Validate that the V&V system integration test designs satisfy the criteria in V&V Activity 8.2, Task 5.</u> b) <u>System integrity levels 2 and 3</u> 1) <u>Verify that the developer's test designs for system integration testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Validate that the developer's system integration test designs satisfy the criteria in V&V activity 8.2, Task 5.</u> c) <u>System integrity level 1</u> <u>There are no system integration V&V test design review requirements.</u>	<u>documents</u> <u>System integration</u> <u>test design</u>	
<u>(6) System Qualification Test Design V&V</u> a) <u>System integrity level 4</u> 1) <u>Design tests for V&V system qualification testing.</u> 2) <u>Continue tracing required by the V&V system qualification test plan. Verify that the V&V system acceptance test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 3) <u>Validate that the V&V system qualification test designs satisfy the criteria in V&V activity 8.2, Task 6.</u> b) <u>System integrity levels 2 and 3</u>	<u>Systems</u> <u>requirements</u> <u>document</u> <u>Architectural design</u> <u>documents</u> <u>System qualification</u> <u>test design</u>	<u>V&V system</u> <u>qualification test</u> <u>design(s)</u> <u>Anomaly report(s)</u>

1) <u>Verify that the developer's test designs for system qualification testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Verify that the developer's system qualification test designs satisfy the criteria in V&V activity 8.2, Task 6.</u> c) <u>System integrity level 1</u> <u>There are no system qualification V&V test design generation requirements.</u>		
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<u>8.3 Activity: Architectural Design V&V (System, Process: Architectural Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(7) System Acceptance Test Design V&V</u> a) <u>System integrity level 4</u> 1) <u>Design tests for V&V system acceptance testing.</u> 2) <u>Continue tracing required by the V&V system acceptance test plan. Verify that the V&V system acceptance test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 3) <u>Validate that the V&V system acceptance test designs satisfy the criteria in V&V activity 8.2, Task 7.</u> b) <u>System integrity levels 2 and 3</u> 1) <u>Verify that the acquirer's test designs for system acceptance testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u>	<u>Systems requirements document</u> <u>Architectural design documents</u> <u>System acceptance test design</u>	<u>V&V system acceptance test design(s)</u> <u>Anomaly report(s)</u>

<u>2) Validate that the acquirer's system acceptance test designs satisfy the criteria in V&V activity 8.2, Task 7.</u> <u>c) System integrity level 1</u> <u>There are no system acceptance V&V test plan generation requirements.</u>		
<u>(8) Hazard Analysis</u> <u>Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</u> <u>a) Identify the potential system hazards.</u> <u>b) Assess the consequences of each hazard.</u> <u>c) Assess the probability of each hazard.</u> <u>d) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).</u>	<u>Architectural design documents</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(9) Security Analysis</u> <u>a) Review the system owner's definition of an acceptable level of security risk.</u> <u>b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/process) have been identified. Include an assessment of the sensitivity of the information/data to be processed.</u> <u>c) Analyze security risks introduced by the system itself as well as those associated with the environment with</u>	<u>Architectural design documents</u> <u>Preliminary threat and risk assessment (TRA)</u> <u>Security analysis report</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly report(s)</u>

<u>which the system interfaces.</u> <u>d) Verify the identified security threats and vulnerabilities</u> <u>are prevented, controlled, or mitigated (any unmitigated</u> <u>threats and vulnerabilities are documented and</u> <u>addressed as part of system operations).</u>		
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<u>8.4 Activity: Implementation V&V (System, Process: Implementation)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
<u>(1) Implementation Strategy Assessment</u>	<u>Implementation</u>	<u>Task report(s) -</u>
<u>a) Verify that the implementation strategy accounts</u> <u>for all implementation procedures (e.g.,</u> <u>technical manuals, organizational policies, or</u> <u>standards), fabrication processes, tools and</u> <u>equipment, implementation tolerances, and</u> <u>verification uncertainties needed to construct the</u> <u>system element.</u>	<u>strategy</u> <u>System element</u> <u>description from the</u> <u>architectural design</u>	<u>Implementation</u> <u>strategy</u> <u>assessment</u> <u>Anomaly</u> <u>report(s)</u>
<u>b) Verify that the implementation strategy defines the</u> <u>limitations and constraints imposed on the system</u> <u>element design solution.</u>		
<u>c) Verify that execution of the implementation strategy</u> <u>will produce a system element that conforms to the</u> <u>system element description in the architectural design.</u>		
<u>(2) System Element Implementation Analysis</u>	<u>Concept</u>	<u>Task report(s) -</u>
<u>a) V&V of system element requirements</u> <u>1) System elements of integrity levels 3 and 4</u>	<u>documentation</u>	<u>System element</u> <u>implementation</u>

i) <u>Verify (e.g., through analysis, reviews, and audits) that integrity level 3 and 4 element requirements trace to system requirements.</u> ii) <u>Verify (e.g., through analysis, reviews, and audits) that the element requirements satisfy the integrity level 3 and 4 system requirements.</u> 2) <u>System elements of integrity level 2</u> <u>Verify (e.g., through reviews and audits) that the element requirements satisfy the level 2 system requirements.</u> 3) <u>System elements of integrity level 1</u> <u>There are no element V&V requirements.</u> b) <u>V&V of system element design</u> 1) <u>System elements of integrity level 4</u> i) <u>Verify that the design of integrity level 4 element is consistent with the system architectural design.</u> ii) <u>Verify that the design of the integrity level 4 element satisfies the requirements of the system element.</u> 2) <u>System elements of integrity levels 2 and 3</u> <u>Verify that the design of each element is consistent with the system architectural design.</u>	<u>System requirements</u> <u>Software requirements specifications)</u> <u>Software interface requirements specifications)</u> <u>Hardware requirements specifications)</u> <u>System architectural design</u> <u>Software design documents(s)</u> <u>Hardware drawings</u> <u>System element(s)</u> <u>Software traceability analysis</u> <u>Hardware traceability analysis</u>	<u>analysis</u> <u>Anomaly report(s)</u>
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<u>8.4 Activity: Implementation V&V (System, Process: Implementation)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>3) System elements of integrity level 1</u> <u>There are no element V&V design requirements.</u> <u>c) V&V of system element implementation</u> <u>1) System elements of integrity level 4</u> <u>Begin preintegration testing using early partial deliveries of system elements (software and hardware) to derive early visibility into potential system integration testing issues and to initiate corrective changes to system element's development to correct or mitigate the potential system integration issue(s).</u> <u>2) System elements of integrity levels 3 and 4</u> <u>Evaluate the system element artifacts (requirements, design, anomaly reports/trends, element performance metrics, schedule compliance, resource usage actuals, and projected resource needs) produced from the software/hardware development life cycle process to perform the following:</u> <u>i) Derive an evolving assessment of each system element's performance (i.e., preview of how well the system element is meeting its requirements) and to recommend corrective actions (i.e., feedback to element development</u>		

<u>process or change to system architecture) to mitigate any projected performance shortfalls.</u> <u>ii) Assess each system element's project schedule performance (i.e., estimated system element schedule completion dates for system element milestone reviews and delivery dates) and to recommend system schedule changes if the system element project completion dates are slipping.</u> <u>iii) Estimate remaining schedule resource requirements (i.e., labor or equipment budget) if performance or schedule shortfalls are projected.</u> <u>3) System elements of integrity levels 1 and 2</u> <u>There are no element implementation V&V requirements.</u> <u>d) V&V of system element integration testing</u> <u>1) System elements of integrity level 4</u> <u>Verify through independently produced objective evidence (e.g., independent test) that integrity level 4 requirements in that element are satisfied by the system element hardware/software integration testing.</u> <u>2) System elements of integrity level 2 and 3</u> <u>Verify through the data produced by the development organization (e.g., reviews, audits) that the integrity levels 2 and 3 requirements in that element are satisfied by the system element hardware/software integration testing.</u>		
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<u>3) System elements of integrity levels 1</u> <u>There are no integration V&V testing requirements.</u> <u>e) V&V of system element qualification testing</u> <u>1) System elements of integrity level 4</u> <u>i) Develop V&V test cases for system element qualification testing.</u> <u>ii) Continue tracing required by the system element V&V test plan.</u>		
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<u>8.4 Activity: Implementation V&V (System, Process: Implementation)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>iii) Verify that the system element V&V test cases conform to project-defined test document purpose, format, and content.</u> <u>iv) Validate that the system element V&V test cases satisfy the criteria in system element V&V qualification test plan generation activity.</u> <u>2) System elements of integrity levels 2 and 3</u> <u>Review the results of the developer's qualification testing.</u> <u>3) System elements of integrity level 1</u> <u>There are no V&V qualification testing requirements.</u> <u>f) V&V of system element acceptance testing</u> <u>1) System elements of integrity level 4</u> <u>i) Plan acceptance V&V testing to validate that the</u>		

<u>system element correctly implements system requirements in an operational environment.</u> <u>ii) Design tests for acceptance V&V testing.</u> <u>iii) Develop test cases for acceptance V&V testing.</u> <u>iv) Perform acceptance V&V testing.</u> <u>v) Analyze test results to validate that the system element satisfies the V&V test acceptance criteria.</u> <u>2) System elements of integrity levels 2 and 3</u> <u>Review the acquirer's acceptance test results to validate that the system element satisfies the test acceptance criteria.</u> <u>3) System elements of integrity level 1</u> <u>There are no system element acceptance V&V test requirements.</u>		
<u>(3) Criticality Analysis</u> <u>a) Review and update the existing criticality levels of the system elements during the implementation process based on the implementation of the software and hardware components.</u> <u>b) Verify that no inconsistent or undesired system element integrity consequences have been introduced due to implementation methods and interfacing technologies that may cause previously assigned system element integrity levels to be raised or lowered.</u>	<u>Criticality task report</u> <u>System element requirements, designs, and implementations</u>	<u>Task report(s) -</u> <u>Criticality analysis</u> <u>Anomaly report(s)</u>

8.4 Activity: Implementation V&V (System, Process: Implementation)		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(4) System Integration Test Case V&V</u> <u>a) System integrity level 4</u> 1) <u>Develop test cases for V&V system integration testing.</u> 2) <u>Continue tracing required by the V&V system integration test plan.</u> 3) <u>Verify that the system integration V&V test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 4) <u>Validate that the V&V system integration test cases satisfy the criteria in V&V activity 8.2, Task 5.</u> <u>b) System integrity levels 2 and 3</u> 1) <u>Verify that the developer's system integration test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Validate that the developer's system integration test cases satisfy the criteria in V&V activity 8.2, Task 5.</u> <u>c) System integrity level 1</u> <u>There are no system integration V&V test case review requirements.</u>	<u>Systems requirements document</u> <u>Architectural design documents</u> <u>System integration test cases</u>	<u>V&V system integration test cases</u> <u>Anomaly report(s)</u>
<u>(5) System Qualification Test Case V&V</u> <u>a) System integrity level 4</u> 1) <u>Develop test cases for V&V system qualification testing.</u> 2) <u>Continue tracing required by the V&V system qualification test plan.</u>	<u>Systems requirements document</u> <u>Architectural design documents</u>	<u>V&V system qualification test cases</u> <u>Anomaly report(s)</u>

3) <u>Verify that the V&V system qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 4) <u>Validate that the V&V system qualification test cases satisfy the criteria in V&V activity 8.2, Task 6.</u> b) <u>System integrity levels 2 and 3</u> 1) <u>Verify that the developer's system qualification test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Validate that the developer's system qualification test cases satisfy the criteria in V&V activity 8.2, Task 6.</u> c) <u>System integrity level 1</u> <u>There are no system qualification V&V test case generation requirements.</u>	<u>System qualification test cases</u>	
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<u>8.4 Activity: Implementation V&V (System, Process: Implementation)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(6) System Acceptance Test Case V&V</u> a) <u>System integrity level 4</u> 1) <u>Develop test cases for V&V system acceptance testing.</u> 2) <u>Continue tracing required by the V&V system acceptance test plan.</u> 3) <u>Verify that the V&V system acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u>	<u>Systems requirements document</u> <u>Architectural design documents</u> <u>System acceptance test cases</u>	<u>V&V system acceptance test cases</u> <u>Anomaly report(s)</u>

<u>4) Validate that the V&V system acceptance test cases satisfy the criteria in V&V activity 8.2, Task 7.</u> <u>b) System integrity levels 2 and 3</u> <u>1) Verify that the acquirer's system acceptance test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>2) Verify that the acquirer's system acceptance test cases satisfy the criteria in V&V activity 8.2, Task 7.</u> <u>c) System integrity level 1</u> <u>There are no system acceptance V&V test requirements.</u>		
<u>(7) Hazard Analysis</u> <u>a) Verify the system elements that implement critical requirements introduce no new hazards.</u> <u>b) Update the hazard analysis.</u> <u>c) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).</u>	<u>Subsystems</u> <u>Hazard analysis report</u>	<u>Task report(s) -</u> <u>Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(8) Security Analysis</u> <u>a) Verify that the system element is implemented in accordance with the system architecture in that it addresses the identified security risks and that the implementation does not introduce new security risks.</u> <u>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system operations).</u>	<u>System architecture</u> <u>Subsystems</u> <u>Security analysis report</u> <u>V&V task results</u>	<u>Task report(s) -</u> <u>Security analysis</u> <u>Anomaly report(s)</u>
<u>(9) Risk Analysis</u> <u>Review and update risk analysis using prior task</u>	<u>Subsystems</u>	<u>Task report(s) -</u> <u>Risk analysis</u>

reports. Provide recommendations to eliminate, reduce, or mitigate the risks.	Hazard analysis report Security analysis report Risk analysis report	Anomaly report(s)
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8.5 Activity: Integration V&V (System, Process: Integration)		
V&V tasks	Required inputs	Required outputs
(1) System Integration Strategy Assessment a) Verify the system integration strategy. b) Verify the constraints on the system requirements and design due to the integration strategy. c) Integration test specification and procedure generation.	System integration strategy System architectural design System requirements	Task report(s) - System integration strategy assessment Anomaly report(s)
(2) System Integration Test Procedure V&V a) System integrity level 4 1) Develop test procedures for V&V system integration testing. 2) Continue tracing required by the V&V system integration test plan. 3) Verify that the V&V system integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]). 4) Validate that the V&V system integration test procedures satisfy the criteria in V&V activity 8.2 Task 5.	Systems requirements document System integration test plan System integration test procedures	V&V system integration test procedures Anomaly report(s)

b) <u>System integrity levels 2 and 3</u> 1) <u>Verify that the developer's system integration test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> 2) <u>Validate that the developer's system integration test procedures satisfy the criteria in V&V activity 8.2, Task 5.</u> c) <u>System integrity level 1</u> <u>There are no system integration V&V test procedure review requirements.</u>		
<u>(3) System Integration Test Execution V&V</u> a) <u>System integrity level 4</u> 1) <u>Perform V&V system integration testing.</u> 2) <u>Use the V&V system integration test results to validate that the system satisfies the test acceptance criteria.</u> 3) <u>Document the results as required by the V&V system integration test plan.</u> 4) <u>Use the V&V system integration test results to validate that the system satisfies the test acceptance criteria.</u> 5) <u>Document discrepancies between the actual and expected test results.</u> b) <u>System integrity levels 2 and 3</u> <u>Use the developer's system integration test results to verify that the system satisfies the test acceptance criteria.</u> c) <u>System integrity level 1</u> <u>There are no system integration V&V test execution review requirements.</u>	<u>Integration sequence and strategy (e.g., test scenario inputs, test procedure sequence)</u> <u>Hardware element(s)</u> <u>Software element(s)</u> <u>Other external system element(s)</u>	<u>Task report(s) - V&V system integration execution assessment</u> <u>Anomaly report(s)</u>

<u>(4) System Qualification Test Procedure V&V</u>	<u>Systems</u>	<u>V&V system</u>
a) <u>System integrity level 4</u>	<u>requirements</u>	<u>qualification</u>
1) <u>Develop test procedures for V&V system qualification testing.</u>	<u>document</u>	<u>test</u>
2) <u>Continue tracing required by the V&V system qualification test plan.</u>	<u>System qualification</u>	<u>procedures</u>
	<u>test plan (V&V or</u>	<u>Anomaly</u>
	<u>developer's)</u>	<u>report(s)</u>

<u>8.5 Activity: Integration V&V (System, Process: Integration)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
3) <u>Verify that the V&V system qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u>	<u>System qualification</u> <u>test procedures</u>	
4) <u>Validate that the V&V system qualification test procedures satisfy the criteria in V&V activity 8.2 Task 6.</u>		
b) <u>System integrity levels 2 and 3</u>		
1) <u>Verify that the developer's system qualification test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u>		
2) <u>Validate that the developer's system qualification test procedures satisfy the criteria in V&V activity 8.2, Task 6.</u>		
c) <u>System integrity level 1</u>		
<u>There are no system qualification V&V test procedure generation requirements.</u>		
<u>(5) System Qualification Test Execution V&V</u>	<u>Stakeholders</u>	<u>Task report(s) -</u>

a) <u>System integrity level 4</u> 1) <u>Perform V&V system qualification testing.</u> 2) <u>Use the V&V system qualification test results to validate that the system satisfies the test acceptance criteria.</u> 3) <u>Document discrepancies between the V&V system qualification test results and the test acceptance criteria.</u> 4) <u>Complete the traceability by documenting the location of the V&V system qualification test results.</u> 5) <u>Analyze test results to validate that the system delivers required services when installed in its operational location and staffed by operators. Additionally, the often less formally expressed but sometimes overriding attitudes, experience, and subjective tests that comprise customer satisfaction also need consideration.</u> 6) <u>Document the results as required by the V&V system qualification test plan.</u> b) <u>System integrity levels 2 and 3</u> <u>Use the developer's system qualification test results to verify that the acceptance criteria are satisfied.</u> c) <u>System integrity level 1</u> <u>There are no system qualification V&V test execution requirements.</u>	<u>requirements document</u> <u>System requirements</u> <u>Concept documentation</u> <u>System qualification test results</u>	<u>V&V system qualification test execution assessment</u> <u>Anomaly report(s)</u>
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<u>8.5 Activity: Integration V&V (System, Process: V& V Integration)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
<u>6) System Acceptance Test Procedure V&V</u> <u>a) System integrity level 4</u> <u>1) Develop test procedures for V&V system acceptance testing. Continue the tracing required by the V&V system acceptance test plan.</u> <u>2) Verify that the V&V system acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>3) Validate that the V&V system acceptance test procedures satisfy the criteria in V&V activity 8.2, Task 7.</u> <u>b) System integrity levels 2 and 3</u> <u>1) Verify that the acquirer's system acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).</u> <u>2) Verify that the acquirer's system acceptance test procedures satisfy the criteria in V&V activity 8.2, Task 7.</u> <u>c) System integrity level 1</u> <u>There are no system acceptance V&V test procedure generation requirements.</u>	<u>Systems</u> <u>requirements</u> <u>document</u> <u>System acceptance test</u> <u>procedures</u>	

<u>8.6 Activity: Transition V&V (System, Process: Transition)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>NOTE—If system transition is taking place by parts then the transition process V&V tasks should be executed for each of the parts.</u>		
<u>(1) Transition Strategy Evaluation</u> <u>a) Verify that the transition strategy is comprehensive.</u> <u>Verify that the transition strategy includes the following:</u> <u>1) All the system parts and the system as a whole (e.g., hardware [HW]/software [SW] functionality and databases) are considered in the transition execution.</u> <u>2) Transition schedule and sequence.</u> <u>3) Identification of transition hardware and software tools, equipment, and instructions.</u> <u>4) Archiving system artifacts, such as documentation and code.</u> <u>5) Impact of interacting systems in terms of transition timing and transition impacts.</u> <u>6) Continuity of capabilities when replacing or upgrading a legacy system.</u> <u>7) Site preparation for installation and legacy system retirement, storage, and/or incorporation.</u> <u>8) Provisions for documentation of the process results.</u> <u>9) A fallback position, in case of transition failure (e.g., going back to the old system) and other risk mitigation considerations.</u> <u>b) Verify that the transition strategy has a defined approach to establishing the system in the operational</u>	<u>Transition strategy</u> <u>Installation package (e.g., user documentation, SRD, SDD, IDD, SRS, IRS, concept documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)</u>	<u>Taskreport(s) - Transition plan evaluation</u> <u>Anomaly report(s)</u>

environment that is consistent with stakeholder requirements (e.g., by running the old and new systems in parallel or by well-defined system characteristics that can be compared)		
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8.6 Activity: Transition V&V (System, Process: Transition)		
V&V tasks	Required inputs	Required outputs
(2) Transition Demonstration Assessment	Transitioning	Taskreport(s) -
a) Validate that acceptance tests defined in the acquirer agreement (as documented in the acceptance test plan) demonstrate satisfactory installation at the specified location(s). Where the exact location or environment of operation is not available, a representative example is selected.	system (operational system under test)	Transition demonstration
b) Verify and validate that the installed system (software and hardware elements) and related site-specific parameters are identical to the final as-verified and as-validated system (software and hardware elements).	User documentation	Anomaly report(s)
	Installation package	
	(e.g., user documentation, SRD, SDD, IDD, SRS, IRS, concept documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)	
	Acquirer agreement	
	Acceptance test plan	

<u>(3) System Acceptance Test Execution V&V</u> a) <u>System integrity level 4</u> 1) <u>Perform V&V system acceptance testing.</u> 2) <u>Use the system V&V acceptance test results to validate that the system satisfies the test acceptance criteria.</u> 3) <u>Document discrepancies between the V&V system acceptance test results and the test acceptance criteria.</u> 4) <u>Complete the traceability by documenting the location of the V&V system acceptance test results.</u> 5) <u>Analyze test results to validate that the system delivers required services when installed in its operational location and staffed by operators. Additionally, the often less formally expressed but sometimes overriding attitudes, experience and subjective tests that comprise customer satisfaction also need consideration.</u> 6) <u>Document the results as required by the V&V system acceptance test plan.</u> b) <u>System integrity levels 2 and 3</u> <u>Use the acquirer's system acceptance test results to verify that the system satisfies the test acceptance criteria.</u> c) <u>System integrity level 1</u> <u>There are no system acceptance V&V test requirements.</u>	<u>Stakeholders requirements document</u> <u>System requirements</u> <u>Concept documentation</u> <u>System acceptance test results</u>	<u>Taskreport(s) - System acceptance V&V test execution assessment</u> <u>Anomaly report(s)</u>
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<u>8.7 Activity: Operation V&V (System, Process: Operation)</u>		
<u>V&V</u> <u>tasks</u>	<u>Required</u> <u>inputs</u>	<u>Required</u> <u>outputs</u>
<u>(1) Operating Procedures Evaluation</u> a) <u>Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.</u> b) <u>Evaluate new or revised constraints or changes in the environment (e.g., operational requirements, platform characteristics, and operating environment) on the system requirements to verify the applicability of the System VVP (see Annex H).</u>	<u>Concept</u> <u>documentation</u> <u>Operating</u> <u>procedures</u> <u>User documentation</u> <u>System VVP</u> <u>Constraints</u> <u>Environmental</u> <u>changes</u>	<u>Task</u> <u>report(s)-</u> <u>Operating</u> <u>procedures</u> <u>evaluation</u> <u>Anomaly</u> <u>report(s)</u> <u>Updated</u> <u>security</u> <u>analysis</u>
<u>(2) Hazard Analysis</u> a) <u>Verify that the operating procedures (e.g., instructions to the operator or user, standard operating procedures, abnormal operating procedures, safety procedures, and emergency operating procedures) and operational environment do not introduce new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system operations).</u> c) <u>Update the hazard analysis.</u>	<u>Operating</u> <u>procedures</u> <u>Hazard analysis</u> <u>report</u>	<u>Task</u> <u>report(s) -</u> <u>Hazard</u> <u>analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(3) Security Analysis</u> a) <u>Verify that no new security risks are introduced due to changes in the operational environment.</u> b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled or mitigated (any unmitigated threats and vulnerabilities are documented and addressed</u>	<u>New constraints</u> <u>Environmental</u> <u>changes</u> <u>Operating</u>	<u>Task Reports</u> <u>-Security</u> <u>analysis</u>

as part of system operations). c) Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk.	procedures Security analysis report	
(4) Risk Analysis a) Review and update risk analysis using prior task reports. b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Installation package Proposed changes Hazard analysis report Security analysis report Rick analysis report Supplier development plans and schedules Operation problem reports V&V task results	Task report(s) - Risk Analysis Anomaly report(s)

8.8 Activity: Maintenance V&V (System, Process: Maintenance)		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) System Maintenance Strategy Assessment</u> <u>a) Verify the system maintenance strategy.</u> <u>b) Verify the constraints on the system requirements and design due to the maintenance strategy.</u>	<u>System maintenance strategy</u> <u>System performance data</u> <u>System requirements</u>	<u>Task report(s) - System Maintenance Analysis</u> <u>Corrective actions assessment</u> <u>Anomaly report(s)</u>
<u>(2) System Maintenance Execution Assessment</u> <u>a) Verify the maintenance operational strategy results by monitoring the system maintenance execution.</u> <u>b) Evaluate the maintenance operational strategy results.</u> <u>c) Stakeholder requirements evaluation.</u> <u>d) Requirements.</u> <u>e) Traceability analysis.</u> <u>f) Criticality analysis.</u> <u>g) Hazard analysis.</u> <u>h) Security analysis.</u> <u>i) Risk analysis.</u> <u>j) Interface analysis.</u> <u>k) Design evaluation.</u> <u>l) System element implementation analysis.</u>	<u>System maintenance strategy (e.g., operational availability requirements, system failure, suspension of services)</u>	<u>Task report(s) - System maintenance execution assessment and quality metrics</u> <u>Anomaly report(s)</u>

<u>m) Integration analysis.</u> <u>n) System test analysis (integration, qualification, and acceptance).</u> <u>o) System installation analysis.</u> <u>p) Operation transition analysis.</u>		
<u>NOTE—System changes are maintenance activities (see 8.8).</u>		

<u>8.9 Activity: Disposal V&V (System, Process: Disposal)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Disposal Plan Evaluation</u>	<u>Concept</u>	<u>Taskreport(s)</u>
a) <u>Verify that system boundaries are defined and all relevant system elements (information, hardware, media, etc.) are identified.</u>	<u>Documentation</u>	<u>- System Description</u>
b) <u>Verify that disposal has been planned to a degree commensurate with the complexity and risk of the disposal and all that identified elements are accounted for in the plan.</u>	<u>Configuration</u>	<u>Taskreport(s)</u>
c) <u>Verify that environmental considerations, all applicable laws, regulations, organizational policies and procedures as well as other constraints are identified and communicated to relevant stakeholders.</u>	<u>Repository</u>	<u>- Constraints</u>
d) <u>Verify that system elements to be stored have available billets and established retention criteria.</u>	<u>Inventory records</u>	<u>1 ask report(s)</u>
	<u>Disposal plan</u>	<u>— Disposal plan</u>
	<u>Applicable laws</u>	<u>Anomaly</u>
	<u>Organizational</u>	<u>report(s)</u>
	<u>policies and</u>	
	<u>Procedures</u>	
	<u>List of constraints</u>	
	<u>Management and</u>	
	<u>technical decisions</u>	

NOTE (for Table 1b)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.

**Table 2 b—Minimum V&V tasks assigned to each integrity level for
system V&V**

V&V Activities	Activity: <u>Stakeholder</u>				Activity: <u>Requirements</u>				Activity: <u>Architectural</u>				Activity: <u>Implementation</u>				Activity: <u>Integration</u>				Activity: <u>Transition</u>				Activity: <u>Operation</u>				Activity: <u>Maintenance</u>				Activity: <u>Disposal</u>			
	<u>Requirements</u>				<u>Analysis</u>				<u>Design V&V</u>				<u>V&V</u>				<u>V&V</u>				<u>V&V</u>				<u>V&V</u>				<u>V&V</u>							
	<u>Definition</u>				<u>V&V</u>				<u>(see 8.3)</u>				<u>(see 8.4)</u>				<u>(see 8.5)</u>				<u>(see 8.6)</u>				<u>(see 8.7)</u>				<u>(see 8.8)</u>				<u>(see 8.9)</u>			
	<u>V&V (see 8.1)</u>				<u>(see 8.2)</u>																															
<u>Integrity Levels</u>	<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>			
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>
<u>Criticality Analysis</u>	X	X	X	X	X	X	X	X		X	X	X		X	X	X																				
<u>Design Evaluation</u>									X	X	X	X																								
<u>Disposal Plan Evaluation</u>																																X	X	X	X	
<u>Hazard Analysis</u>			X	X			X	X		X	X			X	X							X	X													
<u>Implementation Strategy Assessment</u>														X	X	X																				
<u>Interface Analysis</u>		X	X	X		X	X	X		X	X	X		X	X	X	X	X	X		X	X	X													
<u>Operating Procedures Evaluation</u>																								X	X											
<u>Requirements Evaluation</u>					X	X	X	X																												
<u>Risk Analysis</u>			X	X			X	X		X	X			X	X							X	X													
<u>Security Analysis</u>			X	X			X	X		X	X			X	X							X	X													
<u>Stakeholder Requirements Evaluation</u>	X	X	X	X																																
<u>System Acceptance Test Case V&V</u>														X	X	X																				
<u>System Acceptance Test Design V&V</u>										X	X	X																								
<u>System Acceptance Test Execution V&V</u>																					X	X	X													

V&V Activities	Activity: Stakeholder Requirements				Activity: Requirements Analysis				Activity: Architectural Design V&V (see 8.3)				Activity: Implementation V&V (see 8.4)				Activity: Integration V&V (see 8.5)				Activity: Transition V&V (see 8.6)				Activity: Operation V&V (see 8.7)				Activity: Maintenance V&V (see 8.8)				Activity: Disposal V&V (see 8.9)			
	Definition				V&V (see 8.2)																															
	V&V (see 8.1)																																			
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
System Acceptance Test Plan V&V						X	X	X																												
System Acceptance Test Procedure V&V																	X	X	X																	
System Element Implementation Analysis														X	X	X																				
System Integration Strategy Assessment																	X	X	X																	
System Integration Test Case V&V														X	X	X																				
System Integration Test Design V&V											X	X	X																							
System Integration Test Execution V&V																	X	X	X																	
System Integration Test Plan V&V						X	X	X																												
System Integration Test Procedure V&V																	X	X	X																	
System Maintenance Execution Assessment																													X	X	X					

<u>V&V Activities</u>	<u>Activity: Stakeholder Requirements Definition V&V (see 8.1)</u>				<u>Activity: Requirements Analysis V&V (see 8.2)</u>				<u>Activity: Architectural Design V&V (see 8.3)</u>				<u>Activity: Implementation V&V (see 8.4)</u>				<u>Activity: Integration V&V (see 8.5)</u>				<u>Activity: Transition V&V (see 8.6)</u>				<u>Activity: Operation V&V (see 8.7)</u>				<u>Activity: Maintenance V&V (see 8.8)</u>				<u>Activity: Disposal V&V (see 8.9)</u>			
<u>Integrity Levels</u>	<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>				<u>Levels</u>			
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>
<u>System Maintenance Strategy Assessment</u>																																				
<u>System Qualification Test Case V&V</u>													X	X	X																					
<u>System Qualification Test Design V&V</u>									X	X	X																									
<u>System Qualification Test Execution V&V</u>																	X	X	X																	
<u>System Qualification Test Plan V&V</u>									X	X	X																									
<u>System Qualification Test Procedure V&V</u>																	X	X	X																	
<u>Traceability Analysis</u>		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X												
<u>Transition Demonstration Assessment</u>																					X	X	X													
<u>Transition Strategy Evaluation</u>																					X	X	X													

NOTE (for Table 2b)—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of analysis or test. Higher integrity-level implementation requires greater rigor (e.g., formal methods and structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, and comprehensive fault and recovery scenarios) of analysis or test than the lower integrity level implementation.

Table 3b—Optional V&V tasks and suggested applications in System

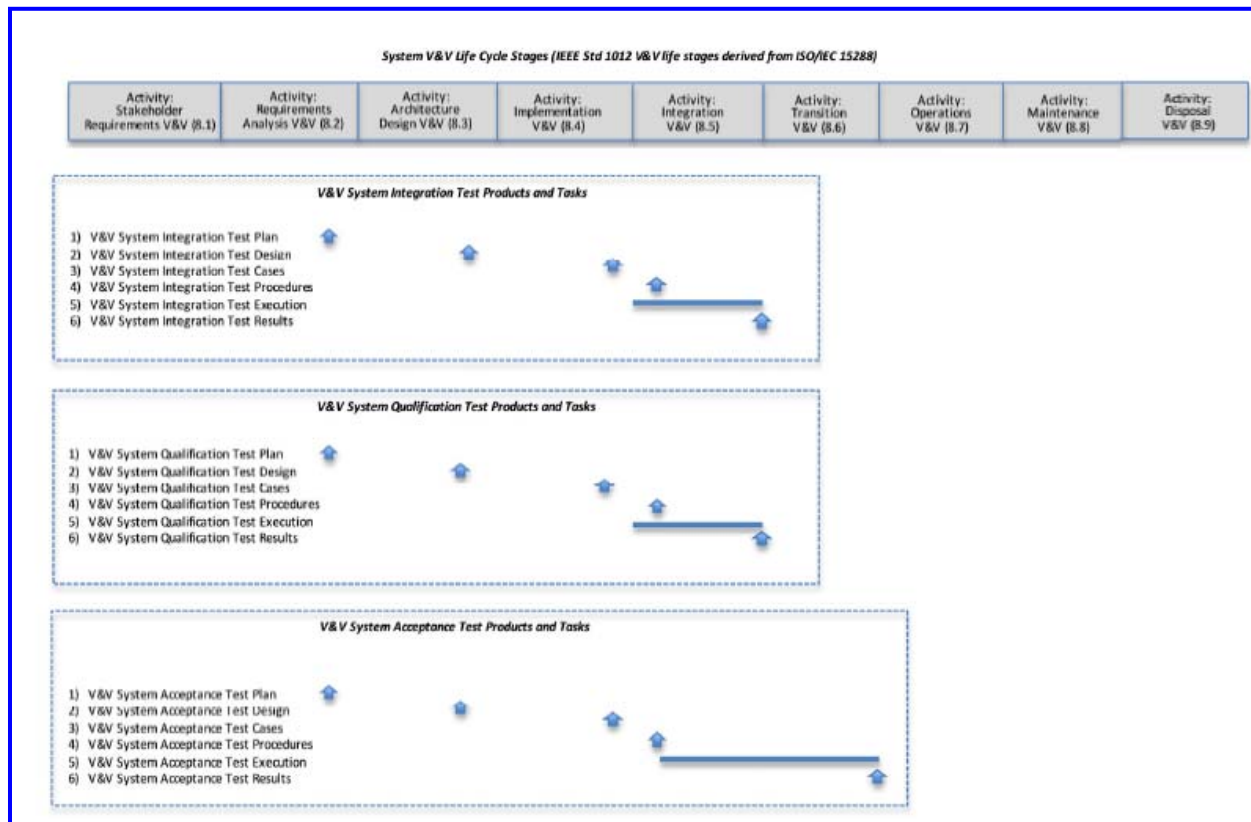
Technical Processes

	<u>Stakeholder Requirements Definition</u>	<u>Requirements Analysis</u>	<u>Architectural Design</u>	<u>Implementation</u>	<u>Integration</u>	<u>Transition</u>	<u>Operation</u>	<u>Maintenance</u>	<u>Disposal</u>
<u>Algorithm analysis</u>		X	X	X				X	
<u>Audit performance</u>		X	X	X	X	X		X	
<u>Audit support</u>		X	X	X	X	X		X	
<u>Control flow analysis</u>		X	X	X				X	
<u>Cost analysis</u>	X	X	X	X	X	X		X	
<u>Database analysis</u>		X	X	X	X	X		X	
<u>Data flow analysis</u>		X	X	X				X	
<u>Disaster recovery plan assessment</u>	X	X	X	X			X	X	X
<u>Distributed architecture assessment</u>	X	X	X					X	
<u>Feasibility study evaluation</u>	X	X	X					X	
<u>Independent risk assessment</u>								X	
<u>Inspection</u>									
<u>Concept</u>	X							X	
<u>Requirements</u>		X						X	
<u>Design</u>			X					X	
<u>Source code</u>									
<u>Test plan</u>		X	X	X	X	X		X	
<u>Test design</u>			X	X	X	X		X	
<u>Test case</u>			X	X	X	X		X	
<u>Operational evaluation</u>							X		
<u>Performance monitoring</u>	X	X	X	X	X	X	X	X	X
<u>Post installation validation</u>							X	X	

Project management oversight support	X	X	X	X	X	X	X	X	X
Proposal evaluation support									
Qualification testing				X		X			
Regression analysis and testing		X	X	X	X	X		X	
Reusability analysis	X	X	X	X				X	
Reuse analysis	X	X	X					X	
Simulation analysis	X	X	X	X	X	X	X	X	X
Sizing and timing analysis		X	X	X	X	X		X	
System software assessment			X	X	X	X	X	X	
Test certification				X	X	X	X	X	X
Test evaluation		X	X	X	X	X	X	X	X
Test witnessing				X	X	X	X	X	X
Training document evaluation		X	X	X	X	X	X	X	X
Usability analysis	X	X	X	X	X	X	X	X	
User documentation evaluation	X	X	X	X	X	X	X	X	
User training				X	X	X	X	X	
V&V tool plan generation									
V&V tool qualification	X	X	X	X	X	X	X	X	
Walkthroughs									
Design			X					X	
Requirements		X						X	

	Stakeholder Requirements	Definition	Requirements Analysis	Architectural Design	Implementation	Integration	Transition	Operation	Maintenance	Disposal
Source code										
Test					X	X			X	
Work breakdown structure evaluation										

[Annex G contains a description of the optional V& V tasks.](#)



(New)

Figure 2b—Summary of system V&V test products and tasks

NOTE 1—All V&V system test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of system V&V test products and tasks across the system life cycle. The system V&V test products (upward arrows) are shown in the system life cycle stages when the products are generated (per IEEE Std 1012-2004 [B5]). System test execution tasks are shown to occur during one or more system life cycle stages as indicated by "activity bars" in the diagram. The life cycle stage (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project-specific needs.

NOTE 3—The V&V activity clauses referenced in the system V&V life cycle stages are IEEE Std 1012 clauses.

9. Software V&V activities

9.1 Activity: Software Concept V&V

The concept activity represents the delineation of a specific implementation solution to solve the user's problem. During the concept activity, the system architecture is selected and system requirements are allocated to hardware, software, and user interface components. The Software Concept V&V activity addresses system architectural design and system requirements analysis. The objective of Software Concept V&V is to verify the allocation of system requirements (primary focus on software with consideration of interactions with hardware and user allocations), validate the selected solution, and assure ~~ensure~~ that no false assumptions have been incorporated in the solution.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Concept V&V tasks described in Table 1c:

- 1) ~~Task:~~ Concept Documentation Evaluation

- 2) ~~Task:~~ Hardware/Software/User Requirements Allocation Analysis
- 3) ~~Task:~~ Traceability analysis
- 4) ~~Task:~~ Criticality Analysis
- 5) ~~Task:~~ Hazard Analysis
- 6) ~~Task:~~ Security Analysis
- 7) ~~Task:~~ Risk Analysis

[The Software Concept V&V tasks listed shall be used to verify and validate the Software Requirements Analysis process outcomes described in ISO/IEC 12207:2008 \[B11\]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Requirements Analysis process outcomes are contained in AnnexL.](#)

9.2 Activity: [Software](#) Requirements V&V

The [Software](#) Requirements V&V activity addresses software requirements analysis of the functional and performance requirements; interfaces external to the software; and requirements for qualification, safety and security, human factors engineering, data definitions, user documentation for the software, installation and acceptance, user operation and execution, and user maintenance. V&V test planning begins during the [Software](#) Requirements V&V activity and spans several V&V activities.

The objective of [Software](#) Requirements V&V is to ~~ensure~~ [assure](#) the correctness, completeness, accuracy, testability, and consistency of the system software requirements.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Requirements V&V tasks described in Table 1c:

- 1) ~~Task:~~ Requirements Evaluation
- 2) ~~Task:~~ Interface Analysis
- 3) ~~Task:~~ Traceability Analysis
- 4) ~~Task:~~ Criticality Analysis
- 5) ~~Task:~~ Software Qualification ~~System~~ Test Plan V&V ~~generation~~
- 6) ~~Task:~~ Software Acceptance Test Plan V&V ~~generation~~
- 7) ~~Task: Configuration management assessment~~
- 8) ~~Task:~~ Hazard Analysis
- 9) ~~Task:~~ Security Analysis
- 10) ~~Task:~~ Risk Analysis

The Software Requirements V&V tasks listed shall be used to verify and validate the Software Requirements Analysis process outcomes, Software Qualification Testing process outcomes, and Software Acceptance Support process outcomes described in ISO/TEC 12207:2008 [BII]. The mappings of IEEE 1012 V&V tasks to ISO/IEC 12207:2008 [BII] Software Requirements Analysis

[process outcomes, Software Qualification Testing process outcomes, and Software Acceptance Support process outcomes are contained in Annex L.](#)

9.3 Activity: [Software](#) Design V&V

In software design, software requirements are transformed into an architecture and a detailed design for each software component. The design includes databases and system interfaces (e.g., hardware, operator/user, software components, and subsystems). The [Software](#) Design V&V activity addresses software architectural design and software detailed design. V&V test planning continues during the [Software](#) Design V&V activity.

The objective of [Software](#) Design V&V is to demonstrate that the design is a correct, accurate, and complete transformation of the software requirements and that no unintended features are introduced.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Design V&V tasks described in Table 1c:

- 1) ~~Task: Software~~ Design Evaluation
- 2) ~~Task:~~ Interface Analysis
- 3) ~~Task:~~ Traceability Analysis
- 4) ~~Task:~~ Criticality Analysis
- 5) ~~Task:~~ [Software](#) Component Test Plan V&V ~~generation~~
- 6) ~~Task:~~ [Software](#) Integration Test Plan V&V ~~generation~~
- 7) ~~Task:~~ [Software](#) Component Test Design V&V ~~generation~~
- 8) ~~Task:~~ [Software](#) Integration Test Design V&V ~~generation~~
- 9) ~~Task:~~ [Software](#) ~~System~~ Qualification Test Design V&V ~~generation~~

- 10) ~~Task:~~ Software Acceptance Test Design V&V ~~generation~~
- 11) ~~Task:~~ Hazard Analysis
- 12) ~~Task:~~ Security Analysis
- 13) ~~Task:~~ Risk Analysis

The Software Design V&V tasks listed shall be used to verify and validate the Software Architectural Design process outcomes, Software Detailed Design process outcomes, Software Integration process outcomes, Software Qualification Testing process outcomes, Software Acceptance Testing process outcomes, and Software Acceptance Support process outcomes described in ISO/IEC 12207:2008 [B1]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Architectural Design process outcomes, Software Detailed Design process outcomes, Software Integration process outcomes, Software Qualification Testing process outcomes, Software Acceptance Testing process outcomes, and Software Acceptance Support process outcomes are contained in Annex L.

9.4 Activity Software Construction ~~Implementation~~ V&V

In software ~~implementation~~ construction, the system design is transformed into code, database structures, and related machine executable representations. The ~~Implementation~~ Software Construction V&V activity addresses software coding and testing, including the incorporation of reused software products.

The objective of Software Construction ~~Implementation~~ V&V is to verify and validate that these transformations are correct, accurate, and complete.

The V&V effort shall perform, as specified in Table 2c for the selected ~~software~~ integrity level, the following Software Construction ~~Implementation~~ V&V tasks described in Table 1c:

- 1) ~~Task:~~ Source Code and Source Code Documentation Evaluation
- 2) ~~Task:~~ Interface Analysis
- 3) ~~Task:~~ Traceability Analysis
- 4) ~~Task:~~ Criticality Analysis
- 5) ~~Task:~~ [Software](#) Component Test Case V&V ~~generation~~
- 6) ~~Task:~~ [Software](#) Integration Test Case V&V ~~generation~~
- 7) ~~Task:~~ [Software Qualification](#) ~~System~~ Test Case V&V ~~generation~~
- 8) ~~Task:~~ [Software](#) Acceptance Test Case V&V ~~generation~~
- 9) ~~Task:~~ [Software](#) Component Test Procedure V&V ~~generation~~
- 10) ~~Task:~~ [Software](#) Integration ~~V&V test procedure generation~~
- 11) ~~Task:~~ [Software Qualification](#) ~~System~~ Test Procedure V&V ~~generation~~
- 12) ~~Task:~~ [Software](#) Component Test Execution V&V
- 13) ~~Task:~~ Hazard Analysis
- 14) ~~Task:~~ Security Analysis

15) ~~Task-~~Risk Analysis

The Software Construction V&V tasks listed shall be used to verify and validate the Software Construction process outcomes, Software Integration process outcomes, Software Qualification Testing process outcomes, and Software Acceptance Support process outcomes described in ISO/IEC 12207:2008 [B11]. The mappings of IEEE 1012 V&V tasks to the IEEE 12207 Software Construction process outcomes, Software Integration process outcomes, Software Qualification Testing process outcomes, and Software Acceptance Support process outcomes are contained in Annex L.

9.5 Activity: Software Integration Test V&V

~~Testing includes software testing, software integration testing, software qualification testing, system integration testing, and system qualification testing. The Test V&V activity and its relationship to the software life cycle are shown in Figure 2.~~ The objective of software integration ~~Test~~ V&V testing is to assure that the software requirements and system requirements allocated to software are validated ~~by execution of integration, system, and acceptance tests~~ as each software component (e.g., unit or module) is incrementally integrated.

The V&V effort shall perform, as specified in Table 2c for the selected ~~software~~ integrity level, the following software integration V&V test tasks described in Table 1c:

1) ~~Task-~~Software Integration Test Execution V&V

- 2) ~~Task:~~ Traceability Analysis
- 3) ~~Task: Acceptance V&V test procedure generation~~
- 4) ~~Task: Integration V&V test execution~~
- 5) ~~Task: System V&V test execution~~
- 6) ~~Task: Acceptance V&V test execution~~
- 7) ~~Task:~~ Hazard Analysis
- 8) ~~Task:~~ Security Analysis
- 9) ~~Task:~~ Risk Analysis

The Software Integration Test V&V tasks listed shall be used to verify and validate the Software Integration process outcomes described in ISO/TEC 12207:2008 [B1]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Integration process outcomes are contained in Annex L.

9.6 Activity: Software Qualification Test V&V

The objective of software qualification V&V testing is to assure that the integrated software product satisfies its requirements. Software qualification (e.g., demonstration, analysis, inspection, or test) is performed on the complete software element.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following

Software Test V&V tasks described in Table 1c:

- 1) Software Qualification Test Execution V&V
- 2) Traceability Analysis
- 3) Hazard Analysis
- 4) Security Analysis
- 5) Risk Analysis

The Software Qualification Test V&V tasks listed shall be used to verify and validate the Software Qualification Testing process outcomes described in ISO/TEC 12207:2008 [B11]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Qualification Testing process outcomes are contained in Annex L.

9.7 Activity: Software Acceptance Test V&V

The objective of software acceptance V&V testing is to assure that the software satisfies its acceptance criteria and to enable the customer to determine whether or not to accept the integrated software product.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Test V&V tasks described in Table 1c:

- 1) Software Acceptance Test Procedure V&V
- 2) Software Acceptance Test Execution V&V
- 3) Traceability Analysis

4) [Hazard Analysis](#)

5) [Security Analysis](#)

6) [Risk Analysis](#)

[The Software Acceptance Test V&V tasks listed shall be used to verify and validate the Software Acceptance Support process outcomes and Software Operation process outcomes described in ISO/IEC 12207:2008 \[B11\]. The mappings of IEEE 1012 V&V tasks to ISO/IEC 12207:2008 \[B11\] Software Acceptance Support process outcomes and Software Operation process outcomes are contained in Annex L.](#)

9.8 Activity: Software Installation and Checkout V&V

In installation and checkout, the software product is installed and tested in the target environment. The [Software](#) Installation and Checkout V&V activity supports the software system installation activities.

The objective of [Software](#) Installation and Checkout V&V is to verify and validate the correctness of the software installation in the target environment.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Installation and Checkout V&V tasks described in Table 1c:

- 1) ~~Task~~-Installation Configuration Audit
- 2) ~~Task~~-Installation Checkout

- 3) ~~Task:~~ Hazard Analysis
- 4) ~~Task:~~ Security Analysis
- 5) ~~Task:~~ Risk Analysis
- 6) ~~Task: V&V final report generation~~

The Software Installation and Checkout V&V tasks listed shall be used to verify and validate the Software Installation process outcomes and Software Acceptance Support process outcomes described in ISO/TEC 12207:2008 [B11]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Installation process outcomes and Software Acceptance Support process outcomes are contained in Annex L.

~~5.5 Process: Operation~~

~~The operation process involves the use of the software system by the end user in an operational environment.~~

9.9 Activity: Software Operation V&V

The Software Operation V&V activity evaluates the impact of changes in the operating environment, assesses the effect on the system of any proposed changes, evaluates operating

procedures for adherence with the intended use, and analyzes risks affecting the user and the system.

The objective of [Software](#) Operation V&V is to evaluate new constraints in the system, assess [the](#) proposed system changes and their impact on the software, and evaluate [the](#) operating procedures for correctness and usability.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Operation V&V tasks described in Table 1c:

- 1) ~~Task~~-Evaluation of New Constraints
- 2) ~~Task~~-Operating Procedures Evaluation
- 3) ~~Task~~-Hazard Analysis
- 4) ~~Task~~-Security Analysis
- 5) ~~Task~~-Risk Analysis

[The Software Operation V&V tasks listed shall be used to verify and validate the Software Operation process outcomes described in ISO/IEC 12207:2008 \[B11\]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Operation process outcomes are contained in Annex L.](#)

9.10 [Activity](#) ~~Process~~ [Software](#) Maintenance [V&V](#)

The [Software](#) Maintenance Process is activated when the software ~~system~~ or associated documentation ~~is must-be~~ changed in response to a need for system maintenance. The [Software](#) Maintenance V&V activity addresses [the following](#) software system [processes](#):

- Modifications (i.e., corrective, adaptive, or perfective changes)
- Migration (i.e., the movement of software to a new operational environment)
- Retirement (i.e., the withdrawal of active support by the operation and maintenance organization, partial or total replacement by a new system, or installation of an upgraded system)

5.6.1 Activity: Maintenance V&V

System modifications may be derived from [the](#) requirements specified to correct software errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the software system shall be treated as development processes and shall be verified and validated [by performing V&V tasks corresponding to the modifications](#) ~~as described in the following:~~

~~—5.1 Process: Management~~

~~—5.4. Process: Development~~

~~Software~~ Integrity-level assignments shall be assessed as described in Clause 5. The integrity-level assignments shall be revised as appropriate to reflect the requirements derived from the maintenance process.

~~For migrating software, the V&V effort shall verify that the migrated software meets the requirements of Clauses 4 and 5.~~

If ~~the~~ software V&V was performed in accordance with this standard, [then](#) the maintenance process shall continue to conform to this standard. If the software was not verified and validated using this

standard and appropriate documentation is not available or adequate, [then](#) the [Software](#) Maintenance V&V effort shall determine whether the missing or incomplete documentation should be generated. In making this determination of whether to generate missing documentation, the minimum V&V requirements of the assigned ~~software~~ integrity level shall be taken into consideration.

The objectives of [Software](#) Maintenance V&V are to assess [the](#) proposed software system changes and their impact on the software, [to](#) evaluate [the](#) anomalies that are discovered during operation, [to](#) assess [the](#) migration requirements, [to](#) assess [the](#) retirement requirements, and [to](#) reperform [the](#) V&V tasks. The proposed changes are assessed by the Proposed/Baseline Change Assessment task of the Management of V&V activity.

The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Maintenance V&V tasks described in Table 1c:

- 1) ~~Task:~~SVVP Revision
- 2) ~~Task:~~Anomaly Evaluation
- 3) ~~Task:~~Criticality Analysis
- 4) ~~Task:~~Migration Assessment
- 5) ~~Task:~~Retirement Assessment
- 6) ~~Task:~~Hazard Analysis
- 7) ~~Task:~~Security Analysis
- 8) ~~Task:~~Risk Analysis
- 9) ~~Task:~~Task Iteration

The Software Maintenance V&V tasks listed shall be used to verify and validate the Software Maintenance process outcomes described in ISO/IEC 12207:2008 [B11]. The mappings of IEEE 1012 V&V tasks to the IEEE 12207 Software Maintenance process outcomes are contained in Annex L.

9.11 Activity: Software Disposal V&V

The purpose of software disposal is to end the existence of a system's software element.

This process ends active support by the operation and maintenance organization, or deactivates, disassembles, and removes the affected software products, consigning them to a final condition and leaving the environment in an acceptable condition. This process destroys or stores system software elements and related products in a sound manner, in accordance with legislation, agreements, organizational constraints, and stakeholder requirements. Where required, it maintains records that may be monitored. The V&V effort shall perform, as specified in Table 2c for the selected integrity level, the following Software Disposal V&V task described in Table 1c:

1) Software Disposal Evaluation

The Software Disposal V&V task listed shall be used to verify and validate the Software Disposal process outcomes described in ISO/IEC 12207:2008 [B11]. The mappings of IEEE 1012 V&V tasks to IEEE 12207 Software Disposal process outcomes are contained in Annex L.

Table 1 c— V&V tasks, inputs, and outputs

9.1 Activity: Concept V&V (Process: Development) (Software, Process: Software Requirements Analysis)		
V&V tasks	Required inputs	Required outputs
(1) Concept Documentation Evaluation a) Validate that the concept documentation satisfies user needs and is consistent with acquisition needs. b) Validate constraints of interfacing systems and constraints or limitations of proposed approach. c) Analyze system requirements and validate that the following satisfy user needs: 1) System functions. 2) End-to-end system performance. 3) Feasibility and testability of the functional requirements. 4) System architecture design. 5) Operation and maintenance requirements and environments. 6) Migration requirements from an existing system where applicable.	Concept documentation System architectural design Supplier development plans and schedules User needs Acquisition needs	Task report(s) - Concept documentation evaluation Anomaly report(s)
(2) Hardware/Software/User Requirements Allocation Analysis Verify the correctness, accuracy, and completeness of the concept requirement allocation to hardware, software, and user interfaces against user needs.	User needs Concept documentation	Task report(s) - Hardware/software/user requirements allocation analysis

a) Correctness Verify that performance requirements (e.g., timing, response time, and throughput) allocated to hardware, software, and user interfaces satisfy user needs. b) Accuracy Verify that the internal and external interfaces specify the interface protocols, interface configuration, and other key performance requirements to demonstrate satisfaction of user requirements. c) Completeness 1) Verify that application specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs. 2) Verify that the user's maintenance requirements for the system are completely specified. 3) Verify that the migration from existing system and replacement of the system satisfy user needs.		Anomaly report(s)
(3) Traceability Analysis a) Identify all system requirements. that will be implemented completely or partially by software. b) Verify that these system requirements are traceable to acquisition needs. c) Start the requirements traceability analysis with system requirements.	Concept documentation	Task report(s) - Traceability analysis Anomaly report(s)
(4) Criticality Analysis a) Determine whether integrity levels are established for	Concept documentation	Task report(s) - Criticality analysis

requirements, detailed functions, hardware components, subsystem, or other partitions. b) Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the system requirements. c) Document the integrity level assigned to individual components (e.g., requirements, detailed functions, hardware components, subsystems, or other partitions). For V&V planning purposes, the system shall be assigned the same integrity level as the highest level assigned to any individual element. d) Verify whether any component can influence individual components assigned a higher integrity level, and if such conditions exist, then assign that component the same higher integrity level.	(system requirements) Developer integrity level assignments	Anomaly report(s)
(5) Hazard Analysis Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following: a) Identify the potential system hazards, b) Assess the consequences of each hazard, c) Assess the probability of each hazard, d) Identify mitigation strategies for each hazard.	Concept documentation	Task report(s) - Hazard analysis Anomaly report(s)
(6) Security Analysis a) Review the system owner's definition of an acceptable level of security risk. b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of	Concept documentation Preliminary TRA	Task report(s) - Security analysis Anomaly report(s)

sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/ process) have been identified. Include an assessment of the sensitivity of the information/data to be processed. c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.		
(7) Risk Analysis a) Identify the technical and management risks, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Concept documentation Supplier development plans and schedules Hazard analysis report Security analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

9.2 Activity: <u>Software</u> Requirements V&V (Process: Development) (<u>Software, Process: Software Requirements Analysis</u>)		
V&V tasks	Required inputs	Required outputs
(1) Requirements Evaluation Evaluate the requirements (e.g., functional, capability, interface, qualification, safety, security, human factors, data definitions, user documentation, installation and acceptance, user operation, and user maintenance) of the	Concept documentation SRS	Task report(s) - Hardware requirements evaluation

hardware requirements specification (HRS) and IRS for correctness, consistency, completeness, accuracy, readability, and testability. The task criteria are as follows: a) Correctness 1) Verify and validate that the software requirements satisfy the system requirements allocated to software within the assumptions, constraints, and operating environment for the system. 2) Verify that the software requirements comply with standards, references, regulations, policies, physical laws, and business rules. 3) Validate the sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other basis. 4) Validate that the flow of data and control satisfy functionality and performance requirements. 5) Validate data usage and format. b) Consistency 1) Verify that all terms and concepts are documented consistently. 2) Verify that the function interactions and assumptions are consistent and satisfy system requirements and acquisition needs. 3) Verify that there is internal consistency between the software requirements and external consistency with the system requirements.	IRS	Anomaly report(s)
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c) Completeness 1) Verify that the following elements are in the SRS or IRS, within the assumptions and constraints of the system: i) Functionality (e.g., algorithms, state/mode definitions, input/output validation, exception handling, reporting and logging). ii) Process definition and scheduling. iii) Hardware, software, and user-interface descriptions. iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security). v) Critical configuration data. vi) System, device, and software control (e.g., initialization, transaction and state monitoring, and self-testing). 2) Verify that the SRS and IRS satisfy specified configuration management procedures. d) Accuracy 1) Validate that the logic, computational, and interface precision (e.g., truncation and rounding) satisfy the requirements in the system environment. 2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws.		
e) Readability		

1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience. 2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols. f) Testability Verify that there are objective acceptance criteria for validating the requirements of the SRS and IRS.		
(2) Interface Analysis Verify and validate that the requirements for hardware interfaces with software, user, operator, other hardware, and other systems are correct, consistent, complete, accurate, and testable. The task criteria are as follows: a) Correctness Validate the external and internal system and software interface requirements. b) Consistency Verify that the interface descriptions are consistent between the SRS and IRS. c) Completeness Verify that each interface is described and includes data format and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security). d) Accuracy Verify that each interface provides information with the required accuracy. e) Testability Verify that there are objective acceptance criteria for validating the interface requirements.	Concept documentation SRS IRS	Task report(s) - Interface analysis Anomaly report(s)
(3) Traceability Analysis	Concept	Task report(s) —

Trace the hardware requirements (SRS and IRS) to system requirements (Concept Documentation) and system requirements to the software requirements. Analyze identified relationships for correctness, consistency, completeness, and accuracy. The task criteria are as follows: a) Correctness Validate that the relationships between each software requirement and its system requirement are correct. b) Consistency Verify that the relationships between the software and system requirements are specified to a consistent level of detail. c) Completeness 1) Verify that every software requirement is traceable to a system requirement with sufficient detail to show conformance to the system requirement. 2) Verify that all system requirements related to software are traceable to software requirements. d) Accuracy Validate that the system performance and operating characteristics are accurately specified by the traced software requirements.	Documentation (System requirements) SRS IRS	Traceability analysis Anomaly report(s)
(4) Criticality Analysis a) Review and update the existing criticality analysis results from the prior Criticality Task Report using the SRS and IRS. b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e.,	Criticality task report SRS IRS	Task report(s) - Criticality analysis Anomaly report(s)

requirement, module, function, subsystem, and other software partition). Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels.		
(5) <u>Software Qualification</u> System Test Plan V&V generation a) Integrity Levels 3 and 4 1) Plan V&V hardware qualification testing to validate software requirements 2) Plan tracing of system requirements to test designs, cases, procedures, and results. 3) Plan documentation of V&V software qualification test designs, cases, procedures, and results. 4) The V&V software qualification test plan shall address the following: i) Conformance to all system requirements (e.g., functional, performance, security, operation, and maintenance) as complete hardware end items in the system environment. ii) Adequacy of user documentation (e.g., training materials and procedural changes). iii) Performance at boundaries (e.g., data and interfaces) and under stress conditions. 5) Verify that the V&V hardware qualification test plan satisfies the following criteria: i) Conformance to project-defined test document purpose, format, and content. (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). ii) Test coverage of system requirements. 6) Validate that the V&V software qualification test plan	Concept documentation (System requirements) SRS IRS User documentation <u>software qualification test plan</u> System test plan	V&V software qualification test plan Anomaly report(s)

satisfies the following criteria: i) Appropriateness of test methods and standards used, ii) Conformance to expected results, iii) Feasibility of system qualification testing. iv) Feasibility and testability of operation and maintenance requirements. b) Software Integrity 1 Level 2 1) Verify that the developer's hardware qualification test plan satisfies the following criteria: i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). ii) Test coverage of system requirements. 2) Validate that the developer's software qualification test plan satisfies the following criteria: i) Appropriateness of test methods and standards used. ii) Conformance to expected results. iii) Feasibility of system qualification testing. iv) Capability to be operated and maintained.		
(6) Software Acceptance Test Plan V&V generation a) Software Integrity Levels 3 and 4 1) Plan V&V software acceptance testing to validate that the software correctly implements system and software requirements in an operational environment. 2) Plan tracing of test requirements to test software acceptance design, cases, procedures, and execution results.	Concept documentation SRS IRS User documentation	V&V hardware acceptance test plan Anomaly report(s)

3) Plan documentation of test tasks and results. 4) The V&V software acceptance test plan shall address the following: i) Conformance to acceptance requirements in the operational environment. ii) Adequacy of user documentation. 5) Verify that the V&V software acceptance test plan satisfies the following criteria: i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] -1998 [B4]). ii) Test coverage of acceptance requirements. 6) Validate that the V&V software acceptance test plan satisfies the following criteria: i) Conformance to expected results. ii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs). b) Software Integrity Level 2 1) Verify that the acquirer's software acceptance test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]). 2) Validate that the acquirer's software acceptance test plan satisfies the following criteria: i) Test coverage of acceptance requirements. ii) Conformance to expected results. iii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs). c) Software Integrity Level 1	Software acceptance test plan	
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There are no software acceptance V&V test requirements.		
(7) Hazard Analysis a) Determine software contributions to system hazards. The hazard analysis shall perform the following: 1) Identify the software requirements that contribute to each system hazard. 2) Validate that the software addresses, controls, or mitigates each hazard.	SRS IRS Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)
(8) Security Analysis a) Determine that the security requirements identified in the SRS and IRS address the security risks introduced by the system concept. b) Verify that the system security requirements will mitigate the identified security risks to an acceptable level.	SRS IRS Preliminary TRA Security analysis report	Task report(s) - Security analysis Anomaly report(s)
(9) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Concept documentation SRS IRS Supplier development plans and schedules Hazard analysis report Security analysis report	Task report(s) - Risk analysis Anomaly report(s)

	Risk analysis report V&V task results	
9.3 Activity: Software Design V&V (Process: Development) (Software, Process: Software Architectural Design and software Detailed Design)		
V&V tasks	Required inputs	Required outputs
(1) Software Design Evaluation Evaluate the design elements (SDD and IDD) for correctness, consistency, completeness, accuracy, readability, testability, and testability. The task criteria are as follows: a) Correctness 1) Verify and validate that the software design satisfies the software requirements. 2) Verify that the software design complies with standards, references, regulations, policies, physical laws, and business rules. 3) Validate the design sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other basis. 4) Validate that the detailed design and its element interactions do not result in unnecessary, unintended, or deleterious consequences. 5) Validate that the flow of data and control satisfy functionality and performance requirements. 6) Validate data usage and format. 7) Assess the appropriateness of design methods and	SRS IRS SDD IDD Design standards (e.g., standards, practices, and conventions)	Task report(s) - design evaluation Anomaly report(s)

standards used.		
b) Consistency 1) Verify that all terms and design concepts are documented consistently. 2) Verify that there is internal consistency between the design elements and external consistency with architectural design. c) Completeness 1) Verify that the following elements are in the SDD, within the assumptions and constraints of the system: i) Functionality (e.g., algorithms, state/mode definitions, input/output validation, exception handling, reporting, and logging). ii) Process definition and scheduling. iii) Hardware, software, and user interface descriptions. iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security). v) Critical configuration data. vi) System, device, and software control (e.g., initialization, transaction and state monitoring, and self-testing). vii) Verify that the SDD and IDD satisfy specified configuration management procedures. d) Accuracy 1) Validate that the logic, computational, and interface precision (e.g., truncation and rounding) satisfy the requirements in the system		

environment. 2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws. e) Readability 1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience. 2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, symbols, and design language, if any. f) Testability 1) Verify that there are objective acceptance criteria for validating each software design element and the system design. 2) Verify that each software design element is testable to objective acceptance criteria.		
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(2) Interface Analysis Verify and validate that the software design interfaces with hardware, user, operator, other software, and other systems for correctness, consistency, completeness, accuracy, and testability. The task criteria are as follows: a) Correctness Validate the external and internal software interface design in the context of system requirements.	Concept documentation System requirements SRS IRS	Task report(s) - Interface analysis Anomaly report(s)
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b) Consistency Verify that the interface design is consistent between the SDD and IDD. c) Completeness Verify that each interface is described and includes data format and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security). d) Accuracy Verify that each interface provides information with the required accuracy. e) Testability Verify that there are objective acceptance criteria for validating the interface design.	SDD IDD	
(3) Traceability Analysis Trace the design elements (SDD and IDD) to the requirements (SRS and IRS), and the requirements to the design elements. Analyze the relationships for correctness, consistency, and completeness. The task criteria are as follows: a) Correctness Validate the relationship between each design element and the software requirement(s). b) Consistency Verify that the relationships between the design elements and the software requirements are specified to a consistent level of detail. c) Completeness 1) Verify that all design elements are traceable from the software requirements. 2) Verify that all software requirements are traceable	SRS SDD IRS IDD	Task report(s) - Traceability analysis Anomaly report(s)

to the design elements.		
(4) Criticality Analysis a) Review and update the existing criticality analysis results from the prior Criticality Task Report using the SDD and IDD. b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e., requirement, module, function, subsystem, other software partition). Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels.	Criticality task report SDD IDD	Task report(s) - Criticality analysis Anomaly report(s)
(5) <u>Software</u> Component Test Plan <u>Assessment</u> generation a) Software Integrity Levels 3 and 4 1) Plan V&V software component testing to validate that the software components (e.g., units and source code modules) correctly implement component requirements. 2) Plan tracing of design requirements to test design, cases, procedures, and results. 3) Plan documentation of test tasks and results. 4) The V&V software component test plan shall address the following: i) Conformance to design requirements. ii) Assessment of timing, sizing, and accuracy. iii) Performance at boundaries and interfaces and under stress and error conditions. iv) Measures of requirements test coverage and	SRS SDD IRS IDD <u>Software</u> component test plan	Component V&V software component test plan Anomaly report(s)

software reliability and maintainability. 5) Verify that the V&V software component test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 829-1998 [B4]). 6) Validate that the V&V software component test plan satisfies the following criteria: i) Traceable to the software requirements and design. ii) External consistency with the software requirements and design.		
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9.3 Activity: Software Design V&V (Process: Development) (Software, Process: Software Architectural Design and software Detailed Design		
V&V tasks	Required inputs	Required outputs
iii) Internal consistency between unit requirements. iv) Test coverage of requirements in each unit. v) Feasibility of software integration and testing. vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs). b) Software Integrity Level 2 1) Verify that the developer's software component test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 829-1998 [B4]).		

2) Validate that the developer's software component test plan satisfies the following criteria: i) Traceable to the software requirements and design. ii) External consistency with the software requirements and design. iii) Internal consistency between unit requirements. iv) Test coverage of units. v) Feasibility of software integration and testing. vi) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs). c) Software Integrity Level 1 There are no <u>software</u> component <u>V&V</u> test requirements.		
(6) <u>Software</u> Integration Test Plan V&V generation a) For software Integrity Levels 3 and 4 1) Plan V&V software integration testing to validate that the software correctly implements the software requirements and design as each software component (e.g., units or modules) is incrementally integrated with each other. 2) Plan tracing of requirements to test design, cases, procedures, and results. 3) Plan documentation of test tasks and results. 4) The Integration V&V software integration test plan shall address the following: i) Conformance to increasingly larger set of functional requirements at each stage of	SRS IRS SDD IDD <u>Software</u> integration test plan	V&V software integration test plan Anomaly report(s)

integration. ii) Assessment of timing, sizing, and accuracy. iii) Performance at boundaries and under stress conditions. iv) Measures of requirements test coverage and software reliability. 5) Verify that the V&V software integration test plan satisfies the following criteria: i) Conformance to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 829-1998 [B4]). 6) Validate that the V&V software integration test plan satisfies the following criteria: i) Traceable to the system requirements. ii) External consistency with the system requirements. iii) Internal consistency. iv) Test coverage of the software requirements.		
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9.3 Activity: Software Design V&V (Process: Development) (Software, Process: Software Architectural Design and software Detailed Design		
V&V tasks	Required inputs	Required outputs
v) Appropriateness of test standards and methods used. vi) Conformance to expected results. vii) Feasibility of software qualification testing. viii) Feasibility of operation and maintenance (e.g.,		

capability to be operated and maintained in accordance with user needs). b) Software Integrity Levels 1 and 2 1) Verify that the developer's software integration test plan conforms to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 2) Validate that the developer's software integration test plan satisfies the following criteria: i) Traceable to the system requirements. ii) External consistency with the system requirements. iii) Internal consistency. iv) Test coverage of the software requirements. v) Appropriateness of test standards and methods. vi) Conformance to expected results. vii) Feasibility of software qualification testing. viii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).		
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(7) <u>Software</u> Component Test Design V&V generation a) Software Integrity Levels 3 and 4 1) Design tests for V&V software component testing. 2) Continue tracing required by the V&V software component test plan. 3) Verify that the V&V software component test designs conform to project-defined test document	SDD IDD User documentation <u>Software</u>	V&V software component test design(s) Anomaly report(s)
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purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 4) Validate that the V&V software component test designs satisfy the criteria in V&V activity 5.4.3 Task 5. b) Software Integrity Level 2 1) Verify that the developer's test designs for software component testing conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 2) Validate that the developer's software component test designs satisfy the criteria in V&V activity 5.4.3, Task 5. c) Software Integrity Level 1 There are no software component V&V test requirements.	component test plans Software component test designs	
(8) Software Integration System Test Design V&V generation a) Software Integrity Levels 3 and 4 1) Design tests for V&V software integration testing. 2) Continue tracing required by the V&V software integration test plan. Verify that the V&V software integration test designs conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 3) Validate that the V&V software integration test designs satisfy the criteria in V&V Activity 5.4.3, Task 6. b) Software Integrity Levels 1 and 2	SDD IDD User documentation Software integration test plans Software	V&V software integration System test design(s) Anomaly report(s)

	integration test designs	
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9.3 Activity: Software Design V&V (Process: Development) (Software, Process: Software Architectural Design and software Detailed Design)		
V&V tasks	Required inputs	Required outputs
1) Verify that the developer's test designs for software integration testing conform to project-defined test document purpose, format, and content. e.g., IEEE Std 829- 2008 [B2] 1998 [B4] . 2) Validate that the developer's software integration system test designs satisfy the criteria in V&V activity 5.4.3, Task 6.		
(9) Software Qualification System Test Design V&V generation a) Software Integrity Levels 3 and 4 1) Design tests for V&V software qualification testing. 2) Continue tracing required by the V&V software qualification test plan. Verify that the V&V software qualification test designs conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 3) Validate that the System V&V software qualification test designs satisfy the criteria in V&V Activity 5.4.2, Task 5 b) Software Integrity Level 1 and 2 1) Verify that the developer's test designs for software qualification system testing conform to project-	SDD IDD User documentation plans Software qualification test Software qualification test designs	V&V software qualification test design(s) Anomaly report(s)

defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 2) Validate that the developer's software qualification system test designs satisfy the criteria in V&V Activity 5.4.2, Task 5.		
(10) SOFTWARE Acceptance Test Design V&V generation a) Software Integrity Levels 3 and 4 1) Design tests for V&V software acceptance testing. 2) Continue tracing required by the V&V software acceptance test plan. Verify that the V&V software acceptance test designs conform to project-defined test document purpose, format, and content (e.g., see IEEE Std 829-1998 [B4]). Software acceptance 3) Validate that the V&V software acceptance test designs satisfy the criteria in V&V Activity 5.4.2 Task 6. b) Software Integrity Level 2 1) Verify that the acquirer's test designs for software acceptance testing conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). Software acceptance 2) Validate that the acquirer's software acceptance test designs satisfy the criteria in V&V activity 5.4.2, Task 6. c) Software Integrity Level 1 There are no hardware acceptance V&V test requirements.	SDD IDD User documentation test plans Software acceptance	V&V hardware acceptance test design(s) Anomaly report(s)
(11) Hazard Analysis a) Verify the logic design and associated data elements that correctly implement the critical requirements introduce no new hazards.	SDD IDD	Task report(s) – Hazard analysis Anomaly

b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations). c) Update the hazard analysis.	Hazard analysis Report	report(s)
9.3 Activity: Software Design V&V (Process: Development) (Software, Process: Software Architectural Design and software Detailed Design)		
V&V tasks	Required inputs	Required outputs
(12) Security Analysis a) Verify that the architecture and detailed design outputs adequately address the identified security requirements. This verification includes both the system itself and security risks introduced as a result of interfacing with external components. b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and software operations).	SDD IDD Security analysis report	Task report(s) Security analysis Anomaly report(s)
(13) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	SDD IDD Supplier development plans and schedules Hazard analysis Report Security analysis Report Risk analysis	Task report(s) –Risk analysis Anomaly report(s)

	report V&V task results	
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9.4 Activity: [Hardware Fabrication](#) V&V **Implementation** ([Hardware](#), Process: [Fabrication Development](#))

V&V tasks	Required inputs	Required outputs
(1) Source Code and Source Code Documentation Evaluation Evaluate the source code components (source code and source code documentation) for correctness, consistency, completeness, accuracy, readability, and testability. The task criteria are as follows: a) Correctness 1) Verify and validate that the source code component satisfies the software design. 2) Verify that the source code components comply with standards, references, regulations, policies, physical laws, and business rules. 3) Validate the source code component sequences of states and state changes using logic and data flows coupled with domain expertise, prototyping results, engineering principles, or other basis. 4) Validate the software code and its interactions with other elements do not result in unnecessary, unintended, or deleterious consequences. 5) Validate that the flow of data and control satisfy functionality and performance requirements. 6) Validate data usage and format.	Source Code SDD IDD Coding standards (e.g., standards, practices, project restrictions, and conventions) User documentation	Task report(s) – Source code and source code documentation evaluation Anomaly report(s)

7) Assess the appropriateness of coding methods and standards. b) Consistency 1) Verify that all terms and code concepts are documented consistently. 2) Verify that there is internal consistency between the source code components. 3) Validate external consistency with the software design and requirements. c) Completeness 1) Verify that the following elements are in the source code, within the assumptions and constraints of the system: i) Functionality (e.g., algorithms, state/mode definitions, input/output validation, exception handling, reporting, and logging). ii) Process definition and scheduling. iii) Hardware, software, and user interface descriptions. iv) Performance criteria (e.g., timing, sizing, speed, capacity, accuracy, precision, safety, and security). v) Critical configuration data. vi) System, device, and software control (e.g., initialization, transaction and state monitoring, defensive programming practices, and self-testing). 2) Verify that the source code documentation satisfies specified <u>coding standards</u> configuration management procedures.		
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d) Accuracy 1) Validate the logic, computational, and interface precision (e.g., truncation and rounding) in the system environment. 2) Validate that the modeled physical phenomena conform to system accuracy requirements and physical laws. e) Readability 1) Verify that the documentation is legible, understandable, and unambiguous to the intended audience. 2) Verify that the documentation defines all acronyms, mnemonics, abbreviations, terms, and symbols. f) Testability 1) Verify that there are objective acceptance criteria for validating each source code component. 2) Verify that each source code component is testable against objective acceptance criteria.		
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9.4 Activity: <u>Hardware Fabrication</u> Implementation —V&V (Hardware, Process: Fabrication Development)		
V&V tasks	Required inputs	Required outputs
(2) Interface Analysis Verify and validate that the software source code interfaces with hardware, user, operator, software, and other systems for correctness, consistency, completeness, accuracy, and testability. The task criteria are as follows: a) Correctness	Concept Documentation System requirements SDD	Task report(s) Interface analysis Anomaly report(s)

Validate the external and internal software interface code in the context of system requirements b) Consistency Verify that the interface code is consistent between source code components and to external interfaces (i.e., hardware, user, operator, and other software). c) Completeness Verify that each interface is described and includes data format and performance criteria (e.g., timing, bandwidth, accuracy, safety, and security). d) Accuracy Verify that each interface provides information with the required accuracy. e) Testability Verify that there are objective acceptance criteria for validating the interface code	IDD Source code User documentation	
(3) Traceability Analysis a) Trace the source code components to corresponding design specification(s), and design specification(s) to source code components. The task criteria are as follows: b) Analyze identified relationships for correctness, consistency, and completeness. 1) Correctness Validate the relationship between the source code components and design element(s). 2) Consistency Verify that the relationships between the source code components and design elements are specified to a consistent level of detail.	SDD IDD Source code	Task report(s) – Traceability analysis Anomaly report(s)

3) Completeness i) Verify that all source code components are traceable from the design elements. ii) Verify that all design elements are traceable to the source code components		
(4) Criticality Analysis a) Review and update the existing criticality analysis results from the prior Criticality Task Report using the source code b) Implementation methods and interfacing technologies may cause previously assigned integrity levels to be raised or lowered for a given software element (i.e., requirement, module, function, subsystem, or other software partition). Verify that no inconsistent or undesired integrity consequences are introduced by reviewing the revised integrity levels. Criticality task report Source code Task report(s) –Criticality analysis Anomaly report(s)	Criticality Task Report Source code	Task report(s) Criticality analysis Anomaly report(s)
(5) Software Component Test Case V&V generation a) Software Integrity Levels 2, 3, and 4 1) Develop test cases for V&V test cases for component testing. software component testing (e.g., statistical sampling, boundary conditions, and code coverage)2) Validate that the developer's hardware component test cases satisfy the criteria in V&V Activity 10.3, Task 5. 2) Continue tracing required by the V&V software component test plan. 3) Verify that the software component test cases	SRS IRS SDD IDD User documentation Software component test design Software component test cases	V&V software component test cases Anomaly report(s)

conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 4) Validate that the software component test cases satisfy the criteria in V&V Activity 5.4.3, Task 5. b) Software Integrity Level 2 1) Verify that the developer's software component test cases conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2] 1998 [B4]). 2) Validate that the developer's software component test cases satisfy the criteria in V&V Activity 5.4.3, Task 5. c) Software Integrity Level 1 There are no software component V&V test requirements.		
(6) Software Integration Test Case V&V generation a) Software Integrity Levels 3, and 4 1) Develop test cases for V&V software integration testing. 2) Continue tracing required by the V&V software integration test plan. 3) Verify that the developer's hardware integration test cases conform to project-defined test document purpose, format, and content (e.g., see IEEE Std 829-1998 [B4]). 4) Validate that the V&V software integration test cases satisfy the criteria in V&V Activity 5.4.3, Task 6.	SRS IRS SDD IDD User documentation Software integration test design Software integration	V&V software integration test cases Anomaly report(s)

b) Software Integrity Level 1 and 2 1) Verify that the developer's software integration test cases conform to project-defined test document purpose, format, and content. 2) Validate that the developer's software integration test cases satisfy the criteria in V&V Activity 5.4.3, Task 6.	test cases	
(7) Software System Qualification Test Case V&V a) Software Integrity Levels 3 and 4 1) Develop test cases for V&V software qualification System testing. 2) Continue tracing required by the V&V software qualification System test plan. 3) Verify that the V&V software qualification system test cases conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 4) Validate that the V&V software qualification test cases satisfy the criteria in V&V Activity 10.2, Task 5. b) Software Integrity Level 1 and 2 1) Verify that the developer's software qualification system test cases conform to project-defined test document purpose, format, and content. 2) Validate that the developer's software qualification system test cases satisfy the criteria in V&V Activity 5.4.2 Task 5.	SRS IRS SDD IDD User documentation Software qualification test design Software qualification test cases	V&V software Acceptance System test cases Anomaly report(s)
(8) Software Acceptance Test Case V&V generation a) Software Integrity Levels 3 and 4 1) Develop test cases for V&V software acceptance	SRS IRS	V&V software acceptance test cases

testing. 2) Continue tracing required by the V&V software acceptance test plan. 3) Verify that the V&V software Acceptance test cases conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 4) Validate that the V&V software acceptance test cases satisfy the criteria in V&V Activity 10.2, Task 6. b) Software Integrity Level 2 1) Verify that the acquirer's software acceptance test cases conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 2) Validate that the acquirer's software acceptance test cases satisfy the criteria in V&V Activity 5.4.2 Task 6. c) Software Integrity Level 1 There are no software acceptance V&V test requirements.	SDD IDD User documentation Software acceptance test design Software acceptance test cases	Anomaly report(s)
(9)Software Component Test Procedures V&V generation a) Software Integrity Levels 2, 3, and 4 1) Develop test procedures for V&V software component testing. 2) Continue tracing required by the V&V software component testplan. 3) Verify that the V&V software component test procedures conform to project-defined test	SRS IRS SDD IDD User documentation Software component	V&V software component test procedures Anomaly report(s)

document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]).	test cases	
4) Validate that the V&V software component test procedures satisfy the criteria in V&V Activity 5.4.3, Task 5.	Software component test procedures	
b) Software Integrity Level 2		
1) Verify that the developer's software component test procedures conform to project-defined test document purpose, format, and content		
2) Validate that the developer's software component test procedures satisfy the criteria in V&V Activity 5.4.3, Task 5.		
c) Software Integrity Level 1		
There are no hardware component V&V test requirements.		

9.4 Activity: Hardware Fabrication Implementation —V&V (Hardware, Process: Fabrication Development)		
V&V tasks	Required inputs	Required outputs
(10) Software Integration Test Procedure V&V	SRS	V &V software integration test procedures
a) Software Integrity Levels 3, and 4		Anomaly report(s)
1) Develop test procedures for V&V software integration testing.	IRS	
2) Continue tracing required by the V&V software integration test plan.	SDD	
3) Verify that the V&V software integration test procedures conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]).	IDD	
	User documentation	
	Software integration	

4) Validate that the V&V software integration test procedures satisfy the criteria in V&V Activity 5.4.3 Task 6. b) Software Integrity Level 1 and 2 1) Verify that the developer's software integration test procedures conform to project-defined test document purpose, format, and content (e.g., see IEEE Std 829-1998 [B4]). 2) Validate that the developer's software integration test procedures satisfy the criteria in V&V Activity 5.4.3, Task 6.	test cases Software integration test procedures	
(11) Software System Qualification Test Procedure V&V generation a) Software Integrity Levels 3 and 4 1) Develop test procedures for V&V software qualification System testing. 2) Continue tracing required by the V&V software qualification System test plan. 3) Verify that the V&V software qualification test procedures conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]). 4) Validate that the V&V software qualification System test procedures satisfy the criteria in V&V Activity 5.4.2 Task 5. b) Software Integrity Level 2 1) Verify that the developer's software qualification System test procedures conform to project-defined test document purpose, format, and content. (e.g., see IEEE Std 829-1998 [B4]).	SRS IRS SDD IDD User documentation Software qualification test cases Software qualification test procedures	V&V software qualification System test procedures Anomaly report(s)

2) Validate that the developer's software qualification System test procedures satisfy the criteria in V&V Activity 5.4.2 Task 5.		
(12) Software Component Test Execution V&V a) Software Integrity Levels 2, 3, and 4 1) Perform V&V software component testing. 2) Analyze test results to validate that software correctly implements the design. 3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents. 4) Document the results as required by the V&V software component test plan. 5) Use the V&V software component test results to validate that the software satisfies the test acceptance criteria. 6) Document discrepancies between the actual and expected test Results b) Software Integrity Level 2 Use the developer's software component test results to validate that the software satisfies the test acceptance criteria. c) Software Integrity Level 1 There are no software component V&V test requirements.	Source code Executable code SDD IDD Software component test plans Software component test procedures Software component test results	Task report(s) – V&V software component test results Anomaly report(s)

(13) Hazard Analysis a) Verify that the implementation and associated data elements correctly implements the critical	Source code SDD	Task report(s) – Hazard analysis Anomaly report(s)
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requirements and introduce no new hazards. b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations). c) Update the hazard analysis.	IDD Hazard analysis Report	
(14) Security Analysis a) Verify that the implementation is completed in accordance with the system design in that it addresses the identified security risks and that the implementation does not introduce new security risks through coding flaws, or compiler error. b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and software operations).	Source code SDD IDD Security analysis report	Task report(s) – Security analysis Anomaly report(s)
(15) Risk Analysis Review and update risk analysis using prior task reports. Provide recommendations to eliminate, reduce, or mitigate the risks.	Source code Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) – Risk analysis Anomaly report(s)

9.5 Activity: [Software Integration](#) Test V&V (~~Process: Development~~) ([Software](#), [Process: Software Integration](#))

V&V tasks	Required inputs	Required outputs
(1) Software Integration Test Execution V&V a) Integrity Levels 2, 3, and 4 1) Perform V&V software integration testing. 2) Analyze test results to verify that the software components are integrated correctly. 3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents. 4) Document the results as required by the V&V software integration test plan. 5) Use the V&V software integration test results to validate that the hardware satisfies the test acceptance criteria. 6) Document discrepancies between the actual and expected test results. b) Software Integrity Level 1 and 2 Use the developer's software integration test results to verify that the software satisfies the test acceptance criteria.	Source code Executable code Software integration test plan Software integration test procedures Software integration test results	Task report(s) – Test results V&V software integration test results Anomaly report(s)
(2) Traceability Analysis Analyze relationships in the V&V Test Plans, Designs, Cases, and Procedures for correctness and completeness. The task criteria are as follows: a) Correctness Verify that there is a valid relationship between the V&V test plans, designs, cases, and procedures. b) Completeness Verify that all V&V test procedures are traceable to the V&V test plans.	V&V test plans V&V test designs V&V test procedures	Task report(s) – Traceability analysis Anomaly report(s)

9.5 Activity: Software Integration Test V&V (Software, Process: Software Integration)		
V&V tasks	Required inputs	Required outputs
(3) Hazard Analysis a) Verify that the test instrumentation does not introduce new hazards. <u>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) Update the hazard analysis.	Source code Executable code Test results Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)
(4) Security Analysis a) Verify that the implemented system does not increase the security risk. <u>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	Source code Executable code Security analysis report	Task report(s) - Security analysis Anomaly report(s)
(5) Risk Analysis a) Review and update risk analysis using prior task reports. b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) Risk analysis Anomaly report(s)

9.6 Activity: Software Qualification Test V&V (Software, Process: Software Qualification Testing)

V&V tasks	Required inputs	Required outputs
(1) System <u>Software Qualification</u> Test Execution V&V a) Software Integrity Levels 3 and 4 1) Perform V&V <u>software qualification</u> testing. 2) Analyze test results to validate that the software satisfies the system requirements. 3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents. 4) Document the results as required by the Acceptance <u>Software</u> V&V <u>software qualification</u> test plan. 5) Use the Acceptance V&V <u>software qualification</u> test results to validate that the software satisfies the test acceptance criteria. 6) Document discrepancies between the actual and expected test results. b) Software Integrity Levels 1 and 2 Use the acquirer's acceptance-developer's software <u>qualification</u> test results to verify that the software satisfies the test acceptance criteria. c) Software integrity level 1 There are no acceptance test requirements.	Source code Executable code User documentation Acceptance <u>Software</u> <u>qualification</u> test plan Acceptance <u>Software</u> <u>qualification</u> test procedures Acceptance <u>Software</u> <u>qualification</u> test results V&V task results	Task report(s) – Test results <u>V&V software qualification test results</u> Anomaly report(s)
(2) Traceability Analysis Analyze the relationships in the V&V <u>software</u> qualification test plans, designs, cases, and procedures for correctness and completeness. The task criteria are as follows:	V&V <u>software</u> <u>qualification</u> test plans V&V <u>software</u> <u>qualification</u> test	Task report(s) – Traceability Analysis Anomaly report(s)

9.6 Activity: Software Qualification Test V&V (Software, Process: Software Qualification Testing)		
V&V tasks	Required inputs	Required outputs
a) Correctness Verify that there is a valid relationship between the V&V software qualification test plans, designs, cases, and procedures. b) Completeness Verify that all V&V software qualification test procedures are traceable to the V&V test plans.	Designs V&V software qualification test cases V&V software qualification test procedures	
(3) Hazard Analysis a) Verify that the test instrumentation does not introduce new hazards. b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations). c) Update the hazard analysis.	Source code Executable code Test results Hazard analysis report	Task report(s) – Hazard analysis Anomaly report(s)
(4) Security Analysis a) Verify that the implemented system does not increase the security risk. b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations).	Source code Executable code Security analysis report	Task report(s) – Security analysis Anomaly report(s)
(5) Risk Analysis	Supplier	Task report(s) –

a) Review and update risk analysis using prior task reports.	development plans	Risk analysis
b) Provide recommendations to eliminate, reduce, or mitigate the risks.	and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Anomaly report(s)

9.7 Activity: Software Acceptance Test V&V (Software, Process: Software Acceptance Support)		
V&V tasks	Required inputs	Required outputs
(1) Software Acceptance Test Procedure V&V	SDD	V&V software
a) Software Integrity Levels 3 and 4		acceptance test
1) Develop test procedures for V&V software acceptance testing.	IDD	procedures
2) Continue the tracing required by the V&V software acceptance test plan.	Source code	Anomaly
3) Verify that the V&V software acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]).	User documentation	report(s)
4) Validate that the V&V software acceptance test procedures satisfy the criteria in V&V Activity 5.4.2, Task 6.	Software acceptance test plan	
b) Software Integrity Level 2	Software	
1) Verify that the developer's acceptance acquirer's software acceptance test procedures conform to project-defined test document purpose, format, and	acceptance test procedures	

content (e.g., IEEE Std 829-2008 [B2]). 2) Validate that the acquirer's software acceptance test procedures satisfy the criteria in V&V Activity 5.4.2, Task 6. c) Software Integrity Level 1 There are no software acceptance V&V test requirements.		
(2) System Software Acceptance Test Execution V&V a) Software Integrity Levels 3 and 4 1) Perform V&V software acceptance testing. 2) Analyze test results to validate that the software satisfies the system requirements. 3) Validate that the test results trace to test criteria established by the test traceability in the V&V software acceptance test planning documents. 4) Document the results as required by the System V&V software acceptance test plan. 5) Use the V&V System software acceptance test results to validate that the software satisfies the V&V test acceptance criteria. 6) Document discrepancies between the actual and expected test results. b) Software Integrity Level 2 Use the developer's—system acquirer's software acceptance test results to verify that the software satisfies the test acceptance criteria. c) Integrity Level 1 There are no software acceptance V&V test requirements.	Source code Executable code User documentation System Software acceptance test plan System Software acceptance test procedures System Software acceptance test results V&V task results	Task report(s) – V&V software acceptance test results Anomaly report(s)

(3) Traceability Analysis Analyze the relationships in the V&V software acceptance test plans, designs, cases, and procedures for correctness and completeness. The task criteria are as follows: a) Correctness Verify that there is a valid relationship between among the V&V software acceptance test plans, designs, cases, and procedures. b) Completeness Verify that all V&V software acceptance test procedures are traceable to the V&V software acceptance test plans.	V&V software acceptance test plans V&V software acceptance test designs V&V software acceptance test procedures	Task report(s) – Traceability Analysis Anomaly report(s)
(4) Hazard Analysis a) Verify that the test instrumentation does not introduce new hazards. b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of the system and software operations). c) Update the hazard analysis.	Source code Executable code Test results Hazard analysis report	Task report(s) – Hazard analysis Anomaly report(s)
(5) Security Analysis a) Verify that the implemented system does not increase the security risk. b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of the system and software operations).	Source code Executable code Security analysis report	Task report(s) – Security Analysis Anomaly report(s)

9.7 Activity: Software Acceptance Test V&V (Software, Process: Software Acceptance Support)		
V&V tasks	Required inputs	Required outputs
(6) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) Risk analysis Anomaly report(s)

9.8 Activity: Software Installation and Checkout V&V (Software, Process: Software Installation Process)		
V&V tasks	Required inputs	Required outputs
1) Installation Configuration Audit a) Verify that all Software products required to correctly install and operate the Software are present in the installation package. b) Validate that all site-dependent parameters or conditions to verify supplied values are correct.	Installation package (e.g., user documentation, HDD, IDD, HRS, IRS, Software drawings, concept documentation, installation procedures, site-specific	Task report(s) Installation Configuration Audit Anomaly report(s)

	parameters, installation tests, and configuration management data)	
(2) Installation Checkout a) Conduct analyses or tests to verify that the installed Software corresponds to the Software subjected to V&V. b) Verify that the Software initializes, executes, and terminates as specified. c) In the transition from one version of Software to the next, validate that the Software can be removed from the system without replaced with the new version without <u>adversely</u> affecting or <u>degrading</u> the functionality of the remaining system components. d) Verify the requirements for continuous operation and service during transition, including requirements for user notification.	User documentation Installation package	Task report(s) - Installation checkout Anomaly report(s)
(3) Hazard Analysis a) Verify that the installation procedures and installation environment does not introduce new hazards. b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and Software operations).</u> c) Update the hazard analysis.	Installation package Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)
(4) Security Analysis	Installation package	Task report(s) -

a) Verify that the installed Software does not introduce new or increased vulnerabilities or security risks to the overall system. b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and Software operations).</u>	User documentation Security analysis report	Security analysis Anomaly report(s)
(5) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Installation package Supplier development plans and schedules Security analysis report Risk analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

9.9 Activity: Software Operation V&V (Process: Operation) (Software , Process : Software Operation)		
V&V tasks	Required inputs	Required outputs
(1) Evaluation of New Constraints Evaluate new constraints (e.g., operational requirements, platform characteristics, and operating environment) on the system or Software requirements to verify the applicability of the VVP. (See NOTE.)	VVP New constraints	Task report(s) - Evaluation of new constraints
(2) Operating Procedures Evaluation Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.	Operating procedures User documentation	Task report(s) - Operating procedures

	Concept documentation	evaluation Anomaly report(s)
(3) Hazard Analysis a) Verify that the operating procedures and operational environment does not introduce new hazards. b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and Software operations).</u> c) Update the hazard analysis.	Operating procedures Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)
(4) Security Analysis a) Verify that no new security risks are introduced due to changes in the operational environment. b) Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk. c) <u>Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and Software operations).</u>	New constraints Environmental changes Operating procedures Security analysis report	Task Reports - Security analysis
(5) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Installation package Proposed changes Hazard analysis report	Task report(s) - Risk analysis Anomaly report(s)

	Security analysis report Risk analysis report Supplier development plans and schedules Operation problem reports V&V task results	
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9.10 Activity: Software Maintenance V&V (Process: Maintenance) (Software, Process: Software Maintenance)		
V&V tasks	Required inputs	Required outputs
(1) SVVP Revision a) Revise the S VVP to conform to approved changes. b) When the development documentation required by this standard is not available, generate a new S VVP and consider the methods in Annex D for deriving the required development documentation.	S VVP Approved changes Installation package Supplier development plans and schedules	Updated S VVP
(2) Anomaly Evaluation Evaluate the effect of Software operation anomalies.	Anomaly report(s)	Task report(s) - Anomaly evaluation
(3) Criticality Analysis a) Determine the software integrity levels for proposed modifications.	Proposed changes Installation package	Task report(s) - Criticality analysis

b) Validate the integrity levels provided by the maintained For V&V planning purposes, the highest software integrity level assigned to the Software shall be the integrity level of the system.	Maintainer integrity levels	Anomaly report(s)
(4) Migration Assessment Assess whether the Software requirements and fabrication address the following: a) Specific migration requirements, b) Migration tools, c) Conversion of Software products and data, d) Software archiving, e) Support for the prior environment, f) User notification.	Installation package Approved changes	Task report(s) - Migration assessment Anomaly report(s)
(5) Retirement Assessment Assess whether the installation package addresses the following: a) Software support, b) Impact on existing systems and data bases, c) Software archiving, d) Transition to a new Software product, e) User notification.	Installation package Approved changes	Task report(s) - Retirement assessment Anomaly report(s)
(6) Hazard Analysis a) Verify that Software modifications correctly implement the critical requirements and introduce no new hazards. b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and Software operations).</u>	Proposed changes Installation package Hazard analysis report	Task report(s) - Hazard analysis Anomaly report(s)

c) Update the hazard analysis.		
(7) Security Analysis a) Verify that proposed changes/updates to the Software do not introduce new or increased security risks to the overall system. <u>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and Software operations).</u>	Proposed changes Installation package Security analysis report	Task reports - Security analysis
(8) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Installation package Proposed changes Hazard analysis report Security analysis report Risk analysis report Supplier development plans and schedules Operation problem reports V&V task results	Task report(s) - Risk analysis Anomaly report(s)

Table 2c—Minimum V&V tasks assigned to each ~~Software~~ integrity level for Software V&V

V&V Activities	Activity: Software Concept V&V (see 9.1)	Activity: Software Requirements V&V (see 9.2)	Activity: Software Design V&V (see 9.3)	Activity: Software Construction V&V (see 9.4)	Activity: Software Integration V&V (see 9.5)	Activity: Software Qualification V&V (see 9.6)	Activity: Software Acceptance V&V (see 9.7)	Activity: Software Installation and Checkout V&V (see 9.8)	Activity: Software Operation V&V (see 9.9)	Activity: Software Maintenance V&V (see 9.10)	Activity: Software Disposal V&V (see 9.11)
Integrity Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels
	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234
Anomaly Evaluation										X	X
Concept Documentation Evaluation	X	X	X								
Criticality Analysis	X	X	X	X	X	X				X	X
Design Evaluation				X	X	X					
Evaluation of New Constraints									X	X	X
Hardware/Software/U ser Requirements Allocation Analysis		X									
Hazard Analysis		X	X	X	X	X	X	X	X	X	X
Installation Checkout								X	X		
Installation Configuration Audit								X	X		
Interface Analysis		X	X	X	X	X					
Migration Assessment										X	X
Operating Procedures									X	X	

V&V Activities	Activity: Software Concept V&V (see 9.1)				Activity: Software Requirements V&V (see 9.2)				Activity: Software Design V&V (see 9.3)				Activity: Software Construction V&V (see 9.4)				Activity: Software Integration V&V (see 9.5)				Activity: Software Qualification V&V (see 9.6)				Activity: Software Acceptance V&V (see 9.7)				Activity: Software Installation and Checkout V&V (see 9.8)				Activity: Software Operation V&V (see 9.9)				Activity: Software Maintenance V&V (see 9.10)				Activity: Software Disposal V&V (see 9.11)				
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	
Evaluation																																													
Requirements Evaluation					X	X	X	X																																					
Retirement Assessment																																													
Risk Analysis		X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X						
Security Analysis		X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X						
Software Acceptance Test Case V&V													X	X	X																														
Software Acceptance Test Design V&V									X	X	X																																		
Software Acceptance Test Execution V&V																								X	X	X																			
Software Acceptance Test Plan V&V					X	X	X																																						
Software Acceptance Test Procedure V&V																								X	X	X																			

V&V Activities	Activity: Software Concept V&V (see 9.1)	Activity: Software Requirements V&V (see 9.2)	Activity: Software Design V&V (see 9.3)	Activity: Software Construction V&V (see 9.4)	Activity: Software Integration V&V (see 9.5)	Activity: Software Qualification V&V (see 9.6)	Activity: Software Acceptance V&V (see 9.7)	Activity: Software Installation and Checkout V&V (see 9.8)	Activity: Software Operation V&V (see 9.9)	Activity: Software Maintenance V&V (see 9.10)	Activity: Software Disposal V&V (see 9.11)
Integrity Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels
	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234
Component Test Case V&V											
Software Component Test Design V&V					X	X	X				
Software Component Test Execution V&V						X	X	X			
Software Component Test Plan V&V					X	X	X				
Software Component Test Procedure V&V						X	X	X			
Software Integration Test Case V&V						X	X	X	X		
Software Integration Test Design V&V					X	X	X	X			
Software						X	X	X	X		

[illegible]

V&V Activities	Activity: Software Concept V&V (see 9.1)	Activity: Software Requirements V&V (see 9.2)	Activity: Software Design V&V (see 9.3)	Activity: Software Construction V&V (see 9.4)	Activity: Software Integration V&V (see 9.5)	Activity: Software Qualification V&V (see 9.6)	Activity: Software Acceptance V&V (see 9.7)	Activity: Software Installation and Checkout V&V (see 9.8)	Activity: Software Operation V&V (see 9.9)	Activity: Software Maintenance V&V (see 9.10)	Activity: Software Disposal V&V (see 9.11)
Integrity Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels
	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234	1234
Execution V&V											
Software Integration Test Plan V&V				X	X	X	X				
Software Integration Test Procedure V&V					X	X	X	X			
Software Qualification Test Case V&V					X	X	X	X			
Software Qualification Test Design V&V				X	X	X	X				
Software Qualification Test Execution V&V							X	X	X	X	
Software Qualification Test Plan V&V			X	X	X	X					
Software Qualification Test					X	X	X	X			

V&V Activities	Activity: Software Concept V&V (see 9.1)				Activity: Software Requirements V&V (see 9.2)				Activity: Software Design V&V (see 9.3)				Activity: Software Construction V&V (see 9.4)				Activity: Software Integration V&V (see 9.5)				Activity: Software Qualification V&V (see 9.6)				Activity: Software Acceptance V&V (see 9.7)				Activity: Software Installation and Checkout V&V (see 9.8)				Activity: Software Operation V&V (see 9.9)				Activity: Software Maintenance V&V (see 9.10)				Activity: Software Disposal V&V (see 9.11)				
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				
V&V	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4					
Source Code and Source Code Documentation Evaluation													X	X	X																														
Task Iteration																																													
Traceability Analysis	X	X	X		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X																		
Transition to New Product																																													
VVP Revision																																													

NOTE (for Table 2c)—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of the analysis or test. Higher integrity level implementation requires greater rigor (e.g., formal methods and structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states), and depth (e.g., abnormal cases, boundary conditions, and comprehensive fault and recovery scenarios) of the analysis or test than the lower integrity level implementation.

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Table 2—Minimum V&V tasks assigned to each software integrity level

Life cycle processes	Process: Acquisition (see 5.2)				Process: Supply (see 5.3)				Process: Development (see 5.4)												Process: Operation (see 5.5)				Process: Maintenance (see 5.6)															
	Activity: Acquisition support V&V (see 5.2.1)				Activity: Planning V&V (see 5.3.1)				Activity: Concept V&V (see 5.4.1)				Activity: Requirements V&V (see 5.4.2)				Activity: Design V&V (see 5.4.3)				Activity: Implementation V&V (see 5.4.4)				Activity: Test V&V (see 5.4.5)				Activity: Installation/checkout V&V (see 5.4.6)				Activity: Operation V&V (see 5.5.1)				Activity: Maintenance V&V (see 5.6.1)			
Software integrity levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels			
	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1				
Acceptance support													X	X	X		X	X	X		X	X	X		X	X	X													
Acceptance V&V test case generation																																								
Acceptance V&V test design generation																	X	X	X																					
Acceptance V&V test execution																																								
Acceptance V&V test plan generation													X	X	X																									
Acceptance V&V test procedure generation																																								
Anomaly evaluation																																								
Component V&V test case generation																					X	X	X																	
Component V&V test design generation																	X	X	X																					
Component V&V test execution																					X	X	X																	

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Table 2—Minimum V&V tasks assigned to each software integrity level (continued)

Life cycle processes	Process: Acquisition (see 5.2)		Process: Supply (see 5.3)		Process: Development (see 5.4)												Process: Operation (see 5.5)		Process: Maintenance (see 5.6)	
	Activity: Acquisition support V&V (see 5.2.1)		Activity: Planning V&V (see 5.3.1)		Activity: Concept V&V (see 5.4.1)		Activity: Requirements V&V (see 5.4.2)		Activity: Design V&V (see 5.4.3)		Activity: Implementation V&V (see 5.4.4)		Activity: Test V&V (see 5.4.5)		Activity: Installation/checkout V&V (see 5.4.6)		Activity: Operation V&V (see 5.5.1)		Activity: Maintenance V&V (see 5.6.1)	
	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels	Levels
Software integrity levels	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
Component V&V test plan generation									X	X										
Component V&V test procedure generation											X	X								
Concept documentation evaluation					X	X														
Configuration management assessment								X												
Contract verification																				
Criticality analysis					X	X	X	X	X	X	X	X							X	X
Hardware/software/user requirements allocation analysis					X															
Hazard analysis					X	X		X	X		X	X			X	X		X	X	
Identify improvement opportunities in the conduct of V&V	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Installation checkout																				
Installation configuration audit															X	X				

Life cycle processes	Process: Acquisition (see 5.2)					Process: Supply (see 5.3)					Process: Development (see 5.4)										Process: Operation (see 5.5)					Process: Maintenance (see 5.6)																								
	Activity: Acquisition support V&V (see 5.2.1)					Activity: Planning V&V (see 5.3.1)					Activity: Concept V&V (see 5.4.1)					Activity: Requirements V&V (see 5.4.2)					Activity: Design V&V (see 5.4.3)					Activity: Implementation V&V (see 5.4.4)					Activity: Test V&V (see 5.4.5)					Activity: Installation/checkout V&V (see 5.4.6)					Activity: Operation V&V (see 5.5.1)					Activity: Maintenance V&V (see 5.6.1)				
Software integrity levels	Levels					Levels					Levels					Levels					Levels					Levels					Levels					Levels					Levels					Levels				
	4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1		4	3	2	1						
Integration V&V test case generation																																																		
Integration V&V test design generation																																																		
Integration V&V test execution																																																		
Integration V&V test plan generation																																																		
Integration V&V test procedure generation																																																		
Interface analysis																																																		
Interface with organizational and supporting processes	X	X				X	X				X	X				X	X				X	X				X	X																							
Management and technical review support	X	X				X	X				X	X				X	X				X	X				X	X																							
Management review of the V&V effort	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Migration assessment																																																		
New constraints evaluation																																																		
Operating procedures evaluation																																																		

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Table 2—Minimum V&V tasks assigned to each software integrity level (continued)

Life cycle processes	Process: Acquisition (see 5.2)	Process: Supply (see 5.3)	Process: Development (see 5.4)												Process: Operation (see 5.5)	Process: Maintenance (see 5.6)				
			Activity: Acquisition support V&V (see 5.2.1)	Activity: Planning V&V (see 5.3.1)	Activity: Concept V&V (see 5.4.1)	Activity: Require- ments V&V (see 5.4.2)	Activity: Design V&V (see 5.4.3)	Activity: Implemen- tation V&V (see 5.4.4)	Activity: Test V&V (see 5.4.5)	Activity: Installation/ checkout V&V (see 5.4.6)	Activity: Operation V&V (see 5.5.1)	Activity: Maintenance V&V (see 5.6.1)								
													Levels				Levels			Levels
Software integrity levels			Levels			Levels			Levels			Levels			Levels					
	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
Planning the interface between the V&V effort and supplier	X	X	X	X	X	X	X	X												
Proposed/baseline change assessment						X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Retirement assessment																			X	X
Risk analysis						X	X		X	X			X	X		X	X		X	X
Scoping the V&V effort	X	X	X	X																
Security analysis					X	X		X	X		X	X		X	X		X	X		
Software design evaluation									X	X	X		X	X						
Software requirements evaluation								X	X	X										
SVVP generation	X	X	X	X	X	X	X	X	X									X	X	X
SVVP revision																		X	X	X
Source code and source code documentation evaluation										X	X	X								
System requirements review	X	X	X	X																
System V&V test case generation										X	X	X	X							

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Table 2—Minimum V&V tasks assigned to each software integrity level (continued)

Life cycle processes	Process: Acquisition (see 5.2)	Process: Supply (see 5.3)	Process: Development (see 5.4)						Process: Operation (see 5.5)	Process: Maintenance (see 5.6)		
	Activity: Acquisition support V&V (see 5.2.1)	Activity: Planning V&V (see 5.3.1)	Activity: Concept V&V (see 5.4.1)	Activity: Require- ments V&V (see 5.4.2)	Activity: Design V&V (see 5.4.3)	Activity: Implemen- tation V&V (see 5.4.4)	Activity: Test V&V (see 5.4.5)	Activity: Installation/ checkout V&V (see 5.4.6)	Activity: Operation V&V (see 5.5.1)	Activity: Maintenance V&V (see 5.6.1)		
Software integrity levels	Levels			Levels			Levels			Levels		
	4	3	2	1	4	3	2	1	4	3	2	1
System V&V test design generation												
System V&V test execution												
System V&V test plan generation												
System V&V test procedure generation												
Task iteration												
Traceability analysis												
V&V final report generation												

Table 3c—Optional V&V tasks and suggested applications in ~~the life cycle~~

Software Technical and Implementation Processes

V&V life cycle activities	<u>Software Requirements</u> Activity: Management of V&V (see 5.1.1)	<u>Software Architectural Design</u> Activity: Acquisition support V&V (see 5.2.1)	<u>Software Detailed Design</u> Activity: Planning (see 5.3.1)	<u>Software Construction</u> Activity: Concept V&V (see 5.4.1)	<u>Software Integration</u> Activity: Requirements V&V	<u>Software Qualification Testing</u> Activity: Design V&V (see 5.4.3)	<u>Software Installation</u> Activity: Implementation V&V	<u>Software Acceptance Support</u> Activity: Test V&V (see 5.4.5)	<u>Software Operation</u> Activity: Installation/checkout V&V	<u>Software Maintenance</u> Activity: Operation V&V (see 5.5.1)	<u>Software Disposal</u> Activity: Maintenance V&V
Algorithm analysis	X	X	X	X	X	X	X			X	X
Audit performance	X	X	X	X	X	X	X	X		X	
Audit support	X	X	X	X	X	X	X	X		X	
Control flow analysis	X	X	X	X	X	X	X			X	X
Cost analysis	X	X	X	X	X	X	X	X	X	X	X
Database analysis	X	X	X	X	X	X	X	X	X	X	X
Data flow analysis	X	X	X	X	X	X	X			X	X
Disaster recovery plan assessment	X	X	X	X	X	X	X		X	X	X
Distributed architecture assessment	X	X		X	X	X				X	X
Feasibility study evaluation	X	X	X	X	X	X				X	X
Independent risk assessment	X	X	X	X	X	X	X	X	X	X	X
Inspection											
Concept				X						X	X
Requirements	X				X					X	X
Design		X	X			X				X	X
Source code				X			X				X
Test plan	X	X	X	X	X	X	X	X		X	X
Test design		X	X	X	X	X	X	X	X	X	X
Test case		X	X	X	X	X	X	X	X	X	X
Operational evaluation									X	X	
Performance monitoring	X	X	X	X	X	X	X	X	X	X	X
Post installation validation									X	X	X

Project management oversight support	X	X	X	X	X	X	X	X	X	X	X
Proposal evaluation support		✗									
Qualification testing				X		X		X	✗		✗
Regression analysis and testing	X	X		X	X	X		✗		X	✗
Reusability analysis	X	X	X	X	✗	✗	✗			X	✗
Reuse analysis	X	X	X	✗	✗	✗				X	✗
Simulation analysis	X	X	X	X	X	X	X	X	X	X	✗
Sizing and timing analysis	X	X	X	X	X	X	X	X		X	✗
System software assessment		X	X	X	X	X	X	X	X	X	✗
Test certification				X	X	X		X	X	X	X
Test evaluation	X	X	X	X	X	X	✗	X	X	X	X
Test witnessing				X	X	X		X	X	X	X
Training document evaluation	X	X	X	X	X	✗		X	X	X	X
Usability analysis	X	X	X	X	X	X	✗	X	X	X	✗
User documentation evaluation	X	X	X	X	X	X	X	X	X	X	✗
User training	✗			X	X	X		X	X	X	✗
V&V tool plan generation	✗	✗	✗								✗
V&V tool qualification	X	X	X	X	X	X	X	X	X	X	
Walkthroughs											
Design		X				✗				X	✗
Requirements	X				✗					X	✗
Source code				X			✗				✗
Test				X	X	X		✗	✗	X	
Work Breakdown Structure Evaluation											

[Annex G contains a description of the optional V&V tasks.](#)

V&V Inputs	Process: SW Requirements Analysis (7.1.2)		Process: SW Requirements Analysis (7.1.2)		Process: SW Architecture Design (7.1.3/7.1.4)		Process: SW Construction (7.1.5)		Process: SW Integration (7.1.6)		Process: SW Qualification Testing (7.1.7)		Process: SW Acceptance Support (6.4.8)		Process: SW Installation (6.4.7)	
	1) Acquisition Needs Concept Documentation Developer Integrity Level Assignment Hazard Analysis Report Preliminary Threat and Risk Assessment Security Analysis Report System Architectural Design Supplier Development Plans and Schedules System Requirements Document User Needs V&V Task Results	1) Concept Documentation 2) Criticality Report 3) Hazard Analysis Report 4) Preliminary Threat and Risk Assessment 5) Risk Analysis Report 6) Security Analysis Report 7) SRS, IRS, SDD, IDD 8) System Requirements 9) Software Test Plan - Qualification (Developer's) 10) System Requirements Document 11) V&V Task Results	1) Concept Documentation 2) Criticality Report 3) Design Standards 4) Hazard Analysis Report 5) Risk Analysis Report 6) Security Analysis Report 7) SRS, SDD, IDD 8) Supplier Development Plans and Schedules 9) System Requirements 10) Software Test Plan - Component Integration - Qualification - Acceptance 11) Software Test Design 12) User Documentation 13) V&V Task Results	1) Coding Standard 2) Concept Documentation 3) Criticality Report 4) Hazard Analysis Report 5) Risk Analysis Report 6) Security Analysis Report 7) SRS, IDD, Source & Executable Code 8) System Requirements 9) SW Component Test Execution Results 10) Software Test Design and Test Cases - Integration - Qualification - Acceptance 11) Software Test Procedures V&V 12) Supplier Development Plans and Schedules 13) User Documentation 14) V&V Task Results	1) Hazard Analysis Report 2) Risk Analysis Report 3) Security Analysis Report 4) Source and Executable Code 5) SW Test Plan, Test Design, Test Cases, Test Procedures - Integration - Qualification - Acceptance 6) SW Integration Test Supplier Development Plans and Schedules 7) V&V Task Results	1) Hazard Analysis Report 2) Risk Analysis Report 3) Security Analysis Report 4) SW Qualification Test Plan, Design, Cases, Procedures & Test Execution Results 5) Source and Executable Code 6) Supplier Development Plans and Schedules 7) V&V Task Results	1) Hazard Analysis Report 2) Risk Analysis Report 3) SDD, IDD 4) Security Analysis Report 5) SW Acceptance Test Plan, Design, Cases, Procedures, & Test Execution Results 6) Source and Executable Code 7) User Documentation 8) V&V Task Results	1) Hazard Analysis Report 2) Installation Package 3) Risk Analysis Report 4) Security Analysis Report 5) Supplier Development Plans and Schedules 6) User Documentation 7) V&V Task Results								
V&V Tasks	Activity: SW Concept V&V (9.1)		Activity: SW Requirements Analysis V&V (9.2)		Activity: SW Design V&V (9.3)		Activity: SW Construction V&V (9.4)		Activity: SW Integration Test V&V (9.5)		Activity: SW Qualification Test V&V (9.6)		Activity: SW Acceptance Test V&V (9.7)		Activity: SW Installation & Checkout V&V (9.8)	
	1) Concept Evaluation 2) Criticality Analysis 3) Hardware/Software/User Requirements Allocation 4) Traceability Analysis 5) Hazard Analysis 6) Security Analysis 7) Risk Analysis	1) Requirements Evaluation 2) Traceability Analysis 3) Interface Analysis 4) Criticality Analysis 5) Software Component Test Plan V&V 6) Software Integration Test Plan V&V 7) Software Component Test Plan V&V 8) Hazard Analysis 9) Security Analysis 10) Risk Analysis	1) Design Evaluation 2) Traceability Analysis 3) Interface Analysis 4) Criticality Analysis 5) Software Component Test Plan V&V 6) Software Integration Test Plan V&V 7) Software Component Test Plan V&V 8) Design V&V 9) Software Integration Test Design V&V 10) Software Qualification Test Design V&V 11) Software Acceptance Test Design V&V 12) Hazard Analysis 13) Risk Analysis	1) Source Code and Documentation Evaluation 2) Traceability Analysis 3) Interface Analysis 4) Criticality Analysis 5) SW Component Test Case V&V 6) SW Integration Test Case V&V 7) SW Qualification Test Case V&V 8) SW Acceptance Test Case V&V 9) SW Component Test Procedure V&V 10) SW Integration Test Procedure V&V 11) SW Qualification Test Procedure V&V 12) SW Component Execution V&V 13) Hazard Analysis 14) Security Analysis 15) Risk Analysis	1) SW Integration Test Execution V&V 2) Traceability Analysis 3) Hazard Analysis 4) Security Analysis 5) Risk Analysis	1) SW Qualification Test Execution V&V 2) Traceability Analysis 3) Hazard Analysis 4) Security Analysis 5) Risk Analysis	1) Software Acceptance Test Procedure V&V 2) Software Acceptance Test Execution V&V 3) Traceability Analysis 4) Hazard Analysis 5) Security Analysis 6) Risk Analysis	1) Installation Configuration Audit 2) Installation Checkout 3) Hazard Analysis 4) Security Analysis 5) Risk Analysis								
V&V Outputs	Activity: SW Concept V&V (9.1)		Activity: SW Requirements Analysis V&V (9.2)		Activity: SW Design V&V (9.3)		Activity: SW Construction V&V (9.4)		Activity: SW Integration Test V&V (9.5)		Activity: SW Qualification Test V&V (9.6)		Activity: SW Acceptance Test V&V (9.7)		Activity: SW Installation & Checkout V&V (9.8)	
	1) Anomaly Report(s) 2) Task Report(s)	1) Anomaly Report(s) 2) Task Report(s) 3) Software Test Plan V&V Qualification - Acceptance	1) Anomaly Report(s) 2) Task Report(s) 3) Software Test Plan V&V Component Integration - Acceptance 4) Software Test Design V&V - Component - Integration - Qualification - Acceptance	1) Anomaly Report(s) 2) Task Report(s) 3) SW Test Case - Component Integration - Qualification 4) SW Test Procedure - Component - Integration - Qualification 5) SW Component Test V&V Execution Report	1) Anomaly Report(s) 2) Task Report(s) 3) SW Test V&V Execution Results Integration	1) Anomaly Report(s) 2) Task Report(s) 3) Software Qualification Test V&V Execution Results	1) Anomaly Report(s) 2) Task Report(s) 3) Software Acceptance Test Procedure V&V 4) Software Acceptance Test V&V Execution Results	1) Anomaly Report(s) 2) Task Report(s)								

(New)

Figure 1c— Summary of software V&V activities and tasks

V&V	V&V Inputs	Process: SW Operation (6.4.9)	Process: SW Maintenance (6.4.10)	Process: SW Disposal (6.4.11)
		1) Concept Documentation 2) Environmental Changes 3) Hazard Analysis Report 4) Installation Package 5) New Constraints 6) Operating Procedures 7) Operation Problem Report(s) 8) Proposed Changes 9) Risk Analysis Report 10) Security Analysis Report 11) Supplier Development Plans and Schedules 12) User Documentation 13) VVP 14) V&V Task Results	1) Approved Changes 2) Anomaly Report(s) 3) Installation Package 4) Hazard Analysis Report 5) Maintainer Integrity Level(s) 6) Operation Problem Report(s) 7) Proposed Changes 8) Risk Analysis Report 9) Security Analysis Report 10) Supplier Development Plans and Schedules 11) VVP 12) V&V Task Results	1) Software Disposal Strategy
	V&V Tasks	Activity: SW Operation V&V (9.9)	Activity: SW Maintenance V&V (9.10)	Activity: SW Disposal V&V (9.11)
V&V		1) Evaluation of New Constraints 2) Operating Procedures Evaluation 3) Hazard Analysis 4) Security Analysis 5) Risk Analysis	1) VVP Revision 2) Anomaly Evaluation 3) Criticality Analysis 4) Migration Assessment 5) Retirement Assessment 6) Hazard Analysis 7) Security Analysis 8) Risk Analysis 9) Task Iteration	1) Software Disposal Evaluation
	Outputs	1) Anomaly Report(s) 2) Task Report(s)	1) Anomaly Report(s) 2) Task Report(s) 3) VVP	1) Anomaly Report(s) 2) Task Report(s)

(New)

NOTE 1—Clause references in the Process Definitions (top graphic bar) are ISO/IEC 12207:2008 [B11] clause numbers (7.1.2 through 7.1.7) and ISO/IEC 15288:2008(E) [B16] clause numbers (6.4.7 through 6.4.11).

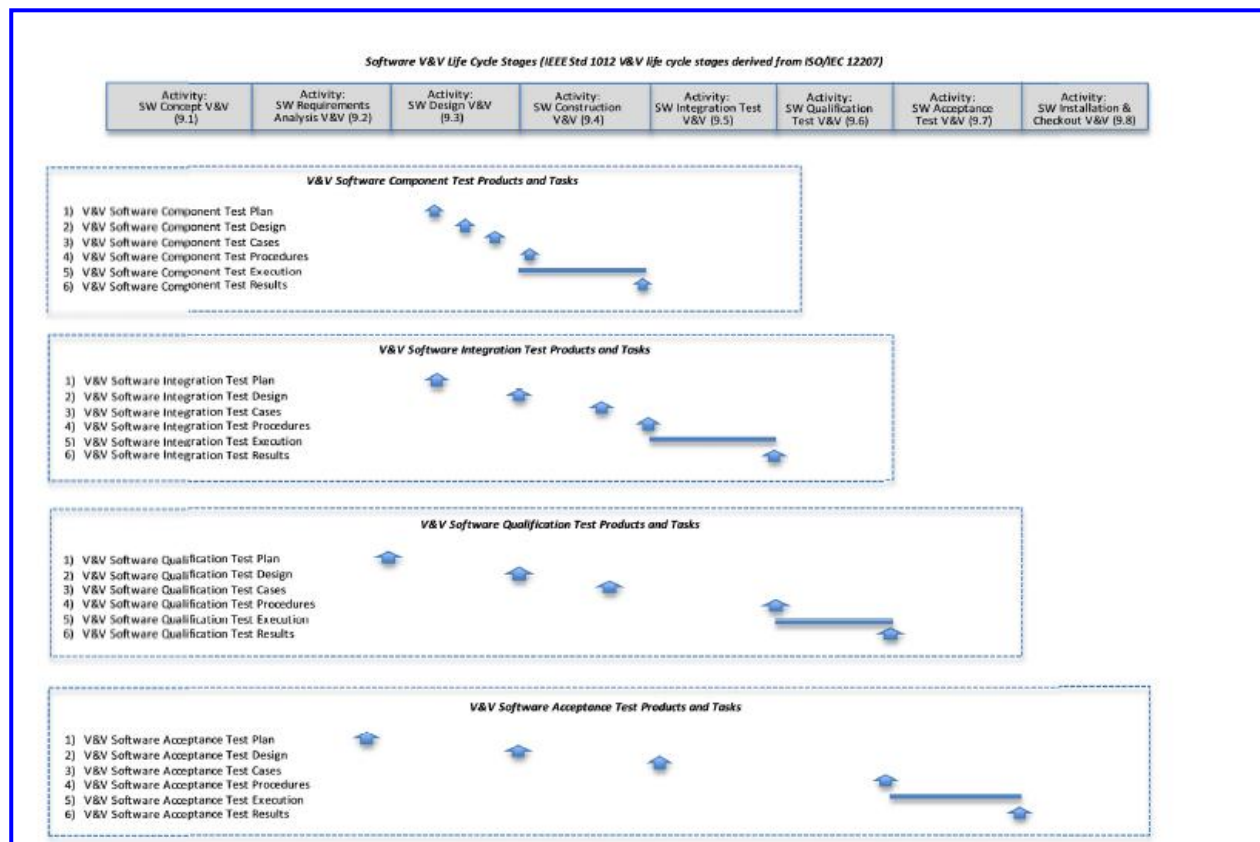
NOTE 2—Clause references in the Activity V&V Definitions (middle graphic bar) are IEEE Std 1012 clause numbers.

NOTE 3—V&V tasks listed in the figure are the minimum required for Integrity Level 4 (highest integrity level).

NOTE 4—Software Acceptance Testing Process supports the Systems Integration Testing Process.

NOTE 5—Software Installation and Checkout Process supports the System Transition Process.

Figure 1c— Summary of software V&V activities and tasks (*continued*)



(New)

NOTE 1—All V&V software test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of software V&V test products and tasks across the software life cycle. The software V&V test products (upward arrows) are shown in the software life cycle stages when the products are generated (per IEEE Std 1012-2004 [B5]). Software test execution tasks are shown to occur during one or more software life cycle stages as indicated by "activity bars" in the diagram. The life cycle stage (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project specific needs.

NOTE 3—The V&V activity clauses referenced in the software V&V life cycle stages are IEEE Std 1012 clauses.

Figure 2c—Summary of software V&V test products and tasks

10. Hardware V&V activities

10.1 Activity: Hardware Concept V&V

The objectives of Hardware Concept V&V are to verify that all system requirements allocated to the hardware components are addressed and to validate that the selected hardware concepts satisfy the system needs (i.e., performance and schedule). The primary focus is on hardware with consideration of the interactions with software and user allocations to verify the allocation of system requirements, validate the selected solution, and assure that no false assumptions have been incorporated in the solution.

During the concept process, different hardware concepts are investigated and trade studies are conducted on each concept before a final concept is selected. These trade studies may involve assessing the performance features of each concept, estimating the cost of parts, determining the manufacturing efficiency, identifying the technology risks, and estimating the schedule to develop the hardware. Hardware models and prototypes may be constructed to conduct these trade studies in conjunction with simulations and analytic analyses. The hardware concept stage may provide preliminary hardware models and prototypes to the systems concept stage to support trade studies for system concept definition. In such cases, the hardware concept stage may start before final hardware requirements are allocated.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Concept V&V tasks described in Table Id:

- a) [Concept Documentation Evaluation](#)
- b) [Hardware/Software/User Requirements Allocation Analysis](#)
- c) [Criticality Analysis](#)
- d) [Traceability Analysis](#)
- e) [Hazard Analysis](#)
- f) [Security Analysis](#)
- g) [Risk Analysis](#)

10.2 Activity: Hardware Requirements V&V

[The hardware requirements process begins after the system requirements have been defined and allocated to each software and hardware element of the system. Given the system requirements and hardware concept architecture, this stage of the hardware life cycle further refines the allocated system requirements into specific requirements for the hardware element. Any constraints or limitations imposed by the hardware requirements on the system performance are identified and verified with the system acquirer that these limitations or constraints are acceptable. V&V test planning begins during the Hardware Requirements V&V activity and spans several V&V activities.](#)

The objectives of the Hardware Requirements V&V are 1) to verify that the hardware requirements correctly, completely, and accurately satisfy the system requirements allocated to the hardware element; and 2) to validate that the hardware requirements, in total, satisfy the system needs.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Requirements V&V tasks described in Table Id:

- a) Requirements Evaluation
- b) Interface Analysis
- c) Criticality Analysis
- d) Traceability Analysis
- e) Hardware Qualification Test Plan V&V
- f) Hardware Acceptance Test Plan V&V
- g) Hazard Analysis
- h) Security Analysis
- i) Risk Analysis

10.3 Activity: Hardware Design V&V

The objective of Hardware Design V&V is to verify that all hardware design components satisfy the hardware requirements specification of the hardware element. After the hardware requirements have been approved, the hardware design process is initiated. Detailed design solutions are developed that will satisfy the specific hardware requirements. Depending on the hardware, the design can take the form of selection of the materials or composites, drawings showing dimensions, circuit diagrams, and other electromechanical descriptions. Often, the hardware element involves subtiered design components from a variety of suppliers. These subtier designs are documented and evaluated as part of the entire hardware element. Hardware Design V&V will validate that the design solution satisfies the system performance, safety, and reliability requirements and that no unintended or undesired consequences are introduced into the system.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Design V&V tasks described in Table Id:

- a) Design Evaluation
- b) Interface Analysis
- c) Traceability Analysis
- d) Criticality Analysis
- e) Hardware Component Test Plan Assessment
- f) Hardware Integration Test Plan Assessment

g) Hardware Component Test Design Assessment

h) Hardware Integration Test Design Assessment

i) Hardware Qualification Test Design V&V

j) Hardware Acceptance Test Design V&V

k) Hazard Analysis

l) Security Analysis

m) Risk Analysis

10.4 Activity: Hardware Fabrication V&V

The objectives of Hardware Fabrication V&V are to verify that the final fabrication elements comply with the hardware design and to validate that each fabricated hardware component satisfies the overall system performance, safety, and reliability requirements. The methods used in this activity include visual analysis/inspection, quality sampling testing, physical measurements, form/fit checks, and complete component analysis/test. After the hardware design has been approved, the hardware fabrication (implementation) process is initiated. During fabrication, the hardware is built from the raw materials, electronic components are selected and integrated onto circuit boards, mechanical elements are shaped and interconnected, manufactured parts are created from molds and milling processes, and other hardware fabrication methods. The Hardware Fabrication V&V determines whether or not deviations have been introduced into the solution as the hardware progressed from the hardware design to actual fabrication. For example, an electronic component selected from a catalog may be similar to the specified part but not identical. The Hardware Fabrication V&V investigates whether the small difference in the final fabrication creates any

limitations or constraints that affect the overall system performance. The hardware components received from subtiered suppliers are validated by either the manufacturer or the supplier before being included in the overall hardware element fabrication.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Fabrication V&V tasks described in Table Id:

- a) Fabricated Component Documentation Evaluation
- b) Interface Analysis
- c) Traceability Analysis
- d) Criticality Analysis
- e) Hardware Component Test Case Assessment
- f) Hardware Integration Test Case Assessment
- g) Hardware Qualification Test Case V&V
- h) Hardware Acceptance Test Case V&V
- i) Hardware Component Test Procedures Assessment
- j) Hardware Integration Test Procedure Assessment
- k) Hardware Qualification Test Procedure V&V
- l) Hardware Component Test Execution Assessment

m) [Hazard Analysis](#)

n) [Security Analysis](#)

o) [Risk Analysis](#)

10.5 Activity: Hardware Integration Test V&V

[The objectives of Hardware Integration Test V&V are to verify that hardware parts conform to the hardware element requirements during the integration process and to validate that the hardware element satisfies the system requirements. The hardware Integration Test process occurs in parallel with hardware fabrication. As fabrication progresses, hardware parts are integrated into components and components are integrated into the hardware element, and a series of tests are conducted to verify and validate conformance with requirements.](#)

[NOTE—In some programs, the hardware Integration Test process determines the efficiency of the fabrication and integration manufacturing steps especially when multiple hardware copies \(hardware assembly line\) are produced.](#)

[The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Integration Test V&V tasks described in Table Id:](#)

a) [Hardware Integration Test Execution Assessment](#)

b) [Traceability Analysis](#)

c) [Hazard Analysis](#)

d) Security Analysis

e) Risk Analysis

10.6 Activity: Hardware Qualification Test V&V

The objectives of Hardware Qualification Test V&V are to verify that the hardware element as tested satisfies the hardware requirements and to validate that the hardware element satisfies system requirements. Hardware qualification (e.g., demonstration, analysis, inspection, or test) is performed on the complete hardware element. However, preliminary qualification tests and prototype (dry run) qualification tests can be performed on parts of the hardware element as they become available from integration testing. Formal hardware qualification testing occurs on the completed hardware element using other available system hardware elements or simulators and test drivers to provide stimuli and responses from system hardware elements that are not available. Hardware Qualification Test V&V validates that no unintended or undesired consequences are created by the hardware element under test.

NOTE—The hardware element used for hardware qualification testing may not be suitable for operational use.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Qualification Test V&V tasks described in Table Id:

a) Hardware Qualification Test Execution V&V

- b) [Traceability Analysis](#)
- c) [Hazard Analysis](#)
- d) [Security Analysis](#)
- e) [Risk Analysis](#)

10.7 Activity: Hardware Acceptance Test V&V

[The objectives of the Hardware Acceptance Test V&V test are to verify that the hardware element meets the acceptance criteria and requirements and to validate that the system requirements allocated to this particular hardware element are satisfied. Hardware acceptance testing occurs with the hardware element working with all other hardware and software elements. These tests provide the data for hardware acceptance, allowing the acquiring organization to formally accept the hardware product.](#)

[The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Acceptance Test V&V tasks described in Table Id:](#)

- a) [Hardware Acceptance Test Procedure V&V](#)
- b) [Hardware Acceptance Test Execution V&V](#)
- c) [Traceability Analysis](#)
- d) [Hazard Analysis](#)
- e) [Security Analysis](#)

- f) Risk Analysis

10.8 Activity: Hardware Transition V&V

The objective of Hardware Transition V&V is to verify and validate the correctness of the hardware installation in the operational environment. In transition, the hardware product is installed and tested in the operational environment. The Hardware Transition V&V activity supports the hardware system installation activities and validates the hardware is capable of delivering its required services.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Transition V&V tasks described in Table Id:

- a) Task: Installation Configuration Audit
- b) Task: Installation Checkout
- c) Task: Hazard Analysis
- d) Task: Security Analysis
- e) Task: Risk Analysis

10.9 Activity: Hardware Operation V&V

The Hardware Operation V&V activity evaluates the impact of changes in the operating environment, assesses the effect on the system of any proposed changes, evaluates operating

procedures for adherence with the intended use, and analyzes the risks affecting the user and the system.

The objective of Hardware Operation V&V is to evaluate new constraints in the system, assess proposed system changes and their impact on the hardware and evaluate operating procedures for correctness and usability.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Operation V&V tasks described in Table Id:

- a) Task: Evaluation of New Constraints
- b) Task: Operating Procedures Evaluation
- c) Task: Hazard Analysis
- d) Task: Security Analysis
- e) Task: Risk Analysis

10.10 Activity: Hardware Maintenance V&V

The Hardware Maintenance Process is activated when the hardware or associated documentation are changed in response to a need for system maintenance. The Hardware Maintenance V&V activity addresses the following:

- Hardware modifications (i.e., corrective, adaptive, or perfective changes)
- Hardware migration (i.e., the movement of hardware to a new operational environment)
- Hardware retirement (i.e., the withdrawal of active support by the operation and maintenance organization, partial or total replacement by a new system, or installation of an upgraded system)

System modifications may be derived from the requirements specified to correct hardware errors (e.g., corrective), to adapt to a changed operating environment (e.g., adaptive), or to respond to additional user requests or enhancements (e.g., perfective). Modifications of the hardware shall be treated as development processes and shall be verified and validated by performing V&V tasks corresponding to the modifications. Integrity-level assignments shall be assessed as described in Clause 5. The integrity-level assignments shall be revised as appropriate to reflect the requirements derived from the maintenance process.

If the hardware V&V was performed in accordance with this standard, then the maintenance process shall continue to conform to this standard. If the hardware was not verified and validated using this standard and appropriate documentation is not available or adequate, then the Hardware Maintenance V&V effort shall determine whether the missing or incomplete documentation should be generated. In making this determination of whether to generate missing documentation, the minimum V&V requirements of the assigned integrity level shall be taken into consideration.

The objectives of Hardware Maintenance V&V are to assess proposed hardware changes and their impact on the system, to evaluate anomalies that are discovered during operation, to assess migration requirements, to assess retirement requirements, and to reperform V&V tasks. The proposed changes are assessed by the Proposed/Baseline Change Assessment task of the Management of V&V activity.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Maintenance V&V tasks described in Table Id:

- a) Task: VVP Revision
- b) Task: Anomaly Evaluation
- c) Task: Criticality Analysis
- d) Task: Migration Assessment
- e) Task: Retirement Assessment
- f) Task: Hazard Analysis
- g) Task: Security Analysis
- h) Task: Risk Analysis
- h) Task: Task Iteration

10.11 Activity: Hardware Disposal V&V

The purpose of Hardware Disposal is to end the existence of a system's hardware element.

This process ends active support by the operation and maintenance organization, or deactivates, disassembles, and removes the affected hardware products, consigning them to a final condition and leaving the environment in an acceptable condition. This process destroys or stores system hardware elements and related products in a sound manner, in accordance with legislation, agreements, organizational constraints, and stakeholder requirements. Where required, it maintains records that may be monitored.

The V&V effort shall perform, as specified in Table 2d for the selected integrity level, the following Hardware Disposal V&V task described in Table Id:

- a) Task: Hardware Disposal Evaluation

Table 1 d–V&V tasks, inputs, and outputs

<u>10.1 Activity: Hardware Concept V&V (Hardware, Process: Hardware Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Concept Documentation Evaluation</u> a) <u>Validate that the concept documentation satisfies user needs and is consistent with acquisition needs.</u> b) <u>Validate constraints of interfacing systems and constraints or limitations of proposed approach.</u> c) <u>Analyze system requirements and validate that the following satisfy user needs:</u> 1) <u>System functions.</u> 2) <u>End-to-end system performance.</u> 3) <u>Feasibility and testability of the functional requirements.</u> 4) <u>System architecture design.</u> 5) <u>Operation and maintenance requirements and environments.</u> 6) <u>Migration requirements from an existing system where applicable.</u>	<u>Concept documentation</u> <u>System architectural design</u> <u>Supplier development plans and schedules</u> <u>User needs</u> <u>Acquisition needs</u>	<u>Task report(s) - Concept documentation evaluation</u> <u>Anomaly report(s)</u>
<u>(2) Hardware/Software/User Requirements Allocation Analysis</u> <u>Verify the correctness, accuracy, and completeness of the concept requirement allocation to hardware, software, and user interfaces against user needs.</u> a) <u>Correctness</u> <u>Verify that performance requirements (e.g., timing, response time, and throughput) allocated to</u>	<u>User needs</u> <u>Concept documentation</u>	<u>Task report(s) - Hardware/software/user requirements allocation analysis</u> <u>Anomaly report(s)</u>

<u>hardware, software, and user interfaces satisfy user needs.</u> b) <u>Accuracy</u> <u>Verify that the internal and external interfaces specify the interface protocols, interface configuration, and other key performance requirements to demonstrate satisfaction of user requirements.</u> c) <u>Completeness</u> 1) <u>Verify that application specific requirements such as functional diversity, fault detection, fault isolation, and diagnostic and error recovery satisfy user needs.</u> 2) <u>Verify that the user's maintenance requirements for the system are completely specified.</u> 3) <u>Verify that the migration from existing system and replacement of the system satisfy user needs.</u>		
(3) Traceability Analysis a) <u>Identify all system requirements.</u> b) <u>Verify that these system requirements are traceable to acquisition needs.</u> c) <u>Start the requirements traceability analysis with system requirements.</u>	<u>Concept documentation</u>	<u>Task report(s) - Traceability analysis</u> <u>Anomaly report(s)</u>

10.1 Activity: Hardware Concept V&V (Hardware, Process: Hardware Requirements Analysis)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(4) Criticality Analysis</u> a) <u>Determine whether integrity levels are established for requirements, detailed functions, hardware components, subsystem, or other partitions.</u> b) <u>Verify that the assigned integrity levels are correct. If integrity levels are not assigned, then assign integrity levels to the system requirements.</u> c) <u>Document the integrity level assigned to individual components (e.g., requirements, detailed functions, hardware components, subsystems, or other partitions). For V&V planning purposes, the system shall be assigned the same integrity level as the highest level assigned to any individual element.</u> d) <u>Verify whether any component can influence individual components assigned a higher integrity level, and if such conditions exist, then assign that component the same higher integrity level.</u>	<u>Concept documentation (system requirements)</u> <u>Developer integrity level assignments</u>	<u>Task report(s) - Criticality analysis</u> <u>Anomaly report(s)</u>
<u>(5) Hazard Analysis</u> <u>Analyze the potential hazards to and from the conceptual system. The analysis shall perform the following:</u> a) <u>Identify the potential system hazards,</u> b) <u>Assess the consequences of each hazard,</u> c) <u>Assess the probability of each hazard,</u> d) <u>Identify mitigation strategies for each hazard.</u>	<u>Concept documentation</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(6) Security Analysis</u> a) <u>Review the system owner's definition of an acceptable level of security risk.</u>	<u>Concept documentation</u> <u>Preliminary TRA</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly</u>

b) Analyze the system concept from a security perspective and assure that potential security risks with respect to confidentiality (disclosure of sensitive information/data), integrity (modification of information/data), availability (withholding of information or services), and accountability (attributing actions to an individual/ process) have been identified. Include an assessment of the sensitivity of the information/data to be processed. c) Analyze security risks introduced by the system itself as well as those associated with the environment with which the system interfaces.		report(s)
(7) Risk Analysis a) Identify the technical and management risks, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Concept documentation Supplier development plans and schedules Hazard analysis report Security analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

<u>10.2 Activity: Hardware Requirements V&V (Hardware, Process: Hardware Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Requirements Evaluation</u> <u>Evaluate the requirements (e.g., functional, capability, interface, qualification, safety, security, human factors, data definitions, user documentation, installation and acceptance, user operation, and user maintenance) of the hardware requirements specification (HRS) and IRS for correctness, consistency, completeness, accuracy, readability, and testability.</u>	<u>Concept</u> <u>documentation</u> <u>HRS</u> <u>IRS</u>	<u>Task report(s) -</u> <u>Hardware</u> <u>requirements</u> <u>evaluation</u> <u>Anomaly</u> <u>report(s)</u>
<u>(2) Interface Analysis</u> <u>Verify and validate that the requirements for hardware interfaces with software, user, operator, other hardware, and other systems are correct, consistent, complete, accurate, and testable.</u>	<u>Concept</u> <u>documentation</u> <u>HRS</u> <u>IRS</u>	<u>Task report(s) -</u> <u>Interface analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(3) Traceability Analysis</u> <u>a) Trace the hardware requirements (HRS and IRS) to system requirements (Concept Documentation) and system requirements to the hardware requirements.</u> <u>b) Analyze identified relationships for correctness, consistency, completeness, and accuracy.</u>	<u>Concept</u> <u>Documentation</u> <u>System</u> <u>requirements</u> <u>HRS</u> <u>IRS</u>	<u>Task report(s) –</u> <u>Traceability</u> <u>analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(4) Criticality Analysis</u> <u>Review and update the existing criticality analysis results from the prior Criticality Task Report using the HRS and IRS.</u>	<u>Criticality task</u> <u>report</u> <u>HRS</u> <u>IRS</u>	<u>Task report(s) -</u> <u>Criticality</u> <u>analysis</u> <u>Anomaly</u> <u>report(s)</u>
<u>(5) Hardware Qualification Test Plan V&V</u> <u>a) Integrity Levels 3 and 4</u>	<u>Concept</u> <u>documentation</u>	<u>V&V hardware</u> <u>qualification test</u>

<u>1) Plan V&V hardware qualification testing to validate hardware requirements (e.g., static, transient, or dynamic loads; interface compatibility; electromagnetic compatibility; electromagnetic interference; thermal, mechanical, electrical, acoustic, environmental [humidity, water, saltwater, etc.], acceleration, vibration, shock, and pressure).</u> <u>2) Plan tracing of system requirements to test designs, cases, procedures, and results.</u> <u>3) Plan documentation of test designs, cases, procedures, and results.</u> <u>4) The V&V hardware qualification test plan shall address the following:</u> <u>i) Conformance to all system requirements (e.g., functional, performance, security, operation, and maintenance) as complete hardware end items in the system environment.</u> <u>ii) Adequacy of user documentation (e.g., training materials and procedural changes).</u> <u>iii) Performance at boundaries (e.g., data and interfaces) and under stress conditions.</u> <u>5) Verify that the V&V hardware qualification test plan satisfies the following criteria:</u> <u>i) Conformance to project-defined test document purpose, format, and content.</u> <u>ii) Test coverage of system requirements.</u>	<u>System requirements</u> <u>HRS</u> <u>IRS</u> <u>User documentation</u> <u>Developer's hardware qualification test plan</u>	<u>plan</u> <u>Anomaly report(s)</u>
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10.2 Activity: Hardware Requirements V&V (Hardware, Process: Hardware Requirements Analysis)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
6) <u>Validate that the V&V hardware qualification test plan satisfies the following criteria:</u> i) <u>Appropriateness of test methods and standards used,</u> ii) <u>Conformance to expected results,</u> iii) <u>Feasibility of system qualification testing,</u> iv) <u>Feasibility and testability of operation and maintenance requirements.</u> <u>b) Integrity Level 2</u> 1) <u>Verify that the developer's hardware qualification test plan satisfies the following criteria:</u> i) <u>Conformance to project-defined test document purpose, format, and content.</u> ii) <u>Test coverage of system requirements.</u> 2) <u>Validate that the developer's hardware qualification test plan satisfies the following criteria:</u> i) <u>Appropriateness of test methods and standards used,</u> ii) <u>Conformance to expected results,</u> iii) <u>Feasibility of system qualification testing,</u> iv) <u>Capability to be operated and maintained.</u> <u>c) Integrity Level 1</u> <u>There are no hardware qualification V&V test requirements.</u>		
<u>(6) Hardware Acceptance Test Plan V&V</u> <u>a) Integrity Levels 3 and 4</u> 1) <u>Plan V&V hardware acceptance testing to validate that the hardware correctly implements system and hardware requirements (e.g., static, transient, or dynamic loads; interface compatibility; electromagnetic</u>	<u>Concept documentation</u> <u>HRS</u> <u>IRS</u> <u>User documentation</u>	<u>V&V hardware acceptance test plan</u> <u>Anomaly report(s)</u>

<u>compatibility; electromagnetic interference; thermal; mechanical; electrical; acoustic; environmental (humidity, water, saltwater, etc.); acceleration; vibration; and shock, pressure) in an operational environment.</u> <u>2) Plan tracing of acceptance test requirements to test design, cases, procedures, and execution results.</u> <u>3) Plan documentation of test tasks and results.</u> <u>4) The V&V hardware acceptance test plan shall address the following:</u> <u>i) Conformance to acceptance requirements in the operational environment.</u> <u>ii) Adequacy of user documentation.</u> <u>5) Verify that the V&V hardware acceptance test plan satisfies the following criteria:</u> <u>i) Conformance to project-defined test document purpose, format, and content.</u> <u>ii) Test coverage of acceptance requirements.</u> <u>6) Validate that the V&V hardware acceptance test plan satisfies the following criteria:</u> <u>i) Conformance to expected results.</u> <u>ii) Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</u>	<u>Hardware acceptance test plan</u>	
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<u>10.2 Activity: Hardware Requirements V&V (Hardware, Process: Hardware Requirements Analysis)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>b) Integrity Level 2</u> 1) <u>Verify that the acquirer's hardware acceptance test plan conforms to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the acquirer's hardware acceptance test plan satisfies the following criteria:</u> i) <u>Test coverage of acceptance requirements,</u> ii) <u>Conformance to expected results.</u> iii) <u>Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</u>		
<u>c) Integrity Level 1</u> <u>There are no hardware acceptance V&V test requirements.</u>		
<u>(7) Hazard Analysis</u> <u>Determine hardware contributions to system hazards. The hazard analysis shall perform the following:</u> a) <u>Identify the hardware requirements that contribute to each system hazard.</u> b) <u>Validate that the hardware addresses, controls, or mitigates each hazard.</u>	<u>HRS</u> <u>IRS</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis Anomaly report(s)</u>
<u>(8) Security Analysis</u> a) <u>Determine that the security requirements identified in the HRS and IRS address the security risks introduced by the system concept.</u> b) <u>Verify that the system security requirements will mitigate the identified security risks to an acceptable</u>	<u>HRS</u> <u>IRS</u> <u>Preliminary TRA Security analysis report</u>	<u>Task report(s) - Security analysis Anomaly report(s)</u>

level.		
(9) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Concept documentation HRS IRS Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

10.3 Activity: Hardware Design V&V (Hardware, Process: Hardware Architectural Design and Hardware Detailed Design)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Design Evaluation</u> <u>Evaluate the design elements (hardware design description [HDD] and IDD) for correctness, consistency, completeness, accuracy, readability, testability, and design margins.</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Design standards (e.g., standards, practices, and conventions)</u>	<u>Task report(s) - Hardware design evaluation</u> <u>Anomaly report(s)</u>
<u>(2) Interface Analysis</u> <u>Verify and validate that the hardware design interfaces with software, user, operator, other hardware, and other systems for correctness, consistency, completeness, accuracy, and testability.</u>	<u>Concept documentation (System requirements)</u> <u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u>	<u>Task report(s) - Interface analysis</u> <u>Anomaly report(s)</u>
<u>(3) Traceability Analysis</u> <u>Trace the design elements (HDD and IDD) to the requirements (HRS and IRS), and the requirements to the design elements. Analyze the relationships for correctness, consistency, and completeness.</u>	<u>HRS</u> <u>HDD</u> <u>IRS</u> <u>IDD</u>	<u>Task report(s) - Traceability analysis</u> <u>Anomaly report(s)</u>
<u>(4) Criticality Analysis</u> <u>Review and update the existing criticality analysis results from the prior Criticality Task Report using the HDD and IDD.</u>	<u>Criticality task report</u> <u>HDD</u> <u>IDD</u>	<u>Task report(s) - Criticality analysis</u> <u>Anomaly report(s)</u>

<u>10.3 Activity: Hardware Design V&V (Hardware, Process: Hardware Architectural Design and Hardware Detailed Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(5) Hardware Component Test Plan Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> 1) <u>Verify that the developer's hardware component test plan conforms to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the developer's hardware component test plan satisfies the following criteria:</u> i) <u>Traceable to the hardware requirements and design.</u> ii) <u>External consistency with the hardware requirements and design.</u> iii) <u>Internal consistency between unit requirements.</u> iv) <u>Test coverage of units.</u> v) <u>Feasibility of hardware integration and testing.</u> vi) <u>Feasibility of operation and maintenance (e.g., capability to be operated and maintained in accordance with user needs).</u> <u>b) Integrity Level 1</u> <u>There are no hardware component V&V test requirements.</u>	<u>HRS</u> <u>HDD</u> <u>IRS</u> <u>IDD</u> <u>Hardware drawings</u> <u>Hardware component test plan</u>	<u>Task report(s) - Hardware component test plan assessment</u> <u>Anomaly report(s)</u>
<u>(6) Hardware Integration Test Plan Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> 1) <u>Verify that the developer's hardware integration test plan conforms to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the developer's hardware integration</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware</u>	<u>Task report(s) - Hardware integration test plan assessment</u> <u>Anomaly report(s)</u>

<u>test plan satisfies the following criteria:</u> <u>i) Traceable to the system requirements,</u> <u>ii) External consistency with the system requirements,</u> <u>iii) Internal consistency,</u> <u>iv) Test coverage of the hardware requirements,</u> <u>v) Appropriateness of test standards and methods,</u> <u>vi) Conformance to expected results,</u> <u>vii) Feasibility of hardware qualification testing,</u> <u>viii) Feasibility of operation and maintenance (e.g.,</u> <u>capability to be operated and maintained in</u> <u>accordance with user needs).</u> <u>b) Integrity Level 1</u> <u>There are no hardware integration V&V test</u> <u>requirements.</u>	<u>drawings</u> <u>Hardware</u> <u>integration test</u> <u>plan</u>	
<u>(7) Hardware Component Test Design Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> <u>1) Verify that the developer's test designs for hardware</u> <u>component testing conform to project-defined test</u> <u>document purpose, format, and content.</u> <u>2) Validate that the developer's hardware component</u> <u>test designs satisfy the criteria in V&V activity 10.3,</u> <u>Task 5.</u> <u>b) Integrity Level 1</u> <u>There are no hardware component V&V test</u> <u>requirements.</u>	<u>HDD</u> <u>IDD Hardware</u> <u>drawings</u> <u>User</u> <u>documentation</u> <u>Hardware test</u> <u>plans</u> <u>Hardware test</u> <u>designs</u>	<u>Task report(s) -</u> <u>Hardware</u> <u>component test</u> <u>design assessment</u> <u>Anomaly</u> <u>report(s)</u>

<u>10.3 Activity: Hardware Design V&V (Hardware, Process: Hardware Architectural Design and Hardware Detailed Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(8) Hardware Integration Test Design Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> 1) <u>Verify that the developer's test designs for hardware integration testing conform to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the developer's hardware integration test designs satisfy the criteria in V&V activity 10.3, Task 6.</u> <u>b) Integrity Level 1</u> <u>There are no hardware integration V&V test requirements.</u>	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware test designs</u>	<u>Task report(s) - Integration test design assessment</u> <u>Anomaly report(s)</u>
<u>(9) Hardware Qualification Test Design V&V</u> <u>a) Integrity Levels 3 and 4</u> 1) <u>Design tests for qualification testing.</u> 2) <u>Continue tracing required by the V&V qualification test plan. Verify that the V&V qualification test designs conform to project-defined test document purpose, format, and content.</u> 3) <u>Validate that the V&V qualification test designs satisfy the criteria in V&V Activity 10.2, Task 5</u> <u>b) Integrity Level 2</u> 1) <u>Verify that the developer's test designs for hardware qualification testing conform to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the developer's hardware qualification</u>	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware test designs</u>	<u>V&V hardware qualification test design(s)</u> <u>Anomaly report(s)</u>

<u>test designs satisfy the criteria in V&V Activity 10.2, Task 5.</u> <u>c) Integrity Level 1</u> <u>There are no hardware V&V qualification test requirements.</u>		
<u>(10) Hardware Acceptance Test Design V&V</u> <u>a) Integrity Levels 3 and 4</u> <u>1) Design tests for V&V hardware acceptance testing.</u> <u>2) Continue tracing required by the V&V hardware acceptance test plan. Verify that the V&V hardware acceptance test designs conform to project-defined test document purpose, format, and content</u> <u>3) Validate that the V&V hardware acceptance test designs satisfy the criteria in V&V Activity 10.2, Task 6.</u> <u>b) Integrity Level 2</u> <u>1) Verify that the acquirer's hardware test designs for acceptance testing conform to project-defined test document purpose, format, and content.</u> <u>2) Validate that the acquirer's hardware acceptance test designs satisfy the criteria in V&V activity 10.2, Task 6.</u> <u>c) Integrity Level 1</u> <u>There are no hardware acceptance V&V test requirements.</u>	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware test designs</u>	<u>V&V hardware acceptance test design(s)</u> <u>Anomaly report(s)</u>

<u>10.3 Activity: Hardware Design V&V (Hardware, Process: Hardware Architectural Design and Hardware Detailed Design)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(11) Hazard Analysis</u> a) <u>Verify the design elements that implement critical requirements introduce no new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) <u>Update the hazard analysis.</u>	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(12) Security Analysis</u> a) <u>Verify that the design element complies with the security features of the architectural design and that the design element does not introduce new security risks.</u> b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	<u>System architecture</u> <u>Subsystems security analysis</u> <u>Hardware drawings</u> <u>Security analysis report</u> <u>V&V task results</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly report(s)</u>
<u>(13) Risk Analysis</u> a) <u>Review and update risk analysis using prior task reports.</u> b) <u>Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Supplier development</u>	<u>Task report(s) -Risk analysis</u> <u>Anomaly report(s)</u>

	plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	
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10.4 Activity: Hardware Fabrication V&V (Hardware, Process: Fabrication)		
V&V tasks	Required inputs	Required outputs
(1) Fabricated Component Documentation Evaluation Evaluate the fabricated component documentation for correctness, consistency, completeness, accuracy, and readability.	Component Documentation HDD IDD Hardware drawings User Documentation	Task report(s) – Component documentation evaluation Anomaly report(s)

<u>10.4 Activity: Hardware Fabrication V&V (Hardware, Process: Fabrication)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(2) Interface Analysis</u> Verify and validate the fabricated component interfaces with software, user, operator, other hardware, and other systems for correctness, consistency, completeness, accuracy, and testability.	<u>Concept</u> <u>Documentation</u> <u>System requirements</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u>	<u>Task report(s) -</u> <u>Interface analysis</u> <u>Anomaly report(s)</u>
<u>(3) Traceability Analysis</u> a) Trace the fabricated component(s) to the design, and the design to the fabricated component(s). b) Analyze identified relationships for correctness, consistency, and completeness.	<u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Component documentation</u>	<u>Task report(s) -</u> <u>Traceability analysis</u> <u>Anomaly report(s)</u>
<u>(4) Criticality Analysis</u> Review and update the existing criticality analysis results from the prior Criticality Task Report using the fabricated component and its associated documentation.	<u>Criticality Task Report</u> <u>Component documentation</u>	<u>Task report(s) -</u> <u>Criticality analysis</u> <u>Anomaly report(s)</u>
<u>(5) Hardware Component Test Case Assessment</u> a) Integrity Levels 2, 3, and 4 1) Verify that the developer's hardware component test cases conform to project-defined test document purpose, format, and content. 2) Validate that the developer's hardware component test cases satisfy the criteria in V&V Activity 10.3, Task 5. b) Integrity Level 1 There are no hardware component V&V test	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware component test cases</u>	<u>Task report(s) -</u> <u>Hardware component test case assessment</u> <u>Anomaly report(s)</u>

requirements.		
(6) Hardware Integration Test Case Assessment	HRS	Task report(s) -
a) Integrity Levels 2, 3, and 4	IRS	Hardware
1) Verify that the developer's hardware integration test cases conform to project-defined test document purpose, format, and content.	HDD	integration test case assessment
2) Validate that the developer's hardware integration test cases satisfy the criteria in V&V Activity 10.3, Task 6.	IDD	Anomaly report(s)
b) Integrity Level 1	Hardware drawings	
There are no hardware integration V&V test requirements.	User documentation	
	Hardware integration test cases	

10.4 Activity: Hardware Fabrication V&V (Hardware, Process: Fabrication)		
V&V tasks	Required inputs	Required outputs
(7) Hardware Qualification Test Case V&V	HRS	V&V hardware
a) Integrity Levels 3 and 4	IRS	acceptance test
1) Develop test cases for V&V hardware qualification testing.	HDD	cases
2) Continue tracing required by the V&V hardware qualification test plan.	IDD	Anomaly report(s)
3) Verify that the V&V hardware qualification test cases conform to project-defined test document purpose, format, and content.	Hardware drawings	
4) Validate that the V&V hardware qualification test cases satisfy the criteria in V&V Activity 10.2, Task 5.	User documentation	
b) Integrity Level 2	Hardware qualification test	

<u>1) Verify that the developer's hardware qualification test cases conform to project-defined test document purpose, format, and content.</u> <u>2) Validate that the developer's hardware qualification test cases satisfy the criteria in V&V Activity 10.2, Task 5.</u> <u>c) Integrity Level 1</u> <u>There are no hardware V&V qualification test requirements.</u>	<u>cases</u>	
<u>(8) Hardware Acceptance Test Case V&V</u> <u>a) Integrity Levels 3 and 4</u> <u>1) Develop test cases for V&V hardware acceptance testing.</u> <u>2) Continue tracing required by the V&V hardware acceptance test plan.</u> <u>3) Verify that the V&V hardware Acceptance test cases conform to project-defined test document purpose, format, and content.</u> <u>4) Validate that the V&V hardware acceptance test cases satisfy the criteria in V&V Activity 10.2, Task 6.</u> <u>b) Integrity Level 2</u> <u>1) Verify that the acquirer's hardware acceptance test cases conform to project-defined test document purpose, format, and content.</u> <u>2) Validate that the acquirer's hardware acceptance test cases satisfy the criteria in V&V Activity 10.2, Task 6.</u> <u>c) Integrity Level 1</u> <u>There are no hardware acceptance V&V test requirements.</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User</u> <u>documentation</u> <u>Hardware</u> <u>acceptance test</u> <u>cases</u>	<u>V&V hardware</u> <u>acceptance test</u> <u>cases</u> <u>Anomaly report(s)</u>

<u>(9) Hardware Component Test Procedures Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> <u>1) Verify that the developer's hardware component test procedures conform to project-defined test document purpose, format, and content.</u> <u>2) Validate that the developer's hardware component test procedures satisfy the criteria in V&V Activity 10.3, Task 5.</u> <u>b) Integrity Level 1</u> <u>There are no hardware component V&V test requirements.</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User</u> <u>documentation</u> <u>Hardware</u> <u>component test</u> <u>cases</u> <u>Hardware</u> <u>component test</u> <u>procedures</u>	<u>Task report(s) -</u> <u>Hardware</u> <u>component test</u> <u>procedures</u> <u>assessment</u> <u>Anomaly report(s)</u>
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<u>10.4 Activity: Hardware Fabrication V&V (Hardware, Process: Fabrication)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(10) Hardware Integration Test Procedure</u> <u>Assessment</u> <u>a) Integrity Levels 2, 3, and 4</u> 1) <u>Verify that the developer's hardware integration test procedures conform to project-defined test document purpose, format, and content.</u> 2) <u>Validate that the developer's hardware integration test procedures satisfy the criteria in V&V Activity 10.3, Task 6.</u> <u>b) Integrity Level 1</u> <u>There are no hardware integration V&V test requirements.</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware integration test cases</u> <u>Hardware integration test procedures</u>	<u>Task report(s) - Hardware integration test procedures assessment</u> <u>Anomaly report(s)</u>
<u>(11) Hardware Qualification Test Procedure V&V</u> <u>a) Integrity Levels 3 and 4</u> 1) <u>Develop test procedures for V&V hardware qualification testing.</u> 2) <u>Continue tracing required by the V&V hardware qualification test plan.</u> 3) <u>Verify that the V&V hardware qualification test procedures conform to project-defined test document purpose, format, and content.</u> 4) <u>Validate that the V&V hardware qualification test procedures satisfy the criteria in V&V Activity 10.2, Task 5.</u>	<u>HRS</u> <u>IRS</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>User documentation</u> <u>Hardware qualification test cases</u> <u>Hardware qualification test</u>	<u>V&V hardware qualification test procedures</u> <u>Anomaly report(s)</u>

b) Integrity Level 2 1) Verify that the developer's hardware qualification test procedures conform to project-defined test document purpose, format, and content. 2) Validate that the developer's hardware qualification test procedures satisfy the criteria in V&V Activity 10.2, Task 5. c) Integrity Level 1 There are no hardware V&V qualification test requirements.	procedures	
(12) Hardware Component Test Execution Assessment a) Integrity Levels 2, 3, and 4 Use the developer's hardware component test results to validate that the hardware component satisfies the test acceptance criteria. b) Integrity Level 1 There are no hardware component V&V test requirements.	Component documentation HDD IDD Hardware drawings Component test plans Hardware component test procedures Hardware component test results	Task report(s) - Hardware component test execution assessment Anomaly report(s)

<u>10.4 Activity: Hardware Fabrication V&V (Hardware, Process: Fabrication)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(13) Hazard Analysis</u> a) <u>Verify that the fabricated component correctly implements the critical requirements and introduce no new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) <u>Update the hazard analysis.</u>	<u>Component documentation</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(14) Security Analysis</u> a) <u>Verify that the fabricated component is completed in accordance with the security features of the system design and that the fabricated component does not introduce new security risks.</u> b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	<u>Component documentation</u> <u>HDD</u> <u>IDD</u> <u>Hardware drawings</u> <u>Security analysis report</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly report(s)</u>
<u>(15) Risk Analysis</u> <u>Review and update risk analysis using prior task reports. Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>Component documentation</u> <u>Hardware drawings</u> <u>Supplier development plans and schedules</u>	<u>Task report(s) -Risk analysis</u> <u>Anomaly report(s)</u>

	Hazard analysis report Security analysis report Risk analysis report V&V task results	
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10.5 Activity: Hardware Integration Test V&V (Hardware, Process: Hardware Integration)		
V&V tasks	Required inputs	Required outputs
(1) Hardware Integration Test Execution Assessment a) Integrity Levels 2, 3, and 4 Use the developer's hardware integration test results to verify that the hardware satisfies the test acceptance criteria. b) Integrity Level 1 There are no hardware acceptance V&V test requirements.	Hardware component specifications Hardware integration test plan Hardware integration test procedures Hardware integration test results	Task report(s) - Hardware integration test execution assessment Anomaly report(s)

<u>10.5 Activity: Hardware Integration Test V&V (Hardware, Process: Hardware Integration)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(2) Traceability Analysis</u> <u>Analyze relationships in the V&V Test Plans, Designs, Cases, and Procedures for correctness and completeness.</u> <u>The task criteria are as follows:</u> <u>a) Correctness</u> <u>Verify that there is a valid relationship between the V&V test plans, designs, cases, and procedures.</u> <u>b) Completeness</u> <u>Verify that all V&V test procedures are traceable to the V&V test plans.</u>	<u>V&V test plans</u> <u>V&V test designs</u> <u>V&V test procedures</u>	<u>Task report(s) - Traceability analysis</u> <u>Anomaly report(s)</u>
<u>(3) Hazard Analysis</u> <u>a) Verify that the test instrumentation does not introduce new hazards.</u> <u>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> <u>c) Update the hazard analysis.</u>	<u>Hardware component specifications</u> <u>Test results</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(4) Security Analysis</u> <u>a) Verify that the implemented system does not increase the security risk.</u> <u>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and</u>	<u>Hardware component specifications</u> <u>Security analysis report</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly report(s)</u>

hardware operations).		
(5) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

<u>10.6 Activity: Hardware Qualification Test V&V (Hardware, Process: Hardware Qualification Testing)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Hardware Qualification Test Execution V&V</u> <u>a) Integrity Levels 3 and 4</u> <u>1) Perform V&V hardware qualification testing.</u> <u>2) Analyze test results to validate that the hardware satisfies the system requirements.</u> <u>3) Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</u> <u>4) Document the results as required by the V&V hardware qualification test plan.</u> <u>5) Use the V&V hardware qualification test results to validate that the hardware satisfies the test acceptance criteria.</u> <u>6) Document discrepancies between the actual and expected test results.</u> <u>b) Integrity Levels 1 and 2</u> <u>Use the developer's hardware qualification test results to verify that the hardware satisfies the test acceptance criteria.</u>	<u>Hardware component specifications</u> <u>Hardware qualification test plan</u> <u>Hardware qualification test procedures</u> <u>Hardware qualification test results</u>	<u>V&V hardware qualification test results</u> <u>Anomaly report(s)</u>
<u>(2) Traceability Analysis</u> <u>Analyze relationships in the V&V Test Plans, Designs, Cases, and Procedures for correctness and completeness.</u> <u>The task criteria are as follows:</u> <u>a) Correctness</u> <u>Verify that there is a valid relationship between the V&V test plans, designs, cases, and procedures.</u>	<u>V&V test plans</u> <u>V&V test designs</u> <u>V&V test procedures</u>	<u>Task report(s) - Traceability analysis</u> <u>Anomaly report(s)</u>

<u>b) Completeness</u> <u>Verify that all V&V test procedures are traceable to the V&V test plans.</u>		
<u>(3) Hazard Analysis</u> <u>a) Verify that the test instrumentation does not introduce new hazards.</u> <u>b) Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> <u>c) Update the hazard analysis.</u>	<u>Hardware component specifications</u> <u>Test results</u> <u>Hazard analysis report</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(4) Security Analysis</u> <u>a) Verify that the implemented system does not increase the security risk.</u> <u>b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	<u>Hardware component specifications</u> <u>Security analysis report</u>	<u>Task report(s) - Security analysis</u> <u>Anomaly report(s)</u>

<u>10.6 Activity: Hardware Qualification Test V&V (Hardware, Process: Hardware Qualification Testing)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(5) Risk Analysis</u> <u>c) Review and update risk analysis using prior task reports.</u> <u>d) Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>Supplier development plans and schedules</u> <u>Hazard analysis report</u> <u>Security analysis report</u> <u>Risk analysis report</u> <u>V&V task results</u>	<u>Task report(s) Risk analysis</u> <u>Anomaly report(s)</u>

<u>10.7 Activity: Hardware Acceptance Test V&V (Hardware, Process: Hardware Acceptance Support)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Traceability Analysis</u> <u>Analyze relationships in the V&V Test Plans, Designs, Cases, and Procedures for correctness and completeness.</u> <u>The task criteria are:</u> <u>a) Correctness</u> <u>Verify that there is a valid relationship between the V&V test plans, designs, cases, and procedures.</u> <u>b) Completeness</u>	<u>V&V test plans</u> <u>V&V test designs</u> <u>V&V test procedures</u>	<u>Task report(s) - Traceability analysis</u> <u>Anomaly report(s)</u>

Verify that all V&V test procedures are traceable to the V&V test plans.		
(2) Hardware Acceptance Test Procedure V&V a) Integrity Levels 3 and 4 1) Develop test procedures for V&V hardware acceptance testing. 2) Continue the tracing required by the V&V hardware acceptance test plan. 3) Verify that the V&V hardware acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]). 4) Validate that the V&V hardware acceptance test procedures satisfy the criteria in V&V Activity 10.2, Task 6. b) Integrity Level 2 1) Verify that the acquirer's hardware acceptance test procedures conform to project-defined test document purpose, format, and content (e.g., IEEE Std 829-2008 [B2]). 2) Validate that the acquirer's hardware acceptance test procedures satisfy the criteria in V&V Activity 10.2, Task 6. c) Integrity Level 1 There are no hardware acceptance V&V test requirements.	HDD IDD Hardware drawings Hardware component specifications User documentation Hardware acceptance test plan Hardware acceptance test procedures	V&V hardware acceptance test procedures Anomaly report(s)

10.7 Activity: Hardware Acceptance Test V&V (Hardware, Process: Hardware Acceptance Support)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
(3) Hardware Acceptance Test Execution V&V a) <u>Integrity Levels 3 and 4</u> 1) <u>Perform V&V hardware acceptance testing.</u> 2) <u>Analyze test results to validate that the hardware satisfies the system requirements.</u> 3) <u>Validate that the test results trace to test criteria established by the test traceability in the test planning documents.</u> 4) <u>Document the results as required by the V&V hardware acceptance test plan.</u> 5) <u>Use the V&V hardware acceptance test results to validate that the hardware satisfies the test acceptance criteria.</u> 6) <u>Document discrepancies between the actual and expected test results.</u> b) <u>Integrity Level 2</u> <u>Use the acquirer's hardware acceptance test results to verify that the hardware satisfies the test acceptance criteria.</u> c) <u>Integrity Level 1</u> <u>There are no hardware acceptance V&V test requirements.</u>	<u>Hardware component specifications</u> <u>User documentation</u> <u>Hardware acceptance test plan</u> <u>Hardware acceptance test procedures</u> <u>Hardware acceptance test results</u>	<u>V&V hardware acceptance test results</u> <u>Anomaly report(s)</u>
(4) Hazard Analysis a) <u>Verify that the test instrumentation does not introduce new hazards.</u> b) <u>Assess the identified mitigation strategies to verify</u>	<u>Hardware component specifications</u> <u>Test results</u>	<u>Task report(s) - Hazard analysis</u> <u>Anomaly report(s)</u>

each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations). c) Update the hazard analysis.	Hazard analysis report	
(5) Security Analysis a) Verify that the implemented system does not increase the security risk. b) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).	Hardware component specifications Security analysis report	Task report(s) - Security analysis Anomaly report(s)
(6) Risk Analysis a) Review and update risk analysis using prior task reports. b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Supplier development plans and schedules Hazard analysis report Security analysis report Risk analysis report V&V task results	Task report(s) - Risk analysis Anomaly report(s)

10.8 Activity: Hardware Transition V&V (Hardware, Process: Hardware Transition)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
1) Hardware Installation Configuration Audit a) <u>Verify that all hardware products required to correctly install and operate the hardware are present in the installation package.</u>	<u>Installation package (e.g., user documentation, HDD, IDD, HRS,</u>	<u>Task report(s)</u> <u>Installation configuration</u>

10.8 Activity: Hardware Transition V&V (Hardware, Process: Hardware Transition)

<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
b) <u>Validate that all site-dependent parameters or conditions to verify supplied values are correct.</u>	<u>IRS, hardware drawings, concept documentation, installation procedures, site-specific parameters, installation tests, and configuration management data)</u>	<u>audit</u> <u>Anomaly report(s)</u>
(2) Hardware Installation Checkout a) <u>Conduct analyses or tests to verify that the installed hardware corresponds to the hardware subjected to V&V.</u> b) <u>Verify that the hardware initializes, executes, and terminates as specified.</u> c) <u>In the transition from one version of hardware to the next, validate that the hardware can be replaced with the new version without adversely affecting or</u>	<u>User documentation</u> <u>Installation package</u>	<u>Task report(s) -</u> <u>Installation checkout</u> <u>Anomaly report(s)</u>

<u>degrading the functionality of the remaining system components.</u> d) <u>Verify the requirements for continuous operation and service during transition, including requirements for user notification.</u>		
(3) Hazard Analysis a) <u>Verify that the installation procedures and installation environment does not introduce new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) <u>Update the hazard analysis.</u>	<u>Installation package</u> <u>Hazard analysis report</u>	<u>Task report(s) -</u> <u>Hazard analysis</u> <u>Anomaly report(s)</u>
(4) Security Analysis a) <u>Verify that the installed hardware does not introduce new or increased vulnerabilities or security risks to the overall system.</u> b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	<u>Installation package</u> <u>User documentation</u> <u>Security analysis report</u>	<u>Task report(s) -</u> <u>Security analysis</u> <u>Anomaly report(s)</u>
(5) Risk Analysis a) <u>Review and update risk analysis using prior task reports.</u> b) <u>Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>Installation package</u> <u>Supplier</u> <u>development plans</u> <u>and schedules</u> <u>Security analysis report</u> <u>Risk analysis report</u> <u>V&V task results</u>	<u>Task report(s) -</u> <u>Risk analysis</u> <u>Anomaly report(s)</u>

<u>10.9 Activity: Hardware Operation V&V (Hardware, Process: Hardware Operation)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Evaluation of New Constraints</u> <u>Evaluate new constraints (e.g., operational requirements, platform characteristics, and operating environment) on the system or hardware requirements to verify the applicability of the VVP. (See NOTE.)</u>	<u>VVP New constraints</u>	<u>Task report(s) -</u> <u>Evaluation of new constraints</u>
<u>(2) Operating Procedures Evaluation</u> <u>Verify that the operating procedures are consistent with the user documentation and conform to the system requirements.</u>	<u>Operating procedures</u> <u>User documentation</u> <u>Concept documentation</u>	<u>Task report(s) -</u> <u>Operating procedures evaluation</u> <u>Anomaly report(s)</u>
<u>(3) Hazard Analysis</u> a) <u>Verify that the operating procedures and operational environment does not introduce new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled (any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) <u>Update the hazard analysis.</u>	<u>Operating procedures</u> <u>Hazard analysis report</u>	<u>Task report(s) -</u> <u>Hazard analysis</u> <u>Anomaly report(s)</u>
<u>(4) Security Analysis</u> a) <u>Verify that no new security risks are introduced due to changes in the operational environment.</u> b) <u>Over time, changes in external interfaces, threats, or technology in general require that an updated security analysis be performed to determine an updated residual risk.</u>	<u>New constraints</u> <u>Environmental changes</u> <u>Operating procedures</u> <u>Security analysis report</u>	<u>Task Reports -</u> <u>Security analysis</u>

c) Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).		
(5) Risk Analysis a) Review and update risk analysis using prior task reports, b) Provide recommendations to eliminate, reduce, or mitigate the risks.	Installation package Proposed changes Hazard analysis report Security analysis report Risk analysis report Supplier development plans and schedules Operation problem reports V&V task results	Task report(s) - Risk analysis Anomaly report(s)

10.10 Activity: Hardware Maintenance V&V (Hardware, Process: Hardware Maintenance)		
V&V tasks	Required inputs	Required outputs
(1) WP Revision a) Revise the VVP to conform to approved changes. b) When the development documentation required by this standard is not available, generate a new VVP and consider the methods in Annex D for deriving the required development documentation.	VVP Approved changes Installation package Supplier development plans and schedules	Updated VVP
(2) Anomaly Evaluation Evaluate the effect of hardware operation anomalies.	Anomaly report(s)	Task report(s) - Anomaly evaluation

<u>(3) Criticality Analysis</u> a) <u>Determine the integrity levels for proposed modifications.</u> b) <u>Validate the integrity levels provided by the maintained For V&V planning purposes, the highest integrity level assigned to the hardware shall be the integrity level of the system.</u>	<u>Proposed changes</u> <u>Installation package</u> <u>Maintainer integrity levels</u>	<u>Task report(s) -</u> <u>Criticality analysis</u> <u>Anomaly report(s)</u>
<u>(4) Migration Assessment</u> <u>Assess whether the hardware requirements and fabrication address the following:</u> a) <u>Specific migration requirements,</u> b) <u>Migration tools,</u> c) <u>Conversion of hardware products and data,</u> d) <u>Hardware archiving,</u> e) <u>Support for the prior environment,</u> f) <u>User notification.</u>	<u>Installation package</u> <u>Approved changes</u>	<u>Task report(s) -</u> <u>Migration assessment</u> <u>Anomaly report(s)</u>
<u>(5) Retirement Assessment</u> <u>Assess whether the installation package addresses the following:</u> a) <u>Hardware support,</u> b) <u>Impact on existing systems and data bases,</u> c) <u>Hardware archiving,</u> d) <u>Transition to a new hardware product,</u> e) <u>User notification.</u>	<u>Installation package</u> <u>Approved changes</u>	<u>Task report(s) -</u> <u>Retirement assessment</u> <u>Anomaly report(s)</u>
<u>(6) Hazard Analysis</u> a) <u>Verify that hardware modifications correctly implement the critical requirements and introduce no new hazards.</u> b) <u>Assess the identified mitigation strategies to verify each hazard is prevented, mitigated, or controlled</u>	<u>Proposed changes</u> <u>Installation package</u> <u>Hazard analysis report</u>	<u>Task report(s) -</u> <u>Hazard analysis</u> <u>Anomaly report(s)</u>

<u>(any unmitigated hazards are documented and addressed as part of system and hardware operations).</u> c) <u>Update the hazard analysis.</u>		
(7) Security Analysis a) <u>Verify that proposed changes/updates to the hardware do not introduce new or increased security risks to the overall system.</u> b) <u>Verify the identified security threats and vulnerabilities are prevented, controlled, or mitigated (any unmitigated threats and vulnerabilities are documented and addressed as part of system and hardware operations).</u>	<u>Proposed changes</u> <u>Installation package</u> <u>Security analysis</u> <u>report</u>	<u>Task reports -</u> <u>Security analysis</u>

<u>10.10 Activity: Hardware Maintenance V&V (Hardware, Process: Hardware Maintenance)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
(8) Risk Analysis a) <u>Review and update risk analysis using prior task reports,</u> b) <u>Provide recommendations to eliminate, reduce, or mitigate the risks.</u>	<u>Installation package</u> <u>Proposed changes</u> <u>Hazard analysis report</u> <u>Security analysis report</u> <u>Risk analysis report</u> <u>Supplier development plans and schedules</u> <u>Operation problem reports</u> <u>V&V task results</u>	<u>Task report(s) -</u> <u>Risk analysis</u> <u>Anomaly report(s)</u>

<u>(9) Task Iteration</u> <u>Perform V&V tasks, as needed, to assure that the following is performed:</u> a) <u>Planned changes are implemented correctly.</u> b) <u>Documentation is complete and current.</u> c) <u>Changes do not cause unacceptable or unintended system behaviors.</u>	<u>Approved changes</u> <u>Installation package</u>	<u>Task report(s)</u> <u>Anomaly report(s)</u>
NOTE—Hardware changes are maintenance activities (see 10.10).		

<u>10.11 Activity: Hardware Disposal V&V (Hardware, Process: Hardware Disposal)</u>		
<u>V&V tasks</u>	<u>Required inputs</u>	<u>Required outputs</u>
<u>(1) Hardware Disposal Evaluation</u> <u>Verify that any constraints specified or implied by the hardware disposal strategy are included in the hardware requirements, including hardware element destruction/storage and recording disposal actions and analysis of disposal impacts on the system. Validate that disposal leaves the system in an agreed-on state.</u>	<u>Hardware disposal strategy</u>	<u>Task report(s) – Hardware Disposal Evaluation</u> <u>Anomaly report(s)</u>

NOTE (for Table 1d)—Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed.

Table 2 d – Minimum V&V tasks assigned to each integrity level for hardware V&V

V&V Activities	Activity: Hardware Concept V&V (see 10.1)				Activity: Hardware Requirements V&V (see 10.2)				Activity: Hardware Design V&V (see 10.3)				Activity: Hardware Fabrication V&V (see 10.4)				Activity: Hardware Integration V&V (see 10.5)				Activity: Hardware Qualification V&V (see 10.6)				Activity: Hardware Acceptance V&V (see 10.7)				Activity: Hardware Transition V&V (see 10.8)				Activity: Hardware Operation V&V (see 10.9)				Activity: Hardware Maintenance V&V (see 10.10)				Activity: Hardware Disposal V&V (see 10.11)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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V&V Activities	Activity: Hardware Concept V&V (see 10.1)				Activity: Hardware Requirements V&V (see 10.2)				Activity: Hardware Design V&V (see 10.3)				Activity: Hardware Fabrication V&V (see 10.4)				Activity: Hardware Integration V&V (see 10.5)				Activity: Hardware Qualification V&V (see 10.6)				Activity: Hardware Acceptance V&V (see 10.7)				Activity: Hardware Transition V&V (see 10.8)				Activity: Hardware Operation V&V (see 10.9)				Activity: Hardware Maintenance V&V (see 10.10)				Activity: Hardware Disposal V&V (see 10.11)				
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	
Test Plan V&V																																													
Hardware Acceptance Test Procedure V&V																																													
Hardware Component Test Case Assessment																																													
Hardware Component Test Design Assessment																																													
Hardware Component Test Execution Assessment																																													
Hardware Component Test Plan Assessment																																													
Hardware Component Test Procedures Assessment																																													
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V&V Activities	Activity: Hardware Concept V&V (see 10.1)				Activity: Hardware Requirements V&V (see 10.2)				Activity: Hardware Design V&V (see 10.3)				Activity: Hardware Fabrication V&V (see 10.4)				Activity: Hardware Integration V&V (see 10.5)				Activity: Hardware Qualification V&V (see 10.6)				Activity: Hardware Acceptance V&V (see 10.7)				Activity: Hardware Transition V&V (see 10.8)				Activity: Hardware Operation V&V (see 10.9)				Activity: Hardware Maintenance V&V (see 10.10)				Activity: Hardware Disposal V&V (see 10.11)			
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4				
Hardware Integration Test Design Assessment												X	X	X	X																													
Hardware Integration Test Execution Assessment															X	X	X	X																										
Hardware Integration Test Plan Assessment												X	X	X	X																													
Hardware Integration Test Procedure Assessment													X	X	X	X																												
Hardware Qualification Test Case V&V													X	X	X	X																												
Hardware Qualification Test Design V&V												X	X	X	X																													
Hardware Qualification Test Execution V&V																	X	X	X	X																								
Hardware Qualification					X	X	X	X																																				

V&V Activities	Activity: Hardware Concept V&V (see 10.1)				Activity: Hardware Requirements V&V (see 10.2)				Activity: Hardware Design V&V (see 10.3)				Activity: Hardware Fabrication V&V (see 10.4)				Activity: Hardware Integration V&V (see 10.5)				Activity: Hardware Qualification V&V (see 10.6)				Activity: Hardware Acceptance V&V (see 10.7)				Activity: Hardware Transition V&V (see 10.8)				Activity: Hardware Operation V&V (see 10.9)				Activity: Hardware Maintenance V&V (see 10.10)				Activity: Hardware Disposal V&V (see 10.11)			
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4				
Test Plan V&V																																												
Hardware Qualification Test Procedure V&V													X	X	X	X																												
Hardware/Software/User Requirements Allocation Analysis				X																																								
Hazard Analysis			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X								
Installation Checkout																											X	X																
Installation Configuration Audit																											X	X																
Interface Analysis						X	X	X		X	X	X		X	X	X																												
Migration Assessment																																												
Operating Procedures Evaluation																																												
Requirements Evaluation						X	X	X	X																																			
Retirement Assessment																																												
Risk Analysis			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X				
Security Analysis			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X			X	X				

V&V Activities	Activity: Hardware e Concept V&V (see 10.1)				Activity: Hardware Requirements V&V (see 10.2)				Activity: Hardware Design V&V (see 10.3)				Activity: Hardware Fabrication V&V (see 10.4)				Activity: Hardware Integration V&V (see 10.5)				Activity: Hardware Qualification V&V (see 10.6)				Activity: Hardware Acceptance V&V (see 10.7)				Activity: Hardware Transition V&V (see 10.8)				Activity: Hardware Operation V&V (see 10.9)				Activity: Hardware Maintenance V&V (see 10.10)				Activity: Hardware Disposal V&V (see 10.11)				
Integrity Levels	Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				Levels				
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4					
Task Iteration																																													
Traceability Analysis	X	X	X		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X		X	X	X																		
Transition to New Product																																													
VVP Revision																																													

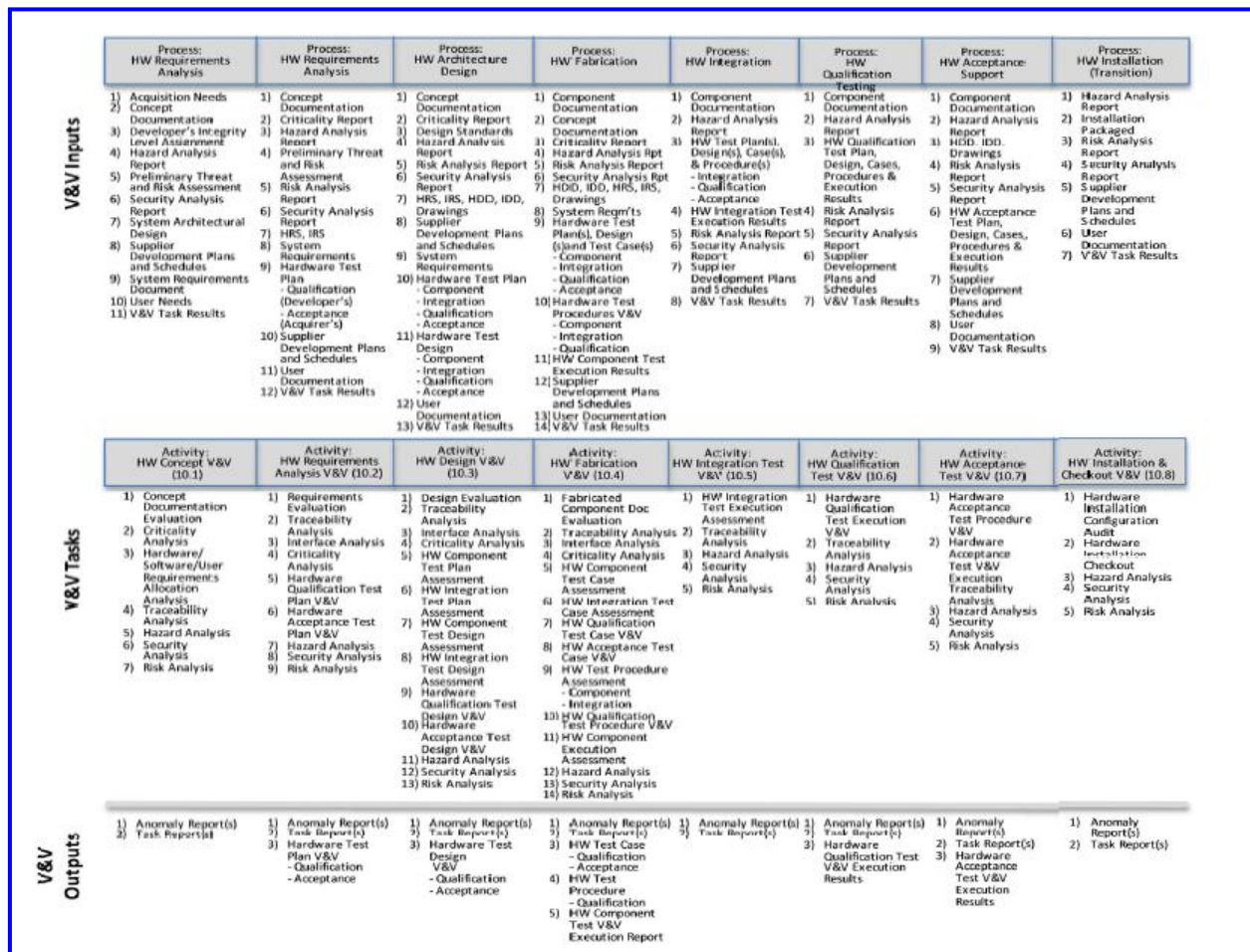
NOTE (for Table 2d)—Whenever a V&V task is selected as a mandatory requirement for multiple integrity levels, the V&V task implementation is dictated by the rigor, intensity, and depth of analysis or test. A higher integrity-level implementation requires greater rigor (e.g., formal methods, structured analysis methods), intensity (e.g., consideration of all system conditions and system environment states) and depth (e.g., abnormal cases, boundary conditions, comprehensive fault, and recovery scenarios) of analysis or test than the lower integrity-level implementation.

**Table 3 d—Optional V&V tasks and suggested applications in Hardware
Technical and Implementation Processes**

	<u>Hardware Requirements Analysis</u>	<u>Hardware Architectural Design</u>	<u>Hardware Detailed Design</u>	<u>Hardware Construction</u>	<u>Hardware Integration</u>	<u>Hardware Qualification Testing</u>	<u>Hardware Installation</u>	<u>Hardware Acceptance Support</u>	<u>Hardware Operation</u>	<u>Hardware Maintenance</u>	<u>Hardware Disposal</u>
<u>Algorithm analysis</u>											
<u>Audit performance</u>	X	X	X	X	X	X	X	X		X	
<u>Audit support</u>	X	X	X	X	X	X	X	X		X	
<u>Control flow analysis</u>	X	X	X	X						X	
<u>Cost analysis</u>	X	X	X	X	X	X	X	X		X	
<u>Database analysis</u>											
<u>Data flow analysis</u>											
<u>Disaster recovery plan assessment</u>	X	X	X	X					X	X	X
<u>Distributed architecture assessment</u>	X	X								X	
<u>Feasibility study evaluation</u>	X	X	X							X	
<u>Independent risk assessment</u>										X	
<u>Inspection</u>											
<u> Concept</u>										X	
<u> Requirements</u>	X									X	
<u> Design</u>		X	X							X	
<u> Source code</u>											
<u> Test plan</u>	X	X	X	X	X	X		X		X	
<u> Test design</u>		X	X	X	X	X		X		X	
<u> Test case</u>		X	X	X	X	X		X		X	
<u>Operational evaluation</u>									X		
<u>Performance monitoring</u>	X	X	X	X	X	X	X	X	X	X	X
<u>Post installation validation</u>									X	X	
<u>Project management oversight support</u>	X	X	X	X	X	X	X	X	X	X	X
<u>Proposal evaluation support</u>											

	<u>Hardware Requirements Analysis</u>	<u>Hardware Architectural Design</u>	<u>Hardware Detailed Design</u>	<u>Hardware Construction</u>	<u>Hardware Integration</u>	<u>Hardware Qualification Testing</u>	<u>Hardware Installation</u>	<u>Hardware Acceptance Support</u>	<u>Hardware Operation</u>	<u>Hardware Maintenance</u>	<u>Hardware Disposal</u>
<u>Qualification testing</u>				X		X		X			
<u>Regression analysis and testing</u>	X	X		X	X	X		X		X	
<u>Reusability analysis</u>											
<u>Reuse analysis</u>											
<u>Simulation analysis</u>	X	X	X	X	X	X	X	X	X	X	X
<u>Sizing and timing analysis</u>	X	X	X	X	X	X	X	X		X	
<u>System software assessment</u>											
<u>Test certification</u>				X	X	X		X	X	X	X
<u>Test evaluation</u>	X	X	X	X	X	X		X	X	X	X
<u>Test witnessing</u>				X	X	X		X	X	X	X
<u>Training document evaluation</u>	X	X	X	X	X	X		X	X	X	X
<u>Usability analysis</u>	X	X	X	X	X	X		X	X	X	
<u>User documentation evaluation</u>	X	X	X	X	X	X		X	X	X	
<u>User training</u>				X	X	X		X	X	X	
<u>V&V tool plan generation</u>											
<u>V&V tool qualification</u>	X	X	X	X	X	X	X	X	X	X	
<u>Walkthroughs</u>											
<u>Design</u>		X								X	
<u>Requirements</u>	X									X	
<u>Source code</u>											
<u>Test</u>				X	X	X				X	
<u>Work Breakdown Structure Evaluation</u>											

Annex G contains a description of the optional V&V tasks.



(New)

Figure 1d— Example of hardware V&V processes, activities, and tasks

	Process: HW Operation	Process: HW Maintenance	Process: HW Disposal
	1) Concept Documentation 2) Environmental Changes 3) Hazard Analysis Report 4) Installation Package 5) New Constraints 6) Operating Procedures 7) Operation Problem Report(s) 8) Proposed Changes 9) Risk Analysis Report 10) Security Analysis Report 11) Supplier Development Plans and Schedules 12) User Documentation 13) VVP 14) V&V Task Results	1) Approved Changes 2) Anomaly Report(s) 3) Installation Package 4) Hazard Analysis Report 5) Maintainer Integrity Level(s) 6) Operation Problem Report(s) 7) Proposed Changes 8) Risk Analysis Report 9) Security Analysis Report 10) Supplier Development Plans and Schedules 11) VVP 12) V&V Task Results	1) Hardware Disposal Strategy
V&V Inputs	Activity: HW Operation V&V (10.9)	Activity: HW Maintenance V&V (10.10)	Activity: HW Disposal V&V (10.11)
	1) Evaluation of New Constraints 2) Operating Procedures Evaluation 3) Hazard Analysis 4) Security Analysis 5) Risk Analysis	1) VVP Revision 2) Anomaly Evaluation 3) Criticality Analysis 4) Migration Assessment 5) Retirement Assessment 6) Hazard Analysis 7) Security Analysis 8) Risk Analysis 9) Task Iteration	1) Hardware Disposal Evaluation
V&V Tasks			
V&V Outputs			
	1) Anomaly Report(s) 2) Task Report(s)	1) Anomaly Report(s) 2) Task Report(s) 3) VVP	1) Anomaly Report(s) 2) Task Report(s)

(New)

NOTE 1—No clause reference are listed in the Process Definitions (top graphic bar) as there are no ISO/IEC or IEEE hardware life cycle definition standards used.

NOTE 2—Clause references in the Activity V&V Definitions (middle graphic bar) are IEEE Std 1012 clause numbers.

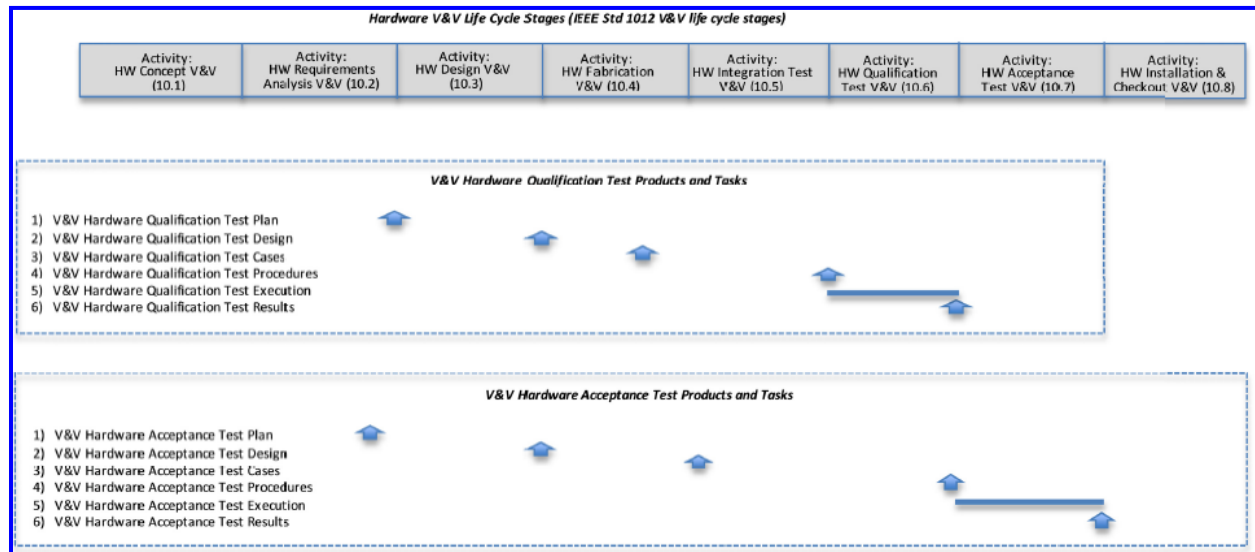
NOTE 3—V&V tasks listed in the figure are the minimum required for Integrity Level 4 (highest integrity level).

NOTE 4—Hardware Acceptance Testing Process supports the Systems Integration Testing Process.

NOTE 5—Hardware Installation and Checkout Process supports the System Transition Process.

Figure 1d—Example of hardware V&V processes, activities, and tasks

(continued)



(New)

NOTE 1—All V&V hardware test products and tasks represent the activities and products required as a minimum for integrity level 4.

NOTE 2—This is an example of the phasing of hardware V&V test products and tasks across the hardware life cycle. The hardware V&V test products (upward arrows) are shown in the hardware life cycle stages when the products are generated (per IEEE Std 1012-2004 [B5]). Hardware test execution tasks are shown to occur during one or more hardware life cycle stages as indicated by "activity bars" in the diagram. The life cycle stage (in which each test product is generated) and phasing of each test product and task can vary from this diagram in accordance with project specific needs.

NOTE 3—The V&V activity clauses referenced in the hardware V&V life cycle stages are IEEE Std 1012 clauses.

Figure 2d—Summary of hardware V&V test products and tasks

11. ~~Software~~ V&V reporting, administrative, and documentation requirements

11.1 V&V reporting requirements

V&V reporting occurs throughout the system, software, or hardware life cycle. The V&V effort shall produce the required outputs listed in Table 1 a through Table Id for each V&V task performed. The format and grouping of the V&V reports may be user defined. The V&V reports shall constitute the ~~software~~ V&V report ~~(SVVR)~~.

The V&V reports shall consist of the following:

- a) V&V task reports. The V&V effort shall document V&V task results and status. A task report may be combined with its associated activity report or with other task reports ~~include~~ . Task reports are generated for the following tasks:
 - 1) Acceptance Support
 - 2) Anomaly Evaluation
 - 3) Concept Documentation Evaluation
 - 4) Configuration Management Assessment
 - 5) Contract Verification
 - 6) Criticality Analysis

- 7) [Design Evaluation](#)
- 8) [Disposal Plan Evaluation](#)
- 9) Evaluation of New Constraints
- 10) [Fabricated Component Documentation Evaluation](#)
- 11) [Hardware Acceptance Test Case V&V](#)
- 12) [Hardware Acceptance Test Design V&V](#)
- 13) [Hardware Acceptance Test Execution V&V](#)
- 14) [Hardware Acceptance Test Plan V&V](#)
- 15) [Hardware Acceptance Test Procedure V&V](#)
- ~~15) Recommendations~~
- 16) [Hardware Component Test Case Assessment](#)
- 17) [Hardware Component Test Design Assessment](#)
- ~~17) Review results~~
- 18) [Hardware Component Test Execution Assessment](#)
- 19) [Hardware Component Test Plan Assessment](#)

20) [Hardware Component Test Procedures Assessment](#)

~~20) Software design evaluation~~

21) [Hardware Disposal Evaluation](#)

22) [Hardware Integration Test Case Assessment](#)

23) [Hardware Integration Test Design Assessment](#)

24) [Hardware Integration Test Execution Assessment](#)

25) [Hardware Integration Test Plan Assessment](#)

26) [Hardware Integration Test Procedure Assessment](#)

27) [Hardware Qualification Test Case V&V](#)

28) [Hardware Qualification Test Design V&V](#)

29) [Hardware Qualification Test Execution V&V](#)

30) [Hardware Qualification Test Plan V&V](#)

31) [Hardware Qualification Test Procedure V&V](#)

32) Hardware/Software/User Requirements Allocation Analysis

33) Hazard Analysis

- 34) Identify Process Improvement Opportunities in the Conduct of V&V
- 35) [Implementation Strategy Assessment](#)
- 36) Installation Checkout
- 37) Installation Configuration Audit
- 38) Interface Analysis
- 39) [Interface with other Processes](#)
- 40) Migration Assessment
- 41) Operating Procedures Evaluation
- 42) [Project Planning Strategy Assessment](#)
- 43) Proposed/[Baseline](#) Change Assessment
- 44) [Requirements Evaluation](#)
- 45) [Requirements Review](#)
- 46) Retirement Assessment
- 47) Risk Analysis
- 48) [Scoping the V&V Effort](#)

- 49) Security [Analysis](#)
- 50) Software ~~requirements evaluation~~ [Acceptance Test Case V&V](#)
- 51) [Software Acceptance Test Design V&V](#)
- 52) [Software Acceptance Test Execution V&V](#)
- 53) [Software Acceptance Test Plan V&V](#)
- 54) [Software Acceptance Test Procedure V&V](#)
- 55) [Software Component Test Case V&V](#)
- 56) [Software Component Test Design V&V](#)
- 57) [Software Component Test Execution V&V](#)
- 58) [Software Component Test Plan V&V](#)
- 59) [Software Component Test Procedure V&V](#)
- 60) [Software Disposal Evaluation](#)
- 61) [Software Integration Test Case V&V](#)
- 62) [Software Integration Test Design V&V](#)
- 63) [Software Integration Test Execution V&V](#)

- 64) [Software Integration Test Plan V&V](#)
- 65) [Software Integration Test Procedure V&V](#)
- 66) [Software Qualification Test Case V&V](#)
- 67) [Software Qualification Test Design V&V](#)
- 68) [Software Qualification Test Execution V&V](#)
- 69) [Software Qualification Test Plan V&V](#)
- 70) [Software Qualification Test Procedure V&V](#)
- 71) Source [Code](#) and [Source Code Documentation Evaluation](#)
- 72) [Stakeholder Requirements Evaluation](#)
- 73) System ~~requirements review~~ [Acceptance Test Case V&V](#)
- 74) [System Acceptance](#) Test ~~results~~ [Design V&V](#)
- 75) [System Acceptance Test Execution V&V](#)
- 76) [System Acceptance Test Plan V&V](#)
- 77) [System Acceptance Test Procedure V&V](#)
- 78) [System Element Implementation Analysis](#)

- 79) [System Integration Strategy Assessment](#)
- 80) [System Integration Test Case V&V](#)
- 81) [System Integration Test Design V&V](#)
- 82) [System Integration Test Execution V&V](#)
- 83) [System Integration Test Plan V&V](#)
- 84) [System Integration Test Procedure V&V](#)
- 85) [System Maintenance Execution Assessment](#)
- 86) [System Maintenance Strategy Assessment](#)
- 87) [System Qualification Test Case V&V](#)
- 88) [System Qualification Test Design V&V](#)
- 89) [System Qualification Test Execution V&V](#)
- 90) [System Qualification Test Plan V&V](#)
- 91) [System Qualification Test Procedure V&V](#)
- 92) Traceability Analysis
- 93) [Transition Demonstration Assessment](#)

94) [Transition Strategy Evaluation](#)

- b) *V&V activity summary reports*. An activity summary report shall summarize the results of V&V tasks performed for the following V&V life cycle activities:

[System V&V](#)

1) Acquisition Support

2) [Supply](#) Planning

3) [Project Planning](#)

~~3) Concept~~

4) [Configuration Management](#)

~~4) Requirements~~

5) [Stakeholder Requirements Definition](#)

~~5) Design~~

6) [Requirements Analysis](#)

7) [Architectural Design](#)

~~7) Test~~

8) Implementation

8) ~~Installation and checkout~~

9) [Integration](#)

10) [Transition](#)

11) Operation

12) Maintenance

13) [Disposal](#)

Software V&V

14) [Software Concept](#)

15) [Software Requirements](#)

16) [Software Design](#)

17) [Software Construction](#)

18) [Software Integration Test](#)

19) [Software Qualification Test](#)

20) [Software Acceptance Test](#)

[21\) Software Installation and Checkout](#)

[22\) Software Operation](#)

[23\) Software Maintenance](#)

[24\) Software Disposal](#)

Hardware V&V

[25\) Hardware Concept](#)

[26\) Hardware Requirements](#)

[27\) Hardware Design](#)

[28\) Hardware Fabrication](#)

[29\) Hardware Integration Test](#)

[30\) Hardware Qualification Test](#)

[31\) Hardware Acceptance Test](#)

[31\) Hardware Transition](#)

[32\) Hardware Operation](#)

[33\) Hardware Maintenance](#)

34) Hardware Disposal

For the operation and maintenance life cycle activities, V&V activity summary reports may be either updates to previous V&V activity summary reports or separate documents.

- c) V&V anomaly reports. The V&V effort shall document in an anomaly report each anomaly it detects.
- d) V&V final report. The V&V final report shall be issued at the ~~end of the installation and checkout activity or at the~~ conclusion of the V&V effort.
- e) Optional V&V reports. The V&V reports may also include optional reports (i.e., special studies reports and other reports). The V&V effort shall document in a special studies report any special V&V studies conducted during the software life cycle. The V&V effort shall document in a report the results of tasks conducted but not defined in the ~~SVVP~~. These other task reports may include, for example, quality assurance results, end-user testing results, safety assessment report, or configuration and data management status results. The title of the report may vary according to the subject matter.

Task report(s), ~~V&V activity summary report(s)~~, and Anomaly report(s) should be provided as feedback to the ~~software~~ development process regarding the technical quality of each ~~software~~ product and process.

11.2 V&V administrative requirements

The V&V administrative requirements shall consist of the following:

- a) Anomaly resolution and reporting policy
- b) Task iteration policy
- c) Deviation policy
- d) Control procedures
- e) Standards, practices, and conventions

These administrative requirements shall be documented in the VVP.

11.3 V&V documentation requirements

11.3.1 V&V test documentation

~~The scope of consists of V&V test documentation and SVVP documentation. The requirements for documentation are described in the following subclauses.~~

V&V test documentation requirements shall include the test plans, designs, cases, procedures, and results for component, integration, system, and acceptance testing developed by the V&V effort. The V&V test documentation shall conform to project-defined test document purpose, format, and content (e.g., ~~see~~ IEEE Std 829 ~~1998 [B4]~~ [2008 \[B2\]](#)). If the V&V effort uses test documentation or test types different from those in this standard (i.e., component, integration, system, [and](#) acceptance), the ~~software~~ V&V effort shall show a mapping of the proposed test documentation and execution to the test items defined in this standard. Test planning tasks defined in Table 1a through Table 1d shall be documented in the test plan, test design(s), test case(s), and test procedure(s).

11.3.2 ~~S~~VVP documentation

The V&V effort shall generate a VVP that addresses the topics described in Clause 12 of this standard. If there is no information pertinent to a topic, [then](#) the VVP shall contain the phrase "This topic is not applicable to this plan" and shall state an appropriate reason for the exclusion. Additional topics may be added to the plan. If some VVP material appears in other documents, the VVP may repeat the material or make reference to the material. The VVP shall be maintained throughout the life of the [system](#), software [or hardware](#).

12. **Software** V&V plan outline

12.1 Overview

The **SVVP** shall contain the content described in this [sub](#)clause. The user of this standard may adopt any format and section numbering system for the SVVP. The SVVP section numbers listed in this clause are provided to assist readability. An example SVVP outline is shown in the following boxed text.

SVVP outline (example)	
1. Purpose	
2. Referenced documents	
3. Definitions	
4. V&V overview	
4.1 Organization	
4.2 Master schedule	
4.3 Software Integrity level scheme	
4.4 Resources summary	
4.5 Responsibilities	
4.6 Tools, techniques, and methods	

5.V&V processes

[5.1 Common V&V Processes, Activities and Tasks](#)

[5.2 System V&V Processes, Activities and Tasks](#)

[5.2.1 Acquisition Support](#)

[5.2.2 Supply Planning](#)

[5.2.3 Project Planning](#)

[5.2.4 Configuration Management](#)

[5.2.5 Stakeholder Requirements Definition](#)

[5.2.6 Requirements Analysis](#)

[5.2.7 Architectural Design](#)

[5.2.8 Implementation](#)

[5.2.9 Integration](#)

[5.2.10 Transition](#)

[5.2.11 Operation](#)

[5.2.12 Maintenance](#)

[5.2.13 Disposal](#)

[5.3 Software V&V Processes, Activities and Tasks](#)

[5.3.1 Software Concept](#)

[5.3.2 Software Requirements](#)

[5.3.3 Software Design](#)

[5.3.4 Software Construction](#)

[5.3.5 Software Integration Test](#)

[5.3.6 Software Qualification Test](#)

[5.3.7 Software Acceptance Test](#)

[5.3.8 Software Installation and Checkout \(Transition\)](#)

[5.3.9 Software Operation](#)

[5.3.10 Software Maintenance](#)

[5.3.11 Software Disposal](#)

[5.4 Hardware V&V Processes, Activities and Tasks](#)

[5.4.1 Hardware Concept](#)

[5.4.2 Hardware Requirements](#)

[5.4.3 Hardware Design](#)

[5.4.4 Hardware Fabrication](#)

[5.4.5 Hardware Integration Test](#)

[5.4.6 Hardware Qualification Test](#)

[5.4.7 Hardware Acceptance Test](#)

[5.4.8 Hardware Transition](#)

[5.4.9 Hardware Operation](#)

[5.4.10 Hardware Maintenance](#)

[5.4.11 Hardware Disposal](#)

~~5. V&V processes~~

~~5.1 Process: Management~~

~~5.1.1 Activity: Management of V&V~~

~~5.2 Process: Acquisition~~

~~5.2.1 Activity: Acquisition support V&V~~

~~5.3 Process: Supply~~

~~5.3.1 Activity: Planning V&V~~

~~5.4 Process: Development~~

~~5.4.1 Activity: Concept V&V~~

~~5.4.2 Activity: Requirements V&V~~

~~5.4.3 Activity: Design V&V~~

~~5.4.4 Activity: Implementation V&V~~

~~5.4.5 Activity: Test V&V~~

~~5.4.6 Activity: Installation and checkout V&V~~

~~5.5 Process: Operation~~

~~5.5.1 Activity: Operation V&V~~

~~5.6 Process: Maintenance~~

~~5.6.1 Activity: Maintenance V&V~~

6. V&V reporting requirements

6.1 Task reports

~~6.2 Activity summary reports~~

6.2 Anomaly reports

6.3 V&V final report

6.4 Special studies reports (optional)

6.5 Other reports (optional)

7. V&V administrative requirements

7.1 Anomaly resolution and reporting

- 7.2 Task iteration policy
- 7.3 Deviation policy
- 7.4 Control procedures
- 7.5 Standards, practices, and conventions

8.V&V test documentation requirements

12.2 **SVVP** section 1: Purpose

The SVVP shall describe the purpose, goals, and scope of the ~~software~~ V&V effort, including waivers from this standard. The ~~SVVP also shall identify the specific software processes and products covered by the software V&V effort.~~ date of issue and status, identification of issuing organization, and identification of approval authority shall be provided.

12.3 **SVVP** section 2: Referenced documents

The SVVP shall identify the compliance documents, documents referenced by the SVVP, and any supporting documents supplementing or implementing the SVVP.

12.4 **SVVP** section 3: Definitions

The SVVP shall define or reference all terms used in the SVVP, including the criteria for classifying an anomaly as a critical anomaly. All abbreviations and notations used in the SVVP also shall be described.

12.5 ~~S~~VVP section 4: V&V overview

~~The SVVP shall describe the organization, schedule, software integrity level scheme, resources, responsibilities, tools, techniques, and methods necessary to perform the software V&V.~~

12.5.1 ~~S~~VVP section 4.1: Organization

The SVVP shall describe the organization of the V&V effort, including the degree of independence required (see Annex C). The SVVP shall describe the relationship of the V&V processes to other processes, such as development, project management, quality assurance, and configuration management. The SVVP shall describe the lines of communication within the V&V effort, the authority for resolving issues raised by V&V tasks, and the authority for approving V&V products. Annex F illustrates a sample organizational interrelationship chart.

12.5.2 ~~S~~VVP section 4.2: Master schedule

The SVVP shall describe the project life cycle and milestones, and [it](#) shall summarize the schedule of V&V tasks and task results as feedback to the development, organizational [project-enabling](#), and supporting processes (e.g., quality assurance and configuration management). V&V tasks should be scheduled to be reperformed according to the task iteration policy.

If the life cycle used in the SVVP differs from the life cycle model in this standard, this section shall describe how all requirements of the standard are satisfied (e.g., by cross referencing to this standard).

12.5.3 ~~S~~VVP section 4.3: ~~Software~~ Integrity level scheme

The SVVP shall describe the agreed ~~upon software~~ integrity level scheme established for the system, software or hardware, and the mapping of the selected scheme to the model used in this standard. The SVVP shall document (by inclusion or by reference to the criticality analysis) the assignment of ~~software~~ integrity levels to individual components (e.g., requirements, detailed functions, subsystems, software modules, ~~subsystems~~, hardware component, or other ~~software partitions components~~), where ~~there are~~ differing ~~software~~ integrity levels are assigned within the program.

12.5.4 ~~S~~VVP section 4.4: Resources summary

The SVVP shall summarize the V&V resources, including staffing, facilities, tools, finances, and special procedural requirements (e.g., security, access rights, and documentation control).

12.5.5 ~~S~~VVP section 4.5: Responsibilities

The SVVP shall identify an overview of the organizational element(s) and responsibilities for V&V tasks.

12.5.6 ~~S~~VVP section 4.6: Tools, techniques, and methods ⁴

The SVVP shall describe documents, hardware and software V&V tools, techniques, methods, and operating and test environment to be used in the V&V processes. Acquisition, training, support, and qualification information for each tool, technology, and method shall be included.

The SVVP should document the measures to be used by V&V (see Annex E) and should describe how these measures support the V&V objectives.

⁴ [The information in this footnote is based on excerpts from ISO/IEC/IEEE 29148-2011 \[B20\].](#)

[Verification and validation methods define how \(including success criteria and closure approach\), where and when each requirement's conformance will be proven. A verification or validation method is associated with each requirement to define activities that yield objective information to prove satisfaction of the requirement. A verification or validation method definition addresses the following content considerations:](#)

- [— How: Identify which method\(s\) will be applied \(e.g., inspection, analysis, demonstration, and test\).](#)
- [— Who: Identify the organization/person with the lead responsibility for performing the method, such as a contractor, subcontractor, vendor, product team, or supplier.](#)
- [— When: Designate a time in the program plan when the method is to be done. This should be an event based and not a calendar date or accomplishment.](#)
- [— Where: Specify any unique venue and environment needed for the method.](#)

[Examples of verification and validation methods to use to obtain the objective evidence that the requirements have been fulfilled are inspection, analysis or simulation, demonstration, and test. The definitions are as follows:](#)

- [— Inspection : An examination of the item against applicable documentation to confirm compliance with requirements.](#)
- [— Analysis \(including modeling and simulation\): Use of analytical data or simulations under defined conditions to show theoretical compliance.](#)
- [— Demonstration: A qualitative exhibition of functional performance, usually accomplished with no or minimal instrumentation or test equipment.](#)
- [— Test: An action by which the operability, supportability, or performance capability of an item is quantitatively verified when subjected to controlled conditions that are real or simulated.](#)

12.6 ~~S~~VVP section 5: V&V processes

The SVVP also shall identify the specific processes and products covered by the V&V effort and the activities and tasks ~~to be performed for each of the V&V processes described in Clause 5 of this standard, and shall document those V&V activities and~~ The SVVP shall contain an overview of the V&V activities and tasks for all software life cycle processes as derived from the Table 2a through Table 2d minimum V&V requirements and the optional tasks selected for the project.

~~The SVVP shall address the following topics for each V&V activity.~~

12.6.1 WP Section 5.1: Common V&V Processes, Activities, and Tasks

The VVP shall identify and describe the Common V&V processes, activities and tasks as identified in Table 1a to be implemented and performed for this V&V project. The minimum common V&V tasks for the selected system integrity level are defined in Table 2a.

12.6.2 WP Section 5.2: System V&V Processes, Activities, and Tasks

The VVP shall identify and describe the System V&V processes, activities, and tasks as identified in Table 1b to be implemented and performed for this V&V project. The minimum system V&V tasks for the selected system integrity level are defined in Table 2b.

12.6.3 WP Section 5.3: Software V&V Processes, Activities, and Tasks

The VVP shall identify and describe the Software V&V processes, activities, and tasks as identified in Table 1c to be implemented and performed for this V&V project. The minimum common V&V tasks for the selected system integrity level are defined in Table 2c.

12.6.4 WP Section 5.4: Hardware V&V Processes, Activities, and Tasks

The VVP shall identify and describe the Hardware V&V processes, activities, and tasks as identified in Table 1d to be implemented and performed for this V&V project. The minimum common V&V tasks for the selected system integrity level are defined in Table 2d.

~~7.5.1 SVVP sections 5.1 through 5.6: Software life cycle³~~

~~The SVVP shall include sections 5.1 through 5.6 for V&V activities and tasks as shown in the SVVP outline (boxed text).~~

~~The SVVP shall address the following eight topics for each V&V activity:~~

~~³Software life cycle V&V sections are 5.1 Process: Management, 5.2 Process: Acquisition, 5.3 Process: Supply, 5.4 Process: Development, 5.5 Process: Operation, and 5.6 Process: Maintenance.~~

~~1) — V&V tasks~~

~~The SVVP shall identify the V&V tasks to be performed. Table 1 describes the minimum V&V tasks, task criteria, and required inputs and outputs. Table 2 specifies the minimum V&V tasks that shall be performed for each software integrity level.~~

~~The minimum tasks for software integrity level 4 are consolidated in graphic form in Figure 1.~~

~~2) — Methods and procedures~~

~~The SVVP shall describe the methods and procedures for each task, including on-line access, and conditions for observation/evaluation of development processes. The SVVP shall define the criteria for evaluating the task results.~~

~~3) — Inputs~~

~~The SVVP shall identify the required inputs for each V&V task. The SVVP shall specify the source and format of each input. The inputs required for the minimum V&V tasks are identified in Table 1. Other inputs may be used. For any V&V activity and task, all of the required inputs and outputs from preceding activities and tasks may be used, but for conciseness, only the primary inputs are listed in Table 1.~~

~~4) — *Outputs*~~

~~The SVVP shall identify the required outputs from each V&V task. The SVVP shall specify the purpose, format, and recipients of each output. The required outputs from each of the V&V tasks are identified in Table 1. Other outputs may be produced.~~

~~The outputs of the management of V&V and of the V&V tasks shall become inputs to subsequent processes and activities, as appropriate (see Clause 6).~~

~~5) — *Schedule*~~

~~The SVVP shall describe the schedule for the V&V tasks. The SVVP shall establish specific milestones for initiating and completing each task, for the receipt and criteria of each input, and for the delivery of each output.~~

~~6) — *Resources*~~

~~The SVVP shall identify the resources for the performance of the V&V tasks. The SVVP shall specify resources by category (e.g., staffing, equipment, facilities, travel, and training). The costs of V&V activities and resources shall be provided or referenced.~~

~~7) *Risks and assumptions*~~

~~The SVVP shall identify the risks (e.g., schedule, resources, or technical approach) and assumptions associated with the V&V tasks. The SVVP shall provide recommendations to eliminate, reduce, or mitigate risks.~~

~~8) *Roles and responsibilities*~~

~~The SVVP shall identify the organizational elements or individuals responsible for performing the V&V tasks.~~

12.7 SVVP section 6: V&V reporting requirements

The SVVP shall specify the purpose, content, format, recipients, and timing of all V&V reports. The V&V reporting requirements are specified in Clause 11.

12.8 SVVP section 7: V&V administrative requirements

12.8.1 General

The SVVP shall describe the anomaly resolution and reporting, task iteration policy, deviation policy, control procedures, and standards, practices, and conventions.

12.8.2 ~~S~~VVP section 7.1: Anomaly resolution and reporting

The SVVP shall describe the method of reporting and resolving anomalies, including the criteria for reporting an anomaly, the anomaly report distribution list, the authority and time lines for resolving anomalies, and the anomaly criticality levels. Classification for software anomalies may be found in IEEEStd 1044TM ~~-1993 [B8~~ [-2009 \[B7\]](#).

12.8.3 ~~S~~VVP section 7.2: Task iteration policy

The SVVP shall describe the criteria used to determine the extent to which a V&V task should be repeated when its input is changed or task procedure is changed. These criteria may include assessments of change, ~~software~~ integrity level, and effects on budget, schedule, and quality.

12.8.4 ~~S~~VVP section 7.3: Deviation policy

The SVVP shall describe the procedures and criteria used to deviate from the plan. The information required for deviations shall include task identification, rationale, and effect on ~~software~~-quality. The SVVP shall identify the authorities responsible for approving deviations.

12.8.5 ~~S~~VVP section 7.4: Control procedures

The SVVP shall identify control procedures applied to the V&V effort. These procedures shall describe how ~~software~~-products and V&V results should be configured, protected, and stored.

These procedures may describe quality assurance, configuration management, data management, or other activities if they are not addressed by other efforts. The SVVP shall describe how the V&V effort shall conform to existing security provisions and how the validity of V&V results shall be protected from unauthorized alterations.

12.8.6 ~~S~~VVP section 7.5: Standards, practices, and conventions

The SVVP shall identify the standards, practices, and conventions that govern the performance of V&V tasks including internal organizational standards, practices, and policies.

12.9 ~~S~~VVP section 8: V&V test documentation requirements

The SVVP shall describe the purpose, format, and content for the following V&V test documents:

- a) Test plan
- b) Test design
- c) Test cases
- d) Test procedures
- e) Test results

The V&V effort may define the format for these documents. IEEE Std 829- ~~1998~~ ~~[B4]~~ [2008](#) [\[B2\]](#) contains sample formats for these test documents.

Annex A

(informative)

Mapping of IEEE ~~Std~~ 1012 V&V activities and tasks

A.1 Mapping of ISO/IEC ~~12207~~ [15288](#) V&V requirements to IEEE ~~Std~~ 1012 V&V activities and tasks

Table A.I shows a mapping of all ISO/IEC ~~12207:1995 [B13]~~ [15288:2008 \[B16\]](#) V&V requirements (i.e., processes, activities, and tasks) to the V&V activities and tasks of this standard.

The first column of Table A.I lists the ISO/IEC ~~12207:1995 [B13]~~ [15288:2008 \[B16\]](#) clause numbers and titles of V&V processes and activities. The second column of Table A.I lists the IEEE Std 1012 ~~2004~~ clauses and tables that address the topics listed in the first column.

Table A.1—Mapping ISO/IEC ~~12207~~ 15288 V&V requirements to IEEE Std 1012 V&V activities and tasks

<u>ISO/IEC 15288 V&V requirements</u>	<u>IEEE 1012 V&V activities and tasks (see corresponding clause/table)</u>
<u>6.1.1.3.d Monitor the agreement [with the supplier]</u>	<u>7.2 Activity: Acquisition Support V&V</u> <u>Task 2 Planning the Interface between the V&V Effort and Supplier</u> <u>Task 3 System Requirements Review</u> <u>7.3 Activity: Supply Planning V&V</u> <u>Task 1 Planning the Interface between the V&V Effort and Supplier</u> <u>Task 2 Contract Verification</u>
<u>6.1.1.3.e. Accept the product or service [from the supplier]</u>	<u>7.2 Activity: Acquisition Support V&V</u> <u>Task 4 Acceptance Support</u>
<u>6.4.1.3c Analyze and maintain stakeholder requirements</u>	<u>5. Integrity Levels</u> <u>8.1 Activity: Stakeholder Requirements Definition V&V</u> <u>Task 1 Stakeholder Requirements Evaluation</u> <u>Task 2 Traceability Analysis</u> <u>Task 3 Criticality Analysis</u> <u>Task 4 Hazard Analysis</u> <u>Task 5 Security Analysis</u> <u>Task 6 Risk Analysis</u>

6.4.2.3.a Define system requirements	<u>5. Integrity Levels</u> <u>8.2 Activity: Requirements Analysis V&V</u> Task 1 Requirements Evaluation Task 2 Interface Analysis Task 3 Traceability Analysis Task 4 Criticality Analysis Task 5 System Integration Test Plan V&V Task 6 System Qualification Test Plan V&V Task 7 System Acceptance Test Plan V&V Task 8 Hazard Analysis Task 9 Security Analysis Task 10 Risk Analysis
6.4.2.3.b Analyze and maintain system requirements	<u>5. Integrity Levels</u> <u>8.3 Activity: Architectural Design V&V</u> Task 3 Traceability Analysis Task 4 Criticality Analysis <u>8.4 Activity: Implementation V&V</u> Task 3 Criticality Analysis
6.4.3.3.b Analyze and evaluate the architecture	<u>5. Integrity Levels</u> <u>8.3 Activity: Architectural Design V&V</u> Task 1 Design Evaluation Task 2 Interface Analysis Task 3 Traceability Analysis Task 4 Criticality Analysis Task 5 System Integration Test Design V&V

	Task 6 System Qualification Test Design V&V Task 7 System Acceptance Test Design V&V Task 8 Hazard Analysis Task 9 Security Analysis Task 10 Risk Analysis
6.4.4.3.b Perform implementation	5. Integrity Levels 8.4 Activity: Implementation V&V Task 1 Implementation Strategy Assessment Task 2 System Element Implementation Analysis Task 3 Criticality Analysis Task 4 Hazard Analysis Task 5 Security Analysis Task 6 Risk Analysis Task 7 System Integration Test Case V&V Task 8 System Qualification Test Case V&V Task 9 System Acceptance Test Case V&V
6.4.5.3.b Perform integration	8.2 Activity: Requirements Analysis V&V Task 5 System Integration Test Plan V&V 8.3 Activity: Architectural Design V&V Task 5 System Integration Test Design V&V 8.4 Activity: Implementation V&V Task 7 System Integration Test Case V&V 8.5 Activity: Integration V&V Task 1 System Integration Strategy Assessment

	Task 2 System Integration Test Procedure V&V Task 3 System Integration Test Execution V&V Task 4 System Qualification Test Procedure V&V Task 5 Qualification Test Execution V&V Task 6 System Acceptance Test Procedure V&V
6.4.6.3.a Plan Verification	7.1 Activity: V&V Management Task 1 VVP Generation
6.4.6.3.b Perform Verification	Clause 1 through Clause 7, Clause 8, Clause 11, Clause 12, Table 1a, Table 1b, Table 2a, Table 2b, Table 3a, and Table 3b, Figure 1a, Figure 1b, Figure 2a, and Figure 2b, all annexes
6.4.7.3.b Perform the transition	8.2 Activity: Requirements Analysis V&V Task 7 System Acceptance Test Design V&V 8.3 Activity: Architectural Design V&V Task 7 System Acceptance Test Design V&V
	8.4 Activity: Implementation V&V Task 9 System Acceptance Test Case V&V 8.5 Activity: Integration V&V Task 6 System Acceptance Test Procedure V&V 8.6 Activity: Transition V&V Task 1 Transition Strategy Evaluation Task 2 Transition Demonstration Assessment

	Task 3 System Acceptance Test Execution V&V
6.4.8.3.a Plan Validation	7.1 Activity: V&V Management Task 1 VVP Generation
6.4.8.3.b Perform Validation	Clauses 1-7,8, 11-12, Tables 1a, 1b, 2a, 2b, 3a and 3b, Figures 1 a, 1b, 2a and 2b, all annexes
6.4.9.3.d Perform operational problem resolution	8.7 Activity: Operation V&V Task 1 Operating Procedures Evaluation Task 2 Hazard Analysis Task 3 Security Analysis Task 4 Risk Analysis
6.4.10.3.b Perform maintenance	8.8 Activity: Maintenance V&V Task 1 System Maintenance Strategy Assessment Task 2 System Maintenance Execution Assessment
6.4.11.3.a Plan Disposal	8.9 Activity: Disposal V&V Task 1 Disposal Plan Evaluation

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
5.1.4.1 Supplier Monitoring V&V	5.2.1 Activity: Acquisition support V&V Task 1 Scoping the V&V effort Task 2 Planning the interface between the V&V effort and supplier Task 3 System requirements review
5.2.4.5(h) and 5.2.5.5 Interfacing with V&V ^a	5.2.1 Activity: Acquisition support V&V Task 2 Planning the interface between the V&V effort and supplier 5.3.1 Activity: Planning V&V Task 1 Planning the interface between the V&V effort and supplier Annex C Definition of IV&V
5.2.6.3 Verification and Validation ^a	All clauses, tables, software V&V figures, and annexes
5.3.2 System Requirements Analysis	5.2.1 Activity: Acquisition support V&V Task 3 System requirements review 5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation Task 4 Traceability analysis
5.5.5 and 5.5.6 Migration and Software Retirement	5.1.1 Activity: Management of V&V Task 2 Proposed/baseline change assessment 5.6.1 Activity: Maintenance V&V

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
6.4.1 Verification Process Implementation ^a	Clause 4 Software integrity levels Clause 6 Software V&V reporting, administrative, and documentation requirements 6.2 V&V Administrative requirements Clause 7 SVVP outline 7.7 SVVP section 7: V&V administrative requirements
6.4.1.1 Criticality of Software to be Verified ^a	Clause 4 Software integrity levels Table B.1 Assignment of software integrity levels Table B.2 Definition of consequences Annex D V&V of reuse software
6.4.1.2 Process for Verification ^a	Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline
6.4.1.3 and 6.4.1.4 Extent and Rigor of Verification ^a	Clause 4 Software integrity levels Table 2 Minimum V&V tasks assigned to each software integrity level Annex C Definition of independent V&V (IV&V)

6.4.1.5 Verification Plan ^a	Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline
6.4.1.6 Problem and Non-conformance Reports ^a	6.2 V&V Administrative requirements 7.7 SVVP section 7: V&V Administrative requirements
6.4.2 Verification	Clause 5 Software V&V processes
6.4.2.1 Contract Verification	5.3.1 Activity: Planning V&V Task 2 Contract Verification
6.4.2.2 Process Verification	5.2 Process: Acquisition 5.3 Process: Supply 5.4 Process: Development

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
6.4.2.3 Requirements Verification	5.2.1 Activity: Acquisition support V&V Task 3 System requirements review 5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation 5.4.2 Activity: Requirement V&V Task 1 Traceability analysis Task 2 Software requirements evaluation Task 3 Interface analysis

	Task 4 Criticality analysis Task 5 System V&V test plan generation Task 6 Acceptance V&V test plan generation Task 8 Hazard analysis Task 9 Security analysis
6.4.2.4 Design Verification	5.4.3 Activity: Design V&V Task 1 Traceability analysis Task 2 Software design evaluation Task 3 Interface analysis Task 4 Criticality analysis Task 5 Component V&V test plan generation Task 6 Integration V&V test plan generation Task 7 Component V&V test design generation Task 8 Integration V&V test design generation Task 9 System V&V test design generation Task 10 Acceptance V&V test design generation Task 11 Hazard analysis Task 12 Security analysis
6.4.2.5 Code Verification	5.4.4 Activity: Implementation V&V Task 1 Traceability analysis Task 2 Source code and source code documentation evaluation Task 3 Interface analysis Task 4 Criticality analysis Task 5 Component V&V test case generation Task 6 Integration V&V test case generation Task 7 System V&V test case generation Task 8 Acceptance V&V test case generation Task 9 Component V&V test procedure generation Task 10 Integration V&V test procedure generation

	Task 11 System V&V test procedure generation Task 12 Component V&V test execution Task 13 Hazard analysis Task 14 Security analysis
6.4.2.6 Integration Verification	5.4.5 Activity: Test V&V Task 3 Integration V&V test execution

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
6.4.2.7 Documentation Verification	5.2.1 Activity: Acquisition support V&V Task 3 Systems requirements review 5.3.1 Activity: Planning V&V Task 2 Contract verification 5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation 5.4.2 Activity: Requirements V&V Task 2 Software requirements evaluation Task 3 Interface analysis 5.4.3 Activity: Design V&V Task 2 Software design evaluation Task 3 Interface analysis 5.4.4 Activity: Implementation V&V Task 2 Source code and source code documentation evaluation

	Task 3 Interface analysis 5.4.6 Activity: Installation and Checkout Task 1 Installation configuration audit 5.5.1 Activity: Operation V&V Task 2 Operating procedures evaluation
6.5.1 Validation Process Implementation^a	Clause 4 Software integrity levels Clause 6 Software V&V reporting, administrative, and documentation requirements 6.2 V&V Administrative requirements Clause 7 SVVP outline 7.7 SVVP section 7: V&V administrative requirements Annex C Definition of independent V&V (IV&V) Annex D V&V of reuse software Annex E V&V measures
6.5.1.1 Criticality of Software to be Validated^a	Clause 4 Software integrity levels Table B.1 Assignment of software integrity levels Table B.2 Definition of consequences

	Annex D V&V of reuse software
6.5.1.2 Process for Validation	Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP Outline

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
6.5.1.3 Extent and Rigor of Validation[‡]	Table 2 Minimum V&V tasks assigned to each software integrity level Annex C Definition of independent V&V (IV&V)
6.5.1.4 Validation Plan[‡]	Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline
6.5.1.5 Problem and Non-conformance Reports[‡]	6.2 V&V Administrative requirements 7.7 SVVP section 7: V&V administrative requirements
6.5.2 Validation	Clause 5 Software V&V processes
6.5.2.1 Validate Test Preparation[‡]	5.4.2 Activity: Requirements V&V Task 5 System V&V Test plan generation Task 6 Acceptance V&V test plan generation 5.4.3 Activity: Design V&V Task 5 Component V&V test plan generation Task 6 Integration V&V test plan generation

	Task 7 Component V&V test design generation Task 8 Integration V&V test design generation Task 9 System V&V test design generation Task 10 Acceptance V&V test design generation 5.4.4 Activity: Implementation V&V Task 5 Component V&V test case generation Task 6 Integration V&V test case generation Task 7 System V&V test case generation Task 8 Acceptance V&V test case generation Task 9 Component V&V test procedure generation Task 10 Integration V&V test procedure generation Task 11 System V& test procedure generation 5.4.5 Activity: Test V&V Task 2 Acceptance V&V test procedure generation
6.5.2.2 Validate Test Traceability ^a	5.4.4 Activity: Implementation V&V Task 12 Component V&V test execution 5.4.5 Activity: Test V&V Task 3 Integration V&V test execution Task 4 System V&V test execution Task 5 Acceptance V&V test execution
6.5.2.3 Validate Test Conduct ^a	5.4.4 Activity: Implementation V&V Task 12 Component V&V test execution 5.4.5 Activity: Test V&V Task 3 Integration V&V test execution Task 4 System V&V test execution Task 5 Acceptance V&V test execution

ISO/IEC 12207 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause/table)
6.5.2.4 Validate Software for Intended Use^a	5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation 5.4.2 Activity: Requirements V&V Task 2 Software requirements evaluation Task 3 interface analysis 5.4.3 Activity: Design V&V Task 2 Software design evaluation Task 3 Interface analysis 5.4.4 Activity: Implementation V&V Task 2 Source code and source code documentation evaluation Task 3 Interface analysis 5.4.5 Activity: Test V&V Task 4 System V&V test execution Task 5 Acceptance V&V test execution
6.5.2.5 Installation Test of Software^a	5.4.6 Activity: Installation and checkout V&V Task 1 Installation configuration audit Task 2 Installation checkout Task 3 Hazard analysis Task 4 Security analysis Task 5 Risk analysis
Annex A Tailoring Process	N/A
Annex B Guidance on Tailoring	N/A
Annex C Guidance on Processes and Organizations	N/A
Annex D Bibliography	N/A
Annex E Basic Concepts of	N/A

ISO/IEC 12207	
Annex F Purpose and Outcomes	Clause 4 Software integrity level Clause 5 Software V&V processes Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline
Annex G	N/A
Annex H	N/A

^aNo ISO/IEC 12207:1995 [B13] clause title was listed. For purposes of this mapping, this standard assigned a clause title to reflect the clause contents.

A.2 Mapping of IEEE Std 1012 V&V activities to ISO/IEC 15288 12207 system software life cycle processes and activities

This standard defines 11 5 Common V&V activities and 9 System V&V activities, as shown in the first column of Table A.2, that are part of the V&V processes. These 11 14 V&V activities correspond to the ISO/IEC 12207:1995 [B13] software 15288:2008(E) [B11] system life cycle processes and activities shown in columns 2 and 3 of Table A.2.

Table A.2—Mapping of IEEE Std 1012 V&V activities to ISO/IEC 12207 15288

IEEE 1012 V&V activities (see corresponding clause)	ISO/IEC 15288 system life cycle	
	Process	Activity
7.1 Activity: V&V Management	N/A	N/A

7.2 Activity: Acquisition Support V&V	6.1.1 Acquisition Process	6.1.1.3.a Prepare for the acquisition 6.1.1.3.c Initiate an agreement 6.1.1.3.d Monitor the agreement 6.1.1.3.e Accept the product or service
7.3 Activity: Supply Planning V&V	6.1.2 Supply Process	6.1.2.3.c Initiate an agreement 6.1.2.3.d Execute the agreement 6.1.2.3.e Deliver and support the product or service
7.4 Activity: Project Planning V&V	6.3.1 Project Planning Process	6.3.1.3. a Define the project 6.3.1.3.b Plan the project resources
7.5 Activity: Configuration Management V&V	6.3.5 Configuration Management Process	6.3.5.3.a Plan configuration management 6.3.5.3.b Perform configuration management
8.1 Activity: Stakeholder Requirements Definition V&V	6.4.1 Stakeholder Requirements Definition Process	6.4.1.3.a Elicit stakeholder requirements 6.4.1.3.b Define stakeholder requirements 6.4.1.3. c Analyze and maintain stakeholder requirements
8.2 Activity: Requirements Analysis V&V	6.4.2 Requirements Analysis Process	6.4.2.3.a Define system requirements 6.4.2.3. Analyze and maintain system requirements
8.3 Activity: Architectural	6.4.3 Architectural Design	6.4.3.3.a Define the architecture

Design V&V	Process	6.4.3.3.b Analyze and evaluate the architecture 6.4.3.3.C Document and maintain the architecture
8.4 Activity: Implementation V&V	6.4.4 Implementation Process	6.4.4.3.a Plan the implementation 6.4.4.3.b Perform implementation
8.5 Activity: Integration V&V	6.4.5 Integration Process	6.4.5.3.a Plan integration 6.4.5.3.b Perform integration
8.6 Activity: Transition V&V	6.4.7 Transition Process	6.4.7.3.a Plan the transition 6.4.7.3.b Perform the transition
8.7 Activity: Operation V&V	6.4.9 Operation Process	6.4.9.3.a Prepare for operation 6.4.9.3.b Perform operational activation and check-out 6.4.9.3.d Perform operational problem resolution
8.8 Activity: Maintenance V&V	6.4.10 Maintenance Process	6.4.10.3.a Plan maintenance 6.4.10.3.b Perform maintenance
8.9 Activity: Disposal V&V	6.4.11 Disposal Process	6.4.11.3.a Plan disposal

IEEE Std 1012 V&V activities (see corresponding subclause)	ISO/IEC 12207 software life cycle	
	Process ^a	Activity
5.2.1 Activity: Acquisition support V&V	Acquisition	—Initiation —RFP (tender) preparation —Contract preparation and update —Supplier monitoring —Acceptance and completion

5.3.1 Activity: Planning V&V	Supply	—Initiation —Preparation of response —Contract —Planning —Execution and control —Review and evaluation —Delivery and completion
5.4.1 Activity: Concept V&V	Development	—Process implementation —Requirements elicitation —System requirements analysis —System architectural design
5.4.2 Activity: Requirements V&V	Development	—Software requirements analysis
5.4.3 Activity: Design V&V	Development	—Software architectural design —Software detailed design
5.4.4 Activity: Implementation V&V	Development	—Software coding and testing
5.4.5 Activity: Test V&V	Development	—Software integration —Software qualification testing —System integration —System qualification testing
5.4.6 Activity: Installation and checkout V&V	Development	—Software installation —Software acceptance support
5.5.1 Activity: Operation V&V	Operation	—Process implementation —Operational testing —System operation —User support
5.6.1 Activity: Maintenance V&V	Maintenance	—Process implementation —Problem and modification analysis —Modification implementation

		—Maintenance review/acceptance —Migration —Software retirement
5.1.1 Activity: Management of V&V	All processes	All activities

^aAnnex F contains description, purposes, and outcomes of the ISO/IEC process model.

A.3 Mapping of ~~IEEE Std 1074~~ [ISO/IEC 12207](#) V&V requirements to IEEE Std 1012 V&V activities and tasks

Table A.3 shows a mapping of all ~~IEEE Std 1074-1997 [B10]~~ [ISO/IEC 12207:2008 \[B11\]](#) V&V requirements (i.e., processes, activities, and tasks) to the V&V activities and tasks of this standard.

[The first column of Table A.3 lists the clause numbers and titles of V&V processes and activities from ISO/IEC 12207:2008 \[B11\]. The second column of Table A.3 lists the clauses and tables from IEEE Std 1012 that address the topics listed in the first column.](#)

Table A.3—Mapping ~~IEEE Std 1074~~ [ISO/IEC 12207](#) V&V requirements to IEEE 1012 V&V activities and tasks

ISO/IEC 12207 V&V requirements	IEEE 1012 V&V activities and tasks (see corresponding clause/table)
6.1.1.3.5 Agreement monitoring	7.2 Activity: Acquisition Support V&V Task 1 Scoping the V&V Effort Task 2 Planning the Interface between the V&V Effort and Supplier Task 3 System Requirements Review Task 4 Acceptance Support
6.1.2.3.4.5(h) and 6.1.2.3.4.10 Interfacing with V&V^a	7.2 Activity: Acquisition Support V&V Task 2 Planning the Interface between the V&V Effort and Supplier 7.3 Activity: Supply Planning V&V Task 2 Planning the Interface between the

	V&V Effort and Supplier Annex C Definition of independent V&V (IV&V)
6.1.2.3.4.14 Verification and Validation⁶	Clause 1 through Clause 7, Clause 9, Clause 11, Clause 12, Table 1a, Table 1c, Table 2a, Table 2c, Table 3a, and Table 3c, Figure 1a, Figure 1c, Figure 2a, and Figure 2c, all annexes
7.2.4.3.1 [Software Verification] Process implementation	5. Integrity Levels 7.1 Activity: V&V Management Task 1 VVP Generation 9.1 Activity: Software Concept V&V Task 2 Criticality Analysis 9.2 Activity: Software Requirements V&V Task 4 Criticality Analysis 9.3 Activity: Software Design V&V Task 4 Criticality Analysis 9.4 Activity: Software Construction V&V Task 4 Criticality Analysis 9.10 Activity: Software Maintenance V&V Task 3 Criticality Analysis Annex C
7.2.4.3.2.1 Requirements verification	9.2 Activity: Software Requirements V&V
7.2.4.3.2.2 Design verification	9.3 Activity: Software Design V&V
7.2.4.3.2.3 Code verification	9.4 Activity: Software Construction V&V

7.2.4.3.2.4 Integration verification	9.3 Activity: Software Design V&V Task 6 Software Integration Test Plan V&V Task 8 Software Integration Test Design V&V 9.4 Activity: Software Construction V&V Task 6 Software Integration Test Case V&V 9.5 Activity: Software Integration Test V&V
7.2.4.3.2.5 Documentation verification	7.5 Activity: Configuration Management V&V
7.2.5.3.1 [Software Validation] Process implementation	5. Integrity Levels 7.1 Activity: V&V Management Task 1 VVP Generation
	9.1 Activity: Software Concept V&V Task 2 Criticality Analysis 9.2 Activity: Software Requirements V&V Task 4 Criticality Analysis 9.3 Activity: Software Design V&V Task 4 Criticality Analysis 9.4 Activity: Software Construction V&V Task 4 Criticality Analysis 9.10 Activity: Software Maintenance V&V Task 3 Criticality Analysis Annex C

7.2.5.3.2 Validation	9.2 Activity: Software Requirements V&V Task 5 Software Qualification Test Plan V&V 9.3 Activity: Software Design V&V Task 5 Software Component Test Plan V&V Task 7 Software Component Test Design V&V Task 9 Software Qualification Test Design V&V 9.4 Activity: Software Construction V&V Task 5 Software Component Test Case V&V Task 7 Software Qualification Test Case V&V Task 9 Software Component Test Procedure V&V Task 11 Software Qualification Test Procedure V&V Task 12 Software Component Test Execution V&V 9.6 Activity: Software Qualification Test V&V
7.2.6.3.2 Project Management Reviews	7.1 Activity: V&V Management Task 5 Management and Technical Review Support

^a [No ISO/IEC 12207:2008 \[B9\] clause title was listed. For purposes of this mapping, this standard assigned a clause title to reflect the clause contents.](#)

^b [See footnote a.](#)

IEEE Std 1074 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
A.1 Project management activity groups A.1.1 Project initiation activities A.1.1.1 Create SLCP A.1.1.2 Perform estimations A.1.1.3 Allocate project resources A.1.1.4 Define metrics	5.2.1 Activity: Acquisition support V&V Task 1 Scoping the V&V effort Task 2 Planning the Interface between the V&V effort and supplier 5.3.1 Activity: Planning V&V Task 1 Planning the interface between the V&V effort and supplier 5.1.1 Activity: Management of V&V Task 1 SVVP Generation 5.6.1 Activity: Maintenance V&V Task 1 SVVP Revision 6.1 V&V reporting requirements 7.6 SVVP section 6: V&V reporting requirements 6.2 Administrative V&V requirements 7.7 SVVP section 7: V&V administrative requirements 6.3.1 V&V Test documentation 7.8 SVVP section 8: V&V test documentation

	requirements 6.3.2 SVVP documentation Clause 7 SVVP outline Annex E V&V measures
A.1.2 Project planning activities A.1.2.1 Plan evaluations A.1.2.2 Plan configuration management A.1.2.3 Plan system transition (if applicable) A.1.2.4 Plan installation A.1.2.5 Plan documentation A.1.2.6 Plan training A.1.2.7 Plan project management A.1.2.8 Plan integration	Clause 5 Software V&V processes All tasks Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline

IEEE Std 1074 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
A.1.3 Project monitoring and control activities A.1.3.1 manage risks A.1.3.2 Manage the project A.1.3.3 Identify SLCP improvement needs A.1.3.4 Retain records A.1.3.5 Collect and analyze metric data	5.1.1 Activity: Management of V&V Task 1 SVVP generation Task 2 Proposed/baseline change assessment Task 3 Management review of the V&V effort Task 4 Management and technical review support Task 5 Interface with organizational and supporting processes

A.2 Pre-development activity groups A.2.1 Concept exploration activities A.2.1.1 Identify ideas or needs A.2.1.2 Formulate potential approaches A.2.1.3 Conduct feasibility studies A.2.1.4 Refine and finalize the idea or need A.2.2 System allocation activities A.2.2.1 Analyze functions A.2.2.2 Develop system architecture A.2.2.3 Decompose system requirements A.2.3 Software importation activities A.2.3.1 Identify imported software requirements A.2.3.2 Evaluate software import sources (if applicable) A.2.3.3 Define software import method (if applicable) A.2.3.4 Import software (if applicable)	5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation Task 2 Criticality analysis Task 4 Traceability analysis Task 5 Hazard analysis Task 6 Security analysis Task 7 Risk analysis 5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation Task 3 Hardware/software/user requirements allocation analysis Clause 5 Software V&V Processes All tasks Annex D V&V of reuse software
A.3 Development activity groups A.3.1 Requirements activities A.3.1.1 Define and develop software requirements A.3.1.2 Define interface requirements A.3.1.3 Prioritize and integrate software requirements	5.4.2 Activity: Requirements V&V Task 1 Traceability analysis Task 2 Software requirements evaluation Task 3 Interface analysis Task 4 Criticality analysis Task 5 System V&V test plan generation Task 6 Acceptance V&V test plan generation

	Task 7 Configuration management assessment Task 8 Hazard analysis Task 9 Security analysis Task 10 Risk analysis
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IEEE Std 1074 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
A.3 Development activity groups (continued) A.3.2 Design activities A.3.2.1 Perform architectural design A.3.2.2 Design database (if applicable) A.3.2.3 Design interfaces A.3.2.4 Perform detailed design A.3.3 Implementation activities A.3.3.1 Create executable code A.3.3.2 Create operating documentation A.3.3.3 Perform integration	5.4.3 Activity: Design V&V Task 1 Traceability analysis Task 2 Software design evaluation Task 3 Interface analysis Task 4 Criticality analysis Task 5 Component V&V test plan generation Task 6 Integration V&V test plan generation Task 7 Component V&V test design generation Task 8 Integration V&V test design generation Task 9 System V&V test design generation Task 10 Acceptance V&V test design generation Task 11 Hazard analysis Task 12 Security analysis Task 13 Risk analysis 5.4.4 Activity: Implementation V&V Task 1 Traceability analysis Task 2 Source code and source code documentation evaluation Task 3 Interface analysis

	Task 4 Criticality analysis Task 5 Component V&V test case generation Task 6 Integration V&V test case generation Task 7 System V&V test case generation Task 8 Acceptance V&V test case generation Task 9 Component V&V test procedure generation Task 10 Integration V&V test procedure generation Task 11 System V&V test procedure generation Task 12 Component V&V test execution Task 13 Hazard analysis Task 14 Security analysis Task 15 Risk analysis 5.4.5 Activity: Test V&V Task 1 Traceability analysis Task 2 Acceptance V&V test procedure generation Task 3 Integration V&V test execution Task 4 System V&V test execution Task 5 Acceptance V&V test execution Task 6 Hazard analysis Task 7 Security analysis Task 8 Risk analysis
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IEEE Std 1074 V&V requirements	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
A.4 Post development activity groups	5.4.6 Activity: Installation and checkout V&V
A.4.1 Installation activities	Task 1 Installation configuration audit
A.4.1.1 Distribute software	Task 2 Installation checkout
A.4.1.2 Install software	Task 3 Hazard analysis
A.4.1.3 Accept software in operational environment	Task 4 Security analysis
	Task 5 Risk analysis
	Task 6 V&V final report generation
A.4.2 Operation and support activities	5.5.1 Activity: Operation V&V
A.4.2.1 Operate the system	Task 1 Evaluation of new constraints
A.4.2.2 Provide technical assistance and consulting	Task 2 Operating procedures evaluation
A.4.2.3 Maintain support request log	Task 3 Hazard analysis
	Task 4 Security analysis
	Task 5 Risk analysis
A.4.3 Maintenance activities	5.6.1 Activity: Maintenance V&V
A.4.3.1 Identify software improvement needs	Task 1 SVVP revision
A.4.3.2 Implement problem reporting method	Task 2 Anomaly evaluation
A.4.3.3 Reapply SLC	Task 3 Criticality analysis
	Task 4 Migration assessment
	Task 5 Retirement assessment
	Task 6 Hazard analysis
	Task 7 Security analysis
	Task 8 Risk analysis
	Task 9 Task iteration
A.4.4 Retirement activities	5.6.1 Activity: Maintenance V&V

A.4.4.1 Notify user A.4.4.2 Conduct parallel operations (if applicable) A.4.4.3 Retire system	Task 1 SVVP revision Task 2 Anomaly evaluation Task 3 Criticality analysis Task 4 Migration assessment Task 5 Retirement assessment Task 6 Hazard analysis Task 7 Security analysis Task 8 Risk analysis Task 9 Task iteration
A.5 Integral activity groups A.5.1 Evaluation activities A.5.1.1 Conduct reviews A.5.1.2 Create traceability matrix A.5.1.3 Conduct audits A.5.1.4 Develop test procedures A.5.1.5 Create test data A.5.1.6 Execute tests A.5.1.7 Report evaluation results	Clause 5 Software V&V processes All tasks Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline
A.5 Integral activity groups A.5.1 Evaluation activities A.5.1.1 Conduct reviews A.5.1.2 Create traceability matrix A.5.1.3 Conduct audits A.5.1.4 Develop test procedures A.5.1.5 Create test data A.5.1.6 Execute tests A.5.1.7 Report evaluation results	Clause 5 Software V&V processes All tasks Clause 6 Software V&V reporting, administrative, and documentation requirements Clause 7 SVVP outline

A.4 Mapping ~~between CMMI of IEEE 1012 V&V tasks activities~~ and to IEEE Std 2207 software life cycle processes and activities

This standard defines 5 Common V&V activities and 11 Software V&V activities, as shown in the first column of Table A.4, which are part of the V&V processes. These 16 V&V activities correspond to the IEEE 12207 software life cycle processes and activities shown in columns 2 and 3 of Table A.4.

~~The following tables provide a mapping between the Software Engineering Process Groups CMMI process areas and the IEEE Std 1012 software V&V tasks.~~

Table A.4—Mapping IEEE Std 1012 CMMI process mapping matrix for requirements V&V activities to ISO/IEC 12207:2008 [B11]

<u>IEEE 1012 V&V activities</u> (see corresponding clause)	<u>IEEE 12207 software life cycle</u>	
	<u>Process</u>	<u>Activity</u>
<u>7.1 Activity: V&V Management</u>	<u>N/A</u>	<u>N/A</u>
<u>7.2 Activity: Acquisition Support V&V</u>	<u>6.1.1 Acquisition Process</u>	<u>6.1.1.3.1 Acquisition preparation</u> <u>6.1.1.3.4 Contract agreement</u> <u>6.1.1.3.5 Agreement monitoring</u> <u>6.1.1.3.6 Acquirer acceptance</u>
<u>7.3 Activity: Supply Planning V&V</u>	<u>6.1.2 Supply Process</u>	<u>6.1.2.3.3 Contract agreement.</u> <u>6.1.2.3.4 Contract execution</u> <u>6.1.2.3.5 Product/service delivery and support</u>
<u>7.4 Activity: Project Planning V&V</u>	<u>6.3.1 Project Planning Process</u>	<u>6.3.1.3.2 Project planning</u>
<u>7.5 Activity: Configuration Management V&V</u>	<u>6.3.5 Configuration Management Process</u>	<u>6.3.5.3.1 Configuration management planning</u> <u>6.3.5.3.2 Configuration management execution</u>
<u>9.1 Activity: Software Concept V&V</u>	<u>6.4.2 System Requirements Analysis Process</u> <u>6.4.3 System Architectural Design Process</u>	<u>6.4.2.3.2 Requirements evaluation</u> <u>6.4.3.3.2 Architectural evaluation</u>
<u>9.2 Activity: Software Requirements V&V</u>	<u>7.1.2 Software Requirements Analysis Process</u>	<u>7.1.2.3.1 Software requirements analysis</u>
<u>9.3 Activity: Software Design</u>	<u>7.1.3 Software Architectural</u>	<u>7.1.3.3.1 Software architectural</u>

V&V	Design Process 7.1.4 Software Detailed Design Process	design 7.1.4.3.1 Software detailed design
9.4 Activity: Software Construction V&V	7.1.5 Software Construction Process	7.1.5.3.1 Software construction
9.5 Activity: Software Integration Test V&V	7.1.6 Software Integration Process	7.1.6.3.1 Software integration
9.6 Activity: Software Qualification Test V&V	7.1.7 Software Qualification Testing Process	7.1.7.3.1 Software qualification testing
9.7 Activity: Software Acceptance Test V&V	6.4.8 Software Acceptance Support Process	6.4.8.3.1 Software acceptance support
9.8 Activity: Software Installation and Checkout V&V	6.4.7 Software Installation Process	6.4.7.3.1 Software installation
9.9 Activity: Software Operation V&V	6.4.9 Software Operation Process	6.4.9.3.1 Preparation for operation 6.4.9.3.2 Operation activation and check-out 6.4.9.3.3 Operational use
9.10 Activity: Software Maintenance V&V	6.4.10 Software Maintenance Process	6.4.10.3.1 Process implementation 6.4.10.3.2 Problem and modification analysis 6.4.10.3.3 Modification implementation 6.4.10.3.4 Maintenance review/acceptance 6.4.10.3.5 Migration
9.11 Activity: Software Disposal V&V	6.4.11 Software Disposal Process	6.4.11.3.1 Software disposal planning

CMMI process groups and areas	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
Process management	
Organizational process focus	5.1.1 Activity: Management of the V&V effort Task 3 Management review of the V&V effort
Organizational process definition	5.1.1 Activity: Management of the V&V effort Task 5 Interface with Organizational and Supporting Processes
Organizational training	N/A
Organizational process performance	5.1.1 Activity: Management of the V&V effort Task 3 Management review of the V&V effort
Organizational innovation & development	5.1.1 Activity: Management of the V&V effort Task 6 Identify improvement opportunities in the conduct of V&V
Project management	
Project planning	5.1.1 Activity: Management of the V&V effort Task 1 SVVP Generation Task 2 Proposed/baseline change assessment 5.2.1 Activity: Acquisition support V&V Task 1 Scoping the V&V Effort Task 2 Planning the interface between V&V effort and supplier Task 3 System requirements review 5.3.1 Activity: Planning V&V Task 1 Planning the interface between V&V effort and supplier 5.4.1 Activity: Concept V&V

	Task 3 Hardware/software/user requirements allocation analysis 5.4.2 Activity: Requirements V&V Task 5 System V&V test plan generation Task 6 Acceptance V&V test plan generation 5.4.3 Activity: Design V&V Task 5 Component V&V test plan generation Task 6 Integration V&V test plan generation Task 7 Component V&V test design generation Task 8 Integration V&V test design generation Task 9 System V&V test design generation Task 10 Acceptance V&V test design generation
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CMMI process groups and areas	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
Quantitative project management	5.1.1 Activity: Management of the V&V effort Task 6 Identify improvement opportunities in the conduct of V&V Annex E V&V measures
Engineering	
Requirements development	5.1.1 Activity: Management of the V&V effort Task 2 Proposed/baseline change assessment 5.4.2 Activity: Requirements V&V Task 2 Software Requirements Evaluation

Requirements management	5.1.1 Activity: Management of the V&V effort Task 4 Management and technical review support Task 5 Interface with organizational and supporting processes 5.4.1 Activity: Concept V&V Task 3 Hardware/software/user requirements allocation analysis Task 4 Traceability analysis 5.4.2 Activity: Requirements V&V Task 1 Traceability Analysis
Technical solution	5.4.1 Activity: Concept V&V Task 1 Concept documentation evaluation Task 3 Hardware/software/user requirements allocation analysis Task 5 Hazard analysis Task 6 Security analysis 5.4.2 Activity: Requirements V&V Task 2 Software requirements evaluation Task 8 Hazard analysis Task 9 Security analysis 5.4.3 Activity: Design V&V Task 2 Software design evaluation Task 3 Interface analysis Task 11 Hazard analysis Task 12 Security analysis 5.4.4 Activity: Implementation V&V Task 2 Source code and source code documentation evaluation Task 3 Interface analysis Task 13 Hazard analysis Task 14 Security analysis

CMMI process groups and areas	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
Product integration	5.4.2 Activity: Requirements V&V Task 1 Traceability analysis Task 3 Interface analysis Task 5 System V&V test plan generation Task 6 Acceptance V&V test plan generation 5.4.3 Activity: Design V&V Task 1 Traceability analysis Task 3 Interface analysis Task 5 Component V&V test plan generation Task 6 Integration V&V test plan generation Task 7 Component V&V test design generation Task 8 Integration V&V test design generation Task 9 System V&V test design generation Task 10 Acceptance V&V test design generation 5.4.4 Activity: Implementation V&V Task 1 Traceability analysis Task 3 Interface analysis Task 5 Component V&V test case generation Task 6 Integration V&V test case generation Task 7 System V&V test case generation Task 8 Acceptance V&V test case generation Task 9 Component V&V test procedure generation Task 10 Integration V&V test procedure generation Task 11 System V&V test procedure generation Task 12 Component V&V test execution

	5.4.5 Activity: Test V&V Task 1 Traceability analysis Task 2 Acceptance V&V test procedure generation Task 3 Integration V&V test execution Task 4 System V&V test execution Task 5 Acceptance V&V test execution 5.4.6 Activity: Installation and checkout V&V Task 1 Installation configuration audit Task 2 Installation checkout
Verification	All clauses
Validation	All clauses

CMMI process groups and areas	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
Support	
Configuration management	5.1.1 Activity: Management of the V&V effort Task 2 Proposed/baseline change assessment Task 6 Identify improvement opportunities in the conduct of V&V 5.4.1 Activity: Concept V&V Task 4 Traceability Analysis 5.4.2 Activity: Requirements V&V Task 1 Traceability analysis Task 7 Configuration management assessment

	5.4.3 Activity: Design V&V Task 1 Traceability analysis 5.4.4 Activity: Implementation V&V Task 1 Traceability analysis 5.4.5 Activity: Test V&V Task 1 Traceability analysis 5.4.6 Activity: Installation & Checkout V&V Task 1 Installation configuration audit
Process & product quality assurance	5.1.1 Activity: Management of the V&V effort Task 6 Identify improvement opportunities in the conduct of V&V Annex E V&V measures Annex F Relationship of V&V to other project responsibilities
Organizational environment for integration	Annex C Definition of independent V&V Annex F Relationship of V&V to other project responsibilities

CMMI process groups and areas	IEEE Std 1012 V&V activities and tasks (see corresponding clause)
Decision analysis & resolution	5.1.1 Activity: Management of the V&V effort Task 2 Proposed/baseline change assessment 5.4.1 Activity: Concept V&V Task 7 Risk analysis 5.4.2 Activity: Requirements V&V Task 10 Risk analysis 5.4.3 Activity: Design V&V Task 13 Risk analysis 5.4.4 Activity: Implementation V&V Task 15 Risk analysis 5.4.5 Activity: Test V&V Task 8 Risk analysis 5.4.6 Activity: Installation and checkout V&V Task 5 Risk analysis 5.5.1 Activity: Operation V&V Task 5 Risk analysis 5.6.1 Activity: Maintenance V&V Task 8 Risk analysis
Causal analysis & resolution	5.1.1 Activity: Management of the V&V effort Task 6 Identify improvement opportunities in the conduct of V&V Annex E V&V measures Annex F Relationship of V&V to other project responsibilities

Annex B

(informative)

A risk-based ~~software~~, integrity-level scheme

~~B.1 A risk-based software integrity level scheme~~

Table B. 1 defines four ~~software~~ integrity levels used for reference purposes by this standard. Table B.2 describes the consequences of ~~software~~ errors for each of the four ~~software~~ integrity levels. There are overlaps between the ~~software~~ integrity levels to allow for individual interpretations of acceptable risk depending on the application.

Table B.1— Assignment of ~~software~~ integrity levels

Software Integrity level	Description
4	An error to a function or system feature that causes the following : — Catastrophic consequences to the system with reasonable, probable, or occasional likelihood of occurrence of an operating state that contributes to the error - or — Critical consequences with reasonable or probable likelihood of occurrence of an operating state that contributes to the error.
3	An error to a function or system feature that causes the following : — Catastrophic consequences with occasional or infrequent likelihood of

	occurrence of an operating state that contributes to the error or – Critical consequences with probable or occasional likelihood of occurrence of an operating state that contributes to the error or – Marginal consequences with reasonable or probable likelihood of occurrence of an operating state that contributes to the error:
2	An error to a function or system feature that causes the following: – Critical consequences with infrequent likelihood of occurrence of an operating state that contributes to the error ST. – Marginal consequences with probable or occasional likelihood of occurrence of an operating state that contributes to the error Or – Negligible consequences with reasonable or probable likelihood of occurrence of an operating state that contributes to the error
1	An error to a function or system feature that causes the following: – Critical consequences with infrequent likelihood of occurrence of an operating state that contributes to the error or – Marginal consequences with occasional or infrequent occurrence of an operating state that contributes to the error or – Negligible consequences with probable, occasional, or infrequent likelihood of occurrence of an operating state that contributes to the error:

Table B.2—Definition of consequences

Consequence	Definition
Catastrophic	Loss of human life, complete mission failure, loss of system security and safety, or extensive financial or social loss.
Critical	Major and permanent injury, partial loss of mission, major system damage, or major financial or social loss.
Marginal	Severe injury or illness, degradation of secondary mission, or some financial or social loss.
Negligible	Minor injury or illness, minor impact on system performance, or operator inconvenience.

Table B.3 illustrates the risk-based scheme shown in Table B.1 and Table B.2. Each cell in the table assigns an ~~software~~ integrity level based ~~on~~ ~~upon~~ the combination of an error consequence and the likelihood of occurrence of an operating state that contributes to the error. Some table cells reflect more than one ~~software~~ integrity level, indicating that the final assignment of the ~~software~~ integrity level can be selected to address the system application and risk mitigation recommendations. For some industry applications, the definition of likelihood of occurrence categories may be expressed as probability figures derived by analysis or from system requirements.

Table B.3—Graphic illustration of the assignment of ~~software~~-integrity levels

Error	Likelihood of occurrence of an operating state that contributes to the error (decreasing order of likelihood)			
Consequence	Reasonable	Probable	Occasional	Infrequent
Catastrophic	4	4	4 or 3	3
Critical	4	4 or 3	3	2 or 1
Marginal	3	3 or 2	2 or 1	1
Negligible	2	2 or 1	1	1

Annex C

(informative)

Definition of independent V&V (IV&V)

IV&V is defined by three parameters: technical independence, managerial independence, and financial independence.

C.1 Technical independence

Technical independence requires the V&V effort to use ~~utilize~~ personnel who are not involved in the development of the system or its elements, ~~software~~. The IV&V effort should formulate its own understanding of the problem and how the proposed system is solving the problem. Technical independence ("fresh viewpoint") is an important method to detect subtle errors overlooked by those too close to the solution.

For system ~~software~~ tools, technical independence means that the IV&V effort uses or develops its own set of test and analysis tools separate from the developer's tools. Sharing of tools is allowable for computer support environments (e.g., compilers, assemblers, and utilities) or for system simulations where an independent version would be too costly. For shared tools, IV&V conducts qualification tests on tools to assure ~~ensure~~ that the common tools do not contain errors that may mask errors in the system ~~software~~ being analyzed and tested. Off-the-shelf tools that have extensive history of use do not require qualification testing. The most important aspect for the use of these tools is to verify the input data used.

C.2 Managerial independence

This requires that the responsibility for the IV&V effort be vested in an organization separate from the development and program management organizations. Managerial independence also means that the IV&V effort independently selects the segments of the software, [hardware](#) and system to analyze and test, chooses the IV&V techniques, defines the schedule of IV&V activities, and selects the specific technical issues and problems to act [on](#). ~~upon~~. The IV&V effort provides its findings in a timely fashion simultaneously to both the development and program management organizations. The IV&V effort [is](#) ~~must be~~ allowed to submit to program management the IV&V results, anomalies, and findings without any restrictions (e.g., without requiring prior approval from the development group) or adverse pressures, direct or indirect, from the development group.

C.3 Financial independence

This requires that control of the IV&V budget be vested in an organization independent of the development organization. This independence prevents situations where the IV&V effort cannot complete its analysis or test or deliver timely results because funds have been diverted or adverse financial pressures or influences have been exerted.

C.4 Forms of independence

The extent to which each of the three independence parameters (technical, managerial, [and](#) financial) is vested in a V&V organization determines the degree of independence achieved.

Many forms of independence can be adopted for a V&V organization. The five most prevalent are [as follows](#): 1) classical, 2) modified, 3) integrated, 4) internal, and 5) embedded. Table C.I illustrates the degree of independence achieved by these five forms.

Table C.1—Forms of IV&V

IV&V form	Technical	Management	Financial
Classical	I	I	I
Modified	I	i	I
Integrated	i	I	I
Internal	i	i	i
Embedded	e	e	e
NOTE—I = rigorous independence; i = conditional independence; e = minimal independence.			

C.4.1 Classical IV&V

Classical IV&V embodies all three independence parameters. The IV&V responsibility is vested in an organization that is separate from the development organization. [The IV&V organization establishes](#) ~~uses~~ a close working relationship with the development organization to [assure](#) ~~ensure~~ that IV&V findings and recommendations are integrated rapidly back into the development process. Typically, classical IV&V is performed by one organization (e.g., supplier) and the development is performed by a separate organization (i.e., another vendor). Classical IV&V is generally required for ~~software~~-integrity level 4 (i.e., loss of life, loss of mission, significant social [loss](#), or financial loss) through regulations and standards imposed on the system development.

C.4.2 Modified IV&V

Modified IV&V is used in many large programs where the system prime integrator is selected to manage the entire system development including the IV&V. The prime integrator selects organizations to assist in the development of the system and to perform the IV&V. In the modified IV&V form, the acquirer reduces its own acquisition time by passing this responsibility to the prime integrator. Because ~~Since~~ the prime integrator performs all or some of the development, the managerial independence is compromised by having the IV&V effort report to the prime integrator. Technical independence is preserved because ~~since~~ the IV&V effort formulates an unbiased opinion of the system solution and uses an independent staff to perform the IV&V. Financial independence is preserved because ~~since~~ a separate budget is set aside for the IV&V effort. Modified IV&V effort would be appropriate for systems with ~~software~~ integrity level 3 (i.e., an important mission and purpose).

C.4.3 Integrated IV&V

This form is focused on providing rapid feedback of V&V results into the development process and is performed by an organization that is financially and managerially independent of the development organization to minimize compromises with respect to independence. The rapid feedback of V&V results into the development process is facilitated by the integrated IV&V organization: working side by side with the development organization, reviewing interim work products, and providing V&V feedback during inspections, walkthroughs, and reviews conducted by the development staff (potential impact on technical independence). Impacts to technical independence are counterbalanced by benefits associated with a focus on interdependence between the integrated IV&V organization and the development organization. Interdependence means that the successes of the organizations are closely coupled, ensuring that they work together in a cooperative fashion.

C.4.4 Internal IV&V

Internal IV&V exists when the developer conducts the IV&V with personnel from within its own organization, although preferably not the same personnel involved directly in the development effort. Technical, managerial, and financial independence are compromised. Technical independence is compromised because the IV&V analysis and test is vulnerable to overlooking errors by using the same assumptions or development environment that masked the error from the developers. Managerial independence is compromised because the internal IV&V effort uses the same common tools and corporate analysis procedures as the development group. Peer pressure from the development group may adversely influence how aggressively the [software system](#) is analyzed and tested by the IV&V effort. Financial independence is compromised because the development group controls the IV&V budget. IV&V funds, resources, and schedules may be reduced as development pressures and needs redirect the IV&V funds into solving development problems. The benefit of an internal IV&V effort is access to staff who know the system and its software. This form of IV&V is used when the degree of independence is not explicitly stated and the benefits of preexisting staff knowledge outweigh the benefits of objectivity.

C.4.5 Embedded V&V

This form is similar to internal IV&V in that it uses personnel from the development organization who should not be involved directly in the development effort. Embedded V&V is focused on ensuring conformance to the development procedures and processes. The embedded V&V organization works side by side with the development organization and attends the same inspections, walkthroughs, and reviews as the development staff (i.e., compromise of technical independence). Embedded V&V is not tasked specifically to assess independently the original solution or conduct independent tests (i.e., compromise of managerial independence). Financial independence is compromised because the V&V staff resource assignments are controlled by the development group. Embedded V&V allows rapid feedback of V&V results into the development process but compromises the technical, managerial, and financial independence of the V&V organization.

Annex D

(informative)

V&V of reuse software

D.1 Purpose

The purpose of this annex is to provide options and suggestions to ~~help the~~ V&V ~~effort~~ [aid](#) verification and validation of reuse software ~~and to overcome the particular challenges associated with reuse software~~. Reuse software can take many forms and could include software from software libraries, custom software developed for other applications, COTS software, software requirements, software designs, or other artifacts from existing software. This annex addresses both 1) reuse software developed and used as part of a reuse process, and 2) reuse software developed and used outside of a reuse process. Figure D.I illustrates V&V activities and tasks for reuse software whether it was developed under a reuse process or outside of a reuse process.

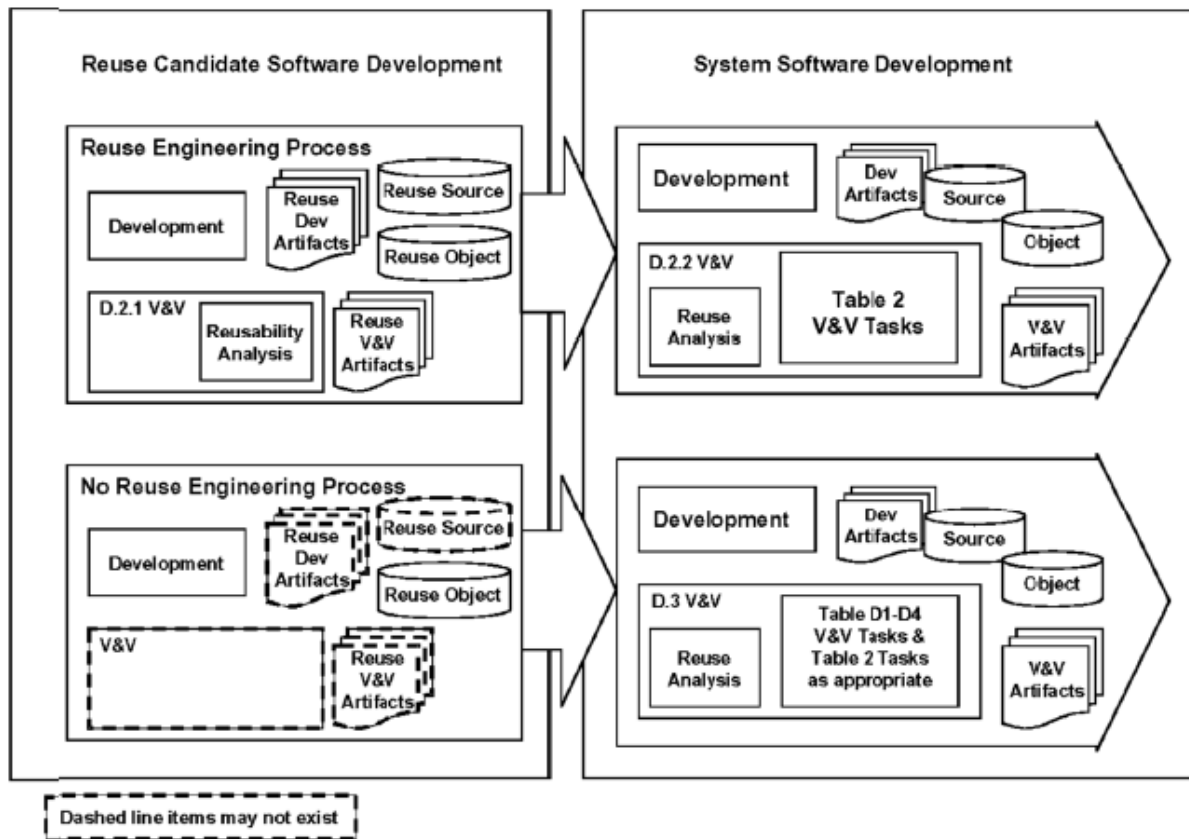


Figure D.1— V&V of reuse software

D.2 V&V of software developed in a reuse process

A structured software reuse process develops assets (e.g., design, code, and documentation) intended for use in multiple contexts. The software reuse processes of IEEE Std 1517- ~~1999 [B11]~~ [2010 \[BIO\]](#) provide a framework for extending the software life cycle processes of ~~IEEE/EIA Std ISO/TEC 12207-0 [B12]~~ [:2008 1996 \[B11\]](#) to include a systematic domain engineering process for software reuse. The domain scope and the domain analysis of an asset provide [the](#) requirements, [the](#) intended use, [the](#) interface parameters, and other information necessary for V&V of the asset, or an understanding of previously performed V&V of the asset.

D.2.1 V&V of assets in development

The V&V effort should analyze the artifacts (e.g., plans, models, [and](#) architecture) of the domain engineering as part of the required V&V tasks. Significant analysis of the domain engineering products should occur during system requirements review, software requirements evaluation, interface analysis, software design evaluation, source code and source code documentation evaluation, and all test planning. The V&V effort ~~must~~ includes the assignment of an ~~software~~ integrity level to the asset, in the context of the domain for its intended use, to determine the minimum V&V tasks to be performed (see Table 2). When planning the V&V effort, the optional task reusability analysis (see Annex G) should be included.

D.2.2 V&V of reused assets

A domain engineering process assures that the information used in developing software systems is identified, captured, and organized so that it can be reused to create new systems within a domain. The V&V effort **must** include the assignment of **software** an integrity level to the asset, in the context of its actual use, to determine the minimum V&V tasks to be performed (see Table 2). When planning the V&V effort, the optional task reuse analysis (see Annex G) should be included.

D.3 V&V of software developed and reused outside of a reuse process

Some software systems are developed, operated, and maintained using software items that were not designed for use in multiple contexts or were not developed as part of a structured software reuse process (e.g., domain engineering products are not available). In these cases, the V&V effort should perform the optional task reuse analysis (see Annex G) to produce inputs for determining the suitability of the reuse candidate software. The V&V effort **must** assign an **software** integrity level to the reuse candidate software to determine the minimum V&V tasks to be performed.

Reused software requires special consideration during the V&V effort when any one of the following is applicable:

- The inputs for a required V&V task are not available for the reused software.
- The reused software was developed as part of a system that is different in function or application from the system where it will be reused.

- The reused software was developed to meet different user needs from the current system.
- The original user needs are unknown.

In some cases, inputs for the V&V tasks may not be available, reducing visibility into the software products and processes. Options and techniques are available to compensate for the lack of inputs. Each technique has varying strengths and weaknesses—consideration should be given to performing multiple techniques to counter the weaknesses of one technique with strengths of another technique when high confidence is demanded. These options are addressed in decreasing order of desirability.

First, substitute tasks. Substitution for Table 1a through Table 1d V&V tasks is permitted if equivalent substitute V&V tasks can be shown to satisfy the same criteria as in Table 1. Two substitution task techniques are suggested in Table D. 1.

Table D.1— Substitution tasks to establish V&V task inputs

Substitution tasks Description: Substitute alternative analysis and test methods in lieu of the IEEE Std 1012 requirement V&V tasks to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.	
Technique 1: Black box testing <i>Black box testing and validation:</i> Execute the reused software against a spectrum of test case inputs and validate the correctness of the output. <i>User's manual analysis:</i> Derive system and software requirements from the user's manual and validate that the black box testing results satisfy the requirements. <i>Limit checks in interfacing software:</i> Add limit checks within all interface software on all data and logical information received from the reused software to assure that no erroneous information is accepted.	Pros — Test results reflect actual target software — Limits catastrophic errors from propagating into interfacing systems — Ability to check the major system and user requirements derived from user's manual — Independent analysis Cons — Inability to detect all test errors if presence of error not observable in black box outputs (e.g., latent errors) — Limit checks difficult to cover all execution scenarios — Limited by the thoroughness of the user's manual

Technique 2: Review developer's <u>quality assurance</u> (QA) <i>Developer's QA results:</i> Review developer's QA results and confirm the evidence of data similar to those that would be generated from V&V tasks. <i>Developer's test results:</i> Review the developer's test results and confirm the evidence of data similar to those that would be generated from V&V tasks. <i>Review of developer's notebook:</i> Review the developer's notebook to derive additional insights and problems with the software during early stages of development. <i>User's manual analysis:</i> Derive system and software requirements from the user's manual and validate that the developer's QA and test results satisfy the requirements.	Pros — Ability to derive insight into the details of the software design and internal performance characteristics — Identification of possible program error characteristics warranting further analysis and testing by other methods — Observations of program performance using test results reflecting actual software execution characteristics Cons — Limited by the scope and extent of the QA analysis and the specific focus of the testing performed — Limited by the thoroughness of the user's manual — Not a totally independent analysis
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Second, use alternative sources of information to perform the V&V tasks in Table 1a through Table 1d and Table 2a through Table 2d of this standard. Three alternative source techniques are suggested in Table D.2.

Table D.2—Substitution tasks to establish V&V task inputs

Alternative sources Description: Use alternative sources of program data to derive conclusions about the correctness, completeness, accuracy, and usability of the reused software.	
Technique 3: Operational history <i>Historical data analysis:</i> Examine and analyze the operational history of the reused software with particular attention to how the software performed in a system with similar characteristics to the new system being proposed. <i>User interviews:</i> Conduct interviews with operational users. Focus data gathering on how the system performed in scenarios and conditions similar to those expected in the new system being proposed.	Pros — Real data of the reused software in an operational environment. — User observations about the performance of the software and its related system. — Software burn-in established and track record of discrepancies recorded. — Independent analysis.

	Cons — Different characteristics, technologies, and user interfaces with new proposed system could cause error not observed in historical system. — Not all interactions recordable or observable in historical systems so data completeness and accuracy are limited. — User observations can be subjective, biased, and error prone, so correctness, completeness, and accuracy may be limited.
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Technique 4: Audit results <i>Developer's interview:</i> Conduct interviews with development team to extract pertinent information about the design and performance characteristics of the reused software. <i>Design walkthrough review analysis:</i> Analyze the design and code walkthrough data to determine how the reused software would interact in the new proposed system. <i>Standard compliance analysis:</i> Review the results of any standards compliance audits to determine that the proper software standards were followed in the construction of the reused software.	Pros — Depending on the thoroughness of the development team's records, good insight into the design and code approaches can be obtained from the interviews. — Historical artifacts of design and code walkthroughs may be of sufficient quality to act as a substitute of the actual source code and design details. Cons — Limited by the thoroughness of the development team's documentation and recollection during interviews. — Not a totally independent analysis.
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Technique 5: Artifacts <i>Product documentation analysis:</i> Review any product documentation to derive artifacts similar to requirements, design, and code [if possible—for example, if a pseudo design language is used] <i>Prior V&V results analysis:</i> Analyze any prior V&V results and develop inferences and extrapolation of the data to the new proposed system.	Pros — Use actual artifacts representing some form of the reused software. — Prior V&V results for an initial basis for formulating additional analysis and testing to conduct to fill in the analysis and testing voids caused by lack of program documentation. — Independent analysis.
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Alternative sources Description: Use alternative sources of program data to derive conclusions about the correctness, completeness, accuracy, and usability of the reused software.	
	Cons — Overstating a conclusion without having a solid basis for knowing the system conditions could lead to faulty conclusions about the suitability in the new proposed system and its different system conditions. — Limited by the thoroughness of the product documentation

Third, use reverse engineering to generate inputs to perform the V&V tasks in Table 1a through Table 1d and Table 2a through Table 2d of this standard. One reverse engineering technique is suggested in Table D.3.

Table D.3—Reverse engineering to establish V&V task inputs

Reverse engineering Description: Reverse engineer requirements, design, and code data to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.	
Technique 3: Reverse compilation <i>Reverse engineer source code:</i> Reverse compile "pseudo source" code from the program object file. Analyze the reverse compiled pseudo code using normal V&V procedures including all the V&V test strategies and methods. <i>Reverse engineer requirements:</i> Derive the system and software requirements from the user's manual. Analyze the requirements using the IEEE 1012 V&V tasks and test the reused software against these reverse engineered requirements.	Pros — Uses the actual code—no hidden or implied data. — Perform all of the IEEE Std 1012 V&V tasks. — Test data reflects actual performance of the reused software under the system conditions of the new proposed system. — Independent analysis. Cons — Time consuming to reverse engineer data. — Pseudocode hard to read.

Fourth, use independent prototyping and comparison of performance results with those of the reuse asset to perform the V&V tasks in Table 1a through Table 1d and Table 2a through Table 2d of this standard. Two independent prototyping and comparison techniques are suggested in Table D.4.

Table D.4—Independent prototyping and comparison to establish V&V task inputs

Independent prototyping and comparison Description: Develop a model (prototype) of the proposed software or use portions of the prior system. Execute test scenarios on the prototype or prior system and compare the test results against the reused software. Analyze the results to generate objective conclusions about the correctness, completeness, accuracy, and usability of the reused software.	
Technique 7: Prototyping <i>Comparison of prototype code:</i> Develop a replication model (prototype) of the function or requirements in a user-friendly language. Execute test cases representing the range of system scenarios on both the model and reuse software. Compare the results and analyze the differences to determine whether the reused software is performing as intended.	Pros — Useful for small functions and sets of requirements. — Easy to diagnose problems in the reused software. — Ability to run a wide range of system scenarios and compare against a benchmark program. — Independent analysis.

	Cons — Cost and time of building the model. — Errors in the model can mask similar errors in the reused software (likelihood of two similar errors generated by two independent sources should be small or unlikely).
Technique 8: Prior system results <i>Comparison with prior system/function:</i> Execute test cases representing the range of system scenarios on the prior system/function and the reused software. Compare the results and analyze the differences to determine whether the reused software is performing as intended.	Pros — Inexpensive to execute test cases on prior system and compare. — Proven track record of performance of the prior system/function establishes a performance baseline. — Differences in execution results leads to other analysis and testing to be conducted. — Independent analysis.

	Cons — Limited in scope to smaller functions. — Ability to instrument prior system/function to extract diagnosis data about interim program steps may be difficult and make it harder to diagnose exact location of a problem. — Any problems hidden in the prior system/function may go undetected or untested in the new proposed system/function (inheritance of errors).
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Lastly, use a combination of the following circumstantial evidence ~~that~~ provides visibility and insight into the reused software to perform the V&V tasks in Table 1a through Table 1d and Table 2a through Table 2d of this standard:

- a) Operational history
- b) Test history
- c) Audit results

- d) User interviews
- e) Engineering judgment
- f) Product documentation
- g) Prior hazard analysis results
- h) Prior V&V results
- i) Software developer's notebook
- j) Design process documentation
- k) Original developers' interviews
- l) Static code analysis results
- m) Standards compliance assessments

If V&V of reused software cannot be accomplished at the appropriate level, [then](#) the items may be used so long as the risk associated with this use is recognized and accounted for in the risk mitigation strategy. The V&V effort should assure that the risks are thoroughly understood, properly documented, and properly tracked under the risk analysis tasks.

Annex E

(informative)

V&V measures

E.1 Introduction

The management of V&V activity uses measures to provide feedback for the continuous improvement of the V&V processes and to evaluate the [system](#) ~~software~~ development processes and products. Trends can be identified and addressed by computing evaluation measures over a period of time. Threshold values of measures should be established and trends should be evaluated to serve as indicators as to whether a process, product, or V&V task has been satisfactorily accomplished. No standard set of measures is applicable for all projects, so the use of measures may vary according to the application domain and software development environment.

No consensus exists on measures for evaluating the quality and coverage of the V&V tasks. IEEE Std 1061TM-1998 [B8] provides a standard definition of [the](#) available software quality measures. Another measure related standard ~~is and guides are~~ IEEE Std 982.1-1988 [B5] and ~~FP-05 Software Measurement [B1]~~2005 [B4].

This standard considers three categories of measures associated with the V&V effort: 1) measures for evaluating anomaly density, 2) measures for assessing V&V effectiveness, and 3) measures for evaluating V&V efficiency.

E.2 Measures for evaluating anomaly density

Anomaly density measures can provide insightful information on the ~~software~~-product quality, the quality of the ~~software~~ system development processes, and the quality of the V&V effort to discover anomalies in the system/~~software~~-hardware and to facilitate correction of the anomalies. Anomaly density measures are influenced by numerous variables (e.g., software complexity, hardware complexity, type of domain, and time-phase application of the V&V processes); consequently, the measures ~~must be~~ are analyzed to gain insight into the interdependencies between the development efforts and the V&V efforts.

If the V&V anomaly density measure value is low, then this suggests that the program development quality is high, that the V&V processes need to be improved, or a combination of both. If the measure value is high, then this suggests that the program development quality is low, that the V&V processes are effective, or a combination of both. Regardless of the measure value, the next step is to evaluate related ~~software~~-program development measures to ~~further~~ clarify and discern the measure trends to determine the need for process improvements.

Anomaly measures and trends can be used to improve the quality of the current project and can be used to improve the planning and execution of V&V processes for future projects with similar characteristics. The measures defined by Equation (E.1), ~~Equation (E.2), Equation (E.3), and~~ through Equation (E.4) are applicable for four system lifecycle ~~software development~~ phases:

$$\text{Requirements anomaly density} = \frac{\# \text{ Requirements anomalies found by V\&V effort}}{\# \text{ Requirements reviewed by V\&V effort}} \quad (\text{E.1})$$

$$\text{Design anomaly density} = \frac{\# \text{ Design statement anomalies found by V\&V effort}}{\# \text{ Design statements reviewed by V\&V effort}} \quad (\text{E.2})$$

$$\text{Code anomaly density} = \frac{\# \text{ Code anomalies found by V\&V effort}}{\# \text{ Code volume reviewed by V\&V effort}} \quad (\text{E.3})$$

$$\text{Implementation anomaly density} = \frac{\# \text{ Implementation anomalies found by V\&V effort}}{\text{Implementation volume reviewed by V\&V effort}} \quad (\text{E.3})$$

E.3 Measures for evaluating V&V effectiveness

Measures associated with V&V effort effectiveness provide quantitative indications that characterize

$$\text{Test anomaly density} = \frac{\# \text{ Test anomalies found by V\&V effort}}{\# \text{ Tests reviewed by V\&V effort}} \quad (\text{E.4})$$

the added benefits of V&V to discover anomalies in system ~~software~~ products and processes. These measures delineate the percentage of the total anomalies found by the V&V effort. The measures are influenced by numerous variables (e.g., complexity ~~software~~), and the measures are ~~must be~~ analyzed to gain insight into the interdependencies between the development efforts and the V&V efforts.

The V&V effectiveness measure values are highly influenced by the degree of parallelism between the software development effort and the V&V effort. Assuming that the efforts are parallel, ~~then~~ a low V&V effectiveness measure value suggests that the ~~software~~ development effort is effective, or that the V&V effort may require improvement, or a combination of both. If the V&V effectiveness measure value is high, then this suggests that the ~~software~~ development processes may require improvement, or that the V&V processes are effective, or that only incremental changes to the V&V processes may be required. Regardless of the measure value, the next step is to evaluate related ~~system software-program~~ development measures to further clarify and discern the measure trends to determine the need for process improvements. The measures defined by Equation (E.5) ~~through~~ Equation ~~(E.6), Equation (E.7), and Equation~~ (E.8) are applicable for four ~~system life cycle software development~~ phases.

$$\text{Requirements V\&V effectiveness} = \frac{\# \text{ Requirement anomalies found by V\&V effort}}{\# \text{ Requirements anomalies found by all sources}} \quad (\text{E.5})$$

$$\text{Design V\&V effectiveness} = \frac{\# \text{ Design statement anomalies found by V\&V effort}}{\# \text{ Design statement anomalies found by all sources}} \quad (\text{E.6})$$

$$\text{Implementation V\&V effectiveness} = \frac{\# \text{ Implementation anomalies found by V\&V effort}}{\# \text{ Implementation anomalies found by all sources}} \quad (\text{E.7})$$

$$\text{Code V\&V effectiveness} = \frac{\# \text{ Code anomalies found by V\&V effort}}{\# \text{ Code anomalies found by all sources}} \quad (\text{E.7})$$

$$\text{Test execution V\&V effectiveness} = \frac{\# \text{ Test anomalies found by V\&V effort}}{\# \text{ Test anomalies found by all sources}} \quad (\text{E.8})$$

E.4 Measures for evaluating V&V efficiency

The measures associated with V&V effort efficiency provide data that characterize the capability of the V&V effort to discover anomalies in software products and processes in the development activity in which they are injected. Maximum benefits are realized when ~~software~~ anomalies are discovered as early as possible in the development life cycle, thereby minimizing rework and development costs. Analysis of these measures, the anomalies, and the causal factors that prevented discovery of the anomaly in the phase in which it was injected can reveal needed improvements in methods, processes, tools, and skills to improve the overall V&V effort.

A low V&V efficiency measure value suggests that the ~~software~~ V&V effort is not discovering anomalies in the earliest possible activity, or that the ~~software~~ development products are immature, or a combination of both. If the V&V efficiency measure value is high, then this suggests that the V&V effort is discovering anomalies in the earliest possible activity, or that the ~~software~~ development products are mature, or a combination of both. Regardless of the measure value, the next step is to evaluate related [system](#) ~~software~~ program development measures to further clarify and discern the measure trends to determine the need for process improvements. The measures defined by Equation (E.9) [through](#) Equation (E.10), ~~Equation (E.11), and Equation~~ 12) are applicable for four [system life cycle](#) ~~software development~~ phases.

$$\text{Requirements V\&V efficiency} = \frac{\# \text{ Requirement anomalies found by V\&V in requirements activity}}{\# \text{ Requirements anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.9})$$

$$\text{Design V\&V efficiency} = \frac{\# \text{ Design statement anomalies found by V\&V in design activities}}{\# \text{ Design statement anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.10})$$

$$\text{Implementation V\&V efficiency} = \frac{\# \text{ Implementation anomalies found by V\&V in implementation activities}}{\# \text{ Implementation anomalies found by all V\&V in all activities}} \times 100\% \quad (\text{E.11})$$

$$V\&V \text{ efficiency} = \frac{\# \text{ Code anomalies found by V\&V in implementation activity}}{\# \text{ Code anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.11})$$

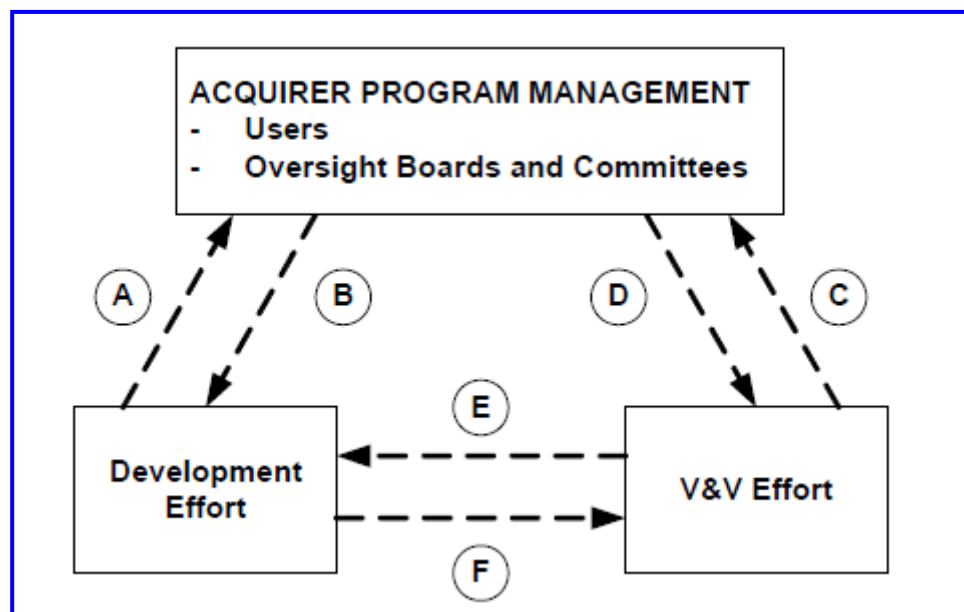
$$\text{Test execution V\&V efficiency} = \frac{\# \text{ Test anomalies found by V\&V in test activities}}{\# \text{ Test anomalies found by V\&V in all activities}} \times 100\% \quad (\text{E.12})$$

Annex F

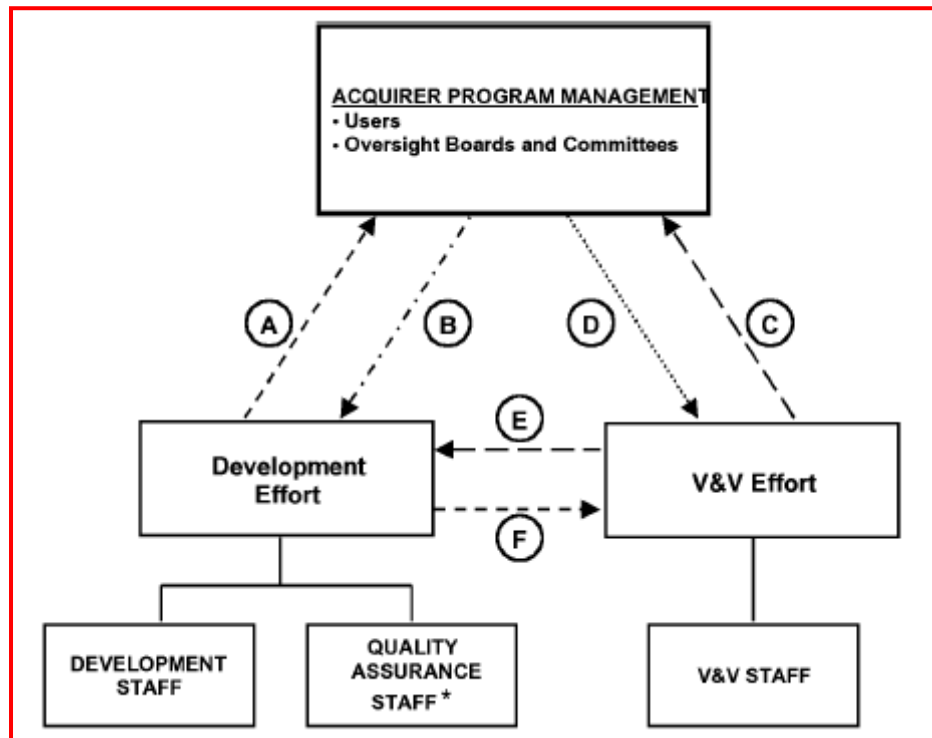
(informative)

Example of V&V ~~organizational~~ relationships to other project responsibilities

Figure F.1 provides an example of ~~organizational~~ relationships among the V&V team, the acquirer, and the supplier, and [it](#) identifies the information and data flows throughout the V&V effort. Many other ~~organizational~~ relationships will work well as long as [the](#) project responsibilities, data flows, and reporting flows are defined and documented.



[\(New\)](#)



~~(Deleted)~~

NOTE—The lines in Figure F.I represent the flow of control and data as follows:

A: Submittal of program documentation (e.g., concept, requirements, design, [and](#) user manuals), source code, program status, program budgets, and developmental plans and schedules.

B: Approval, denial, and recommendations on development issues and deliverables listed in A.

C: Submittal of **SVVP**, V&V task results, anomaly reports, activity reports, and other special reports.

D: Approval, denial, and recommendations on V&V issues and deliverables listed in C.

E: Submittal of V&V task results, anomaly reports, activity reports, and special reports as directed by the acquirer program management.

F: Submittal of program documentation (e.g., concept, requirements, design, user manuals, special reports, source code, and program schedules).

~~*: The quality assurance staff may report directly to the Quality Assurance Director rather than through the development organization.~~

Figure F.1—Relationship of V&V to other project responsibilities

Annex G

(informative)

Optional V&V tasks

Algorithm analysis. Verify the correct implementation of algorithms, equations, mathematical formulations, or expressions. Rederive any significant algorithms and equations from basic principles and theories. Compare against established references or proven~~past~~ historical data. Validate the algorithms, equations, mathematical formulations, or expressions with respect to the system and software requirements. Assure ~~Ensure~~ that the algorithms and equations are appropriate for the problem solution. Validate the correctness of any constraints or limitations, such as rounding, truncation, expression simplifications, best-fit estimations, and nonlinear solutions imposed by the algorithms and equations.

Audit performance. Provide an independent assessment of whether a software process and its products conform to applicable regulations, standards, plans, procedures, specifications, and guidelines. Audits may be applied to any software process or product at any development stage. Audits may be initiated by the supplier, the acquirer, the developer, or other involved party such as a regulatory agency. The initiator of the audit selects the audit team and determines the degree of independence required. The initiator of the audit and the audit team leader establish the purpose, scope, plan, and reporting requirements for the audit.

The auditors collect sufficient evidence to decide whether the software processes and products meet the evaluation criteria. They identify major deviations~~z~~; assess risk to quality, schedule, and cost; and report their findings. Examples of processes that could be audited include configuration management practices, use of software tools, degree of integration of the various software engineering disciplines particularly in developing an architecture, security issues, training, and project management.

Audit support. Provide technical expertise to the auditors on request. They may represent the acquirer at audit proceedings and may assist in the V&V of remedial activities identified by the audit.

Control flow analysis. Assess the correctness of the software by diagramming the logical control. Examine the flow of the logic to identify missing, incomplete, or inaccurate requirements. Validate whether the flow of control among the functions represents a correct solution to the problem.

Cost analysis. Evaluate the cost status of the development processes. Compare budgeted costs against actual costs. Correlate cost expenditures with technical status and schedule progress. Identify program risks if actual costs indicate behind schedule and over cost estimates.

Database analysis. Evaluation of database design as part of a design review process could include [the following](#):

- Physical limitations analysis. Identify the physical limitations of the database, such as maximum number of records, maximum record length, largest numeric value, smallest numeric value, and maximum array length in a data structure and compare them to designed values.
- Index versus storage analysis. Analyze the use of multiple indexes compared to the volume of stored data to determine whether the proposed approach meets the requirements for data retrieval performance and size constraints.
- Data structures analysis. Some database management systems have specific data structures within a record, such as arrays, tables, and date formats. Review the use of these structures for potential impact on requirements for data storage and retrieval.

- Backup and disaster recovery analysis. Review the methods employed for backup against the requirements for data recovery and system disaster recovery, and identify deficiencies.

Dataflow analysis. Evaluation of data flow diagrams as part of a design review process could include [the following](#):

- Symbology consistency check. The various methods used to depict data flow diagrams employ very specific symbology to represent the actions performed. Verify that each symbol is used consistently.
- Flow balancing. Compare the output data from each process to the data inputs and the data derived within the process to [assure](#) ~~ensure~~ the data are available when required. This process does not- specifically examine timing or sequence considerations.
- Confirmation of derived data. Examine the data derived within a process for correctness and- format.
- Data designed to be entered into a process by operator action should be confirmed to [assure](#) ~~ensure~~ availability.
- Keys to index comparison. Compare the data keys used to retrieve data from data stores within a process to the database index design to confirm that no invalid keys have been used and the uniqueness properties are consistent.

Disaster recovery plan assessment. Verify that the disaster recovery plan is adequate to restore critical operation of the system in the case of an extended system outage. The disaster recovery plan should include the following:

- Identification of the disaster recovery team and a contact list .
- Recovery operation procedures
- Procedure for establishing an alternative site including voice and data communications, mail, and support equipment.
- Plans for replacement of computer equipment .
- Establishment of a system backup schedule.
- Procedures for storage and retrieval of software, data, documentation, and vital records off-site .
- Logistics of moving staff, data, documentation, etc.

Distributed architecture assessment. Assess the distribution of data and processes in the proposed architecture for feasibility, timing conflicts, availability of telecommunications, cost, backup and restore features, downtime, system degradation, and provisions for installation of software updates.

Exploratory testing. Perform exploratory testing (i.e., simultaneous learning, test design, and execution rather than scripted testing) on the system, software, or hardware.

Feasibility study evaluation. Verify that the feasibility study is correct, accurate, and complete. Validate that all logical and physical assumptions (e.g., physical models, business rules, [and](#) logical processes), constraints, and user requirements are satisfied.

Independent risk assessment Conduct an independent risk assessment on any aspect of the software project and report on the findings. Such risk assessments will be primarily from a system perspective. Examples of risk assessment include: appropriateness of the selected development methodology or tools for the project; and quality risks associated with proposed development schedule alternatives.

Inspection. Inspect the software products to detect defects in the product at each selected development stage to assure the quality of the emerging software. The inspection process may consist of multiple steps for the segregation of the inspection functions of the following:

- Inspection planning
- Product overview
- Inspection preparation
- Examination meeting
- Defect rework
- Resolution follow-up

An inspection is performed by a small team of peer developers and includes but is not led by the author. The inspection team usually consists of three to six persons, and in some cases includes personnel from the test group, quality assurance, or V&V. The participants assume specific roles to find, classify, report, and analyze defects in the product. Each type of inspection is specifically defined by its intended purpose, required entry criteria, defect classification, checklists, exit criteria, designated participants, and its preparation and examination procedures. Inspections do not debate engineering judgments, suggest corrections, or educate project members; they detect anomalies and problems and verify their resolution by the author.

Inspection (concept). Validate that the system architecture and requirements satisfy customer needs. Verify that the system requirements are complete and correct, and that omissions, defects, and ambiguities in the requirements are detected.

Inspections (design). Verify that the design can be implemented, is traceable to the requirements, ~~and that~~ all interface and procedural logic is complete and correct, and ~~that~~ omissions, defects, and ambiguities in the design are detected.

Inspections (requirements). Validate that the requirements meet customer needs and can be implemented. Verify that they are complete, traceable, testable, and consistent so that omissions, defects, and ambiguities in the requirements are detected.

Inspection (source code). Verify that the source code implementation is traceable to the design, ~~and that~~ all interfaces and procedural logic are complete and correct, and ~~that~~ omissions, defects, and ambiguities in the source code are detected.

Inspection—Test case (component, integration, system, acceptance). Verify that the (component, integration, system, acceptance) test plan has been followed accurately, that the set of component test cases is complete, and that all component test cases are correct.

Inspection—Test design (component integration, system, acceptance). Verify that the (component, integration, system, acceptance) test design is consistent with the test plan, and that the test design is correct, complete, and readable.

Inspection—Test plan (component integration, system, acceptance). Verify that the scope, strategy, resources, and schedule of the (component, integration, system, acceptance) testing process have been completely and accurately specified, that all items to be tested and all required tasks to be performed have been defined, and to [assure](#) ~~ensure~~ that all personnel and resources necessary to perform the testing have been identified.

Operational evaluation. Assess the deployment readiness and operational readiness of the software. Operational evaluation may include examining the results of operational tests, audit reviews, and anomaly reports. This evaluation verifies that the software is [as follows](#):

- At a suitable point of correctness for mass production of that software .
- Valid and correct for site-specific configurations.

Performance monitoring. Collect information on the performance of software under operational conditions. Determine whether system and software performance requirements are satisfied. Performance monitoring is a continual process and may include evaluation of the following items:

- Database transaction rates to determine the need to reorganize or re-index the database .
- [Central processing unit](#) (CPU) performance monitoring for load balancing.
- Direct access storage utilization.

- Network traffic to [assure](#) ~~ensure~~ adequate bandwidth.
- Critical outputs of a system (e.g., scheduled frequency, expected range of values, scheduled system reports, [and](#) reports of events).

Post installation validation. Execute a reference benchmark or periodic test for critical software when reliability is crucial or there is a possibility of software corruption. By automatically or manually comparing results with the established benchmark results, the system can be validated prior to each execution of the software. When pre-use benchmark testing is impractical, such as for real time, process control, and emergency-use software, a periodic test, conducted at a predetermined interval, can be used to [assure](#) ~~ensure~~ continued reliability.

Project management oversight support Assess project development status for technical and management issues, risks, and problems. Coordinate oversight assessment with the acquirer and development organization. Evaluate project plans, schedules, development processes, and status. Collect, analyze, and report on key project measures.

Proposal evaluation support ~~Participate~~ Participate in the development organization source selection process. Develop proposal evaluation factors and assessment criteria. Independently evaluate development organization proposals to assess conformance to the statement of work and performance requirements.

Qualification testing. Verify that all software requirements are tested according to qualification testing requirements demonstrating the feasibility of the software for operation and maintenance. Conduct, as necessary, any tests to verify and validate the correctness, accuracy, and completeness of the qualification testing results. Document the qualification test results together with the expected qualification test results. Planning for qualification testing may begin during the Requirements V&V activity.

Regression analysis and testing. Determine the extent of V&V analyses and tests ~~to that must be~~ repeated when changes are made to any previously examined software products. Assess the nature of the change to determine the potential ripple or side effects and the impacts on other aspects of the system. Rerun test cases based on changes, error corrections, and impact assessment; to detect errors spawned by software modifications.

Reusability analysis. Verify that the artifacts (products) of the domain engineering process conform to project-defined purpose, format, and content (e.g., IEEE Std 1517-2010 [B10]. ~~1999 [B11]~~). Verify that the domain models and domain architecture are correct, consistent, complete, and accurate, and that they conform to the domain engineering plan. Analyze the asset (software item intended for reuse) to verify that the asset is consistent with the domain model and domain architecture.

Reuse analysis. Analyze the developer's documentation to verify that the original domain of the candidate reuse software will satisfy the domain of the new system (e.g. ~~software~~, integrity level, user needs, operating environment, safety, security, and interfaces). If the developer has performed no domain analysis, perform domain analysis (~~see~~ IEEE Std 1517-2010 [B10] ~~1999 [B11]~~) to compare the original domain and the new domain of the candidate reuse software. Verify that the developer's reuse plan dispositions and documents all domain differences.

Simulation analysis. Use a simulation to exercise the software or portions of the software to measure the performance of the software against predefined conditions and events. The simulation can take the form of a manual walkthrough of the software against specific program values and inputs. The simulation can also be another software program that provides the inputs and simulation of the environment to the software under examination. Simulation analysis is used to examine critical performance and response time requirements or the software's response to abnormal events and conditions.

Sizing and timing analysis. Collect and analyze data about the software functions and resource utilization to determine whether system and software requirements for speed and capacity are satisfied. The types of software functions and resource utilization issues include, but are not limited to [the following](#):

- CPU load
- Random access memory and secondary storage (e.g. disk, tape) utilization
- Network speed and capacity
- Input and output speed

Sizing and timing analysis is started at software design and iterated through acceptance testing.

System software assessment. Assess system software (e.g., operating system, computer [-aided](#) software engineering tools, database management system, repository, telecommunications software, [and](#) graphical user interface) for feasibility, impact on performance and functional requirements, maturity, supportability, adherence to standards, developer's knowledge of and experience with the system software and hardware, and software interface requirements.

Test certification. Certify the test results by verifying that the tests were conducted using baseline requirements, a configuration control process, and repeatable tests, and by witnessing the tests. Certification may be accomplished at a software configuration item level or at a system level.

Test evaluation. Evaluate the [developer's](#) tests for requirements coverage and test completeness. Assess coverage by assessing the extent of the software exercised. Assess test completeness by determining whether the set of inputs used during test are a fair representative sample from the set of all possible inputs to the software. Assess whether test inputs include boundary condition inputs, rarely encountered inputs, and invalid inputs. For some software-[z](#) it may be necessary to have a set of sequential or simultaneous inputs on one or several processors to test the software adequately.

Test witnessing. Monitor the fidelity of test execution to the specified test procedures and witness the recording of test results. When a test failure occurs, the testing process can be continued by: (1) implementing a "work around" to the failure, (2) inserting a temporary code patch, or (3) halting the testing process and implementing a software repair. In all cases, assess the test continuation process for test process breakage (e.g., some software is not tested or a patch is left in place permanently), adverse impact on other tests, and loss of configuration control. Regression analysis and testing should be done for all the software affected by the test failure.

Training documentation evaluation. Evaluate the training materials and procedures for completeness, correctness, readability, and effectiveness.

Usability analysis. Verify that stakeholder needs and interests are considered during development, operation, and maintenance process activities. The analysis will assure ~~ensure~~ that: human-centered design activities are performed; human factors and ergonomics considerations are incorporated into the design, potential adverse effects on human health and safety are addressed in the design; and user needs are satisfied in a manner that supports user effectiveness and efficiency.

User documentation evaluation. Evaluate the user documentation for its completeness, correctness, and consistency with respect to requirements for user interface and for any functionality that can be invoked by the user. The review of the user documentation for its readability and effectiveness should include representative end users who are unfamiliar with the software. Employ the user documentation in planning an acceptance test that is representative of the operational environment.

User training. Assure that the user training includes rules that are specific to the administrative, operational, and application aspects, as well as the ~~and~~ industry standards for that system. This training should be based on the technical user documentation and procedures provided by the manufacturer of the system. The organization responsible for the use of the system should be responsible for providing appropriate user training.

V&V tool plan generation. Prepare a plan that describes the tools needed to support the V&V effort. The plan includes a description of each tool's performance, required inputs and associated tools, outputs generated, need date, and cost of tool purchase or development. The tool plan should also describe test facilities and integration and system test laboratories supporting the V&V effort. The scope and rigor of the V&V effort as defined by the selected ~~software~~ integrity level should be considered in defining the performance required of each tool.

V&V tool qualification. Verify the features and usability of a V&V tool to assure that the tool is functioning properly and it does not mask errors that it was designed to find. Validate correct operation, limitations, and workarounds for known problems and deficiencies.

Walkthrough. Participate in the evaluation processes in which development personnel lead others through a structured examination of a product. ~~Assure~~ ~~Ensure~~ that the participants are qualified to examine the products and are not subject to undue influence. See specific descriptions of the requirement walkthrough, design walkthrough, source code walkthrough, and test walkthrough.

Walkthrough (design). Participate in a walkthrough of the design and updates of the design to ~~assure~~ ~~ensure~~ completeness, correctness, technical integrity, and quality.

Walkthrough (requirements). Participate in a walkthrough of the requirements specification to ~~assure~~ ~~ensure~~ that the software requirements are correct, unambiguous, complete, verifiable, consistent, modifiable, traceable, testable, and usable throughout the life cycle.

Walkthrough (source code). Participate in a walkthrough of the source code to ~~assure~~ ~~ensure~~ that the code is complete, correct, maintainable, ~~and~~ free from logic errors; ~~also ensure the code~~ conforms to coding standards and conventions, and will operate efficiently.

Walkthrough (test). Participate in a walkthrough of the test documentation to assure ~~ensure~~ that the planned testing is correct and complete, and that the test results will be correctly analyzed.

Work Breakdown Structure (WBS) Evaluation. Verify that the WBS represents all of the project scope and captures all deliverables, including internal, external, and interim deliverables. Verify that the WBS decomposes the project scope into a set of deliverables that comprehensively defines the work to be performed. If a WBS dictionary exists, then verify that the WBS dictionary contains a brief definition of the project scope, each WBS component is included, each WBS component has a list of associated activities, and each major milestone is included.

Annex H

(informative)

Environmental factors considerations

H.1 Introduction

The performance of system, software, and hardware V&V should be planned and executed to account for environment factors that will influence the life cycle stages of the system—from concept development through disposal. These environmental factors may drive and constrain activities and tasks in each of the V&V life cycle processes; therefore, all environmental factors should be assessed to determine relevance and influence on the V&V activities and tasks to be performed.

H.2 In the agreement processes

Environmental factor requirements and constraints should be defined during the Acquisition Process and specified for the Supply Process. These processes define the activities necessary to establish an agreement between two organizations. When preparing for the acquisition, the strategy development should consider the system, software, and hardware attributes, requirements, and/or constraints to verify that appropriate environmental factor requirements are included in the request for proposal/quotation and the resulting contract. The V&V for this stage should perform the following:

- Assess the *need definition* to verify that all environmental factors affecting system design, development, production, operation, and disposal throughout the life cycle are clearly and comprehensively stated.

- Review the *solution approach selection process* (such as analysis of alternatives) to verify that environmental factors are adequately incorporated into the decision process and criteria.

- Review the *acquisition documentation* for adequacy and sufficiency of definition/specification for the system operating environment(s), system nonoperating environment(s), development environments), and maintenance environments) to assure that the resultant program or contract delivers a system that satisfies the defined need. The request for proposal/quotation should be reviewed to verify that all pertinent environmental requirements are incorporated.

- Review the *acceptance documentation* to assess planned and specified testing environment(s) to assure that the testing will provide full spectrum, operationally representative tests to determine whether the system is properly designed and produced for operation per the need definition.

- Assess the *supplier qualifications* to verify that selected supplier has the industrial capability to design, develop, produce, and support the program in accordance with acquisition documentation requirements. This assessment should include development, testing, production, and repair environments, processes, and tools.

- Assess the *supplier's proposal* to verify that system design and development, testing and manufacturing approaches, and processes adequately and sufficiently address all environmental requirements of the acquisition documentation's request for proposal/quotation.

- Assess the *contract documentation* to verify that environmental requirements reflect defined need and the supplier's proposed solution with clear and binding terms.

H.3 In the organizational project-enabling processes

Environmental considerations should be assessed to verify that the processes for infrastructure management, human resource management, and quality management adequately address the ability of the acquirer and supplier to provision, execute, monitor, assess, maintain, and improve processes and products that satisfy the system, software, and hardware needs over the full range of environmental requirements and constraints. The V&V for this stage should perform the following:

- Assess the *supplier's infrastructure* to be employed in the development and production of the system, software and hardware to verify the environmental requirements and constraints will be maintained throughout the acquisition contract. This assessment should include development, testing, production and repair environments, processes and tools.
- Assess the *human resources* assigned to the contract to verify that they have the necessary education, experience, training and if applicable, certifications for the activities and tasks to which they are assigned. Assess the supplier's strategy and plans to verify that qualified human resources will be assigned to the project throughout the contract life.

- Assess the *quality management strategy* for the project to verify that environmental requirements and constraints for the infrastructure and the product (system, software, and hardware) will be monitored and maintained compliant throughout the contract life.

H.4 In the project processes

The project processes should address environmental requirements and constraints in the project plan development, execution, and monitoring. The V&V for this stage should perform the following:

- Review the project technical management plan to verify that environmental factors will be included in technical reviews throughout the project.
- Assess the quality management plan to verify that environmental requirements and constraints for the infrastructure and the product (system, software, and hardware) will be monitored and maintained compliant throughout the contract life.
- Review the project performance measures to verify that environmental factors will be collected throughout the contract life.
- Verify that project assessments will include environmental factors on a continuing basis.
- Verify that the configuration management process will include adequate controls for environmental requirements and constraints affecting infrastructure, facilities, manufacturing processes, and tools.

H.5 In the technical processes

The V&V should assess environmental factors throughout the technical processes as well as perform the following:

- Assess that all environmental requirements and constraints included in the acquirer's request for proposal and in the resulting contract with the supplier are completely specified in the project requirements documents and are appropriately allocated to the system elements, software, and hardware.

- Assess the architectural documents to verify that the environmental requirements and constraints from the project requirements documents have incorporated into the architectural design.

- Monitor the implementation process, the integration process, and the development qualification testing to verify that the system elements, software, and hardware conform to and perform throughout the complete range of specified environmental conditions. Review the test environments, tools, test plans/procedures, and results to assure the design is adequately tested to verify that the system will meet performance in all operating and nonoperating environment requirements of the contract. Review anomaly clearing procedures and retest procedures for adequacy and sufficiency. Specific test environments to be monitored and assessed include the following:

- 1) Functional
- 2) Integration
- 3) System
- 4) Qualification
- 5) Manufacturing (parts, subassembly, process sampling, etc.)
- 6) Certification and accreditation testing (interoperability, architecture and standards, security, etc.)
- 7) Operational test and evaluation

- Review the *transition plans* to verify the environmental requirements and constraints are adequately addressed.
- Assess the *effectiveness and suitability* of design, procedures, training, and support by reviewing operational use data, operational failure data, and user feedback to determine whether environmental factors are adversely impacting system performance, safety, or security or whether requested changes could improve system performance, safety, of security.
- Perform *evaluations of new constraints* to determine whether changing environments are impacting or have the potential to impact system performance, safety, or security in an adverse way.
- Assess *operating procedures* to determine whether changes in operational procedures or operational use profiles are impacting or have the potential to impact system performance, safety, or security adversely.

- Review metrics from the *maintenance processes*, tools, staffing and controls to verify predictable and consistent results are achieved, adverse trends are identified, and corrective action/process improvements are incorporated when warranted. Review parts/component/COTS substitution and technical refresh processes to verify that environmental considerations are incorporated in the decision process and criteria. Review repair/upgrade kits and procedures intended for operational site location incorporation to verify environmental factors are adequately and sufficiently addressed.

- Assess plans for *disposal of the system elements* to verify that adequate provisions are delineated for retirement of the elements and subsequent storage, destruction, or disposal to preclude or contain hazardous and environmentally destructive materials and to prevent hazardous conditions.

Annex I

(informative)

V&V of system, software, and hardware integration

1.1 Introduction

Disciplines required to develop software elements in digital systems are distinctly different from those required to develop the hardware components of digital systems in many important aspects. Developing software for a digital system requires knowledge of software engineering principles, the software language(s) to be used, the limitations of the language(s), and the processes by which the software requirements can be translated into a set of functions that meet a user's requirements. Optimally, some members of the software development team also should have system domain expertise to understand the system requirements allocated to software. Developing the hardware on which system software is to operate requires knowledge of electrical engineering, computer engineering, hardware performance characteristics, printed circuit board layout, equipment qualification, and similar considerations. Managing these separate disciplines and integrating the disciplines into a unified development process requires its own set of project management disciplines.

Given the diverse disciplines involved in the development of digital systems, misconceptions about the relationships amongst the system, software, and hardware should not be unexpected. These differences in specializations must be resolved to prevent incompatibilities arising between the three categories comprising a digital system (i.e., system, software, and hardware). Consequently, in addition to verifying and validating system, software, and hardware requirements, V&V organizations should verify and validate system, software, and hardware integration.

This annex provides examples of system, software, and hardware integration issues that should be addressed in the phases of the system development lifecycle. The integration considerations discussed in this annex are not the full set of integration considerations a system developer should explore. Instead, the integration considerations are intended to encourage system developers to view the system development effort holistically instead of thinking of the software and hardware development efforts as tasks to be performed in isolation.

The annex provides summaries of digital system faults that were introduced during system development as a result of failing to identify potential adverse integration effects between the systems, software and hardware. The annex then provides in Table I.I examples of integration considerations that should be addressed in the following lifecycle phases:

- Acquisition
- Requirements (stakeholder and analysis)
- Architectural design
- Implementation
- Integration

1.2 Examples of system failures caused by integration issues

Despite the individual skills of the specialists developing a system, there have been events in which integration of the software and hardware into the system did not result in desired system performance reliability goals. The following events histories highlight the necessity for tightly coupling the system development effort with the software and hardware development efforts.

1.2.1 Year 2000 System Integration Issue

Hardware memory limitation constraints in conjunction with the historical period in which early date/time-usage systems were developed (e.g., 1970) encouraged system developers to represent calendar year dates with two digits instead of four (e.g., 70 instead of 1970) when calculating elapsed time between two events. A contributing premise (when even considered) was that the system being developed would likely be updated or replaced before the year 2000 (i.e., 00) could affect elapsed time/date calculations over the transition period between December 31, 1999 and January 1, 2000.

In validating the software during development, developers used ranges of dates that did not include dates beyond December 31, 1999 (rhetorically, why would they?). The software was validated using what was then reasonable dates and performed as required after installation. The system hardware, which was relatively primitive by today's standards of digital equipment capabilities, also performed in accordance with the system hardware requirements. This was not unexpected, as the software requirements were developed with the hardware limitations in mind.

System performance considerations of the effect of dates transitioning from "99" to "00" arose circa 1998 to 1999 as industries realized the degree to which digital systems had been integrated into critical digital assets. The nuclear industry, as well as other industries, began an intensive effort to identify date-related dependencies in critical systems. This industry-wide effort required significant expenditures of resources and funds to remediate critical systems, implement software revisions, and install new equipment.

The issue in this case is not attributed to software or hardware deficiencies; the software and hardware, when integrated, performed as required during the initially planned performance period. The issue in this case was in not identifying the full set of parameters that constrained the system operating space. The system temporal limits were defined for a date/time period that did not include dates in the following century.

The lesson from this example is that the full extent of a system boundary must be verified when developing software and hardware requirements. In the case of the year 2000 example, the system boundary should have included both physical and temporal constraints. For those cases in which hardware limitations forced software requirements limits, it is not enough that software integrated with hardware perform in accordance with software and hardware requirements; a verification process must include the system boundaries that completely constrain the system performance parameters.

1.2.2 System architecture integration issues

In the late 1990s, a consortium developed system requirements for a digital safety system function to be installed at several industrial sites. The system developer was selected for this development effort on the basis of its extensive experience developing high-integrity digital systems for the defense industry.

The consortium developed requirements for a master-slave system architecture to enhance system self-testing capabilities and thereby improve digital safety system availability. This type of architecture, which is common in high-integrity systems in many industries, used a duplicate (i.e., slave) system to check the operability of the master system functions. The assumption was that identical outputs from the master and slave system functions implied the system was operating as required. If the two output signals were different, then the system was not operating as required, and a system reset would be needed to alert the operators that either the master channel was inoperable or the slave channel was inoperable (or perhaps both were inoperable).

To verify that the master-slave calculations in this system were identical when the system was operating, the master system processor synchronized the slave processor to verify that the input signal values were the same in both microprocessor channels. This synchronization required the slave processor to prioritize safety function checking calculations over internal self-testing functions such that self-testing functions were placed on hold until the checking calculations were completed. The self-testing functions then resumed as the highest priority function in the slave processor.

Both the master processor channel and the slave processor channel performed channel self-testing to verify the two channels were operable. These self-testing functions operated cyclically such that the system was continuously tested when it was not performing safety function processing. Both channels used watchdog timers to verify their respective processor did not have an error that would prevent the channel from performing every self-testing function. If the self-testing was not completed within a set period, then the watchdog timer would reset the offending channel and thereby alert the operators that a channel was inoperable.

After the system was installed in several facilities, sporadic watchdog timer-initiated resets occurred in the slave processor channel. In response to these channel resets, the channel was required to be declared inoperable until the system could be placed back in service. The system developer was tasked with determining the cause of the sporadic watchdog timer resets.

After extensive root cause analyses, the system developer determined the fault to be in the priority baton passing function built into the microprocessor. This microprocessor fault only arose when a high priority task acquired the priority baton from a lower priority task before the lower priority task was completed. Upon completion of the high priority task in the Slave processor (i.e., the safety function processing in the Slave processor that was initiated by the Master processor), the Slave microprocessor sometimes would not release the priority baton back to the interrupted self-testing function. This caused the self-testing function to fail to meet the watchdog timer restriction, and thereby caused the Slave channel watchdog timer to reset the channel microprocessor. Approximately 10 months were required for the system developer to identify and correct this fault.

While the system developer had extensive experience developing systems using the microprocessor, that experience did not translate into the master-slave architecture. Interestingly, if the developer had reviewed publically available failure history reports from the microprocessor vendor's Website at the start of the project, the priority baton passing error would have been discovered during the hardware acquisition lifecycle phase instead of in the operations lifecycle phase.

This event illustrates the necessity for verifying that the hardware acquisition requirements and the software requirements (both internal to the hardware and imposed by software on the hardware) are compatible with the system architecture.

1.3 System, software, and hardware interaction issues

The following example interaction issues provide a starting point for identifying issues that should be addressed during the system development process:

- Software size relative to hardware capability
- Hardware compatibility with software type (e.g., C, Hardware Design Language, etc.)
- System timing constraints on software design and hardware capabilities
- System architecture and software architecture relative to hardware capabilities
- Capability to validate software in an environment representative of the hardware architecture

- Hardware restrictions relative to system environment (e.g., equipment qualification issues, etc.)
- Software integrity requirements relative to hardware architecture
- System maintainability relative to hardware acquisition constraints
- Software architecture relative to system maintainability (e.g., use of object-oriented structure or an arcane structure that becomes outdated over time, thereby limiting the number of personnel qualified to perform corrective, adaptive, or perfective changes on the system)
- Hardware availability over the projected lifetime of the system.

This list of potential interaction issues is not exhaustive. The purpose of the list is to instill in system developers the realization that digital system performance issues can arise from interactions among software, hardware, and the system.

Table I.I graphically summarizes the potential interactions among the system, software, and hardware for the example issues described for each system development lifecycle phase in which the software and hardware are developed relatively individually. The cells in Table I.I marked with an "X" correspond to the system, software, and hardware potential contributors to interactions corresponding to the lifecycle phases in which the potential interactions should be addressed during the development effort.

The purpose of Table I.I is to provide a sample framework system developers can use to summarize potential interaction issues identified or anticipated during lifecycle development activities. By identifying these interactions in a graphical format, the developer can refine the scope of verification and validation activities to address system, software and hardware interaction issues specifically. Other interaction concerns may need to be added to the list of examples depending on specific system architecture requirements.

Table I.1 – Examples of integration V&V issues

<u>Interaction issue</u>	<u>System development lifecycle phases (ISO/IEC 15288:2008 [B16D]</u>														
	<u>Acquisition</u>			<u>Requirements</u>			<u>Arch. design</u>			<u>Implementation</u>			<u>Integration</u>		
	<u>Interactions to be addressed</u>														
	<u>Sys</u>	<u>SW</u>	<u>HW</u>	<u>Sys</u>	<u>SW</u>	<u>HW</u>	<u>Sys</u>	<u>SW</u>	<u>HW</u>	<u>Sys</u>	<u>SW</u>	<u>HW</u>	<u>Sys</u>	<u>SW</u>	<u>HW</u>
<u>Software size relative to hardware capability</u>	X	X	X	X	X	X		X	X		X	X		X	X
<u>Hardware compatibility with software language and structure</u>		X	X		X	X		X	X		X	X		X	X
<u>System timing relative to software design and hardware capabilities</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
<u>System and software architecture relative to hardware capabilities</u>	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
<u>Capability to validate software in environment representative of the hardware architecture</u>		X	X	X	X	X		X	X		X	X		X	X
<u>Hardware restrictions</u>	X		X	X		X	X		X	X		X	X		X

relative to system environment																
Software integrity relative to hardware architecture		<u>X</u>	<u>X</u>	<u>X</u>	<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>	
System maintainability relative to hardware acquisition	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	
System maintainability relative to software architecture	<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>		
Hardware availability relative to projected system lifetime	<u>X</u>		<u>X</u>	<u>X</u>		<u>X</u>										

Annex J

(informative)

Hazard, security, and risk analyses

J.1 Hazard analysis

V&V often uses the results of hazard analyses to target V&V activities in a manner that provides greater assurance that the controls for hazards work as needed and that the system does not contribute to a hazardous condition by behaving in an undesirable way. Generally, one or more additional conditions need to exist or additional events need to occur in conjunction with the existence of the hazard in order for an accident or mishap with consequences adverse to safety to result. These additional events enable the hazard to proceed to the adverse consequence. Hazards analysis involves the application of systematic and replicable methods to identify and understand hazards, as well as to characterize the risk of mishaps that involve hazards.

The hazard analysis is a systems engineering activity that will account for the system design, operational conditions, system physical constraints, and regulations to identify hazardous conditions that could lead to adverse consequences. Once the end hazardous conditions are identified, the hazard analysis typically uses a fault tree analysis approach to identify the contributors or combination of contributors (within the bounds of required system fault tolerance) to reaching the hazardous condition. The fault tree is a symbolic logic diagram showing the cause-effect relationship between a top undesired event (failure) and one or more contributing causes. It is a type of logic tree that is developed by deductive logic from a top undesired event to all subevents that must occur to cause it. The contributing causes may be the result of hardware or software faults, human actions (e.g., procedures), or hostile environmental conditions. The hazard analysis is repeated in each lifecycle phase and accounts for further elaboration of designs, changes to intended system use and operations, and the emergence of new hazardous conditions. The hazard analysis may be performed by any organization within the project such as systems engineering, reliability, safety, or V&V. In any case, V&V reviews the hazard analysis for completeness and usability, and it assures that stated hazards and contributors are clearly identified to sufficient detail to affect engineering and mitigation activities properly and to develop V&V plans and evaluation criteria.

Not all hazards represent the same risk. Systems engineers, operations experts, and reliability engineers may all contribute to evaluating the magnitude of risk that a hazard may present. The risk based integrity level scheme presented in Annex B is suitable as well for the assessment of risk associated with a hazard.

As a result of the hazard analysis, the project will define requirements for the control or mitigation of hazards (as well as re-design the system to reduce risk exposure). These requirements may be in the form of system capabilities, redundancy and fault tolerance requirements, use of engineering standards, or operational procedures. The traceability of these requirements and associated design elements to the specific hazard and the hazard's risk magnitude will serve to determine the V&V integrity level for the requirement or system element and the level of V&V rigor needed.

Specific V&V activities that may be appropriate for "critical" requirements necessary to control, mitigate, or prevent hazards may include the following activities:

- a) Traceability of critical requirements through the lifecycle to verify implementation.
- b) Evaluation of potential hazard contributors to validate that critical requirements are complete and are appropriate for the system operational need.
- c) Evaluation of architectures and designs to determine whether hazard mitigation functions meet required capabilities and whether additional mitigation strategies are needed.
- d) Application of verification methods (analyses, inspections, demonstrations, or tests) that are to a hazardous state are possible. These verification methods may include the following:

- 1) Statistical analyses to determine whether the probability of reaching a contributing condition is within acceptable levels. This may include random testing, Monte Carlo analysis, stress testing, duration testing, or mathematical models.
- 2) Inspections to verify that hazard causes are not present and that hazard controls are implemented as specified. This may also include inspections that regulatory or engineering standards have been followed.
- 3) Demonstrations in an operational setting to show that operational and human factor based hazard controls are reasonable and effective.
- 4) Tests to verify that specific hazard controls (physical, procedural, and automated controls) are capable of identifying the faults or hazard contributors and isolating them from propagating into a hazardous condition.

The objective for the V&V of hazards is to perform a sufficient set of V&V tasks for all contributors such that the likelihood of reaching a hazardous condition is known to a desired level of confidence. The V&V integrity levels and their use in tailoring V&V activities as described in this standard provides for more rigor as the integrity level increases. Additionally, the use of assurance cases provides for the evidence to satisfy claims that certain desired properties or behaviors exist or undesired properties or behaviors do not exist as necessary to prevent a specific hazardous condition.

J.2 Security analysis

The objective of security analysis performed by the V&V effort is to verify that the system-required threat controls and safeguards are correctly implemented and to validate that they provide the desired levels of protection of system vulnerabilities. A vulnerability is a flaw or weakness in system security procedures, design, implementation, or internal controls that could be exercised (accidentally triggered or intentionally exploited) and result in a security breach or a violation of security policy. The desired levels of protection and security policies are unique to the stakeholder organization.

The V&V security analysis is performed in parallel with project system security risk assessments that generally include the following activities:

- Identification of threats. These threats may be natural (inclement weather, earthquakes, etc.), human (unintended or malicious), or environmental (chemical leak and power loss).
- An identification of system vulnerabilities. Vulnerabilities may include items such as system physical exposure, inability to recognize an intrusion, security perimeters that can be compromised, or accessibility of sensitive computing systems or data.
- Evaluation of controls needed to prevent threats from exercising a potential vulnerability.
- Evaluation of the likelihood of a threat exercising a potential vulnerability according to current or planned controls.

- Evaluation of the impact of a security breach or security policy violation. Impacts may include items such as human safety, financial loss, social loss, environmental mission loss, or business continuity.

The system security risk assessment creates security threat controls to mitigate security risk to levels acceptable to the stakeholders. The security control requirements may be in the form of system functional capabilities, physical perimeter safeguards, redundancy, backup and recovery, or operational procedures. Much like the hazard analysis, the traceability of these requirements and associated design elements to the specific security impact severity will serve to determine the V&V integrity level for the requirement or system element and the level of V&V rigor needed.

Specific V&V activities that may be appropriate for "critical" security requirements necessary to control threats and exposure to vulnerabilities may include the following activities:

- a) Traceability of critical requirements through the life cycle to verify implementation.
- b) Evaluation of potential threat sources and vulnerabilities to validate that critical security requirements are complete and are appropriate for the system operational need.
- c) Evaluation of architectures and designs to determine whether security functions meet required capabilities, whether additional threat controls are needed, and whether design changes are needed to remove vulnerabilities.

d) Application of verification methods (analyses, inspections, demonstrations, or tests) that are intended to determine whether plausible threats can exploit vulnerabilities. These verification methods may include the following:

- 1) Statistical analyses to determine whether the probability of breaching a security control is within acceptable levels. This may include simulations or mathematical models (e.g., for encryption methods).
- 2) Inspections to verify that security controls are implemented as specified. This may also include inspections that regulatory or policy standards have been followed.
- 3) Demonstrations in an operational setting to show that security controls are reasonable and effective.
- 4) Tests to verify that specific security controls (physical, procedural, and automated controls) cannot be breached. For IT systems, this may also include vulnerability scanning and penetration testing.

J.3 Risk analysis

J.3.1 Risk analysis objectives

The objective of risk analyses performed by the V&V effort is to identify technical and engineering risks that have a measureable possibility of negative consequences to either the operation of the system or the successful development of the system. The risk analysis activity includes the following:

- Identifying the initiating events, hazards, threats, or situations that create risks.
- Estimating the probability of occurrence, the consequences for each risk, and the expected timing of the risk.
- Evaluating each risk or defined combination of risks against its applicable threshold, generate alternatives to treat risks above their risk thresholds, and make recommendations for treatment based on a priority order.

Risk analysis shall be performed continuously throughout the lifecycle.

J.3.2 Risk analysis context

The V&V effort risk analysis activity falls within a broader risk management approach. For example, the risk analysis process steps of identify, estimate, and evaluate described J.3.3 follows the risk analysis process activity within the broader risk process described in ISO/TEC 16085-2006 [B18]. The broader risk management activities of managing the project risk portfolio, risk treatment, or risk monitoring can be found in ISO/TEC 16085-2006 [B18].

Risks identified by the V&V effort often require management and mitigation by other project organizations. Accordingly, the V&V Plan should have clear provisions for communicating and escalating V&V identified risks to the organization with the responsibility to perform risk treatment.

Some risks of concern to the V&V effort are tightly coupled with hazard analyses and integrity levels. The severity of the consequence of a system operational risk will likely not exceed the worst case severity of a system hazard or integrity level. Risks are often directly coupled to hazards. The likelihood that a hazard is reachable may be too high and cause re-engineering of the system to add hazard controls (or security controls for security hazards). There may be risk that a particular hazard control is unreliable or not verifiable. Both cases are of interest to the V&V effort. The hazard analysis is driven by the system design and operating environment. Probabilistic risk assessments evaluate the likelihood of reaching the hazardous condition and risk management determines whether the probabilities exceed acceptable thresholds and require treatment. The V&V effort may identify risks when the probability of reaching a hazardous condition is perceived by the V&V effort to be beyond levels of acceptability for the system and there is no plan to treat the risk. The V&V effort may also identify a risk when there is inadequate evidence that hazard controls provide the desired behavior/function or when there is uncertainty with regard to the conditions that may lead to a hazard. The same hazard based risk analysis concepts apply to security-based risk analyses where undesired conditions may result from uncontrolled vulnerabilities or inadequate protection from threats.

J.3.3 V&V risk analysis

J.3.3.1 Risk identification

Various approaches to identifying risks should be used. These approaches may include the use of risk questionnaires, taxonomies, brainstorming, scenario analysis, lessons learned, and prototyping or other knowledge acquisition approaches. Repeatable identification processes may be used to aid in the capture of lessons learned. Where possible, events, hazards, threats, or situations that can create risks should be identified to aid future risk treatment. Risks not identified are implicitly accepted. Risk categories should be used consistently for effective communication to stakeholders. Risks that are related may be combined for ease of analysis, monitoring, and treatment. System or software anomalies, reports on measures, and other indicators should be continuously reviewed as sources for risks.

NOTE—IEEE Std 1044-2009 [B7] provides useful information regarding anomaly classification. IEEE Std 982.1-2005 [B4] provides useful information regarding software measures related to reliability. ISO/IEC 15939-2007 [B17] provides a measurement process that can be used to help identify and characterize risks.

The V&V risk analysis is complementary to the risk analyses performed by the project management and developer organizations. The focus for V&V effort risk analysis is driven by the visibility the V&V effort has into life cycle products and is often concerned with the following types of risks:

- a) Risk that the system, as specified, will not meet stakeholder needs and requirements.
- b) Risk that the system may exhibit undesired behavior.
- c) Risk that engineering and integration plans will not enable accomplishment of V&V objectives through system integration/buildup.
- d) Risk that the acceptability of system or system elements cannot be determined.
- e) Risk that artifact quality or defect trends will impair accomplishment of project milestones or system operational objectives.
- f) Risk that the likelihood of reaching a defined hazardous condition is too high.
- g) Risk that security threats protections are not adequate.
- h) Risk that security vulnerabilities are not adequately ameliorated.
- i) Risk that the system is not maintainable.

Less frequently, but in some cases, the V&V effort may focus on programmatic risk such as cost and schedule performance.

J.3.3.2 Risk estimation

The probability of occurrence and consequences of each risk identified shall be estimated. Estimates may be either quantitative or qualitative. The stakeholders should define which risks will be evaluated using a qualitative scale and which will be evaluated using a quantitative scale. The scale(s) used for estimating risk probability and consequences shall be used consistently. The descriptive and measurement uncertainty inherent in the scale used should be described in the risk management plan. The level of confidence in a risk's estimate should be captured in its risk state.

J.3.3.3 Risk evaluation

Each risk shall be evaluated against its risk thresholds. Risks should be evaluated independently, in combination, and along with their interactions with system and enterprise risks. Risks should be evaluated against the project risk threshold to assure that a combination of risks, while below their individual thresholds, does not unacceptably place the project as a whole at risk. Different techniques may be used to evaluate the risks, such as decision trees, scenario planning, game theory, probabilistic analysis, and linear programming.

Risks shall be placed in a priority ordering where the ordering criteria are determined by the stakeholders. Priority may be based on when the risk is anticipated to become a problem, the risk exposure, risk-related measures, or some other consistent criteria. Various treatment alternatives to addressing risk should be considered to reduce or eliminate risks. For each risk that is above its risk threshold, recommended treatment strategies such as eliminating the risk, reducing its probability of occurrence or severity of consequence, or accepting the risk shall be defined and documented in a risk action request such as that found in Annex B. Contingency plans should be developed for all risks above their thresholds. Measures indicating the effectiveness of the treatment alternatives shall also be defined. The risks, their recommended treatments, and measures of risk treatment effectiveness shall be communicated to the stakeholders for approval, rejection, or modification.

Annex K

(informative)

Example of assigning and changing the system integrity level of "supporting system functions"

The integrity level of system functions that are designated as critical functions do not change during the system lifecycle unless authorized by the authorized system acquirer or regulatory organization. Noncritical system functions may become critical if the system is configured such that the originally noncritical functions can modify or alter critical data or can create an improper condition or system state to exist causing critical system functions to take incorrect actions. We will refer to these system functions as "critical supporting system functions." Any of the "critical supporting system functions" can lower their assigned system integrity level during development stages by the selection of technology or design/implementation techniques.

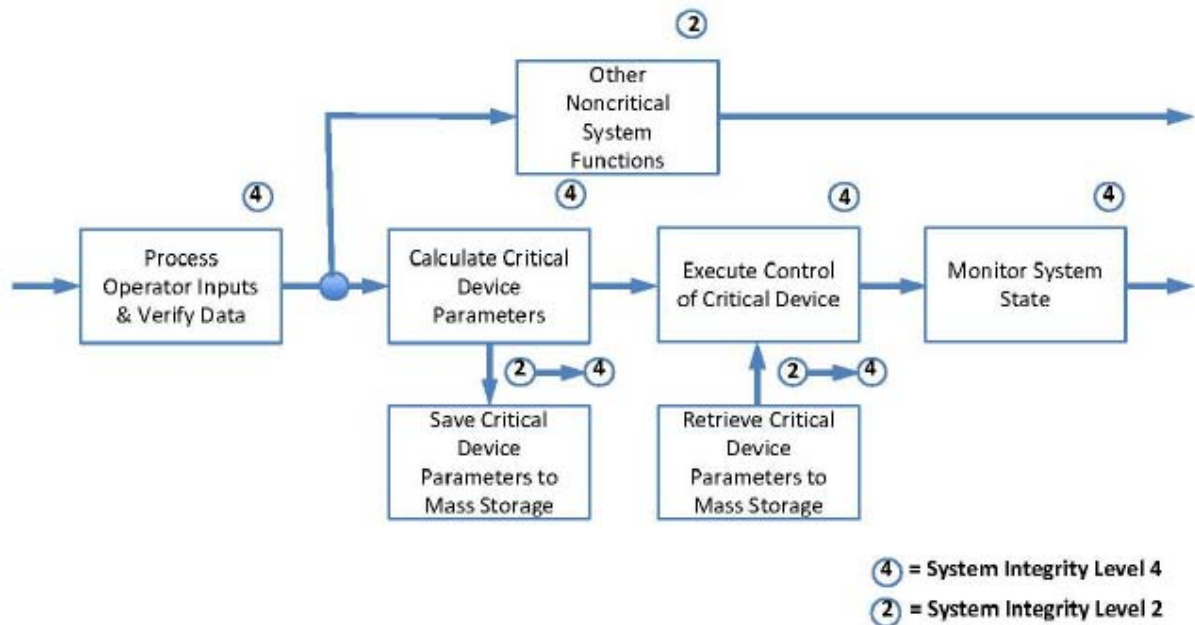
The V&V criticality analysis task performed at each development stage verifies the criticality of each system function and verifies that the correct system integrity level has been assigned. It is in this V&V task that a determination can occur where a "critical supporting system function" from a previous stage can be lowered or raised due to a technology selection or design/implementation technique being used by the subsequent development stage. The following two diagrams illustrate both the raising and the later lowering of system integrity levels for a supporting system function.

The example is a functional control flow diagram of the function of controlling a critical device. During the design stage of development, the control flow shows the receipt of device control inputs from an operator and critical device control parameters being calculated from the operator inputs. Those critical parameters are stored in a mass storage device and then retrieved by the critical device controller for execution of control of the critical device. Initial system integrity-level assignment from the criticality analysis task designates the following system functions at the highest system criticality level 4 (in a four level system integrity scheme):

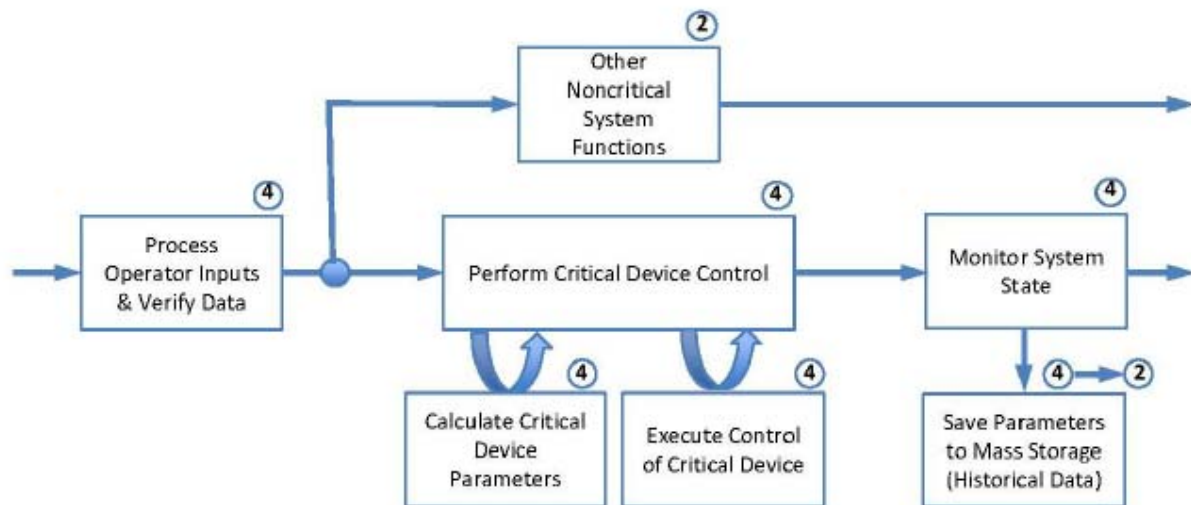
- Process operator inputs and verify data
- Calculate critical device parameters
- Execute control of critical device
- Monitor system

Supporting system functions in the diagram are initially designated as low criticality and thus are assigned a system integrity level 2. However, the criticality analysis of the control flow logic diagram in the design stage clearly indicates that the supporting system functions of "Save" and "Retrieve" critical device parameters from the mass storage device have the possibility of modifying or altering the critical parameters. Thus, these supporting system functions are raised to the high system integrity level 4 so that the level 4 verification and validation tasks are conducted.

However, during the subsequent implementation stage (shown in the second diagram), the developer has adopted an implementation technique whereby a "critical device controller" module is used to invoke the "calculate critical device parameters" and "execute control of critical devices" as subroutine calls. The critical parameters are passed parameters in the subroutine calls and thus eliminate the need to store and retrieve the critical data from a mass storage device. The storage of data on the mass storage device is later performed only for historical record keeping, and thus, the system integrity level is lowered from level 4 to level 2 being that it is no longer a critical execution function.



Design Stage Functional Diagram Illustrating the Raising System Criticality Level for Critical Supporting System Functions ② → ④



Implementation Stage Functional Diagram Illustrating the Lowering of System Integrity Level for Supporting System Functions ④ → ②

(New)

Annex L

(informative)

Mapping of ISO/IEC/IEEE 15288 and IEEE 12207 process outcomes to V&V tasks

Table L.1 – Mapping of ISO/IEC/IEEE 15288 process outcomes to V&V tasks

<u>15288 Stakeholder Requirements Definition Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) The required characteristics and context of use of services and operational concepts are specified.</u>	<u>Activity 8.1 Stakeholder Requirements Definition V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u>
<u>b) The constraints on a system solution are defined.</u>	<u>Activity 8.1 Stakeholder Requirements Definition V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u> <u>Task 4: Hazard Analysis</u> <u>Task 5: Security Analysis</u> <u>Task 6: Risk Analysis</u>
<u>c) Traceability of stakeholder requirements to stakeholders and their needs is achieved.</u>	<u>Activity 8.1 Stakeholder Requirements Definition V&V</u> <u>Task 2: Traceability Analysis</u>
<u>d) The stakeholder requirements are defined.</u>	<u>Activity 8.1 Stakeholder Requirements Definition V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u>
<u>e) Stakeholder requirements for validation are identified.</u>	<u>Activity 8.1 Stakeholder Requirements Definition V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u> <u>Task 3: Criticality Analysis</u>

<u>15288 Requirements Analysis Process</u> <u>Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
a) <u>The required characteristics, attributes, and functional and performance requirements for a product solution are specified.</u>	<u>Activity 82 Requirements Analysis V&V</u> <u>Task 1: Requirements Evaluation</u> <u>Task 2: Interface Analysis</u>
b) <u>Constraints that will affect the architectural design of a system and the means to realize it are specified.</u>	<u>Activity 82 Requirements Analysis V&V</u> <u>Task 1: Requirements Evaluation</u> <u>Task 2: Interface Analysis</u> <u>Task 8: Hazard Analysis</u> <u>Task 9: Security Analysis</u> <u>Task 10: Risk Analysis</u>
c) <u>The integrity and traceability of system requirements to stakeholder requirements is achieved.</u>	<u>Activity 82 Requirements Analysis V&V</u> <u>Task 3: Traceability Analysis</u> <u>Task 4: Criticality Analysis</u>
d) <u>A basis for verifying that the system requirements are satisfied is defined.</u>	<u>Activity 82 Requirements Analysis V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u> <u>Task 5: System Integration Test Plan V&V</u> <u>Task 6: System Qualification Test Plan V&V</u> <u>Task 7: System Acceptance Test Plan V&V</u>

<u>15288 Architectural Design Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) An architectural design baseline is established.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 1: Design Evaluation</u> <u>Task 2: Interface Analysis</u>
<u>b) The implementable set of system element descriptions that satisfies the requirements for the system is specified.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 1: Requirements Evaluation</u> <u>Task 2: Interface Analysis</u> <u>Task 8: Hazard Analysis</u> <u>Task 9: Security Analysis</u> <u>Task 10: Risk Analysis</u>
<u>c) The interface requirements are incorporated into the architectural design solution.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 2: Interface Analysis</u>
<u>d) The traceability of architectural design to system requirements is established.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 3: Traceability Analysis</u> <u>Task 4: Criticality Analysis</u>
<u>e) A basis for verifying the system elements is defined.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 1: Stakeholder Requirements Evaluation</u> <u>Task 4: Criticality Analysis</u>
<u>f) A basis for integration of system elements is established.</u>	<u>Activity 8.3 Architectural Design V&V</u> <u>Task 5: System Integration Test Design V&V</u> <u>Task 6: System Qualification Test Design V&V</u> <u>Task 7: System Acceptance Test Design V&V</u>

<u>15288 Implementation Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) An implementation strategy is defined.</u>	<u>Activity84Implementation V&V</u> <u>Task 1: Implementation Strategy Assessment</u>
<u>b) Implementation technology constraints on the design are identified.</u>	<u>Activity84Implementation V&V</u> <u>Task 4: Hazard Analysis</u> <u>Task 5: Security Analysis</u> <u>Task 6: Risk Analysis</u>
<u>c) A system is realized.</u>	<u>Activity84Implementation V&V</u> <u>Task 2: System Element Implementation Analysis</u> <u>Task 3: Criticality Analysis</u> <u>Task 7: System Integration Test Case V&V</u> <u>Task 8: System Qualification Test Case V&V</u> <u>Task 9: System Acceptance Test Case V&V</u>
<u>d) A system element is packaged and stored in accordance with an agreement for its supply.</u>	<u>Activity84Implementation V&V</u> <u>Task 2: System Element Implementation Analysis</u>

<u>15288 Integration Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) A system integration strategy is defined.</u>	<u>Activity8.5Integration V&V</u> <u>Task 1: System Integration Strategy Assessment</u>
<u>b) Unavoidable constraints of integration that influence requirements are defined.</u>	<u>Activity8.5Integration V&V</u> <u>Task 2: System Integration Test Procedures V&V</u> <u>Task 3: System Integration Test Execution V&V</u> <u>Task 4: System Qualification Test Procedures V&V</u> <u>Task 5: System Qualification Test Execution V&V</u> <u>Task 6: System Acceptance Test Procedure V&V</u>
<u>c) A system capable of being verified against the specified requirements from architectural design is assembled and integrated.</u>	<u>Activity8.5Integration V&V</u> <u>Task 2: System Integration Test ProceduresV&V</u> <u>Task 3: System Integration Test Execution V&V</u> <u>Task 4: System Qualification Test Procedures V&V</u> <u>Task 5: System Qualification Test Execution V&V</u> <u>Task 6: System Acceptance Test Procedure V&V</u>
<u>d) Nonconformances due to integration actions are recorded.</u>	<u>Activity8.5Integration V&V</u> <u>Task 2: System Integration Test Procedures V&V</u> <u>Task 3: System Integration Test Execution V&V</u> <u>Task 4: System Qualification Test Procedures V&V</u> <u>Task 5: System Qualification Test Execution V&V</u> <u>Task 6: System Acceptance Test Procedure V&V</u>

<u>15288 Transition Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) A system transition strategy is defined.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 1: Transition Strategy Evaluation</u>
<u>b) A system is installed in its operational location.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 2: Transition Demonstration Assessment</u>
<u>c) A system, when operated, is capable of delivering specified services.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 3: System Acceptance Test Execution V&V</u>
<u>d) The configuration as installed is recorded.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 2: Transition Demonstration Assessment</u>
<u>e) Corrective action reports are recorded.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 2: Transition Demonstration Assessment</u> <u>Task 3: System Acceptance Test Execution V&V</u>
<u>f) A service is sustainable by enabling systems.</u>	<u>Activity 8.6 Transition V& V</u> <u>Task 2: Transition Demonstration Assessment</u> <u>Task 3: System Acceptance Test Execution V&V</u>

<u>15288 Operation Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
<u>a) An operation strategy is defined.</u>	<u>Activity 8.7 Operation V& V</u> <u>Task 1: Operating Procedures Evaluation</u>
<u>b) Services that meet stakeholder requirements are delivered.</u>	<u>Activity 8.7 Operation V& V</u> <u>Task 2: Operating Procedures Evaluation</u> <u>Task 3: Hazard Analysis</u> <u>Task 4: Security Analysis Task 5: Risk Analysis</u>

c) <u>Approved corrective action requests are satisfactorily completed.</u>	<u>Activity 8.7 Operation V& V</u> <u>Recursive application of prior stage V&V tasks to show objective evidence that corrective action requests are satisfactorily completed.</u>
d) <u>Stakeholder satisfaction is maintained.</u>	<u>Activity 8.7 Operation V& V</u> <u>Recursive application of prior stage V&V analysis, evaluation, assessments, or test of any stakeholder requirements to verify and validate that stakeholder satisfaction is maintained.</u>

<u>15288 Maintenance Process Outcomes</u>	<u>Accomplished by 1012 V&V Tasks</u>
a) <u>A maintenance strategy is developed.</u>	<u>Activity 88Maintenance V& V</u> <u>Task 1: System Maintenance Strategy Assessment</u>
b) <u>Maintenance constraints are provided as inputs to requirements.</u>	<u>Activity 88Maintenance V& V</u> <u>Task 1: System Maintenance Strategy Assessment</u> <u>Task 2: System Maintenance Execution Assessment</u>
c) <u>Replacement system elements are made available.</u>	<u>Activity 88Maintenance V& V</u> <u>Task 2: System Maintenance Execution Assessment</u>
d) <u>Services meeting stakeholder requirements are sustained.</u>	<u>Activity 88Maintenance V& V</u> <u>Task 2: System Maintenance Execution Assessment</u>
e) <u>The need for corrective design changes is</u>	<u>Activity 88Maintenance V& V</u>

reported.	Task 2: System Maintenance Execution Assessment
f) Failure and lifetime data are recorded.	Activity 88Maintenance V& V Task 2: System Maintenance Execution Assessment

15288 Disposal Process Outcomes	Accomplished by 1012 V&V Tasks
a) A system disposal strategy is defined.	Activity 89Disposal V&V Task 1: Disposal Plan Evaluation
b) Disposal constraints are provided as inputs to requirements.	Activity 89Disposal V&V Task 1: Disposal Plan Evaluation
c) The system elements or waste products are destroyed, stored, reclaimed or recycled.	Activity 89Disposal V&V Task 1: Disposal Plan Evaluation
d) The environment is returned to its original or an agreed state.	Activity 89Disposal V&V Task 1: Disposal Plan Evaluation
e) Records allowing knowledge retention of disposal actions and the analysis of long-term hazards are available.	Activity 89Disposal V&V Task 1: Disposal Plan Evaluation

Table L.2— Mapping of IEEE 12207 Process Outcomes to V&V Tasks

<u>IEEE 12207 Software Requirements Analysis</u> <u>Process Outcomes</u>	<u>1012 V&V Tasks Mapped to Outcomes</u>
<u>a) The requirements allocated to the software elements of the system and their interfaces are defined.</u>	<u>Activity 9.1: Software Concept V& V</u> <u>Task 1: Concept Documentation Evaluation</u> <u>Task 3: Hardware/Software/User Allocation Analysis</u> <u>Task 5: Hazard Analysis</u> <u>Task 6: Security Analysis</u> <u>Task 7: Risk Analysis</u> <u>Activity 9.2: Software Requirements V& V</u> <u>Task 1: Requirements Evaluation</u> <u>Task 2: Interface Analysis</u> <u>Task 7: Hazard Analysis</u> <u>Task 8: Security Analysis</u> <u>Task 9: Risk Analysis</u>
<u>b) Software requirements are analyzed for correctness and testability.</u>	<u>Activity 9.1: Software Concept V& V</u> <u>Task 1: Concept Documentation Evaluation</u> <u>Task 3: Hardware/Software/User Allocation Analysis</u> <u>Task 5: Hazard Analysis</u> <u>Task 6: Security Analysis</u> <u>Task 7: Risk Analysis</u> <u>Activity 9.2: Software Requirements V& V</u> <u>Task 1: Requirements Evaluation</u> <u>Task 2: Interface Analysis</u> <u>Task 5: Software Qualification Test Plan V&V</u> <u>Task 6: Software Acceptance Test Plan V&V</u>

	Task 7: Hazard Analysis Task 8: Security Analysis Task 9: Risk Analysis
c) The impact of software requirements on the operating environment is understood.	Activity 9.1: Software Concept V&V Task 1: Concept Documentation Evaluation Task 3: Hardware/Software/User Allocation Analysis Task 5: Hazard Analysis Task 6: Security Analysis Task 7: Risk Analysis Activity 9.2: Software Requirements V&V Task 1: Requirements Evaluation Task 2: Interface Analysis Task 5: Software Qualification Test Plan V&V Task 6: Software Acceptance Test Plan V&V Task 7: Hazard Analysis Task 8: Security Analysis Task 9: Risk Analysis
d) Consistency and traceability are established between the software requirements and system requirements.	Activity 9.1: Software Concept V&V Task 4: Traceability Analysis Activity 9.2: Software Requirements V&V Task 3: Traceability Analysis
e) Prioritization for implementing the software requirements is defined.	Activity 9.1: Software Concept V&V Task 2: Criticality Analysis Activity 9.2: Software Requirements V&V Task 4: Criticality Analysis
f) The software requirements are approved and updated as needed.	Activity 9.1: Software Concept V&V Task 1: Concept Documentation V&V Task 2: Criticality Analysis Task 3: Hardware/Software/User

	Requirements Allocation Analysis Task 4: Traceability Analysis Task 5: Hazard Analysis Task 6: Security Analysis Task 7: Risk Analysis Activity 9.2: Software Requirements V&V Task 1: Requirements Evaluation Task 2: Interface Analysis Task 3: Traceability Analysis Task 4: Criticality Analysis Task 5: Software Qualification Test Plan V&V Task 6: Software Acceptance Test Plan V&V Task 7: Hazard Analysis Task 8: Security Analysis Task 9: Risk Analysis
g) Changes to the software requirements are evaluated for cost, schedule, and technical impact.	Activity 9.1: Software Concept V&V Task 1: Concept Documentation V&V Task 2: Criticality Analysis Task 3: Hardware/Software/User Requirements Allocation Analysis Task 4: Traceability Analysis Task 5: Hazard Analysis Task 6: Security Analysis Task 7: Risk Analysis Activity 9.2: Software Requirements V&V Task 1: Requirements Evaluation Task 2: Interface Analysis Task 3: Traceability Analysis Task 4: Criticality Analysis Task 5: Software Qualification Test Plan V&V

	Task 6: Software Acceptance Test Plan V&V Task 7: Hazard Analysis Task 8: Security Analysis Task 9: Risk Analysis
h) The software requirements are baselined and communicated to all affected parties.	N/A

IEEE 12207 Software Architectural Design Process Outcomes	1012 V&V Tasks Mapped to Outcomes
a) A software architectural design is developed and baselined that describes the software items that will implement the software requirements.	Activity 9.3 Software Design V& V Task 1: Design Evaluation Task 4: Criticality Analysis Task 11: Hazard Analysis Task 12: Security Analysis Task 13: Risk Analysis
b) Internal and external interfaces of each software item are defined.	Activity 9.3 Software Design V& V Task 1: Design Evaluation Task 2: Interface Analysis Task 3: Traceability Analysis
c) Consistency and traceability are established between software requirements and software design.	Activity 9.3 Software Design V& V Task 1: Design Evaluation Task 3: Traceability Analysis Task 9: Software Qualification Test Design V&V Task 10: Software Acceptance Test Design V&V

<u>IEEE 12207 Software Detailed Design Process Outcomes</u>	<u>1012 V&V Tasks Mapped to Outcomes</u>
<u>a) A detailed design of each software component describing the software units to be built is developed.</u>	<u>Activity 9.3 Software Design V& V</u> <u>Task1:Design Evaluation</u> <u>Task4:Criticality Analysis</u> <u>Task11: Hazard Analysis</u> <u>Task12: Security Analysis</u> <u>Task13: Risk Analysis</u>
<u>b) External interfaces of each software unit are defined.</u>	<u>Activity 9.3 Software Design V& V</u> <u>Task1:Design Evaluation</u> <u>Task 2:Interface Analysis</u> <u>Task3:Traceability Analysis</u>
<u>c) Consistency and traceability are established between detailed design and the requirements and architectural design.</u>	<u>Activity 9.3 Software Design V& V</u> <u>Task1:Design Evaluation</u> <u>Task3:Traceability Analysis</u> <u>Task5:Software Component Test Plan V&V</u> <u>Task6:Software Integration Test Plan V&V</u> <u>Task7:Software Component Test Design V&V</u> <u>Task8:Software Integration Test Design V&V</u> <u>Task9:Software Qualification Test Design V&V</u> <u>Task10: Software Acceptance Test Design V&V</u>

<u>IEEE 12207 Software Construction Process Outcomes</u>	<u>1012 V&V Tasks Mapped to Outcomes</u>
<u>a) Verification criteria are defined for all software units against their requirements.</u>	<u>Activity 9.4 Software Construction V& V</u> <u>Task 1: Source Code and Source Code Documentation Evaluation</u> <u>Task 2: Interface Analysis</u>

	Task 4: Criticality Analysis Task 5: Software Component Test Case V&V Task 9: Software Component Test Procedures V&V Task 12: Software Component Test Execution V&V Task 13: Hazard Analysis Task 14: Security Analysis Task 15: Risk Analysis
b) Software units defined by design are produced.	Activity 9.4 Software Construction V& V Task 1: Source Code and Source Code Documentation Evaluation Task 2: Interface Analysis Task 5: Software Component Test Case V&V Task 9: Software Component Test Procedures V&V Task 12: Software Component Test Execution V&V
c) Consistency and traceability are established between the software units and the requirements and design.	Activity 9.4 Software Construction V& V Task 3: Traceability Analysis
d) Verification of the software units against the requirements and the design is accomplished.	Activity 9.4 Software Construction V& V Task 1: Source Code and Source Code Documentation Evaluation Task 2: Interface Analysis Task 3: Traceability Analysis Task 5: Software Component Test Case V&V Task 6: Software Integration Test Case V&V Task 7: Software Qualification Test Case V&V Task 8: Software Acceptance Test Case V&V

	Task 9: Software Component Test Procedures V&V Task 10: Software Integration Test Procedure V&V Task 11: Software Qualification Test Procedure V&V Task 12: Software Component Test Execution V&V
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<u>IEEE 12207 Software Integration Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) An integration strategy is developed for software units consistent with software design and the prioritized software requirements.</u>	<i>Activity 9.3 Software Design V& V</i> Task 6: Software Integration Test Plan V&V Task 8: Software Integration Test Design V&V <i>Activity 9.4 Software Construction V& V</i> Task 10: Software Integration Test Procedure V&V <i>Activity 9.5 Software Integration V& V</i> Task 1: Traceability Analysis
<u>b) Verification criteria for software items are developed that ensure compliance with the software requirements allocated to the item.</u>	<i>Activity 9.3 Software Design V& V</i> Task 6: Software Integration Test Plan V&V Task 8: Software Integration Test Design V&V <i>Activity 9.4 Software Construction V& V</i> Task 10: Software Integration Test Procedure V&V <i>Activity 9.5 Software Integration V& V</i> Task 1: Traceability Analysis Task 3: Hazard Analysis Task 4: Security Analysis Task 5: Risk Analysis
<u>c) Software items are verified using the defined criteria.</u>	<i>Activity 9.5 Software Integration V& V</i> Task 2: Software Integration Test Execution V&V
<u>d) Software items defined by the integration strategy are produced.</u>	<i>Activity 9.5 Software Integration V& V</i> Task 2: Software Integration Test Execution V&V
<u>e) Results of integration testing are recorded.</u>	<i>Activity 9.5 Software Integration V& V</i> Task 2: Software Integration Test Execution V&V
<u>f) Consistency and traceability are established between software design and software items.</u>	<i>Activity 9.4 Software Integration V& V</i> Task 1: Traceability Analysis
<u>g) A regression strategy is developed and applied for reverifying software items when a change</u>	<i>Activity 7.1 V&V Management</i> Task 3: Proposed/Baseline Change Assessment

in software units (including associated requirements, design, and code) occur.	
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IEEE 12207 Software Qualification Testing	1012 V&V Tasks Mapped to Outcomes
a) Criteria for the integrated software is developed that demonstrates compliance with the software requirements.	Activity 9.2 Software Requirements V& V Task 5: Software Qualification Test Plan V&V Activity 9.3 Software Design V& V Task 9: Software Qualification Test Design V&V Activity 9.4 Software Construction V&V Task 7: Software Qualification Test Case V&V Task 11: Software Qualification Test Procedures V&V
b) Integrated software is verified using the	Activity 9.6 Software Qualification Test V& V
c) Test results are recorded.	Activity 9.6 Software Qualification Test V& V
d) A regression strategy is developed and applied for retesting the integrated software when a change in software items is made.	Activity 7.1 V&V Management Task 3: Proposed/Baseline Change Assessment

<u>IEEE 12207 Software Installation Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) A software installation strategy is developed.</u>	<u>Activity 9. S Software Installation and Checkout V&V</u> <u>Task1:Installation Configuration Audit</u> <u>Task 2:Installation Checkout</u>
<u>b) Criteria for software installation are developed that demonstrate compliance with the software installation requirements.</u>	<u>Activity 9. S Software Installation and Checkout V&V</u> <u>Task1:Installation Configuration Audit</u> <u>Task 2:Installation Checkout</u> <u>Task 3:Hazard Analysis</u> <u>Task 4:Security Analysis</u> <u>Task 5:Risk Analysis</u>
<u>c) The software product is installed in the target environment.</u>	<u>Activity 9. S Software Installation and Checkout V&V</u> <u>Task1:Installation Configuration Audit</u> <u>Task 2:Installation Checkout</u>
<u>d) Readiness of the software product for use in its intended environment is assured.</u>	<u>Activity 9. S Software Installation and Checkout V&V</u> <u>Task1:Installation Configuration Audit</u> <u>Task 2:Installation Checkout</u> <u>Task 3:Hazard Analysis</u> <u>Task 4:Security Analysis</u> <u>Task 5:Risk Analysis</u>

<u>IEEE 12207 Software Acceptance Support</u> <u>Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) The product is completed and delivered to the acquirer.</u>	<u>Activity 9.7 Software Acceptance Test V& V</u> <u>Task 2: Software Acceptance Test Execution V&V</u>
<u>b) Acquirer acceptance tests and reviews are supported.</u>	<u>Activity 9.2 Software Requirements V& V</u> <u>Task 6: Software Acceptance Test Plan V&V</u> <u>Activity 9.3 Software Design V& V</u> <u>Task 10: Software Acceptance Test Design V&V</u> <u>Activity 9.4 Software Construction V&V</u> <u>Task 8: Software Acceptance Test Case V&V</u> <u>Activity 9.7 Software Acceptance Test V& V</u> <u>Task 1: Software Acceptance Test Procedure V&V</u> <u>Task 2: Software Acceptance Test Execution V&V</u> <u>Task 3: Traceability Analysis</u> <u>Task 4: Hazard Analysis</u> <u>Task 5: Security Analysis</u> <u>Task 6: Risk Analysis</u>
<u>c) The product is put into operation in the customer's environment.</u>	<u>Activity 9.5 Software Installation and Checkout V&V</u> <u>Task 1: Installation Configuration Audit</u> <u>Task 2: Installation Checkout</u>
<u>d) Problems detected during acceptance are identified and communicated to those responsible for resolution.</u>	<u>Activity 9.7 Software Acceptance Test V& V</u> <u>Task 2: Software Acceptance Test Execution V&V</u>

<u>IEEE 12207 Software Operation Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) An operation strategy is defined.</u>	<u>Activity 9.9 Software Operation V& V</u> <u>Task 2: Operating Procedures Evaluation</u>
<u>b) Conditions for correct operation of the software in its intended environment are identified and evaluated.</u>	<u>Activity 9.9 Software Operation V& V</u> <u>Task 1: Evaluation of New Constraints</u> <u>Task 2: Operating Procedures Evaluation</u> <u>Task 3: Hazard Analysis</u> <u>Task 4: Security Analysis</u> <u>Task 5: Risk Analysis</u>
<u>c) The software is tested and determined to operate in its intended environment.</u>	<u>Activity 9.7 Software Acceptance Test V& V</u> <u>Task 2: Software Acceptance Test Execution V&V</u>
<u>d) Software is operated in its intended environment.</u>	<u>Activity 9.9 Software Operation V& V</u> <u>Task 1: Evaluation of New Constraints Task</u> <u>2: Operating Procedures Evaluation</u> <u>Task 3: Hazard Analysis</u> <u>Task 4: Security Analysis</u> <u>Task 5: Risk Analysis</u>
<u>e) Assistance and consultation is provided to the customers of the software product in accordance with the agreement.</u>	<u>N/A</u>

<u>IEEE 12207 Software Maintenance Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) A maintenance strategy is developed to manage the modification and migration of products according to the release strategy.</u>	<u>Activity 9.10 Software Maintenance V&V</u> <u>Task 1: VVP Revision</u> <u>Task 3: Criticality Analysis</u> <u>Task 4: Migration Assessment</u> <u>Task 5: Retirement Assessment</u>
<u>b) The impact of changes to the existing system on organization, operations, or interfaces is identified.</u>	<u>Activity 9.10 Software Maintenance V&V</u> <u>Task 2: Anomaly Evaluation</u> <u>Task 3: Criticality Analysis</u> <u>Task 4: Migration Assessment</u> <u>Task 5: Retirement Assessment</u> <u>Task 6: Hazard Analysis</u> <u>Task 7: Security Analysis</u> <u>Task 8: Risk Analysis</u> <u>Task 9: Task Iteration</u>
<u>c) Affected system and software documentation is updated as needed.</u>	<u>Activity 9.10 Software Maintenance V&V</u> <u>Task 9: Task Iteration</u>
<u>d) Modified products are developed with associated tests that demonstrate that requirements are not compromised.</u>	<u>Activity 9.10 Software Maintenance V&V</u> <u>Task 9: Task Iteration</u>
<u>e) Product upgrades are migrated to the customer's environment.</u>	<u>Activity 9.10 Software Maintenance V&V</u> <u>Task 9: Task Iteration</u>
<u>f) The system software modification is communicated to all affected parties.</u>	<u>N/A</u>

<u>IEEE 12207 Software Disposal Process Outcomes</u>	<u>IEEE 1012 V&V Tasks Mapped to Outcomes</u>
<u>a) A software disposal strategy is defined.</u>	<u>Activity 9.11 Software Disposal V& V</u> <u>Task 1: Software Disposal V&V</u>
<u>b) Disposal constraints are provided as inputs to requirements.</u>	<u>Activity 9.11 Software Disposal V& V</u> <u>Task 1: Software Disposal V&V</u>
<u>c) The system's software elements are destroyed or stored.</u>	<u>N/A</u>
<u>d) The environment is left in an agreed-on state.</u>	<u>Activity 9.11 Software Disposal V& V</u> <u>Task 1: Software Disposal V&V</u>
<u>e) Records allowing knowledge retention of disposal actions and any analysis of long-term impacts are available.</u>	<u>Activity 9.11 Software Disposal V& V</u> <u>Task 1: Software Disposal V&V</u>

Annex M

(informative)

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⁸The IEEE publications are available from the Institute of Electrical and Electronics Engineers, (<http://standards.ieee.org/>).Inc., ~~445~~ ~~Hoes Lane, Piscataway, NJ 08854, USA~~ (~~<http://standards.ieee.org/>~~).

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